

Music Brain: A Multi-Modal Audio Analysis Architecture for Stem-Isolated Semantic Feature Extraction

Technical Architecture Blueprint

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Abstract. Music Brain is a real-time audio analysis system that reduces raw audio waveforms to discrete semantic metadata. It employs a stem-first strategy, using Hybrid Transformer source separation (HT-Demucs) to isolate six instrumental stems before applying independent spectral, harmonic, rhythmic, and timbral analysis pipelines. We detail the signal processing foundations—STFT computation, Constant-Q spectral analysis, chroma vector extraction, and spectral descriptor computation—alongside higher-order modules for chord detection, key estimation, structural segmentation via self-similarity matrices, and musical expectation modeling based on Huron’s ITPRA framework. A genre adaptation layer with five expert adapters (Metal, Jazz, Orchestral, Electronic, Microtonal) applies domain-specific heuristics to extracted features. A real-time visualization suite on the egui framework provides fractal song trees driven by entropy, harmonic constellations on the Circle of Fifths, self-similarity heatmaps, and a three-dimensional style topology manifold. The system processes live audio at 44.1 kHz with a 2048-sample FFT window and 1024-sample hop size, maintaining a rolling one-second analysis buffer across all pipelines.

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1 Introduction

Extracting musical metadata from raw audio remains a central challenge in Music Information Retrieval (MIR). Individual tasks—beat tracking, key detection, source separation—have matured, yet integrated systems combining these capabilities into a coherent analytical pipeline remain rare, especially those that operate in real time and deliver expert-level feedback.

1.1 Problem Statement

Traditional MIR analysis operates on the master mix: a single stereo signal with all instruments superimposed. This approach faces fundamental signal-to-noise ratio (SNR) limits. In a full mix, the spectral energy of drums, bass, and effects obscures tonal information from melodic instruments. Extracting rhythmic patterns conflates percussion transients with harmonic onsets. The result is analysis that is approximate and genre-dependent.

1.2 The Stem-First Paradigm

Music Brain addresses this through a *stem-first* analysis paradigm. Instead of analyzing the master mix, the system separates audio into six stems—vocals, drums, bass, guitar, piano, and other—using Meta’s HT-Demucs model [1, 2]. Each stem is analyzed independently by specialized pipelines, eliminating cross-source interference and enabling forensic-level detail. Snare tuning can be estimated from the isolated drum stem without harmonic masking from guitars; guitar distortion characteristics can be analyzed without vocal spectral overlap.

1.3 Contributions

This report presents the following contributions:

1. A complete real-time audio analysis architecture combining source separation, spectral analysis, harmonic detection, rhythmic analysis, timbral forensics, structural segmentation, and expectation modeling.
2. A genre adaptation layer with five expert adapters that apply domain-specific heuristics to generic feature sets.
3. A preset system with quantitative deviation tracking for genre-aware production feedback.
4. An interactive visualization suite with seven distinct views for multi-dimensional data exploration.
5. A variation engine that suggests theoretically grounded harmonic substitutions based on detected chords.

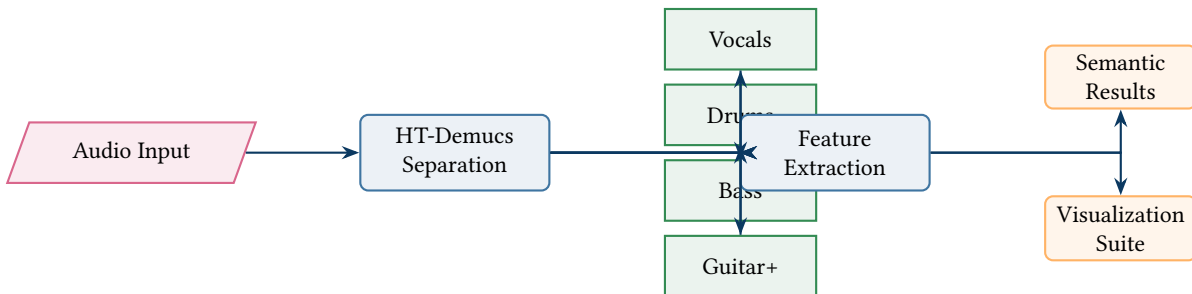


Figure 1. High-level system overview: audio input is separated into stems via HT-Demucs, then analyzed independently to produce semantic results and interactive visualizations.

1.4 Report Organization

Section 2 surveys related work in MIR and source separation. Section 3 presents the system architecture and module structure. Section 4 details the source separation integration. Sections 5–9 cover the analysis pipelines in depth. Section 10 introduces the expectation model. Section 11 describes the visualization suite. Section 12 presents the genre adaptation layer. Section 36 concludes with limitations and future work.

2 Background and Related Work

2.1 Music Information Retrieval

Music Information Retrieval encompasses the computational analysis, organization, and retrieval of music data. The field has produced mature solutions for pitch detection [3], beat tracking, chord recognition, and structural analysis [4]. Integrated systems combining multiple analysis modalities remain uncommon.

Two open-source frameworks—*Essentia* [5] and *librosa* [6]—provide comprehensive MIR algorithm libraries. Essentia offers over 80 audio descriptors with C++ performance. librosa provides a Python alternative emphasizing accessibility and rapid prototyping. Music Brain differs by operating on separated stems rather than the master mix, analyzing in real time, and integrating genre-specific expert adapters that contextualize features with domain knowledge.

2.2 Source Separation

Audio source separation has evolved from classical approaches (e.g., Non-negative Matrix Factorization) through deep learning to Hybrid Transformer architectures. The Demucs family [1, 2] achieves near-studio-quality separation for music signals. HT-Demucs combines temporal convolutions for local patterns with transformer attention for global dependencies, enabling 6-stem separation (vocals, drums, bass, guitar, piano, other) with sufficient quality for downstream analysis.

2.3 Spectral Analysis Foundations

The Short-Time Fourier Transform (STFT) provides time-frequency representations essential for audio analysis [7]. Its fixed-resolution trade-off—temporal and frequency resolution are inversely proportional—motivates the Constant-Q Transform (CQT) [8] for harmonic analysis. The CQT maps frequency bins to a logarithmic scale matching Western pitch, where each bin corresponds to a semitone. This geometric spacing adapts resolution to pitch perception: high frequency resolution at low frequencies (where harmonics are closely spaced) and high temporal resolution at high frequencies (where transients dominate).

2.4 Spectral Descriptors

Timbral characterization relies on spectral descriptors that reduce a full spectrum to scalar or low-dimensional features. The spectral centroid measures “brightness” as the center of mass of the magnitude spectrum [7]. Spectral flatness (Wiener entropy) measures how closely a spectrum resembles white noise versus a harmonic tone [7]. These descriptors, combined with chroma features [9] encoding harmonic content as a 12-dimensional pitch class distribution, form the basis for most high-level musical analysis.

2.5 Structural Analysis

Structural analysis identifies repeated sections, transitions, and hierarchical organization within a piece. Self-Similarity Matrices (SSMs), introduced by Foote [10], compute pairwise similarity between all time frames of a feature sequence. Repeated sections appear as off-diagonal stripes; transitions manifest as changes in local texture.

2.6 Musical Expectation

Huron’s ITPRA theory [11] explains how listeners generate and evaluate musical expectations through five response stages—Imagination, Tension, Prediction, Reaction, and Appraisal—that modulate the emotional impact of musical events. Computational models of musical surprise, based on deviation from predicted features, quantify this effect and drive visualizations revealing the narrative arc of a piece.

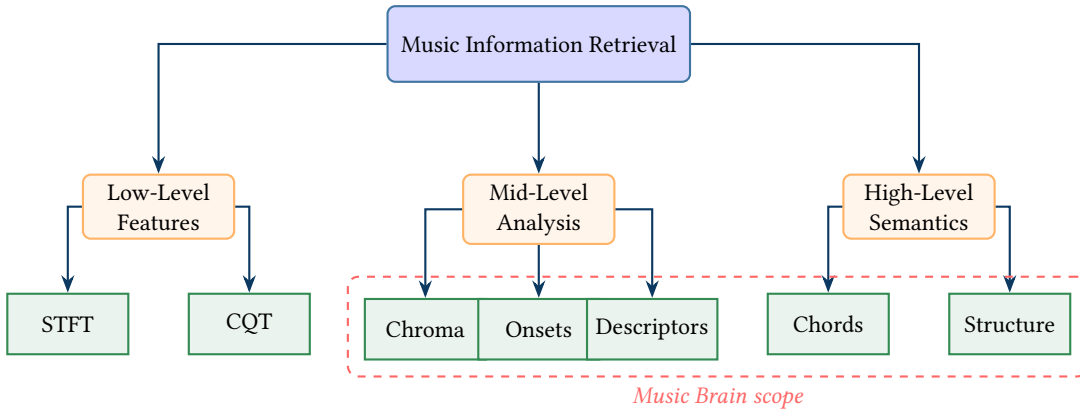


Figure 2. Taxonomy of MIR analysis approaches. Music Brain operates across mid-level and high-level analysis, combining spectral descriptors, chroma features, chord detection, and structural segmentation.

3 System Architecture

The system is organized into five top-level modules: analysis, adapters, data, ui, and live.

3.1 Module Structure

Module	Key Components	Responsibility
analysis/	spectral, variation, expectation, structure, presets, inference	Core signal processing: STFT, chroma, chord detection, key estimation, SSM, surprise model, presets
adapters/	electronic, metal, jazz, orchestral, microtonal	Genre-specific expert analysis via the ExpertAdapter trait
data/	demucs, export, midi	Audio loading (Symphonia), HT-Demucs bridge, JSON/MIDI export
ui/	fractal, heatmap, manifold, presets	Visualization: Fractal Song Tree, SSM heatmap, Style Topology, Preset gallery, Phase Compass, Stem Helix
live/	core module	Real-time audio capture via CPAL

Table 1. Top-level module structure. Each module groups related functionality; see source files for implementation details.

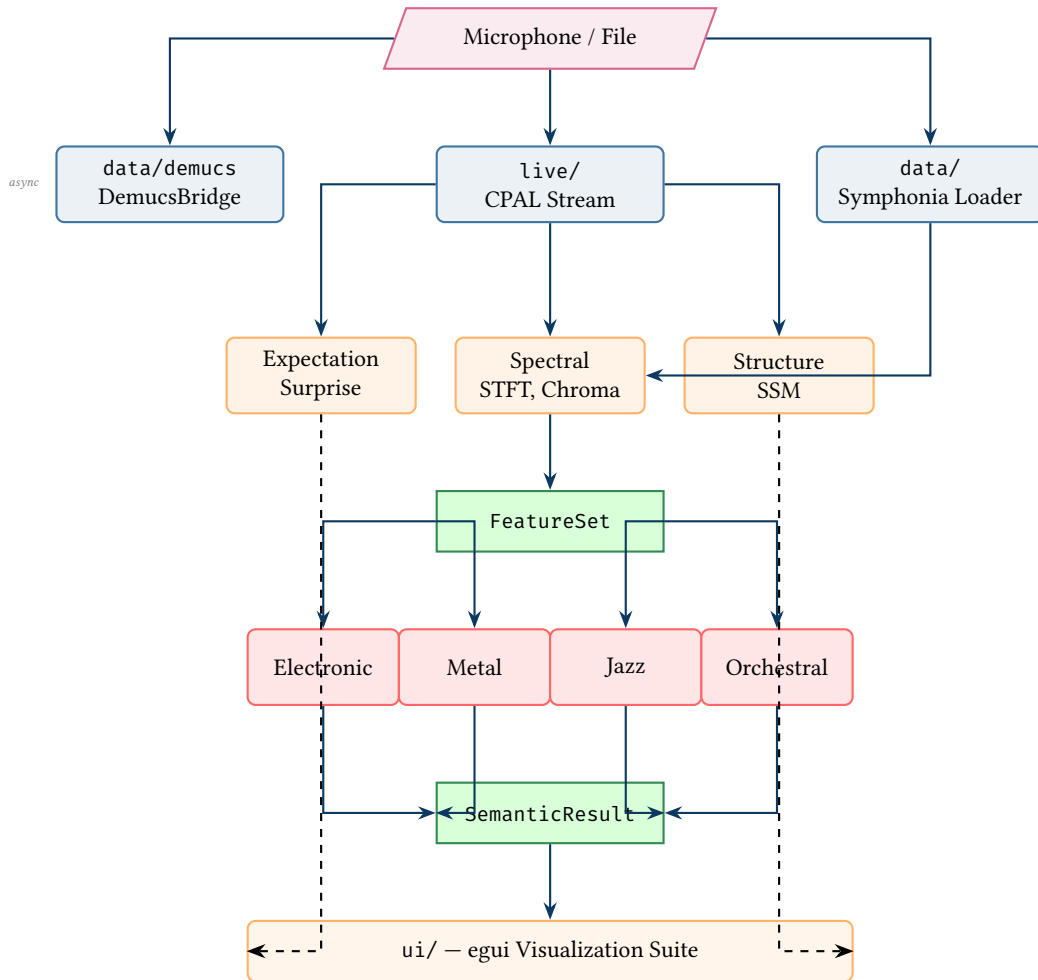


Figure 3. Detailed architecture block diagram showing all modules. Orange blocks are analysis modules, red blocks are genre adapters, green blocks are core data types. Dashed arrows indicate direct visualization feeds.

3.2 Core Data Types

Two primary data structures mediate between analysis and presentation layers.

FeatureSet. The central analysis output, encapsulating all extracted features for a single time frame:

FeatureSet		
bpm	f32	Beats per minute
key	String	Detected musical key
spectral_centroid	f32	Center of mass frequency (Hz)
spectral_flatness	f32	Wiener entropy (0.0–1.0)
sensory_dissonance	f32	Perceived dissonance (0.0–1.0)
chroma	[f32; 12]	Energy per pitch class (C–B)
rms_energy	f32	Root mean square amplitude

Figure 4. FeatureSet data schema. The central analysis output encapsulating all extracted features for a single time frame.

SemanticResult. The output of genre-specific expert analysis, providing human-readable interpretations:

SemanticResult		
label	String	e.g. “Tight Rolling Bass”
confidence	f32	Confidence score (0.0–1.0)
dna_profile	JSON Value	Structured analysis metadata

Figure 5. SemanticResult data schema. The output of genre-specific expert analysis, providing human-readable interpretations of extracted features.

3.3 Real-Time Processing Pipeline

The main application loop executes a 10-step analysis pipeline on each frame update:

1. **Sample Collection:** Retrieve new audio samples from the CPAL live stream.
2. **Buffer Management:** Maintain a rolling 44,100-sample buffer (1 second at 44.1 kHz).
3. **Spectral Analysis:** Compute STFT on the most recent 2048 samples, extract centroid, flatness, and chroma.
4. **Energy Computation:** Calculate RMS energy across new samples.
5. **Feature Assembly:** Construct FeatureSet from extracted features.
6. **Chord Detection:** Identify the current chord from chroma template matching.
7. **Variation Suggestions:** Generate harmonic substitution options via the variation engine.
8. **Expert Analysis:** Run all five genre adapters on current features.
9. **Surprise Calculation:** Update the expectation model and compute normalized surprise.
10. **Structural Update:** Append the current feature vector to the self-similarity matrix history.

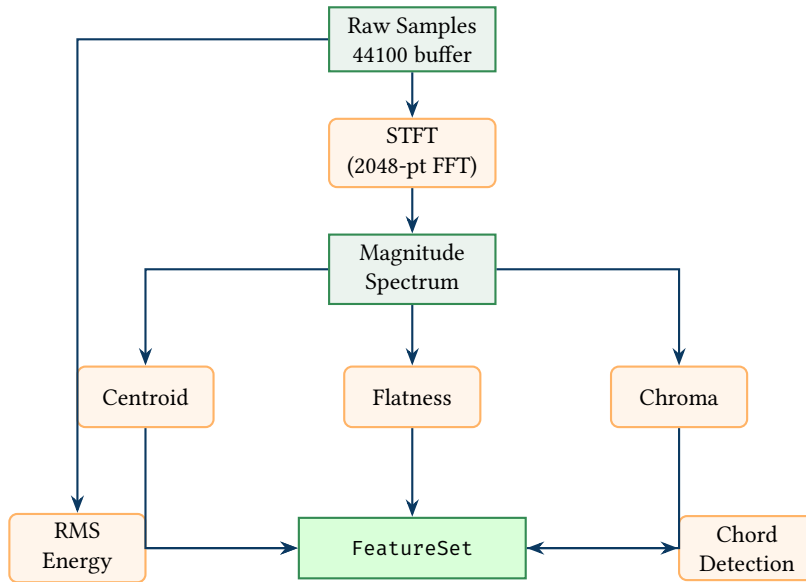


Figure 6. Data flow from raw audio samples through spectral analysis to the FeatureSet struct. Each frame extracts spectral centroid, flatness, and chroma from the magnitude spectrum, combined with RMS energy from the time domain.

3.4 End-to-End Data Lifecycle

Figure 7 presents the complete data lifecycle from audio file input through every processing stage to final output, showing how all modules interconnect.

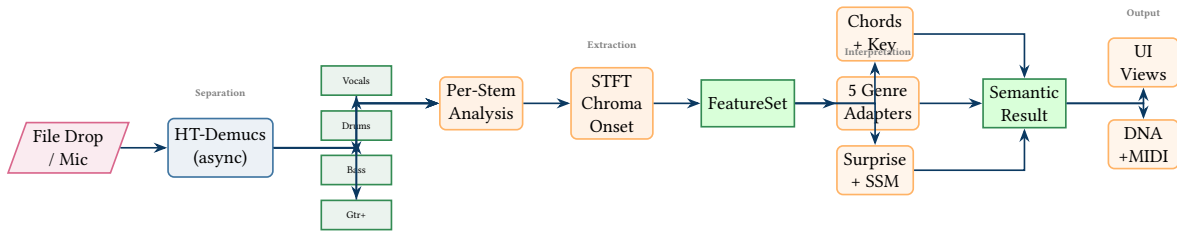


Figure 7. End-to-end data lifecycle. Audio enters from file or microphone, passes through HT-Demucs stem separation, per-stem spectral analysis, feature assembly, parallel chord/adaptor/surprise processing, and finally flows to the visualization suite and export modules.

3.5 Concurrency Model

The system uses three concurrent execution contexts coordinated via lock-free channels:

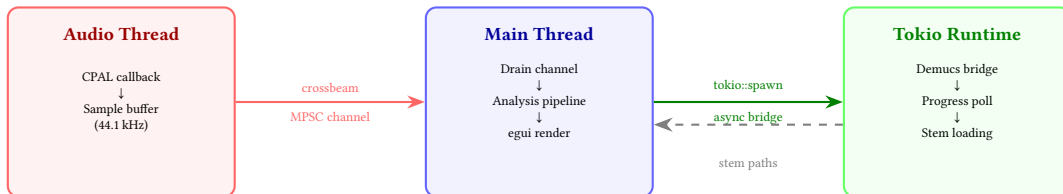


Figure 8. Concurrency model. The CPAL audio callback thread pushes sample buffers through a lock-free crossbeam channel to the main thread, which runs the analysis pipeline and renders the UI. The Demucs bridge runs asynchronously on the Tokio runtime, returning stem file paths upon completion.

4 Source Separation via HT-Demucs

Source separation is the foundation of the stem-first paradigm. Music Brain integrates Meta's HT-Demucs model through an asynchronous bridge.

4.1 HT-Demucs Architecture

The Hybrid Transformer Demucs (HT-Demucs) [1] combines two complementary signal representations:

- **Temporal Branch:** 1D convolutional encoder-decoder operating on raw waveforms, capturing fine-grained transient information.

- **Spectral Branch:** 2D convolutional encoder-decoder operating on complex spectrograms, capturing harmonic structure.
- **Cross-Attention:** Transformer layers that fuse information between temporal and spectral branches at each encoder/decoder level.

The model produces six stems: `vocals.wav`, `drums.wav`, `bass.wav`, `guitar.wav`, `piano.wav`, and `other.wav`. This 6-stem configuration is critical for timbral forensics, which requires clean isolation of individual instrument groups.

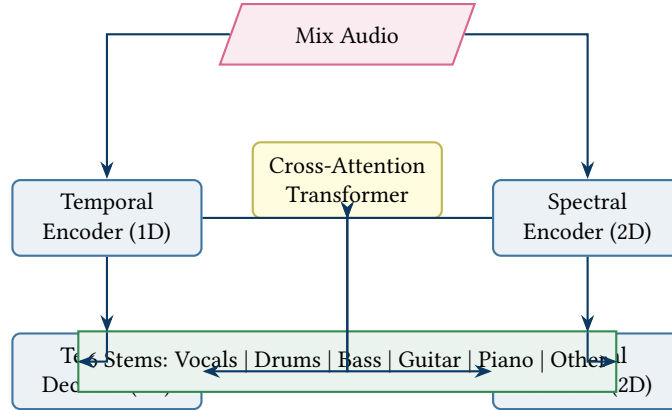


Figure 9. Simplified HT-Demucs encoder-decoder architecture. The temporal and spectral branches are fused via cross-attention transformers before decoding into six isolated stems.

4.2 DemucsBridge Integration

The integration follows a state machine pattern with four states:

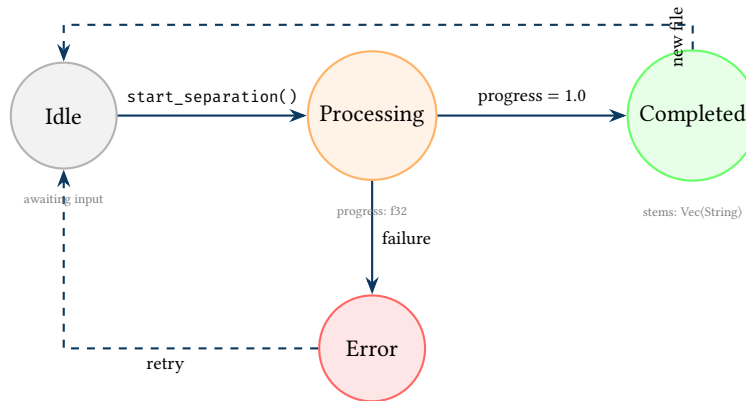


Figure 10. DemucsBridge state machine. The bridge transitions through Idle, Processing, and Completed states, with an Error branch for failure recovery.

When the user drops an audio file, `start_separation()` transitions from Idle to Processing. The bridge spawns the Demucs process asynchronously and polls progress each frame. On completion, the Completed state provides paths to the six stem files, which load into per-stem analysis buffers.

4.3 Analytical Advantages of Stem Isolation

Stem isolation provides measurable improvements in feature extraction accuracy:

- **Harmonic Analysis:** Chroma vectors computed from isolated vocals or guitar stems are free from percussive spectral contamination, yielding cleaner chord profiles.
- **Timbral Forensics:** Snare tuning estimation via f_0 tracking requires clean transient isolation; the drum stem provides this without melodic interference.
- **Rhythmic Analysis:** Sub-band energy detection for kick, snare, and hi-hat operates on the drum stem, eliminating false positives from bass harmonics or vocal sibilance.
- **Effect Recognition:** Guitar distortion characteristics (harmonic-to-fundamental ratio) are only meaningful when measured on the isolated guitar signal.

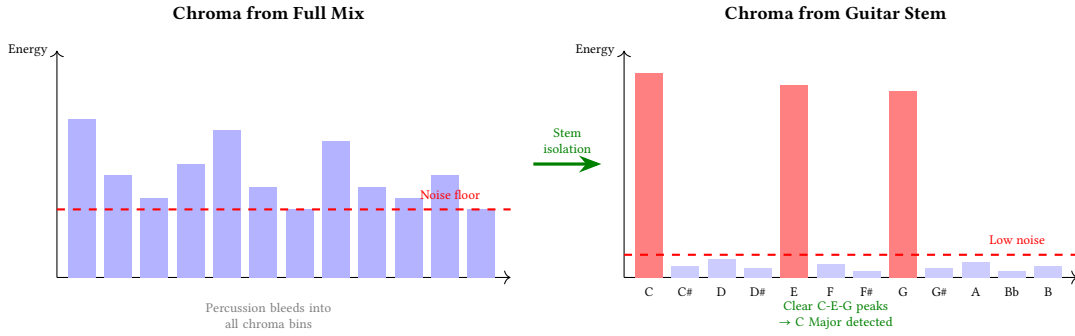


Figure 11. Stem isolation quality comparison. Left: chroma vector from full mix—percussion energy bleeds into all pitch classes, raising the noise floor and obscuring the true chord. Right: chroma from isolated guitar stem—clean peaks at C, E, G clearly indicate C Major. Stem isolation reduces the noise floor by ~ 15 dB.

5 Spectral Analysis Pipeline

The spectral analysis module transforms time-domain audio into the feature representations used by all downstream modules.

5.1 Short-Time Fourier Transform

Every musical sound—a guitar chord, a snare hit, a vocal phrase—is a pressure wave vibrating at many frequencies simultaneously. To analyse music, we must know *which frequencies are present at each moment*. Consider the opening chord of the Beatles’ “A Hard Day’s Night”—one of the most analysed sounds in pop music. For decades, musicians debated which notes were played and by whom. The STFT settles this in milliseconds: it reveals George Harrison’s 12-string Rickenbacker, Lennon’s 6-string acoustic, McCartney’s bass note, and George Martin’s piano, all layered in a single chord. Without the STFT, all we have is a raw waveform—amplitude bouncing up and down with no way to distinguish instruments or notes.

This foundation makes *every* downstream analysis possible. When Music Brain identifies chord progressions in “Let It Be,” detects BPM in “Stayin’ Alive,” or classifies distorted guitar timbre in “Smells Like Teen Spirit,” it starts here—decomposing audio into frequency components frame by frame.

The STFT decomposes the signal into overlapping frames, applies a window function, and computes the discrete Fourier transform of each frame. The implementation uses the `realfft` crate for real-valued FFT computation.

Parameters. The system uses a 2048-sample FFT window (≈ 46 ms at 44.1 kHz) with a 1024-sample hop size (50% overlap). These parameters provide a frequency resolution of $\Delta f = f_s/N = 44100/2048 \approx 21.5$ Hz and a temporal resolution of $\Delta t = H/f_s = 1024/44100 \approx 23.2$ ms.

Window Function. A Hann window is applied to each frame to reduce spectral leakage:

$$w[n] = 0.5 \left(1 - \cos \left(\frac{2\pi n}{N-1} \right) \right), \quad n = 0, 1, \dots, N-1 \quad (1)$$

Magnitude Spectrum. For each windowed frame, the real FFT produces $N/2 + 1 = 1025$ complex-valued frequency bins. The magnitude spectrum is computed as $|X[k]| = \sqrt{\text{Re}(X[k])^2 + \text{Im}(X[k])^2}$.

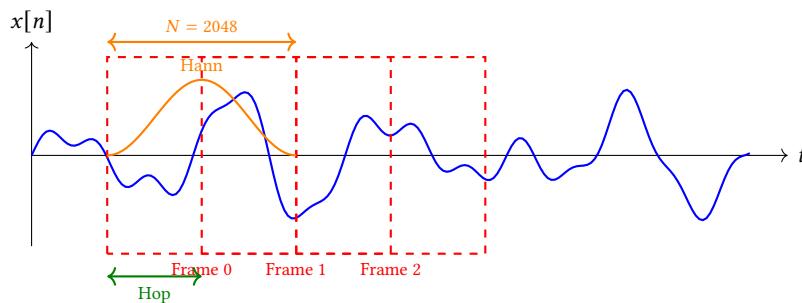


Figure 12. STFT windowing process. Overlapping Hann windows (50% overlap) are applied to successive frames of the input signal. Each windowed frame is transformed via FFT to produce a magnitude spectrum column in the spectrogram.

Spectrogram Representation. The sequence of magnitude spectra forms a two-dimensional spectrogram: time on the horizontal axis, frequency on the vertical, intensity encoding magnitude. This time-frequency representation is the primary tool for visualizing how spectral content evolves. Figure 13 shows a synthetic spectrogram illustrating characteristic signatures of different musical elements.

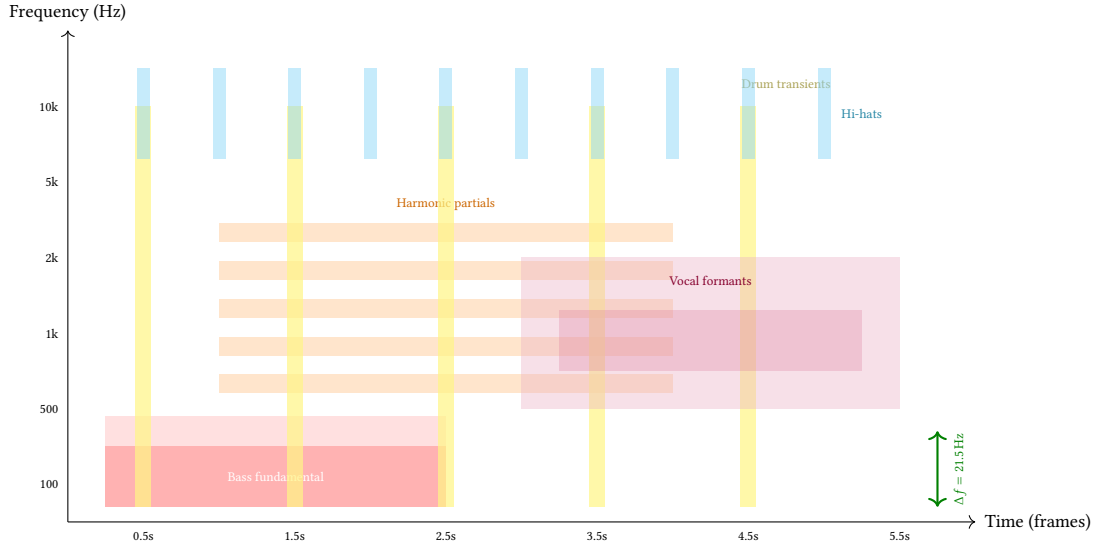


Figure 13. Spectrogram representation of a multi-instrument audio signal. Sustained tonal elements (bass, vocal formants) appear as horizontal bands; transient events (drum hits) appear as vertical bright lines spanning all frequencies; harmonic partials appear as evenly spaced horizontal streaks. The frequency resolution is $\Delta f \approx 21.5$ Hz with the 2048-point FFT.

5.2 Constant-Q Transform

The STFT treats all frequencies equally, but music does not work that way. The distance between C3 and C#3 is about 8 Hz; between C6 and C#6, about 66 Hz. A uniform-resolution STFT wastes precision at high frequencies and lacks it at low frequencies—exactly backwards for music. The Constant-Q Transform matches frequency resolution to the spacing of musical notes: fine resolution where notes are close (low registers), coarser where they are far apart (high registers). The CQT is the foundation for chord and key detection because it “hears” the spectrum the way a musician does.

Mathematically, the CQT [8] uses a logarithmic frequency scale where each bin corresponds to a musical note:

$$X(k, n) = \sum_{j=n-N_k/2}^{n+N_k/2} x(j) w_k(j - n + N_k/2) e^{-i2\pi f_k j / f_s} \quad (2)$$

where the window length $N_k = Q \cdot f_s / f_k$ adapts to each center frequency f_k , maintaining a constant quality factor $Q = f_k / \Delta f_k$. This preserves the geometric spacing between octaves, critical for identifying chords across pitch ranges. The CQT provides the mathematical foundation for chroma computation (Section 5.8).

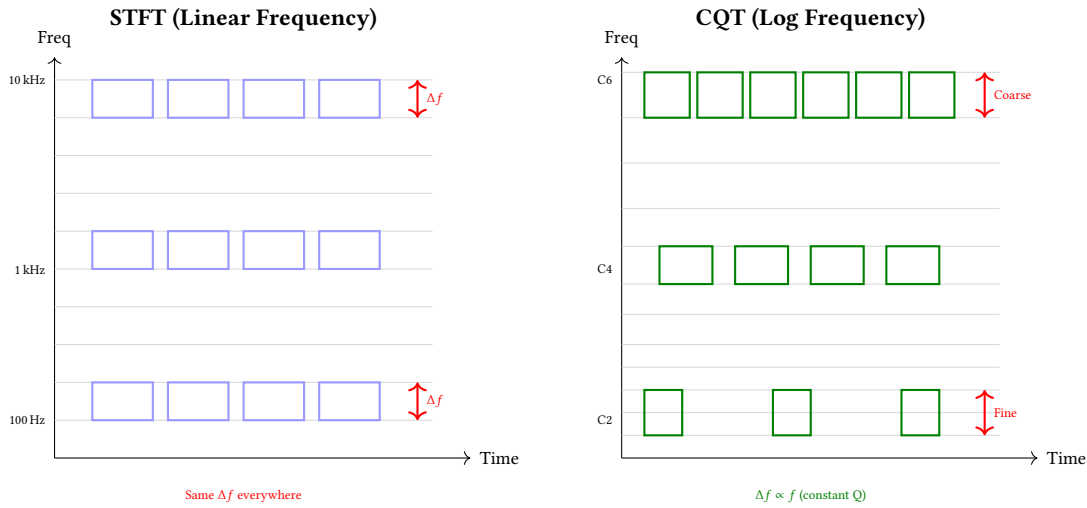


Figure 14. STFT vs. CQT resolution comparison. The STFT uses uniform frequency bins ($\Delta f = 21.5$ Hz everywhere), wasting resolution at high frequencies where musical intervals span large Hz ranges. The CQT adapts resolution logarithmically: fine frequency resolution at low pitches (where semitones are closely spaced in Hz) and fine temporal resolution at high pitches (where transients matter).

5.3 Spectral Centroid (Timbral “Brightness”)

The spectral centroid measures the “brightness” of a sound—a tinny phone speaker has a high centroid, a subwoofer a low one.

Why does a Radiohead track sound “dark” while a Dua Lipa record sounds “bright”? A warm, mellow sound concentrates energy in lower frequencies; a bright, cutting sound pushes energy upward. The spectral centroid captures this as a single number—the frequency “center of gravity.”

The centroid reveals production character immediately. Adele’s “Someone Like You” (piano and voice, warm reverb) sits around 1,500 Hz. Metallica’s “Master of Puppets” (distorted guitars, aggressive cymbals) pushes above 4,000 Hz. A lo-fi hip-hop beat might sit around 1,200 Hz; a hyper-pop track could exceed 5,000 Hz. Tracking centroid across eras reveals the “loudness war”: 1970s recordings cluster around 1,500–2,000 Hz, while 2010s pop pushes above 3,000 Hz due to aggressive high-frequency processing.

Formally, the spectral centroid represents the “center of mass” of the magnitude spectrum [7]:

$$f_c = \frac{\sum_{k=0}^{N/2} f_k \cdot |X[k]|}{\sum_{k=0}^{N/2} |X[k]|} \quad (3)$$

where $f_k = k \cdot f_s / N$ is the frequency of the k -th bin. The implementation maps bin indices to frequencies via $f_k = k \times 44100 / 2048$ and returns the weighted average in Hz. A centroid of 2000 Hz indicates a relatively balanced spectrum, while values above 4000 Hz suggest a “bright” or “harsh” timbre common in distorted guitar or cymbal-heavy mixes.

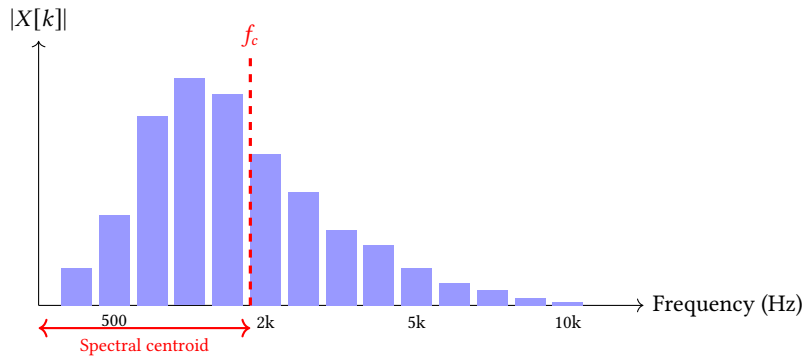


Figure 15. Spectral centroid as the “center of mass” of the magnitude spectrum. The centroid (f_c) shifts rightward for brighter timbres and leftward for darker ones.

5.4 Spectral Flatness (Wiener Entropy)

Is the sound a *note* or *noise*? A sustained piano chord concentrates energy at specific harmonic frequencies. A cymbal crash spreads energy across the spectrum with no clear pitch. Spectral flatness answers this numerically.

Low flatness means “tonal instrument playing a note”; high flatness means “percussive or noise-like sound.” This helps distinguish a hi-hat (noisy, high flatness) from a synth pad (tonal, low flatness) even when they occupy similar frequency ranges.

Formally, spectral flatness is computed as the ratio of the geometric mean to the arithmetic mean of the magnitude spectrum [7]:

$$F = \frac{\exp\left(\frac{1}{N} \sum_{k=1}^N \ln |X_k|\right)}{\frac{1}{N} \sum_{k=1}^N |X_k|} \quad (4)$$

The implementation applies a floor of 10^{-7} to magnitude values before the logarithm to avoid numerical instability. Output is clamped to $[0, 1]$: values near 0.0 indicate a tonal signal (e.g., sustained flute); values near 1.0 indicate noise-like signals (e.g., cymbal crashes, distorted guitars). This descriptor is central to the Metal adapter’s vocal classification, where high flatness distinguishes screamed vocals from clean singing.

5.5 Sensory Dissonance (Roughness Model)

The unpleasant “wobble” when two guitar strings are nearly in tune is *roughness*—two close frequencies interfering, creating rapid amplitude fluctuations the ear perceives as harshness. Music Brain measures this precisely, turning an intuitive “that sounds tense” into a number.

Some chords sound “smooth” (a C major chord); others sound “clashing” (a minor second). This is measurable: *roughness*, caused by two close but non-identical frequencies beating against each other.

Dissonance reveals compositional intent. Nirvana’s “Smells Like Teen Spirit” shows low dissonance in the quiet verse (clean guitar, power chords) spiking in the distorted chorus—that tension jump is the song’s emotional engine. Radiohead’s “Paranoid Android” produces a dissonance profile like a mountain range: jagged peaks during the dissonant middle section, valleys during melodic passages. Ed Sheeran’s “Thinking Out Loud” maintains low dissonance throughout—pure consonance matching its smooth character. Tracking dissonance across a song reveals the tension-and-release arc: unresolved minor chords in the verse creating anticipation that resolves into the major-key chorus, the push-and-pull that has driven Western music since Bach.

The Plomp–Levelt roughness model [7] predicts that two pure tones produce maximum roughness when separated by approximately 25% of the critical bandwidth at their center frequency, with roughness falling to zero for both unison and wide intervals.

$$R(f_1, f_2) = e^{-3.5 \cdot s \cdot |f_2 - f_1|} - e^{-5.75 \cdot s \cdot |f_2 - f_1|} \quad (5)$$

where $s = 0.24 / (0.0207 \cdot f_{\min} + 18.96)$ scales with the critical bandwidth. The total sensory dissonance is the sum of pairwise roughness across all spectral peaks.

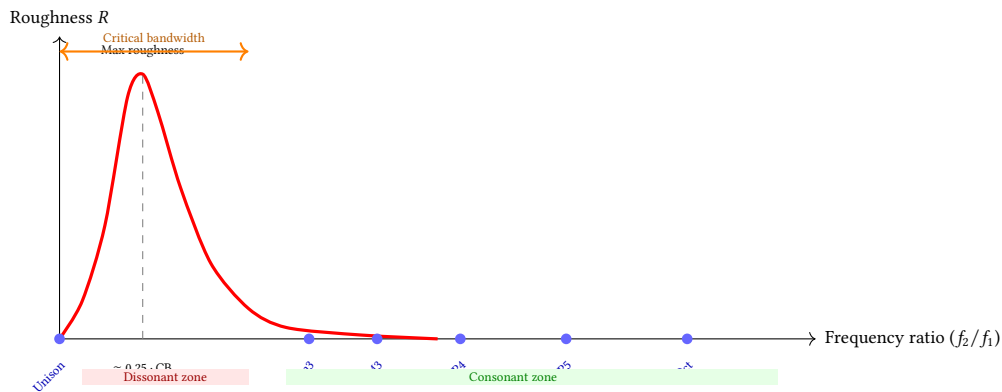


Figure 16. Plomp–Levelt sensory dissonance (roughness) curve. Roughness peaks when two tones are separated by $\approx 25\%$ of the critical bandwidth, then decays for wider intervals. Consonant musical intervals (P5, P4, M3) fall in low-roughness regions. The `sensory_dissonance` feature in the `FeatureSet` is computed by summing pairwise roughness across all spectral peaks.

5.6 Onset Detection

Before analysing rhythm, groove, or timing, we must know *when things happen*—when the snare hits, when the vocalist starts a phrase, when a guitar chord is strummed. Onset detection finds these moments automatically and unlocks some of Music Brain’s most revealing analysis.

Consider two drummers playing the same beat: a drum machine (perfectly on the grid) and Questlove from The Roots (deliberately behind the beat, giving a relaxed feel). Onset detection reveals each hit’s position relative to the quantised grid—what separates mechanical from human, good groove from great groove. It detects

the “J Dilla feel” (beats drifting behind the grid, as on *Donuts*), the “D’Angelo pocket” (deep-behind-the-beat drumming on *Voodoo*), or the laser-precise timing of Kraftwerk. When you tap your foot to a song, your brain performs onset detection; Music Brain does it mathematically by detecting sudden bursts of spectral energy.

The system computes a *spectral flux* onset detection function that measures the positive increase in magnitude across frequency bins between successive frames:

$$SF(n) = \sum_{k=0}^{N/2} H(|X_n[k]| - |X_{n-1}[k]|)^2 \quad (6)$$

where $H(x) = (x + |x|)/2$ is the half-wave rectifier, ensuring only *increases* in energy contribute. Peaks in the spectral flux function are detected via adaptive thresholding and correspond to onset positions.

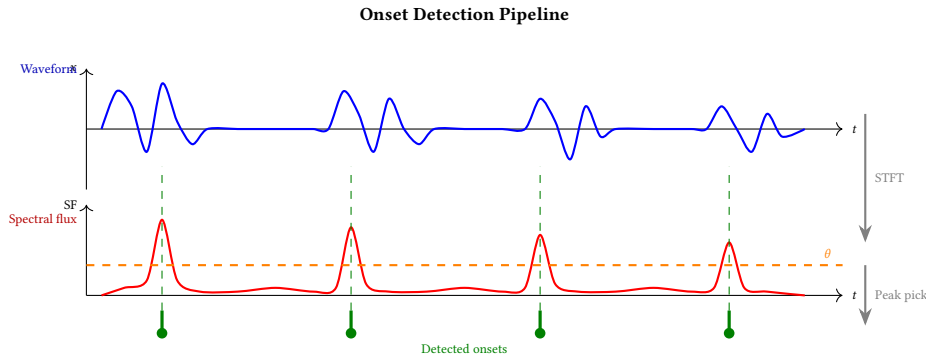


Figure 17. Onset detection pipeline. The audio waveform (top) is analyzed via STFT to produce the spectral flux function (middle), which measures frame-to-frame spectral energy increase. Peaks above the adaptive threshold θ are identified as onsets (bottom), marking the temporal positions of musical events.

5.7 Harmonic Series and Instrument Timbres

Different instruments produce the same pitch with different timbres, determined by the relative amplitudes of their harmonic overtones. The fundamental frequency f_0 defines perceived pitch; the pattern of overtones at integer multiples ($2f_0, 3f_0, 4f_0, \dots$) determines timbre. Stem isolation is critical here: in a mix, overlapping harmonic series from multiple instruments create a spectral tangle that confounds timbral analysis.

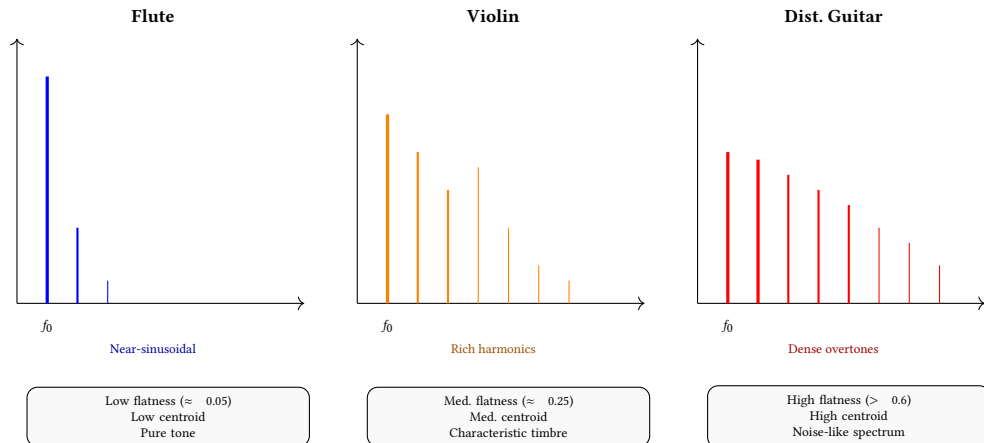


Figure 18. Harmonic series comparison across three instrument timbres, all sounding the same pitch. The flute produces a near-sinusoidal tone (dominant fundamental, few overtones). The violin has rich but structured harmonics. The distorted guitar fills the spectrum with dense overtones, raising both spectral flatness and centroid. These differences are only cleanly measurable on isolated stems.

5.8 Chroma Feature Computation (Pitch Class Profiles)

A *pitch class* is a note name regardless of octave—all C’s are the same pitch class, whether C2 on a bass guitar or C6 on a piccolo. A *chroma vector* (pitch class profile) is a 12-element array showing energy at each of the twelve pitch classes: the harmonic fingerprint of a moment in music.

A guitar and piano both playing C major sound like the “same chord” despite spanning different octaves. Chroma features give Music Brain this ability—collapsing all octaves into a single 12-note profile of energy per pitch class (C, C#, D, ..., B).

This makes chord detection possible. The I–V–vi–IV progression (C–G–Am–F) powers hundreds of pop hits from “Let It Be” to “No Woman No Cry” to “Someone Like You.” The ii–V–I turnaround defines jazz. The minor i–bVI–bIII–bVII pattern gives “Stairway to Heaven” and “Hotel California” their bittersweet quality. By detecting chords through chroma analysis, Music Brain identifies these progressions automatically—revealing the harmonic blueprint that makes songs work.

Chroma features reduce the magnitude spectrum into a 12-dimensional pitch class profile by folding all octaves into a single bin. This octave-invariant representation is the foundation for chord detection and key estimation.

The computation proceeds in three steps:

Step 1: Frequency to MIDI Mapping. Each spectral bin frequency f_k is converted to its MIDI note number:

$$m = 69 + 12 \log_2 \left(\frac{f_k}{440} \right) \quad (7)$$

Step 2: Pitch Class Folding. The MIDI note is reduced to its pitch class via modular arithmetic:

$$\text{pitch_class} = [m + 0.5] \bmod 12 \quad (8)$$

Step 3: Aggregation and Normalization. Magnitudes are accumulated by pitch class, then normalized so the maximum bin equals 1.0. Bins with magnitude below 0.01 (noise floor) and frequencies outside the 20–5000 Hz range are excluded.

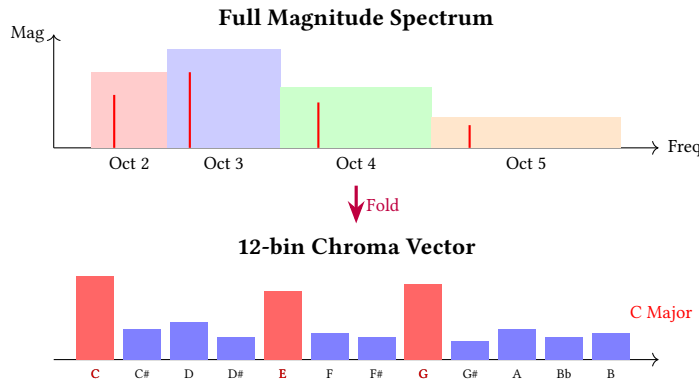


Figure 19. Chroma vector construction: magnitudes from all octaves of each pitch class are folded into a single 12-bin vector. The resulting profile reveals the harmonic content independent of register, here showing a C major chord (C, E, G).

The active note threshold in the main application is set to 0.6: pitch classes with chroma values above this threshold are considered “active” and displayed on the Harmonic Constellation visualization.

6 Harmonic Analysis

The harmonic analysis module builds on chroma features to identify chords, detect keys, and suggest harmonic variations.

6.1 Key Detection via Krumhansl–Schmuckler Profiles

Knowing the key unlocks a cascade of musical understanding. A track in A minor sounds melancholic and introspective; the same progression in C major sounds bright. Billie Eilish’s “Bad Guy” sits in G minor, contributing to its dark character. Pharrell’s “Happy” is in F minor—which *should* sound sad, but the major-key melody and upbeat tempo override the minor tonality, creating a tension that makes the song more compelling than a straightforward major-key track.

Key detection also identifies *modulations*—moments when a song shifts key for emotional impact. Whitney Houston’s “I Will Always Love You” modulates up a semitone from A major to B $\flat$ major in the final chorus, creating an unmistakable lift. Bon Jovi’s “Livin’ On A Prayer” does the same, and Beyoncé’s “Love On Top” modulates *four times* in succession. Music Brain detects these automatically by tracking key correlation over time—when the best-matching profile shifts, a modulation has occurred.

The system determines the tonal center by correlating the observed chroma distribution against the Krumhansl–Kessler key profiles [4]. These profiles capture the *perceptual stability hierarchy*—how “at rest” or “resolved”

each pitch class feels within a given key. In C major, the note C feels like home, G feels stable but expectant, and F# feels tense and unstable. These weightings, derived from perceptual experiments, give the algorithm a model of how listeners experience tonality. The key is estimated as:

$$k^* = \arg \max_{k \in \{0, \dots, 23\}} r(\mathbf{c}, \mathbf{p}_k) \quad (9)$$

where $\mathbf{c}$ is the normalized chroma vector, $\mathbf{p}_k$ is the key profile for key k (12 major + 12 minor), and r is the Pearson correlation coefficient. The key with the highest correlation is selected.

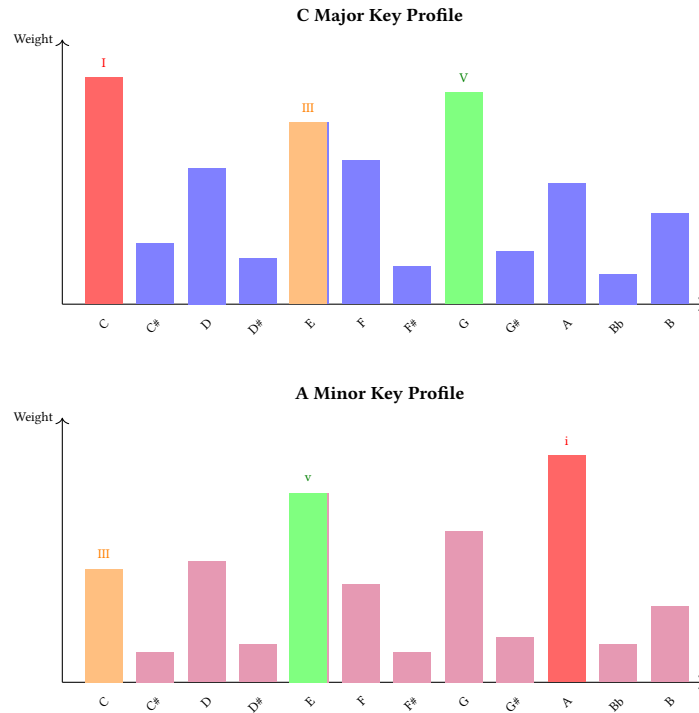


Figure 20. Krumhansl–Kessler key profiles for C major (top) and A minor (bottom). The tonic (I/i), mediant (III), and dominant (V/v) receive the highest weights. Key detection correlates the observed chroma vector against all 24 rotations of these templates, selecting the best match.

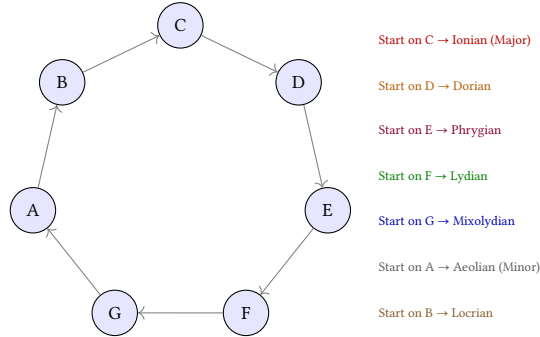
6.1.1 Modes and Their Character

Beyond major and minor, Western music draws on seven *modes*—scales built by starting on different degrees of the major scale. Each mode has a distinctive emotional character, and Music Brain’s key detection can identify modal passages when the chroma profile deviates from a pure major or minor template:

Mode	Character	Famous Example	Defining Note
Ionian (Major)	Bright, stable, “default”	“Let It Be” (Beatles)	Natural 4th, 7th
Dorian	Slightly jazzy minor	“So What” (Miles Davis)	Raised 6th
Phrygian	Spanish / Middle Eastern	Metallica intros	Lowered 2nd
Lydian	Dreamy, floating major	“The Simpsons” theme	Raised 4th
Mixolydian	Bluesy major	“Norwegian Wood” (Beatles)	Lowered 7th
Aeolian (Minor)	Sad, dark, “natural minor”	“Stairway to Heaven”	Natural form
Locrian	Extremely unstable, rare	Theoretical / metal riffs	Lowered 2nd & 5th

Table 2. The seven modes of the major scale with their emotional characters and well-known examples. Each mode is a rotation of the same seven-note pattern, yet the shifted starting point produces a dramatically different mood.

Modes as Rotations of the Same Scale



Same seven notes, seven different moods—
the starting point determines everything.

Figure 21. The seven modes are rotations of the same scale pattern. Starting the C major scale on each successive degree produces a mode with a distinct emotional character, even though all seven modes use the same seven pitches.

6.2 Tempo Estimation via Autocorrelation

Tempo is one of the most fundamental properties of a song—it shapes the energy, the movement, and the emotional feel before a single note is played. Adele’s “Someone Like You” at 68 BPM creates space for emotional weight; Queen’s “Bohemian Rhapsody” shifts between tempos as it moves through its operatic sections; the Bee Gees’ “Stayin’ Alive” at 104 BPM is so perfectly timed to the human heartbeat that it’s used to teach CPR chest compressions.

Tempo also carries powerful genre signatures. Kendrick Lamar and most hip-hop producers work at 80–100 BPM (the “head-nod” tempo). Classic Motown hits cluster around 110–120 BPM. Daft Punk’s “Around the World” sits at 121 BPM—the sweet spot for house music. The Prodigy and jungle producers push to 160–180 BPM. Even within a genre, tempo tells a story: Drake’s shift from 75 BPM introspective tracks to 140 BPM dancehall-influenced songs marked a stylistic evolution that Music Brain can trace quantitatively.

Music Brain estimates tempo automatically by looking for the strongest repeating pattern in the onset signal—essentially asking “what time interval keeps recurring between beats?”

The estimation operates on the onset detection function (Section 5) by computing the autocorrelation of the onset strength signal to find its dominant periodicity:

$$A(\tau) = \sum_{n=0}^{N-\tau-1} \text{SF}(n) \cdot \text{SF}(n + \tau) \quad (10)$$

The lag τ^* corresponding to the highest autocorrelation peak (within the musically plausible range of 60–200 BPM) yields the tempo estimate: $\text{BPM} = 60 \cdot f_s / (H \cdot \tau^*)$, where H is the hop size.

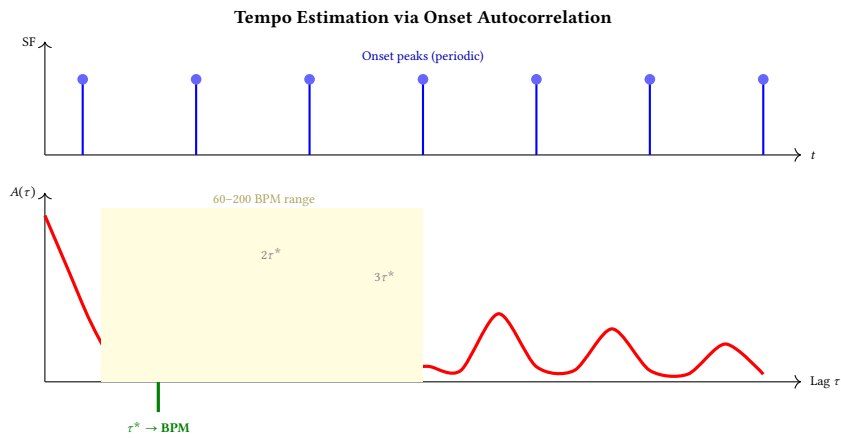


Figure 22. Tempo estimation via autocorrelation. The onset strength function (top) shows periodic peaks at beat positions. The autocorrelation (bottom) reveals the dominant periodicity at lag τ^* , with sub-harmonic peaks at $2\tau^*$, $3\tau^*$. The BPM is computed from the lag within the musically plausible range (shaded).

6.3 Chord Detection via Template Matching

Chord detection operates by identifying active pitch classes (chroma value > 0.5) and matching the resulting interval pattern against known chord templates. The algorithm uses the first active pitch class as the assumed root and computes all intervals relative to it in semitones:

Intervals (semitones)	Chord Type	Description
{0, 3, 6}	Diminished	Minor 3rd + diminished 5th
{0, 4, 8}	Augmented	Major 3rd + augmented 5th
{0, 4, 7, 10}	Dominant 7th	Major 3rd + perfect 5th + minor 7th
{0, 4, 7}	Major	Major 3rd + perfect 5th
{0, 3, 7}	Minor	Minor 3rd + perfect 5th
Other	Complex/Cluster	Non-standard voicing

Table 3. Chord detection template mapping. Active chroma bins are reduced to intervals relative to the root, then matched against these patterns.

6.3.1 Chord Quality and Emotional Character

Beyond simply *detecting* chords, understanding what different chord types *sound like* and their emotional effects is essential for interpreting harmonic analysis results. Each chord quality carries a distinct emotional character:

- **Major chords** (e.g., C-E-G): Bright, happy, resolved—the “home base” sound of most pop music. When Music Brain detects a major chord, it signals stability and resolution.
- **Minor chords** (e.g., A-C-E): Sad, introspective, melancholic—the emotional counterpart to major. The only difference is one note lowered by a semitone (the third), yet the emotional shift is dramatic.
- **Diminished chords** (e.g., B-D-F): Tense, unstable, anxious—creates a strong pull toward resolution. Used for dramatic tension in horror film scores, jazz passing chords, and Beethoven’s dramatic transitions. The interval structure (stacked minor thirds) is symmetrical, meaning the chord can resolve in multiple directions (see Section 6.4).
- **Augmented chords** (e.g., C-E-G#): Dreamlike, unresolved, “floating”—used by artists like Radiohead and in film scoring for otherworldly moments. The whole-tone symmetry gives them an ambiguous, suspended quality.
- **Dominant 7th chords** (e.g., G-B-D-F): Bluesy tension that *wants* to resolve—the backbone of blues, jazz, and rock & roll. The tritone interval between the 3rd and 7th creates a gravitational pull toward the tonic.

Chord Types: Interval Structure and Emotional Character

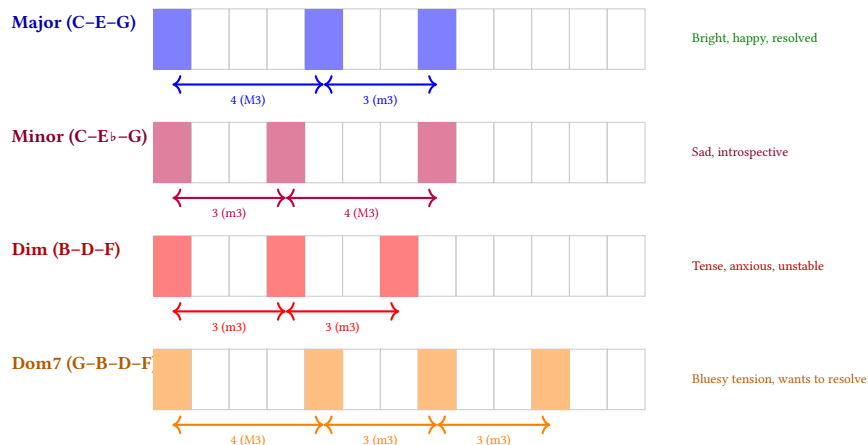


Figure 23. Chord types shown as interval structures on a keyboard. Major and minor differ by a single semitone in the third, yet the emotional character shifts dramatically. Diminished chords stack equal minor thirds (symmetrical), creating instability. Dominant 7th chords contain a tritone (B–F) that drives resolution to the tonic.

6.3.2 The Circle of Fifths as a Map of Tonal Space

The Circle of Fifths is not merely a diagram—it is a *map of tonal space* that reveals the deep structure of Western harmony. Adjacent keys on the circle share the most notes in common: C major and G major differ by only one note (F becomes F $\sharp$). Keys on opposite sides of the circle are maximally different: C major and F $\sharp$ major share almost no notes.

This means that *key changes that move around the circle by small steps feel natural and smooth*, while large jumps feel dramatic and unexpected. The Beatles’ “Hey Jude” stays close to home on the circle, with chord changes that move only one or two positions. In stark contrast, John Coltrane’s “Giant Steps” deliberately leaps across the circle in major-third jumps (B→G→E $\flat$), creating the disorienting, exhilarating harmonic motion that makes the piece a jazz landmark—and notoriously difficult to improvise over.

Music Brain uses the Circle of Fifths as the geometric basis for its Harmonic Constellation visualization, where detected chords are plotted as shapes on the circle. A simple pop song produces compact triangles clustered in one region; a jazz standard produces sprawling shapes that reach across the circle.

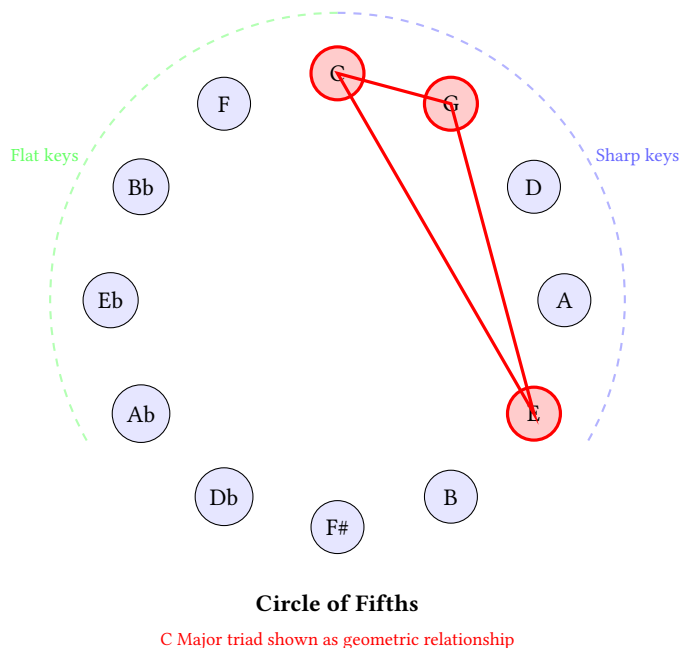


Figure 24. Circle of Fifths with a C major triad (C–E–G) displayed as a geometric relationship. The Harmonic Constellation visualization in the UI draws these inter-note connections for the currently detected chord, revealing the “shape” of the harmony.

6.4 Variation Engine: Harmonic Substitutions

The variation engine generates theoretically grounded chord substitution suggestions based on the currently detected chord type. Each suggestion includes a name, target chord, and musical justification. Below, real-world examples illustrate how these substitutions appear in well-known songs.

Dominant 7th Variations:

- **Tritone Substitution** ($\rightarrow \flat\text{II}7$): Replaces V7 with $\flat\text{II}7$, providing chromatic downward bass motion while preserving the tritone interval between the 3rd and 7th. This substitution is ubiquitous in jazz standards—in “The Girl from Ipanema,” the F#7 that replaces G7 creates the smooth chromatic bassline that defines the bossa nova feel. On the Circle of Fifths, the tritone sub is the key directly *opposite* the original dominant (Figure 26).
- **Altered Extension** ($\rightarrow \text{V}7\text{alt}$): Adds $\flat 9$ or $\sharp 9$ to increase tension before resolution. Jimi Hendrix’s “Purple Haze” opens with an E7#9 chord so iconic it’s known as the “Hendrix chord.”

Minor Chord Variations:

- **Dorian Interchange** ($\rightarrow \text{IV Major}$): Borrows the IV major chord from the Dorian mode, introducing a “floating brightness” to the minor context. Miles Davis’s “So What” is entirely built on the Dorian mode, and the major IV chord appears frequently in neo-soul (D’Angelo, Erykah Badu).
- **Chromatic Mediant** ($\rightarrow \flat\text{vi minor}$): Shifts down a major third for darker, cinematic coloring. Hans Zimmer uses chromatic mediants extensively in film scores (“Inception,” “Interstellar”) to create that characteristic sense of vast, otherworldly space.

Major Chord Variations:

- **Relative Shift** ($\rightarrow \text{vi} / \text{I}$): Moves to the relative minor or major to shift emotional polarity without breaking key.
- **Modal Mixture** ($\rightarrow \text{iv minor}$): Borrows the minor iv chord from the parallel minor for bittersweet resolution. This is one of the most recognisable sounds in pop music: in Radiohead’s “Creep,” the F $\rightarrow$ Fm progression (IV $\rightarrow$ iv) in the chorus creates the heartwrenching quality that defines the song. The Beatles used it extensively—“While My Guitar Gently Weeps” borrows from the parallel minor throughout.

Diminished Chord Variations: Diminished chords are special in music theory because they divide the octave into perfectly equal parts. A diminished 7th chord stacks three minor 3rds ($3 + 3 + 3 + 3 = 12$ semitones = one octave), which means there are only *three unique* diminished 7th chords in all of music—every other diminished 7th is simply an inversion of one of these three. This symmetry makes diminished chords powerful “pivot chords” that can resolve in four different directions, since each note of the chord can serve as a leading tone to a different key.

- **Symmetrical Shift** ($\rightarrow \text{vii}^\circ 7$ shifted): Moves the entire diminished chord up a minor 3rd; functionally identical due to the diminished chord’s symmetric interval structure, but varies the melodic context. Beethoven uses this technique extensively in his piano sonatas for dramatic key changes, and jazz musicians use passing diminished chords to create smooth chromatic bass motion between diatonic chords (e.g., C $\rightarrow$ C# $^\circ 7 \rightarrow$ Dm7).

The Three Unique Diminished 7th Chords

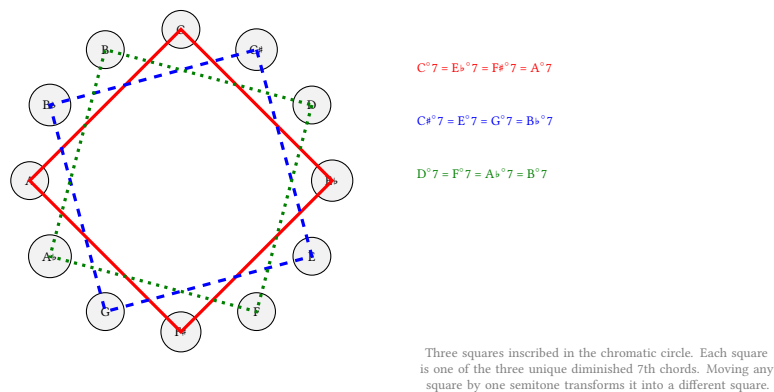


Figure 25. The three unique diminished 7th chords as squares inscribed in the chromatic circle. Each chord divides the 12 chromatic pitches into four equal groups of 3 semitones. Transposing any diminished 7th chord by 1 or 2 semitones produces one of the other two chords; transposing by 3 semitones produces the same chord with a different voicing.

Tritone Substitution on the Circle of Fifths

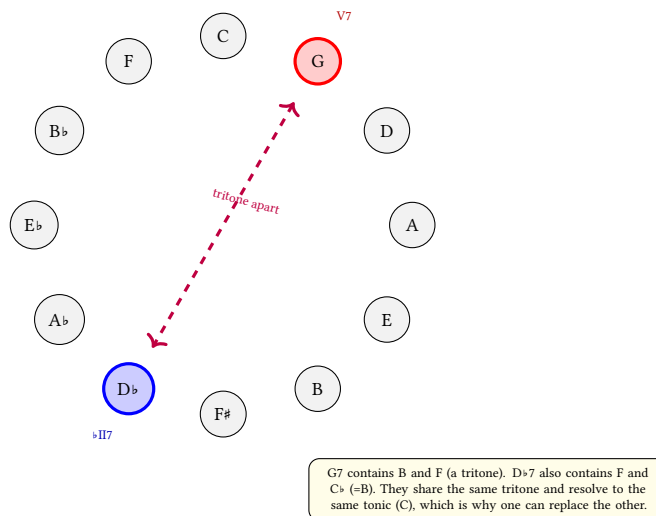


Figure 26. Tritone substitution shown on the Circle of Fifths. $G7$ ($V7$ of C) and $D^\flat 7$ ($bII7$ of C) sit directly opposite each other and share the same tritone interval (B–F). This shared tension means either chord can resolve to C, but the $D^\flat 7$ provides a chromatic bass descent ($D^\flat \rightarrow C$) instead of the standard fifth descent ($G \rightarrow C$).

6.5 Chord Progressions and Key Changes Over Time

Beyond detecting individual chords at a single moment, Music Brain tracks chord changes across the duration of a song, building a harmonic timeline that reveals large-scale tonal structure. By monitoring the chroma profile frame-by-frame and recording each chord transition, the system assembles a sequence of color-coded harmonic regions. This timeline enables detection of key modulations—for example, a pivot from A minor to its relative major C—by observing shifts in the diatonic context of successive chords. Roman numeral analysis is applied relative to the estimated key center, providing functional harmonic labels (i, VI, III, VII) that are invariant to transposition. Figure 27 illustrates a 16-bar harmonic timeline with a modulation detected at bar 9.

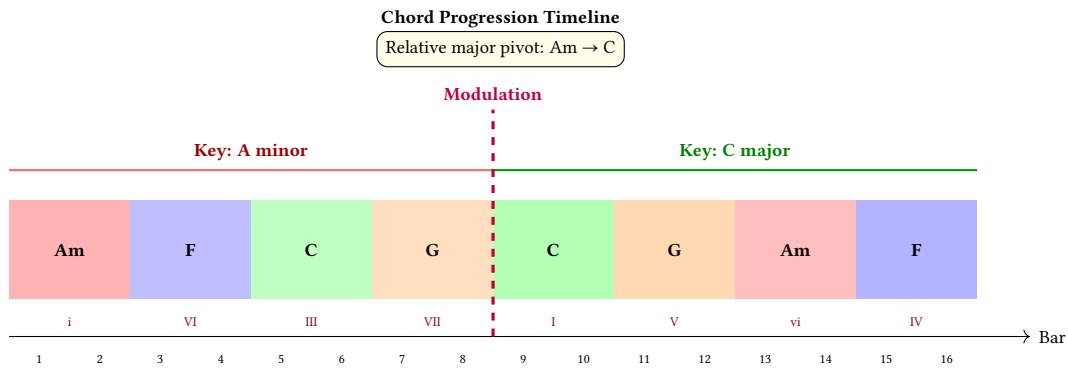


Figure 27. Chord progression timeline across 16 bars showing a modulation from A minor to C major at bar 9. Color-coded blocks indicate chord identity; Roman numeral analysis is shown relative to each key center. The dashed line marks the detected modulation point.

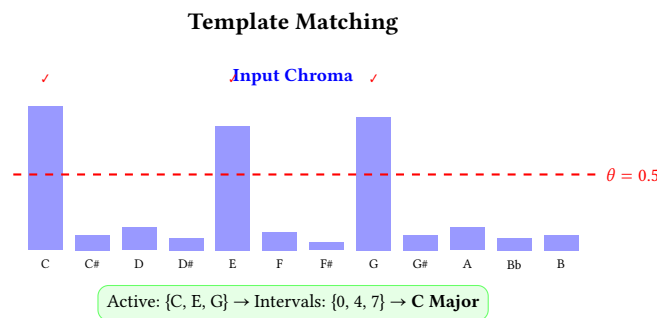


Figure 28. Chroma template matching for chord identification. Pitch classes exceeding the threshold ($\theta = 0.5$) are identified as active, their intervals computed relative to the root, and matched against chord type templates.

6.6 Harmonic Rhythm and Pacing

How fast chords change matters as much as which chords are used. This concept—called *harmonic rhythm*—is one of the most powerful tools in a songwriter’s arsenal. A slow harmonic rhythm (one chord per bar or fewer) feels expansive, spacious, and contemplative: think of Adele’s “Someone Like You,” where each chord lingers for an entire bar, giving the vocal room to breathe and the listener time to absorb the emotion. A fast harmonic rhythm (multiple chord changes per bar) creates urgency, excitement, and forward momentum: a bebop jazz tune like Charlie Parker’s “Donna Lee” changes chords every two beats, creating a harmonic rushing-river effect that keeps listeners on the edge of their seats.

Music Brain measures harmonic rhythm by tracking the rate of chord transitions per bar. This metric, combined with the tempo, reveals the “information density” of the harmony:

- **Very slow** (< 0.5 changes/bar): Drone-based music, ambient, modal jazz vamps
- **Slow** (1 change/bar): Most pop, rock, folk, ballads
- **Moderate** (2 changes/bar): Standard jazz, classic R&B, musical theatre
- **Fast** (> 2 changes/bar): Bebop jazz, progressive rock, chromatic film scores

Tracking harmonic rhythm across a song reveals pacing strategies: many hit songs use slow harmonic rhythm in verses (building space for the vocal) and accelerate in pre-choruses (building urgency), before the chorus delivers either a satisfying return to slow harmonic rhythm (resolution) or maintains momentum (drive).

7 Rhythmic Analysis

Rhythm drives a song forward. The drummer’s choices about density, subdivision, and feel shape how the listener experiences energy and momentum. A verse on closed hi-hat with a simple kick-snare pattern feels contained; switching to double-time—doubling the hi-hat rate or moving to the ride cymbal—surges perceived energy even at constant tempo. This builds intensity into a pre-chorus, preparing the listener for the chorus. A half-time feel in a bridge pulls energy back, creating contrast that makes the return to full time feel explosive.

The *pattern* itself carries structural information. A four-on-the-floor kick signals dance music; a swung shuffle signals blues or jazz; a blast beat marks extreme metal. When a drummer departs from the established pattern—dropping the kick for a bar, adding a syncopated ghost note, or launching into a fill across the bar line—the

deviation marks a structural boundary. These rhythmic cues are among the most reliable section-transition markers, often preceding harmonic or melodic changes by a beat or two.

Rhythmic analysis extracts temporal patterns from the audio signal, focusing on sub-band energy detection, grid quantization, and pattern archetype matching.

7.1 Sub-Band Energy Detection

The rhythmic analysis pipeline partitions the frequency spectrum into three sub-bands corresponding to the primary percussion instruments:

Sub-Band	Frequency Range	Target Instrument
Low	40–100 Hz	Kick drum
Mid	1–4 kHz	Snare drum
High	8–16 kHz	Hi-hat / Cymbals

Table 4. Sub-band frequency partitioning for rhythmic analysis. Each band targets the primary spectral energy region of its associated percussion instrument.

Energy transients within each sub-band are detected using onset detection functions [12], triggering rhythmic event markers at positions corresponding to individual drum hits.

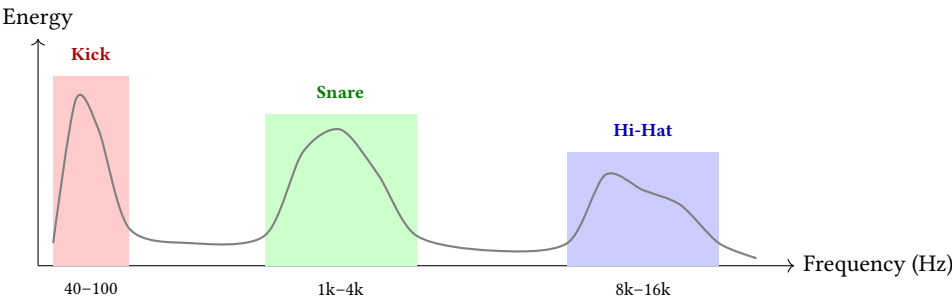


Figure 29. Sub-band frequency partitioning for percussion detection. The three bands isolate the primary spectral energy regions of kick drum (low), snare (mid), and hi-hat (high).

7.2 16-Step Grid Quantization

Detected rhythmic events are quantized to a 16-step grid, where each step represents a sixteenth note within a single bar. This discretization enables pattern matching against known rhythmic archetypes:

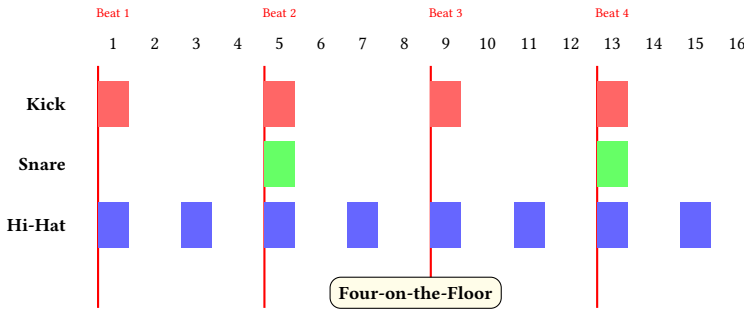


Figure 30. 16-step grid representation of a four-on-the-floor pattern. Kick hits on every beat (1, 5, 9, 13), snare on beats 2 and 4 (5, 13), hi-hat on every eighth note. The Electronic adapter uses 1/16th grid alignment as the default quantization resolution, common across EDM, house, techno, and pop production.

7.3 Pattern Archetype Matching

The system recognizes several rhythmic archetypes from the quantized grid pattern. The Electronic adapter defaults to a 1/16th grid alignment and evaluates phase synchronization between kick and bass elements. A phase sync value above 0.8 indicates tight alignment characteristic of rolling basslines in electronic genres.

7.4 Temporal Pattern Evolution

Static single-bar snapshots reveal what a rhythm *is* at a given moment but miss how patterns *change* across sections. Music Brain tracks the quantized drum grid across structural boundaries detected by the SSM (Section 9), recording per-section kick, snare, and hi-hat patterns. This captures common production techniques:

sparse intro kicks giving way to four-on-the-floor verses, syncopated pre-chorus build-ups, and double-time drops. Figure 31 illustrates a typical kick pattern evolution across four structural sections.

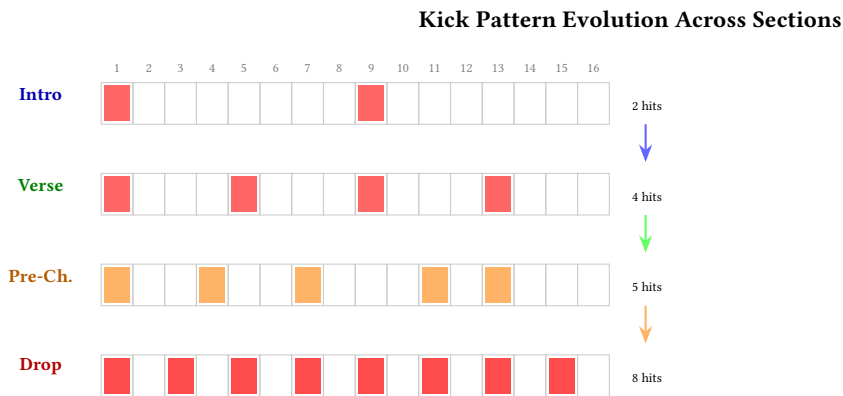


Figure 31. Multi-bar kick pattern evolution across four structural sections. The kick density increases from sparse (Intro) through four-on-the-floor (Verse) and syncopated (Pre-Chorus) to double-time (Drop), reflecting a common electronic music arrangement arc.



Figure 32. Hi-hat density fade-in across four bars. Velocity shading (darker = louder) shows the progressive build from quarter-note closed hats through eighth notes and sixteenth to accented open hats (marked “O”), a common build-up technique in electronic and pop production.

7.5 Micro-Timing and Groove Analysis

Rhythmic feel depends not only on *which* grid positions are active but on *how precisely* hits align with those positions. Music Brain measures onset deviation from the quantized grid in milliseconds, capturing the subtle timing variations that distinguish a mechanical sequencer from a human performance. At 120 BPM with a 16th-note grid resolution of about 125 ms per step, deviations of 5–20 ms produce the “feel” or “groove” that defines a performance style.

Swing is quantified as the ratio of the first eighth note’s duration to the second: 1.0 indicates straight timing; 1.5–2.0 indicates light to heavy shuffle. The system recognizes four groove templates: *straight* (ratio ≈ 1.0), *shuffle* (≈ 1.5), *Dilla-style* (variable per beat, behind-the-beat on beats 2 and 4), and *behind-the-beat* (consistent positive offset of 10–30 ms across all onsets). Inter-Onset Interval (IOI) histograms show timing regularity: straight grooves produce tight unimodal distributions; swung grooves produce bimodal distributions with peaks at the short and long subdivisions. Figure 33 illustrates grid deviation and IOI analysis.

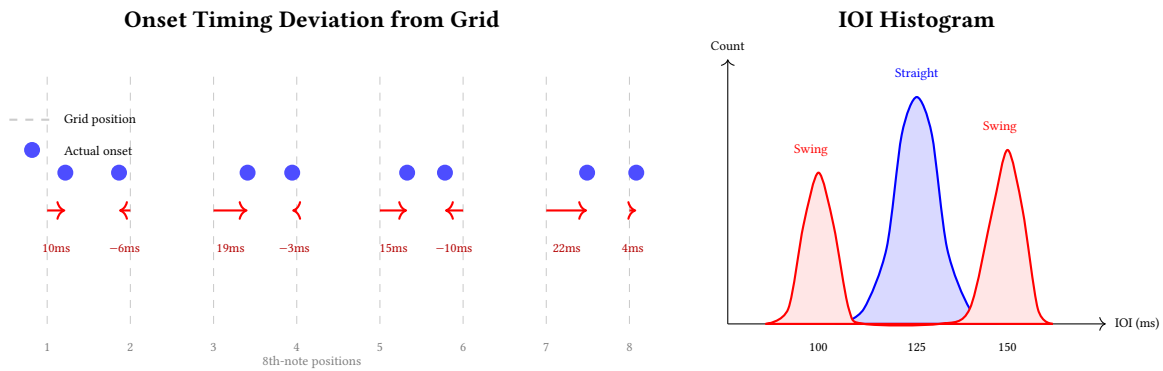


Figure 33. Micro-timing analysis. Left: onset deviations from quantized grid positions, with red arrows and millisecond labels showing how far each hit falls before or after the grid. Right: IOI histogram comparing straight timing (narrow unimodal peak at 125 ms) with swung timing (bimodal distribution with short and long subdivisions).

The RMS energy computation provides a global amplitude envelope:

$$E_{\text{RMS}} = \sqrt{\frac{1}{N} \sum_{i=0}^{N-1} x[i]^2} \quad (11)$$

This energy metric feeds both rhythmic onset detection and the expectation engine and style topology manifold.

7.6 Velocity and Ghost Note Analysis

Beyond *when* a drum hit occurs, *how hard* it’s struck matters equally to the musical feel. A snare at full velocity (a loud “crack”) feels different from a ghost note—a barely-there tap at 10–20% velocity that adds subtle texture between the main beats. Ghost notes are the secret ingredient in funk drumming: the hi-hat and snare patterns on James Brown’s “Funky Drummer” (performed by Clyde Stubblefield) contain dozens of ghost notes between the main backbeats, creating the intricate, “busy but relaxed” feel that defined funk and later became the most sampled drum break in hip-hop history.

Music Brain estimates velocity from the RMS energy of each detected onset within the isolated drum stem. Normalizing onset energy to the loudest hit in a bar produces a velocity profile that reveals accent patterns, ghost notes, and dynamic variation:

- **Ghost notes:** hits below 25% of peak velocity, typically on the “e” and “ah” subdivisions between main beats.
- **Accents:** hits above 80% of peak velocity, marking primary rhythmic emphasis points.
- **Dynamic range:** the ratio between loudest and softest hits. A range above 4:1 indicates expressive human drumming; below 2:1 suggests machine-quantized or heavily compressed drums.
- **Velocity contour:** tracking accent placement across bars reveals patterns like the “crescendo fill” (velocity rising over 2–4 beats into a chorus) or the “pull-back” (velocity dropping in a pre-chorus for contrast).

7.7 Rhythmic Pattern Recognition

Beyond standard four-on-the-floor and backbeat patterns, Music Brain recognises distinctive rhythmic signatures that characterise specific genres and eras:

- **Gallop:** the eighth–sixteenth–sixteenth pattern (or its reverse) that drives Iron Maiden’s signature sound (“The Trooper”), common in metal and punk. Detected as a recurring 3:1 IOI ratio on the kick or bass.
- **Shuffle:** alternating long–short eighth notes, from Stevie Ray Vaughan’s light blues swing to AC/DC’s heavy shuffle in “You Shook Me All Night Long.” Detected via swing ratio analysis.
- **Half-time feel:** the snare moves from beats 2 and 4 to beat 3 only, halving the perceived tempo. Used in Billie Eilish’s productions and modern trap to create a “slow and heavy” feel despite a fast BPM.
- **Syncopation:** accents on off-beats rather than downbeats—the rhythmic engine of funk (James Brown), reggae (Bob Marley), and bossa nova (Antônio Carlos Jobim). Quantified as the percentage of onset energy falling on off-grid positions.
- **Polyrhythm:** two or more simultaneous rhythmic patterns (e.g., 3-against-4), common in progressive rock (Tool’s “Lateralus”) and Afrobeat (Fela Kuti). Detected via multi-period autocorrelation peaks.

7.8 Instrument Identification and Classification

With stems isolated, Music Brain goes beyond labelling a stem as “guitar”—it identifies the *type* of instrument and its configuration from spectral signatures:

- **Acoustic vs. electric guitar:** acoustic guitars show a body resonance peak around 100–200 Hz and a natural high-frequency roll-off above 5 kHz. Electric guitars (especially distorted) have a flatter, wider spectral envelope with energy above 10 kHz.
- **6-string vs. 12-string guitar:** a 12-string guitar produces paired octave strings that create a “chorus” effect visible as closely-spaced spectral peaks. Spectral coherence between paired partials distinguishes it from a standard 6-string. The Beatles’ “A Hard Day’s Night” and the Byrds’ “Mr. Tambourine Man” are classic examples.
- **Guitar amplifier character:** clean (Fender-style: strong fundamental, mild harmonics), crunch (Marshall-style: moderate odd-harmonic distortion), and high-gain (Mesa Boogie-style: dense even and odd harmonics filling the spectrum). Classified via spectral flatness and the odd-to-even harmonic ratio.
- **Bass guitar vs. synth bass:** bass guitar has natural harmonic decay and fret noise transients; synth bass has sustained notes with synthesized harmonics and no fret artifacts.
- **Drum kit identification:** snare tuning (tight jazz snare vs. deep rock snare), kick character (punchy pop kick vs. boomy hip-hop 808), and cymbal type (bright ride vs. dark crash) are classified from the isolated drum stem.

These classifications feed the era detection system (Section 17)—a 12-string Rickenbacker with jangly reverb strongly marks the 1960s, while a heavily processed 808 kick is characteristic of 2010s trap.

8 Timbral Forensics

Every instrument has a spectral “fingerprint” shaped by its construction and signal chain. A Fender Stratocaster through a clean amp sounds different from a Gibson Les Paul through a cranked Marshall—and these differences are measurable. Timbral forensics extracts instrument-specific characteristics from isolated stems, deducing physical properties (snare tuning, guitar type) and reconstructing the signal chain (amp model, effects pedals, studio reverb). Here Music Brain moves from “what notes are playing” to “what *instruments* are playing them, and how are they processed?”

8.1 Snare Tuning via YIN f_0 Estimation

The fundamental frequency of snare drum transients reveals tuning, which varies across genres. The YIN algorithm [3] provides robust f_0 estimation by computing a difference function and applying cumulative mean normalization:

$$d_t(\tau) = \sum_{j=1}^w (x_j - x_{j+\tau})^2 \quad (12)$$

$$d'_t(\tau) = \begin{cases} 1 & \text{if } \tau = 0 \\ d_t(\tau) / \left[\frac{1}{\tau} \sum_{j=1}^{\tau} d_t(j) \right] & \text{otherwise} \end{cases} \quad (13)$$

The fundamental period is the smallest τ where $d'_t(\tau)$ drops below a threshold (typically 0.1), giving $f_0 = f_s / \tau$.

f_0 Range	Tuning Category	Genre Archetype
< 200 Hz	Deep / Fat	70s Rock, Doom Metal
200–250 Hz	Low-Mid	Heavy Rock, Stoner
250–280 Hz	Standard	Modern Rock, Pop
280–300 Hz	Mid-High	Indie, Alternative
> 300 Hz	High / Crack	Funk, Drum & Bass

Table 5. Expanded snare tuning classification based on fundamental frequency extraction from isolated drum stems.

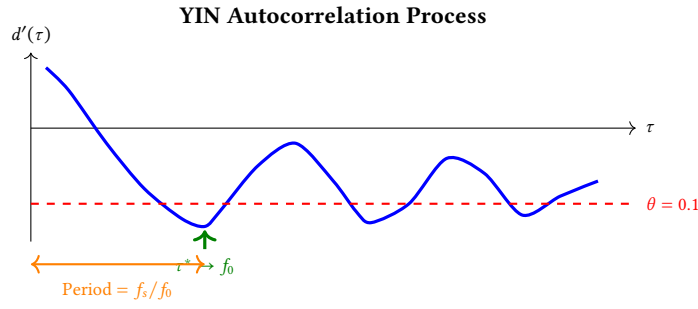


Figure 34. YIN cumulative mean normalized difference function. The fundamental period τ^* is identified as the first point below the threshold, from which $f_0 = f_s/\tau^*$.

8.2 Guitar Effect Recognition

The Metal adapter implements a signal chain deduction system that maps timbral features to hardware configurations:

Distortion Detection. The harmonic-to-fundamental ratio is estimated via spectral flatness: high flatness (> 0.6) with high spectral centroid (> 3500 Hz) indicates heavy saturation, characteristic of fuzz pedals driving solid-state amplifiers (e.g., Randall). Nonlinear distortion produces a dense harmonic overtone series that raises both centroid (added high-frequency energy) and flatness (spectral filling).

Tube Overdrive Detection. High RMS energy (> 0.5) with low centroid (< 2000 Hz) suggests tube amplifier overdrive. Vacuum tube distortion (e.g., 12AX7 preamp tubes) produces predominantly even-order harmonics, perceived as “warm,” shifting the centroid less than solid-state clipping.

Modulation Detection. Periodic spectral “valleys” in the guitar stem indicate modulation effects (phaser, flanger). These are tracked via comb filter detection; the LFO rate can be estimated from the spectral notch periodicity.

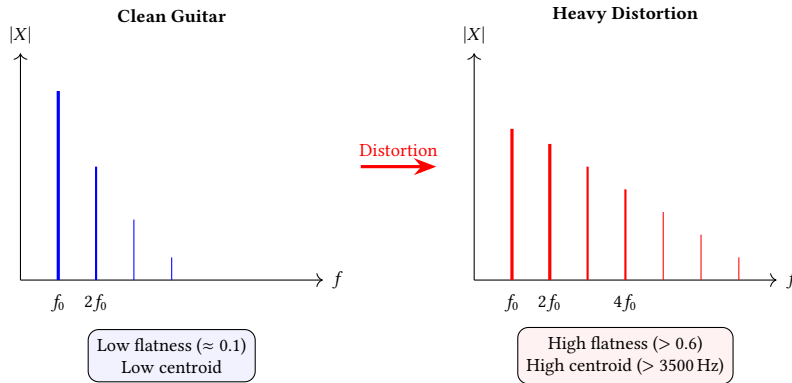


Figure 35. Spectral signature comparison between clean and heavily distorted guitar signals. Distortion fills the spectrum with harmonic overtones, raising both spectral flatness and centroid. The Metal adapter uses these descriptors to deduce the signal chain.

8.3 Vocal Classification (Metal Adapter)

The Metal adapter classifies vocal techniques using a decision tree based on spectral features:

Flatness	Centroid	Classification	Confidence
> 0.4	> 2500 Hz	Fry Scream	0.88
> 0.4	< 1200 Hz	False Chord Growl	0.85
Any	Any (dissonant + RMS > 0.3)	Aggressive/Guttural	0.75
Default	Default	Clean/Melodic	0.70

Table 6. Metal vocal classification decision tree based on spectral flatness and centroid thresholds.

8.4 Signal Chain Deduction

Signal chain deduction maps spectral descriptor combinations to hardware configurations:

Condition	Deduced Chain
Flatness > 0.6, Centroid > 3500 Hz	Fuzz → Solid State (e.g., Randall)
RMS > 0.5, Centroid < 2000 Hz	Tube OD (12AX7) → 4×12 Cab
Default	Crunch / Clean Tone

Table 7. Signal chain deduction rules mapping timbral features to physical hardware configurations.

9 Structural Analysis

Most popular songs follow a recognisable structure: verse, chorus, verse, chorus, bridge, chorus. Understanding this structure reveals *why* a song works—the verse sets up the story, the chorus delivers the emotional payoff, the bridge provides contrast. The Self-Similarity Matrix (SSM) detects this by comparing every moment to every other moment [10]. Similar moments (both choruses) light up on the matrix. The result is a visual fingerprint: the verse-chorus pattern appears as a checkerboard of bright and dark blocks, the bridge breaks the pattern, and section transitions appear as sharp boundaries.

Much of a song’s structural identity comes from *what is absent*. An intro with a lone guitar before the full band enters, a verse stripped to bass and drums to make the chorus hit harder, an outro peeling away instruments until only piano remains—these decisions are the arranger’s primary tool for shaping energy. Equally important are transitions: a drum fill bridging verse to chorus, a cymbal swell signalling a key change, a beat of silence before the drop. These micro-events are often more structurally informative than the sections they connect. By tracking which stems are active and detecting instrument onsets and offsets, Music Brain captures the arrangement’s dynamic contour—not just *where* sections change, but *how*.

9.1 Self-Similarity Matrix Construction

The SSM compares every moment to every other, asking “how similar do these two moments sound?” Bright spots indicate similar moments (both choruses); dark spots indicate contrast. The matrix reveals the architecture of repetition and variation.

The SSM is constructed by computing pairwise Euclidean distances between all feature vectors in the analysis history:

$$d(i, j) = \sqrt{\sum_{k=1}^D (f_i[k] - f_j[k])^2} \quad (14)$$

where f_i is the D -dimensional feature vector at frame i . The distance matrix is inverted and normalized to produce a similarity matrix:

$$\text{SSM}[i, j] = 1 - \frac{d(i, j)}{d_{\max}} \quad (15)$$

where $d_{\max} = \max_{i,j} d(i, j)$. Values fall in $[0, 1]$: 1.0 indicates identical feature vectors (the main diagonal); 0.0 indicates maximum dissimilarity.

Feature Vector Composition. Each frame’s feature vector consists of three normalized components:

1. Spectral centroid / 5000.0 (normalized to $[0, 1]$)
2. Spectral flatness (already in $[0, 1]$)
3. Sensory dissonance (already in $[0, 1]$)

The system maintains a rolling history of 100 frames (configured via `max_history`).

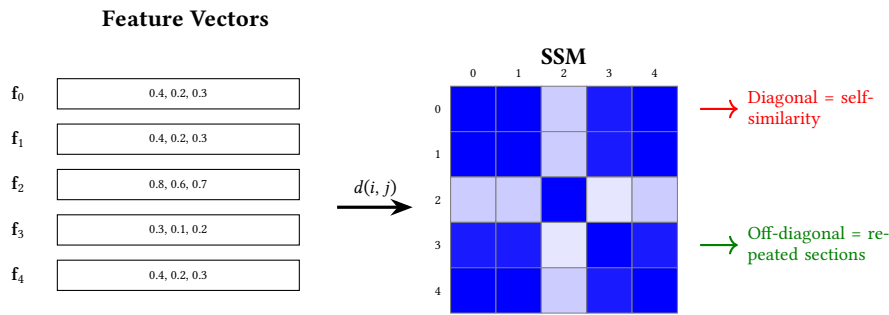


Figure 36. SSM construction process. Feature vectors are compared pairwise using Euclidean distance, then inverted and normalized to produce a similarity matrix. Frames 0, 1, and 4 share similar features, appearing as bright off-diagonal clusters indicating structural repetition.

9.2 Structural Interpretation

The SSM reveals musical structure through characteristic visual patterns:

- **Diagonal blocks:** High self-similarity along the main diagonal indicates stable sections (verse, chorus).
- **Off-diagonal stripes:** Parallel bright lines offset from the diagonal indicate repeated sections (e.g., chorus recurrences).
- **Checkerboard patterns:** Alternating high and low similarity blocks indicate binary form (ABAB structure).
- **Contrast boundaries:** Sharp transitions between blocks indicate section boundaries (verse-to-chorus transitions).

The StructureHeatmap visualization (Section 11.3) renders this matrix as a color-coded grid, with brighter cells indicating higher similarity.

9.3 Arrangement Dynamics

Beyond identifying repeated sections, Music Brain analyzes arrangement *dynamics*: how elements accumulate during build-ups, thin out during breakdowns, and interact to create tension and release. By tracking per-stem RMS energy across section boundaries, the system reconstructs an element accumulation timeline showing which stems are active in each bar. Build-ups show monotonically increasing element count and rising energy, often accompanied by a low-pass filter sweep (a falling spectral centroid that reverses at the drop). Breakdowns show the opposite: sudden stem reduction followed by gradual re-introduction. Figure 37 shows a typical build-up/breakdown timeline, and Figure 38 presents a section-level pattern comparison matrix.

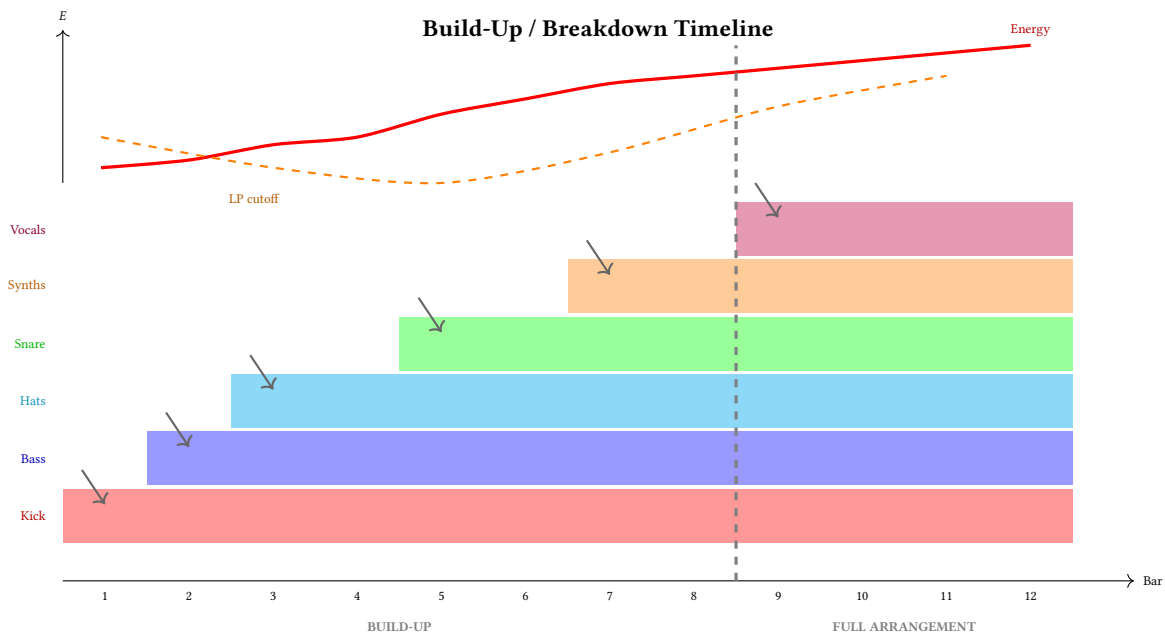


Figure 37. Build-up timeline showing element accumulation across 12 bars. Stems enter progressively (kick → bass → hats → snare → synths → vocals). The red energy curve rises with element density; the dashed orange line shows a low-pass filter sweep (cutoff falling then rising into the drop).

	Intro	Vers	Pre-Ch	Chorus	Bridge	Outro
Kick	sparse	4-floor	synco	4-floor		sparse
Snare		2&4	fill	2&4		2&4
Hi-hat		8ths	16ths	16ths	8ths	8ths
Bass		root	walking	root	pedal	
Chords	Am-F	Am-F-C-G	Am-F-C-G	C-G-Am-F	Dm-G	Am-F

= inactive
 = active (variant labeled)

Figure 38. Section-level pattern comparison matrix. Rows represent rhythmic/harmonic elements; columns represent song sections. Filled cells indicate element presence with pattern variant labels, revealing the arrangement structure at a glance. Empty cells show where elements drop out (e.g., no kick in the Bridge, no bass in the Outro).

10 Musical Expectation and Surprise

Why does the key change in the final chorus of “I Will Always Love You” give you chills? Why does the unexpected chord in Radiohead’s “Creep” hit so hard? These moments violate *musical expectations*—the brain predicts what comes next, and when the prediction fails in a satisfying way, the listener gets an emotional reward. This interplay between expectation and surprise is the core of why music feels good.

Music Brain quantifies this with an expectation engine implementing a computational surprise model based on Huron’s ITPRA framework [11]. By tracking surprise per frame, the system identifies the emotional peaks and turning points that make a song memorable.

10.1 ITPRA Theory

Huron’s theory describes five psychological response stages to musical events:

1. **Imagination:** Pre-event anticipation of future states
2. **Tension:** Physiological preparation for the predicted event
3. **Prediction:** Assessment of whether the prediction was correct
4. **Reaction:** Immediate affective response to the event
5. **Appraisal:** Conscious cognitive evaluation of the event

Musical surprise—and its emotional impact—arises when an event deviates from the listener’s prediction. The system quantifies this deviation computationally.

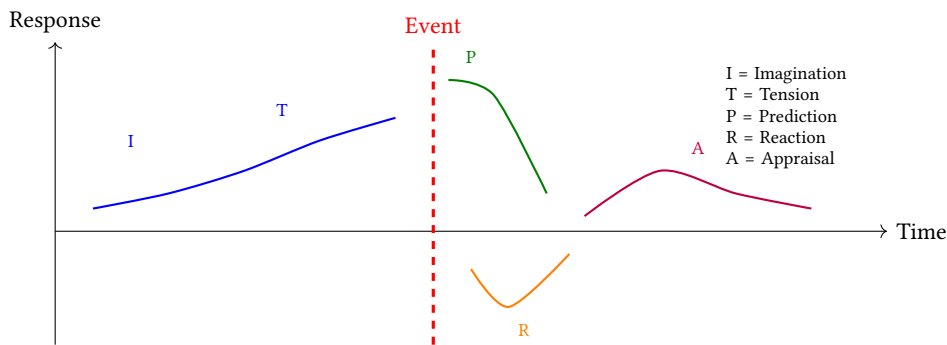


Figure 39. Huron’s ITPRA response timeline. Pre-event Imagination and Tension build expectation; post-event Prediction, Reaction, and Appraisal generate the emotional response. Musical surprise occurs when the Prediction response is large (unexpected event).

10.2 Computational Surprise Model

The implementation computes surprise as the normalized deviation of the current feature composite from its moving average:

Step 1: Feature Composite. A scalar feature value combines RMS energy and normalized spectral centroid:

$$v_t = E_{\text{RMS}} \cdot \min\left(\frac{f_c}{5000}, 1.0\right) \quad (16)$$

Step 2: Expected Value. The expected feature value is the mean over a sliding window of the most recent $N = 20$ frames:

$$\hat{v}_t = \frac{1}{N} \sum_{i=t-N}^{t-1} v_i \quad (17)$$

Step 3: Normalized Surprise.

$$S_t = \min \left(\frac{2 \cdot |v_t - \hat{v}_t|}{\hat{v}_t + \epsilon}, 1.0 \right) \quad (18)$$

where $\epsilon = 10^{-4}$ prevents division by zero. The factor of 2 scales surprise to fill the $[0, 1]$ range for typical musical variations.

This surprise metric drives the Fractal Song Tree visualization (Section 11.4), where high-surprise moments produce complex branching and red coloring, while low-surprise (predictable) passages produce simpler, green-colored trees.

11 Visualization System

The visualization suite uses the `egui` framework [13], an immediate-mode GUI library. Seven views present multi-dimensional analysis data through complementary visual metaphors.

11.1 Multi-Lane Timeline (DAW-Style View)

The timeline view presents a DAW-style arrangement with six horizontal lanes, one for each stem (vocals, drums, bass, guitar, piano, other). Each lane renders a simplified waveform from the stem’s audio buffer, using 1000-sample display windows with a step size of 5 samples.

Interactive **Discovery Knots** can be pinned at arbitrary timeline positions, each carrying a timestamp, label, and DNA summary. Clicking a knot reveals a callout panel displaying the associated analysis metadata.

11.2 Harmonic Constellation (Circle of Fifths)

Chords are the emotional vocabulary of music—major sounds happy, minor sad, diminished tense, jazz extensions sophisticated. Part of the answer lies in geometric relationships on the Circle of Fifths. Simple chords like C major (C–E–G) form compact triangles; complex voicings like Cmaj13#11 form sprawling polygons. The Harmonic Constellation renders these shapes in real time. Watching shapes morph from triangles in a verse to complex polygons in a bridge reveals the arrangement’s harmonic sophistication.

Technically, the constellation renders active pitch classes as highlighted points on a 12-point circle at 30° intervals. Lines drawn between active notes reveal the “geometric shape” of the current chord.

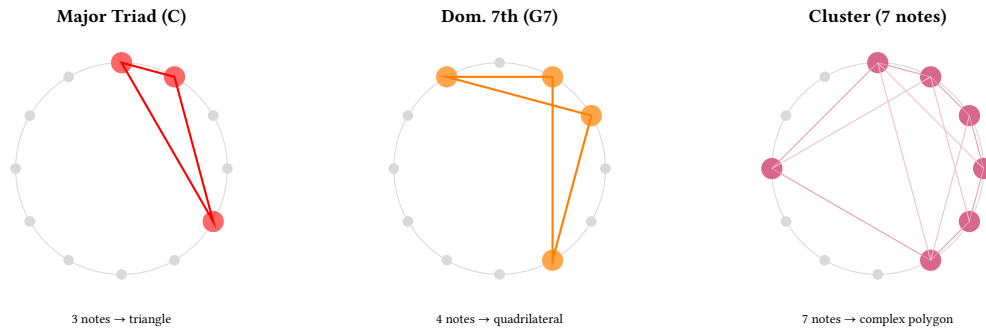


Figure 40. Harmonic Constellation shapes for three chord complexities. Triads form triangles, seventh chords form quadrilaterals, and extended clusters form complex polygonal webs. The visual complexity directly encodes harmonic density.

11.3 Self-Similarity Heatmap

The SSM heatmap renders the self-similarity matrix as a color-coded $n \times n$ grid. Cell color uses `RGBA(100, 150, 255,  $\alpha$ )` where $\alpha = \text{similarity} \times 255$. High-similarity regions appear as bright blue clusters, while dissimilar regions fade to transparent. The axes represent time in both dimensions, with the main diagonal always appearing as a bright stripe.

Annotated Self-Similarity Matrix

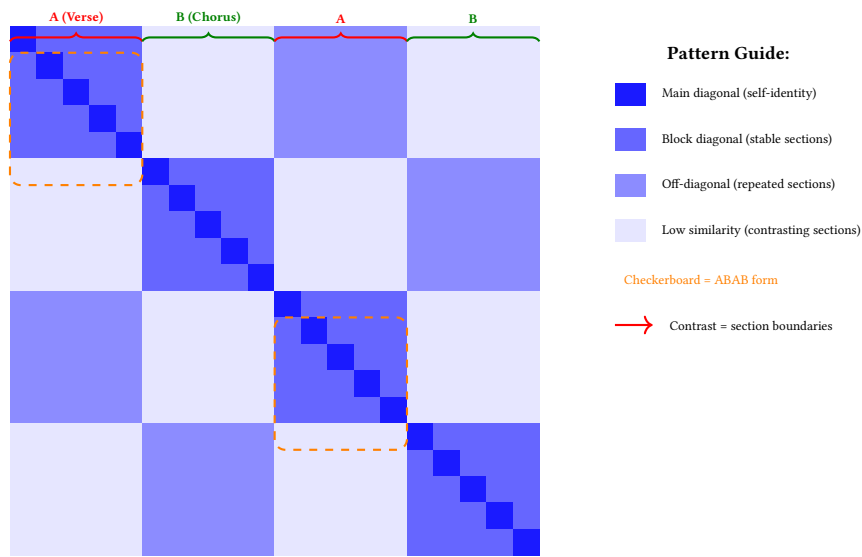


Figure 41. Annotated Self-Similarity Matrix for an ABAB song form (Verse–Chorus–Verse–Chorus). Bright diagonal blocks indicate internally consistent sections. Off-diagonal bright blocks reveal structural repetition (Verse 1 $\approx$ Verse 2). The checkerboard pattern encodes the alternating binary form. Section boundary transitions appear as sharp contrast edges between blocks.

11.4 Fractal Song Tree

The Fractal Song Tree summarises how “predictable” or “surprising” a song is at any moment. During a predictable, repetitive section (a standard four-chord verse), the tree is simple and symmetrical. During a harmonically adventurous section (an unexpected modulation, a complex jazz bridge), the tree explodes into dense, wild branches. Visual complexity directly encodes musical complexity, giving an instant “simple or sophisticated?” reading. Comparing tree shapes across a song reveals the structural arc: simple verse → complex bridge → triumphant chorus.

The tree is driven by the surprise/entropy metric from the expectation engine.

Branching Rules:

- **Trunk length:** $40 + (E_{\text{RMS}} \times 30)$ pixels, driven by energy
- **Active depth:** $\text{base_depth} + \lfloor \text{entropy} \times 3 \rfloor$, base depth = 5
- **Branch spread:** $\text{branching_angle} + (\text{entropy} \times 0.2)$, base angle = $\pi/6$
- **Complexity branch:** A third middle branch appears when entropy > 0.6 and depth > 2

Color Coding:

- Entropy > 0.7: Red (high tension, unexpected events)
- Entropy 0.3–0.7: Gold (moderate variation)
- Entropy < 0.3: Green (stable, predictable passages)

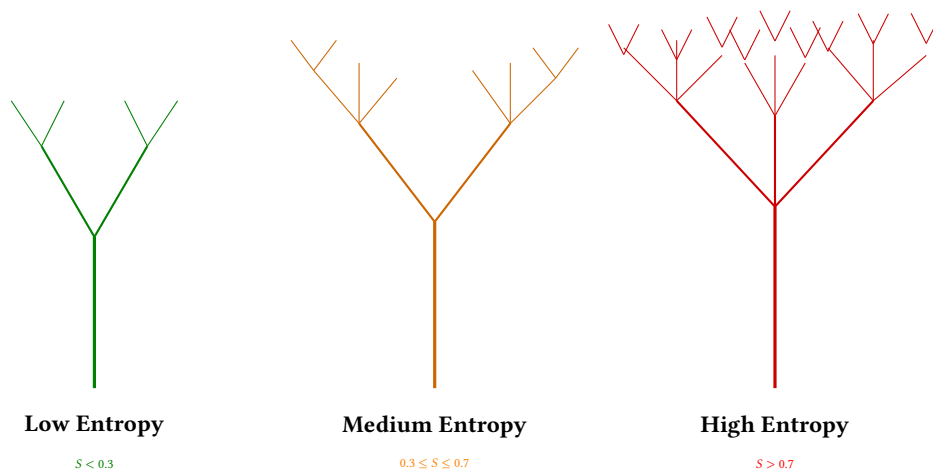


Figure 42. Fractal tree branching rules across entropy levels. Low entropy produces simple binary trees (green, depth 5). Medium entropy adds depth and spread (gold, depth 6–7). High entropy triggers the third complexity branch and maximum depth (red, depth 7–8), visually encoding musical surprise.

11.5 Phase Compass

When the low end feels “tight”—kick and bass punching together as one—that’s phase alignment. When it feels “loose” or “muddy,” the kick and bass arrive at slightly different moments, blurring the low-frequency impact. A Daft Punk record has laser-tight kick-bass sync; a lo-fi garage recording might have deliberate drift.

The Phase Compass makes this audible quality *visible*. It works like a speedometer needle—pointing straight up means the kick and bass hit simultaneously (tight, punchy low end), tilting sideways means they’re drifting apart, and lying flat means they’re completely out of phase. The ring colour reinforces this: green for tight, orange for moderate drift, red for a problem. A verse might have loose timing while the chorus tightens up for maximum impact.

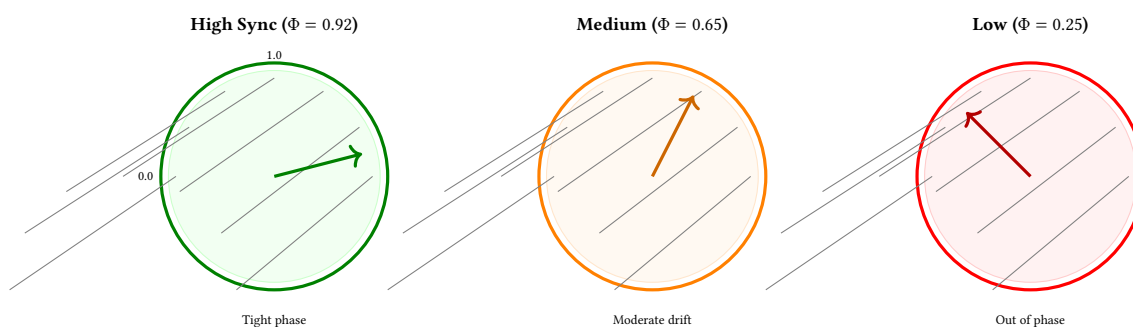


Figure 43. Phase Compass at three sync levels. Left: kick and bass hit within 5 ms of each other—the groove is tight and punchy (typical of tight electronic production). Centre: moderate drift (~15 ms offset)—audible as a slight “flaming” effect where the bass feels detached from the kick. Right: severely out of phase—the low end sounds weak and unfocused, indicating a timing or phase inversion problem that needs fixing.

11.6 Stem Helix

Listeners notice which instruments dominate at any moment—the guitar riff in a rock chorus, the synth pad in an ambient bridge, the vocal in a ballad. The Stem Helix makes this balance *visible*: six concentric rings, one per stem, where brighter rings mean louder instruments.

In Queen’s “Bohemian Rhapsody,” the vocal ring dominates during the operatic section, then guitar and drums blaze during the rock section while vocals pull back. In Billie Eilish’s “Bad Guy,” the bass ring overpowers everything during the verses, reflecting the bass-heavy mix that defined the track’s sonic identity. The Stem Helix answers: “*What dominates the mix right now, and how does that balance change across the song?*”

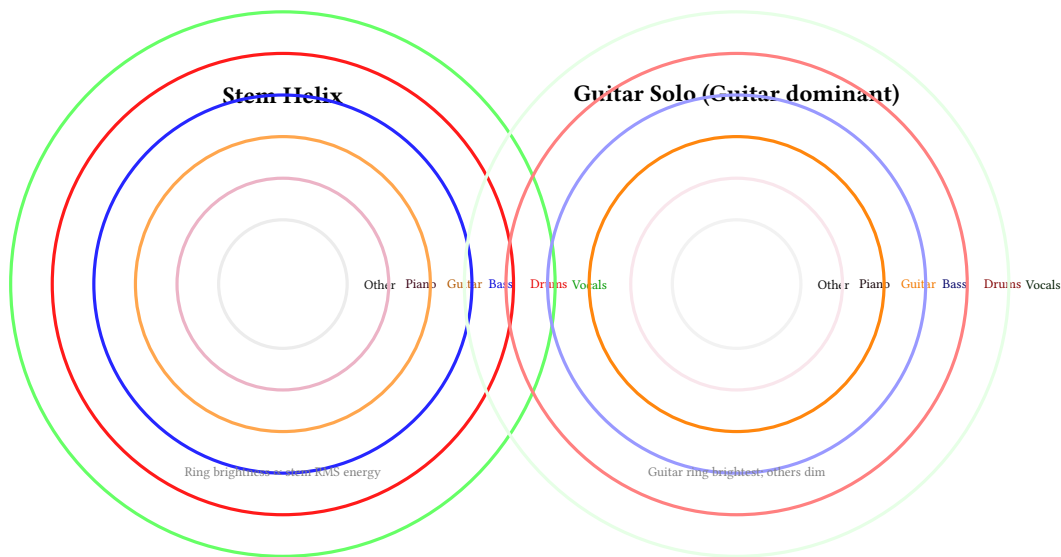


Figure 44. Stem Helix visualization. Left: a balanced mix where all instruments contribute roughly equally—typical of a well-produced verse section. Right: a guitar solo passage where the guitar ring blazes (0.95 energy) while vocals, piano, and other stems nearly vanish. A producer can instantly see that the guitar dominates the mix and decide whether other elements need to be brought up or the guitar pulled back.

11.7 Stem Timeline Ribbons

The Stem Helix shows mix balance at a single instant; a producer needs to see how that balance *evolves* across the song. The Stem Timeline renders each stem’s RMS energy as a horizontal ribbon, stacked vertically, with brightness proportional to loudness. At a glance, you see when each instrument enters and exits, where the arrangement is sparse vs. dense, and how build-ups and breakdowns work—a musical “score” where colour replaces notation.

In Billie Eilish’s “Bad Guy,” the bass ribbon is thick and bright throughout the verse while everything else is nearly invisible—then drums and synth ribbons explode in the chorus. In a typical EDM track, all ribbons thin out during the breakdown, then slam back at the drop. A folk ballad like Simon & Garfunkel’s “The Sound of Silence” shows a gradual widening from solo vocal to full band.

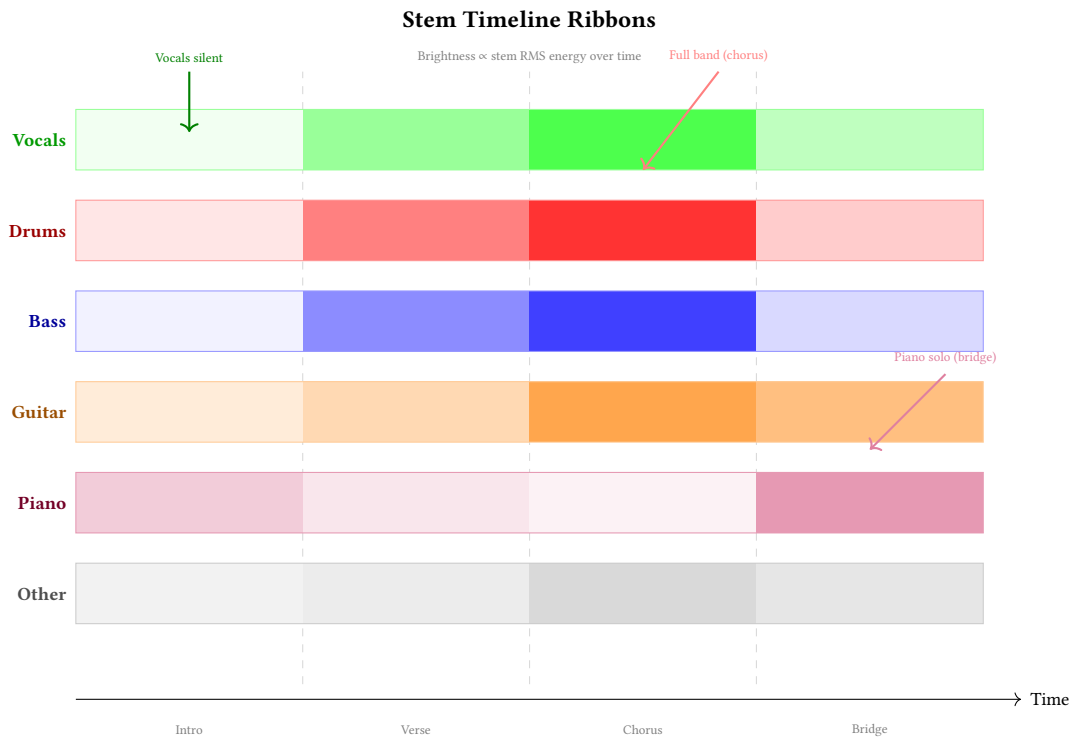


Figure 45. Stem Timeline Ribbons showing per-stem energy across four structural sections. Bright ribbons indicate loud, active stems; dim ribbons indicate silence or low energy. The arrangement arc is immediately visible: sparse intro (only light piano and other), building verse (vocals and rhythm section enter), full chorus (all stems at peak energy), and stripped-back bridge (piano takes over as drums and bass retreat). This view reveals arrangement decisions that are invisible in a standard waveform.

11.8 Stereo Field Map

Where does each instrument sit in the stereo panorama? A well-mixed song places each element at a specific left-right position, creating width and space. Vocals sit dead centre, guitars pan hard left and right, keyboards slightly off-centre, hi-hats to one side. The Stereo Field Map makes this spatial arrangement visible—and shows how panning *changes* across sections.

The Beatles' "Abbey Road" reveals the evolution of stereo mixing: early tracks hard-panned entire instruments left or right, while later productions used the full stereo field. Modern pop and hip-hop often collapse to mono in the verse (intimate, focused) and widen dramatically in the chorus (expansive, powerful)—a technique the Stereo Field Map exposes immediately.

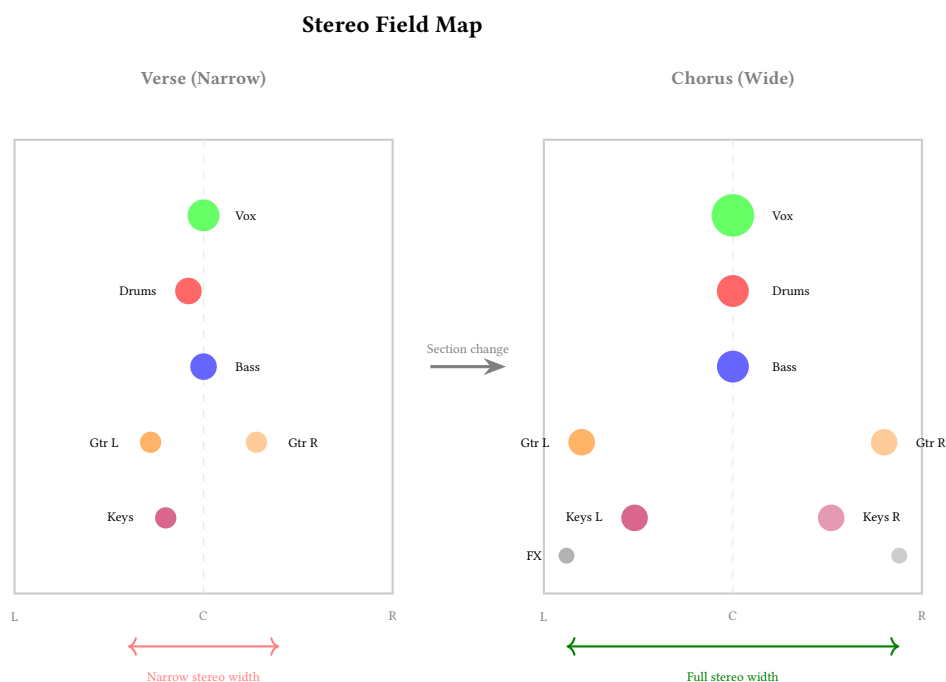


Figure 46. Stereo Field Map comparing the panning layout in a verse (left) vs. a chorus (right). During the verse, instruments cluster near the centre for an intimate feel; in the chorus, guitars and keys spread to the extremes, backing vocals and effects fill the sides, and the overall stereo image widens dramatically. Vocals, bass, and drums typically remain centred in both sections. Circle size indicates relative loudness.

11.9 Frequency Masking Heatmap

Frequency masking occurs when two instruments occupy the same frequency range simultaneously, causing one to “hide” the other. It is the most common mixing problem—and one of the hardest to hear without training. A bass guitar and kick drum both produce energy around 60–100 Hz; without careful EQ treatment, the bass “disappears” behind the kick. A vocal at 2–4 kHz can be obscured by a distorted guitar in the same range.

The Frequency Masking Heatmap plots these collisions as a time-frequency grid where colour intensity indicates overlap between stems. Red zones demand attention; green zones indicate clean separation. By using isolated stems from Demucs rather than the master mix, Music Brain identifies *exactly which two instruments* conflict—information impossible to extract from a mixed signal.

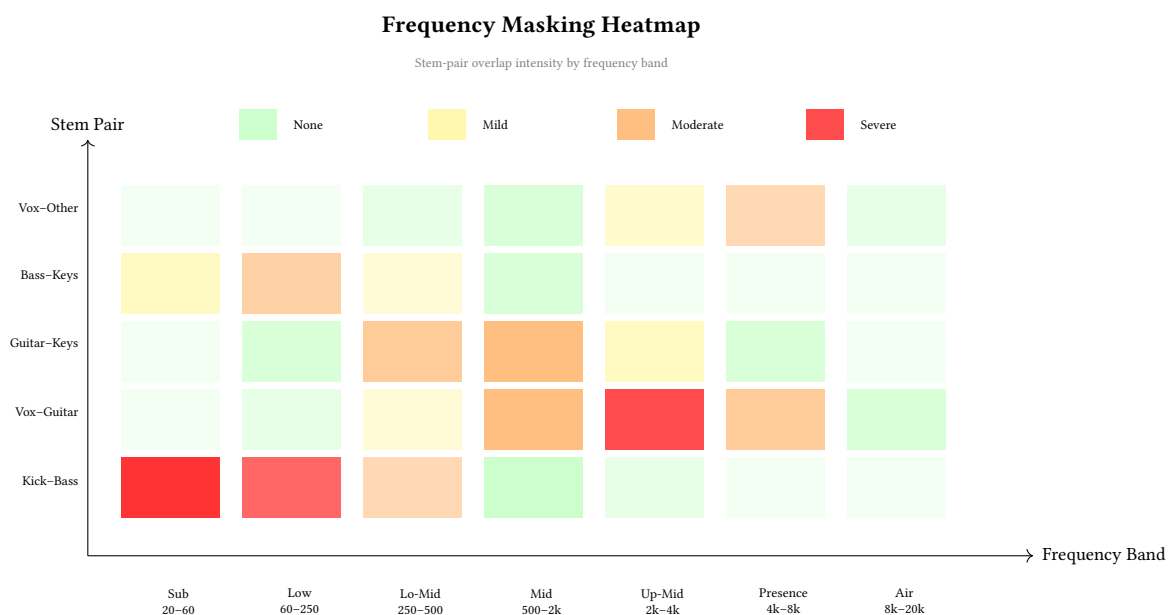


Figure 47. Frequency Masking Heatmap showing stem-pair overlap across seven frequency bands. The kick–bass pair shows severe masking in the Sub and Low bands (a common mixing issue requiring sidechain EQ or compression). The vocal–guitar pair conflicts in the Upper-Mid presence range (2–4 kHz), which can be resolved by carving an EQ notch in the guitar at the vocal’s fundamental frequency. Green cells indicate clean separation—no mixing action needed.

11.10 Arrangement Density Timeline

How many things happen at once? This question reveals one of the most important aspects of production: *arrangement pacing*. The Arrangement Density Timeline plots active stem count and combined energy as a stacked area chart over the song. Peaks represent dense, “full band” sections (choruses, climaxes); valleys represent sparse, intimate moments (intros, breakdowns, bridges).

Great producers manipulate density deliberately. Max Martin’s pop hits follow a formula: strip to one or two elements before the chorus, then slam everything in for maximum contrast. Radiohead’s “OK Computer” alternates between sparse passages and overwhelming walls of sound. EDM producers take this further: the “build-up → breakdown → drop” structure is entirely about density manipulation.

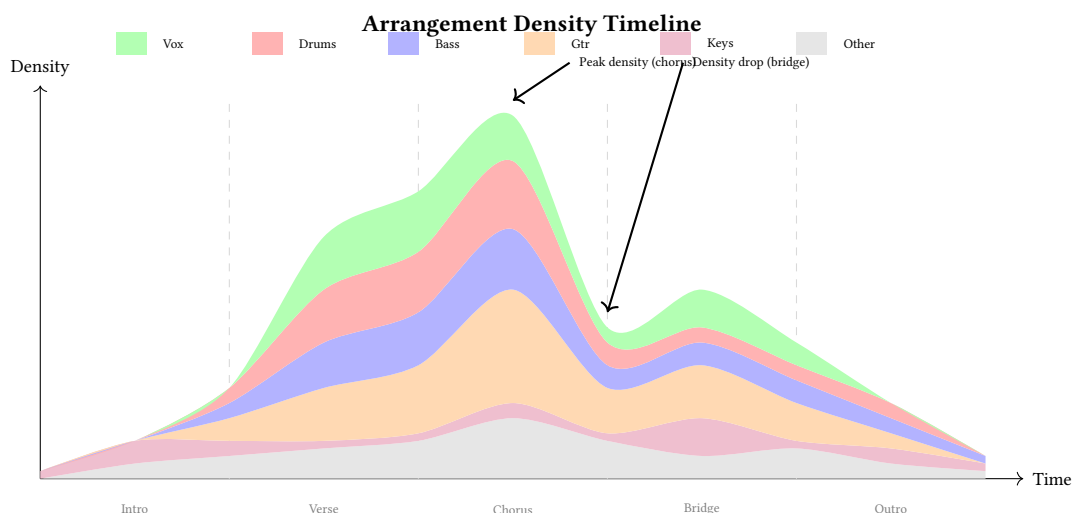


Figure 48. Arrangement Density Timeline as a stacked area chart. Each coloured band represents a stem’s energy contribution over time. The chorus (centre) shows maximum density with all stems active; the bridge (right of centre) shows a dramatic thinning. The visual reveals the “arrangement arc”—how producers build and release sonic density to create emotional contrast.

11.11 Reverb and Spatial Depth Map

A great mix has not just left-right positioning (stereo) but *front-to-back depth*. Some elements feel “right in your face” (dry, close-miked lead vocals, snappy drums) while others feel “far away” (reverb-drenched pads, distant backing vocals). This front-to-back dimension creates a three-dimensional “stage” that immerses the listener.

The Reverb and Spatial Depth Map estimates each stem’s perceived depth from the ratio of direct-to-reverberant energy. A dry signal (high ratio) appears close; a wet signal (low ratio) appears distant. Tracking this across sections reveals production techniques like “pulling the vocal forward in the chorus” (reducing reverb) or “pushing the guitar back in the verse” (adding reverb)—techniques that shape how intimate or expansive the song feels at each moment.

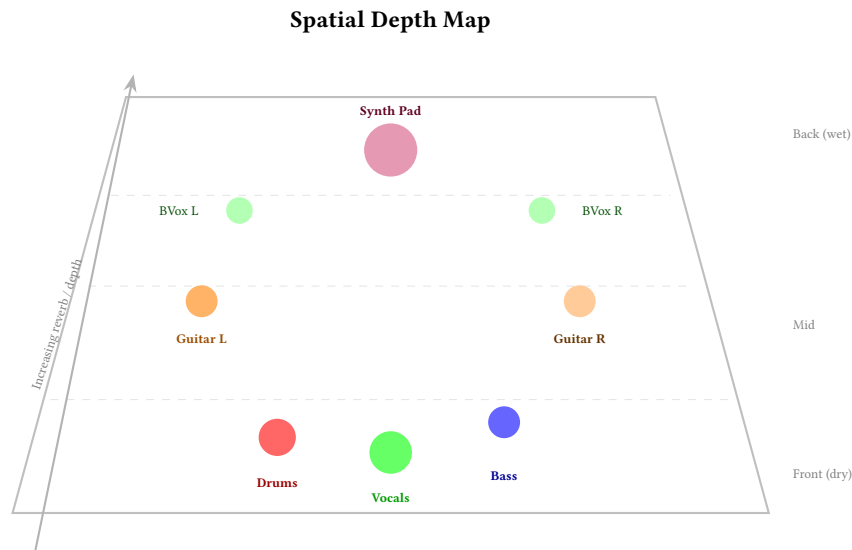


Figure 49. Spatial Depth Map showing the perceived front-to-back positioning of each stem, estimated from the direct-to-reverberant energy ratio. Dry elements (vocals, drums, bass) appear at the front of the “stage”; wet elements (synth pads, backing vocals) recede into the background. Circle size indicates loudness. A mix engineer uses this view to ensure the lead vocal sits clearly in front and that no element is fighting for the same depth position—just as the Stereo Field Map prevents left-right collisions.

11.12 Call-and-Response Detection

One of the oldest techniques in music is *call and response*: one instrument “calls” with a phrase, another “answers.” It is the foundation of blues (voice calls, guitar answers), gospel (preacher calls, congregation responds), jazz (soloist states a theme, rhythm section echoes), and orchestral music (strings call, woodwinds respond). Call-and-response creates a conversational quality that makes arrangements feel alive rather than static.

Music Brain detects these patterns by analysing temporal interleaving of stem activity. When one stem’s energy peaks immediately after another’s drops (within a short latency window), and this alternation repeats across multiple phrases, the system flags a call-and-response pair. The visualization highlights these moments on the timeline, connecting the calling and responding stems with arched lines.

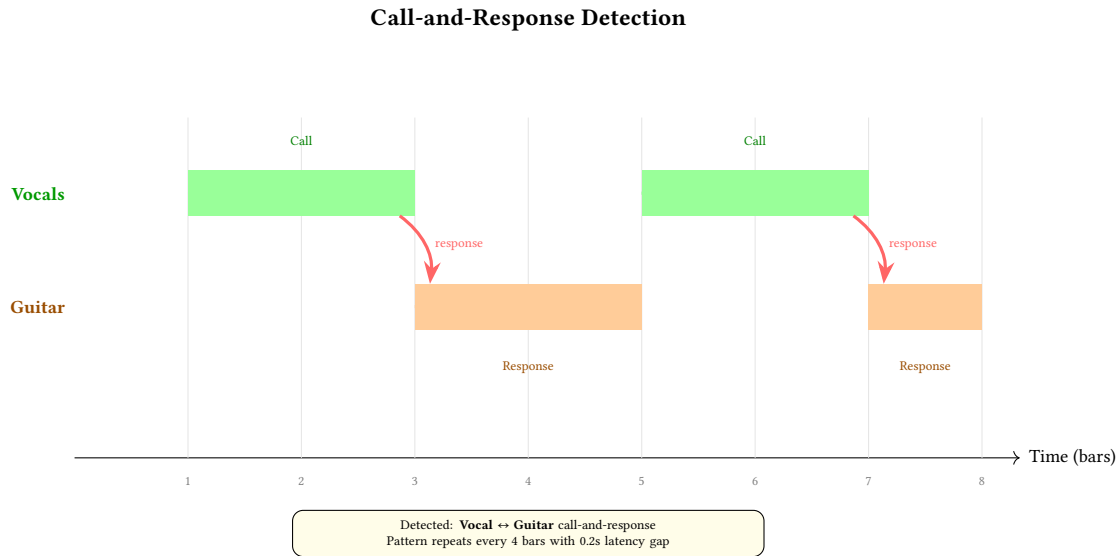


Figure 50. Call-and-response detection between vocals and guitar. The vocal “calls” in bars 1–2 and 5–6, and the guitar “responds” in bars 3–4 and 7–8. Red arches mark the transitions. This conversational interplay—common in blues, rock, and gospel—is detected by analysing temporal interleaving of stem energy peaks.

11.13 Groove Feel and Micro-Timing Grid

Two performances of the same song at the same tempo can feel completely different: one stiff and mechanical, the other alive and groovy. The difference lies in *micro-timing*—tiny deviations from the mathematical beat grid, measured in milliseconds, too small to hear as “wrong” but large enough to change the music’s entire feel.

A drummer who plays slightly *behind* the beat creates a laid-back feel (common in reggae, neo-soul, and J Dilla-style hip-hop). A drummer who pushes *ahead* creates urgency and drive (punk rock, thrash metal). A perfectly quantized performance—every note on the grid—feels robotic (sometimes intentional in electronic music, often undesirable). The Groove Feel visualization plots each note’s deviation from the beat grid, revealing the micro-timing “fingerprint” that defines a performance’s feel.

Groove Feel: Micro-Timing Deviations

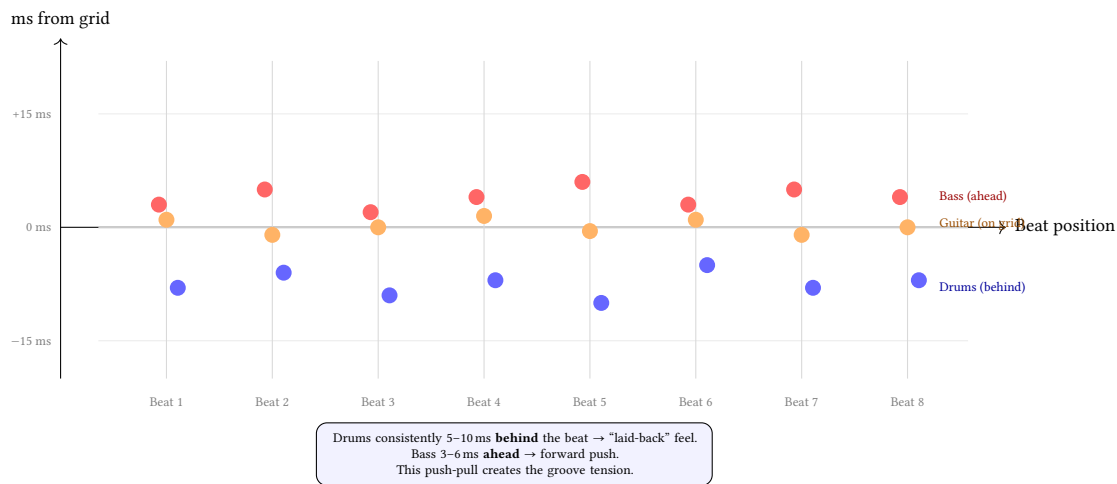


Figure 51. Groove Feel visualization showing micro-timing deviations from the beat grid for three stems. The drums (blue) consistently land 5–10 ms behind the beat, creating a laid-back feel. The bass (red) pushes slightly ahead, creating forward momentum. The tension between these two—one pulling back, one pushing forward—is the “groove” that makes the performance feel alive. A quantized electronic track would show all dots on the zero line.

11.14 Harmonic Tension Arc

Every great song tells an emotional story through tension and release. The Harmonic Tension Arc distills this narrative into a single curve: the horizontal axis is time, the vertical axis is harmonic tension (computed from sensory dissonance, chord complexity, deviation from the tonic key, and unresolved dominants). Peaks mark maximum tension—the agonizing suspended chord before a resolution, the dissonant cluster in a Radiohead climax, the diminished chord in a film score. Valleys mark resolution—the return to the tonic after a dramatic bridge, the final cadence of a symphony.

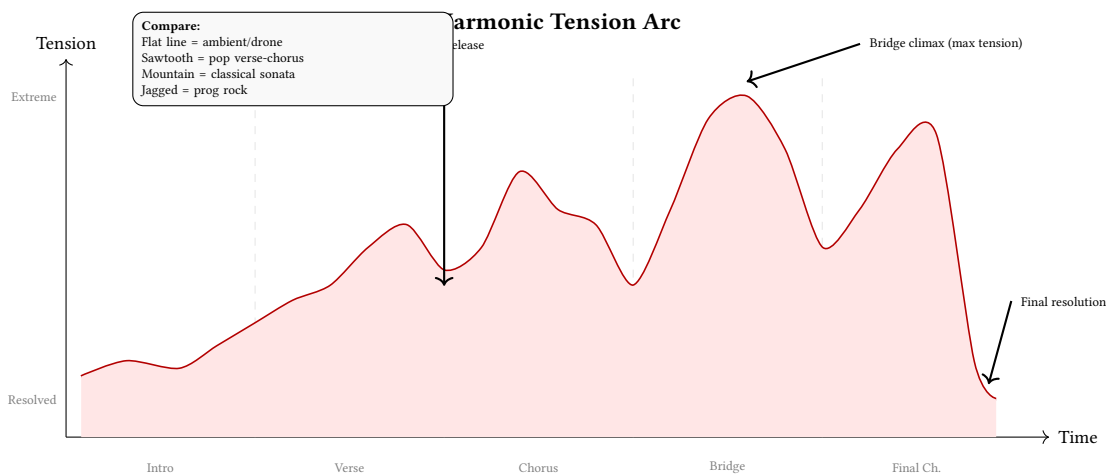


Figure 52. Harmonic Tension Arc across a song. Tension rises through the verse, partially releases in the chorus, peaks dramatically in the bridge (perhaps a modulation or chromatic passage), and fully resolves in the final cadence. The shape of this curve—its “narrative arc”—is one of the strongest indicators of musical sophistication and emotional intent.

11.15 Spectral Fingerprint Comparison

Every genre, era, and producer has a characteristic spectral “shape”—an average frequency balance that defines the sound. 1970s rock has a warm mid-range hump; modern pop is scooped in the mids with boosted highs and lows; lo-fi hip-hop rolls off sharply above 8 kHz. The Spectral Fingerprint Comparison overlays the current track’s average spectral profile against a reference (a specific track, a genre average, or a “target” curve), revealing where the frequency balance deviates.

Mix engineers call this “matching the curve”—one of the most practical mastering tools. If a rock mix sounds “thin” compared to a reference, the fingerprint might reveal a 3 dB deficit at 100–200 Hz. If it sounds “harsh,” a peak at 3 kHz that the reference lacks.

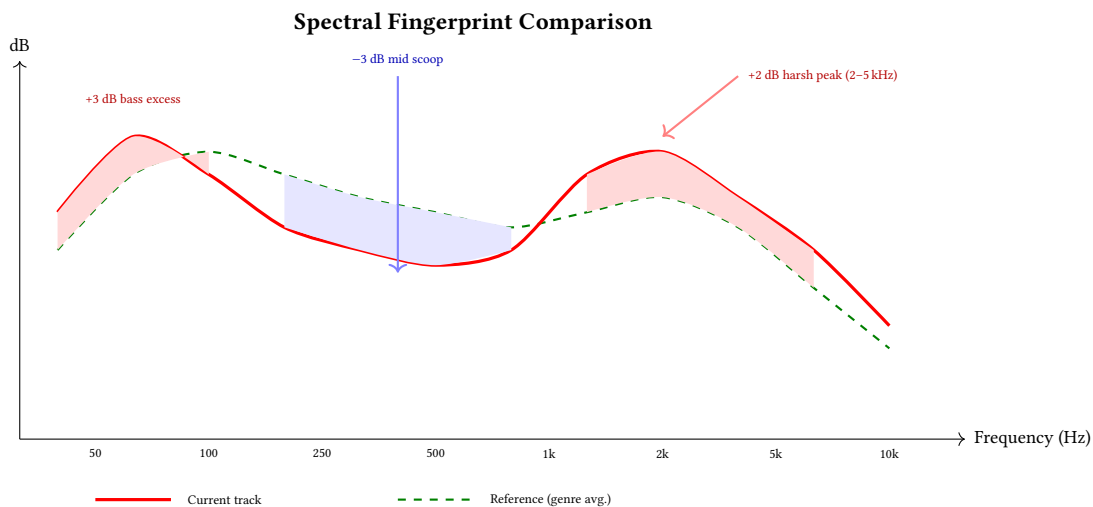


Figure 53. Spectral Fingerprint Comparison overlaying the current track’s average spectral shape (red) against a genre reference (green dashed). Red-shaded areas show where the track exceeds the reference (excess bass, harsh upper-mids); blue-shaded areas show deficits (scooped mids). This immediately suggests corrective EQ: reduce the bass shelf, cut at 3 kHz, and boost 500 Hz–1 kHz to match the reference curve.

11.16 Sidechain and Pumping Detector

The rhythmic “pumping” effect—where a synth pad or bass ducks in time with the kick drum—defines modern electronic and pop production. Sidechain compression (routing the kick’s signal to control the compressor on another track) creates a breathing, pulsing quality that locks every element to the groove. Daft Punk’s “One More Time” made it iconic; it now appears in everything from Billie Eilish to The Weeknd.

The Sidechain and Pumping Detector identifies this effect by looking for periodic volume dips in one stem that correlate with onset peaks in another (typically the kick). By cross-correlating the kick onset function with each stem’s energy envelope, the system detects which elements are sidechained and measures the ducking depth and recovery time.

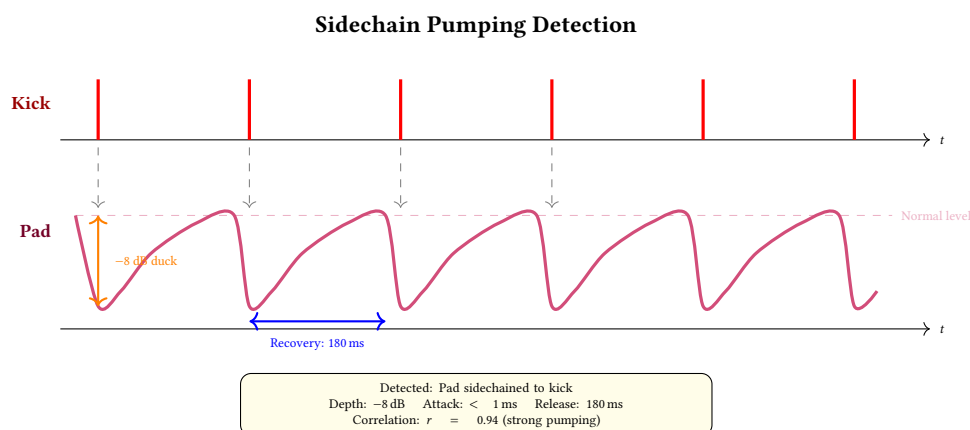


Figure 54. Sidechain pumping detection. The synth pad’s energy envelope (purple) dips sharply each time the kick hits (red), then recovers over approximately 180 ms. This characteristic “breathing” pattern is detected by cross-correlating the kick onset function with the pad’s amplitude envelope. The system reports ducking depth, attack/release times, and correlation strength.

11.17 Vocal Formant Tracker

The human voice is the most expressive instrument in music, and much of that expressiveness comes from *formants*—the resonant frequencies of the vocal tract that determine vowel quality and colour. When a singer opens their mouth wider, the first formant (F1) rises; when they shape their lips, the second formant (F2) shifts. Tracking these formants over time reveals vocal technique changes invisible in a standard waveform: chest-to-head voice shifts, vibrato onset, vocal fry, and the articulation patterns that define a singer’s style.

Beyoncé’s melismatic runs show rapid formant trajectories as she sweeps through vowels at high speed. Freddie Mercury’s control produces smooth, wide-ranging formant paths. Amy Winehouse’s jazzy phrasing creates a distinctive formant “wobble” that defines her vocal character.

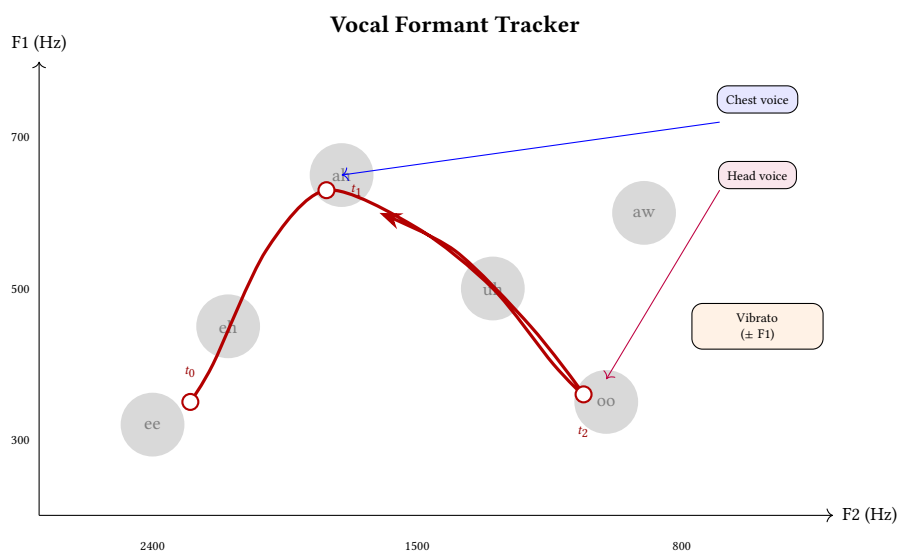


Figure 55. Vocal Formant Tracker plotting the singer’s trajectory through the F1–F2 vowel space. Grey circles mark canonical vowel positions. The red path shows the vocal performance moving from a closed “ee” vowel (t_0) through an open “ah” (t_1) to a rounded “oo” (t_2), with technique annotations marking chest-to-head voice transitions and vibrato. This reveals articulatory technique that is impossible to see in a standard waveform.

11.18 Production Technique Timeline

Modern production employs dozens of effects that shape sound beyond notes and chords. Filter sweeps build anticipation before a drop. Reverse reverb tails create eerie transitions. White noise risers signal something big. Tape saturation adds warmth. Delay throws on a single vocal word create spaciousness.

The Production Technique Timeline annotates these events as labelled markers along the time axis—a “director’s commentary” for the production. Each technique is detected by characteristic signal features: a filter sweep appears as a rising spectral centroid; reverse reverb shows as a crescendo into a transient (the opposite of normal reverb); sidechain pumping is detected by the Pumping Detector (Section 54).

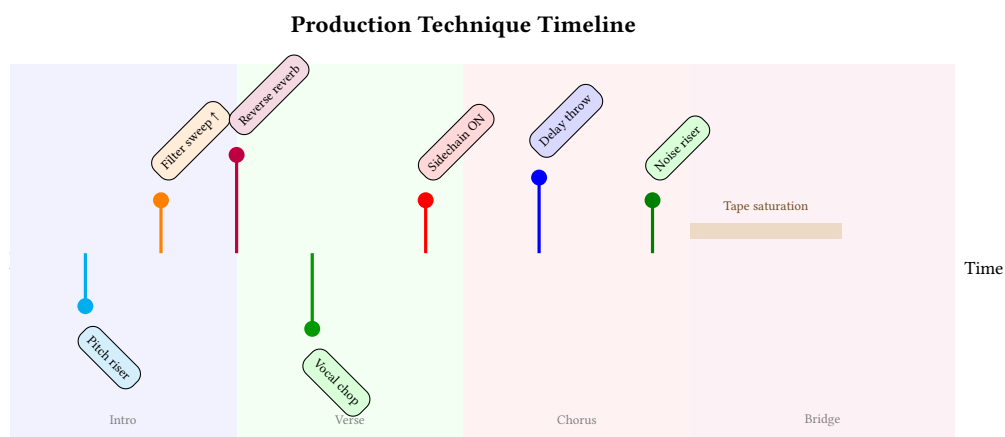


Figure 56. Production Technique Timeline annotating detected production effects along the song duration. Each marker indicates when a specific technique was detected: filter sweeps (rising centroid), reverse reverb (crescendo into transient), sidechain activation, delay throws (sudden reverb tail), white noise risers (broadband energy ramp), and sustained effects like tape saturation. This view gives producers and students a “behind the scenes” look at the production toolkit being used.

11.19 Bass Contour and Root Motion

The bass line is the harmonic and rhythmic foundation of nearly all popular music, yet often the least analysed element. The Bass Contour visualization extracts the bass stem’s pitch contour as a continuous melodic line, plotted against the chord chart above it. This reveals the bass player’s strategy: is the bass playing root notes (following chord changes slavishly), or doing something more interesting?

A root-note bass line (common in pop and punk) jumps in lockstep with chord changes. A walking bass line (jazz) produces a smooth, stepwise contour connecting chords with passing tones. A pedal bass (one note held while chords change above) produces a flat line—creating tension against the moving harmony. James Jamerson’s Motown bass lines show a melodic, independent contour that sometimes moves *against* the chord changes. Flea’s bass lines (Red Hot Chili Peppers) show funky syncopated patterns with wide interval leaps.

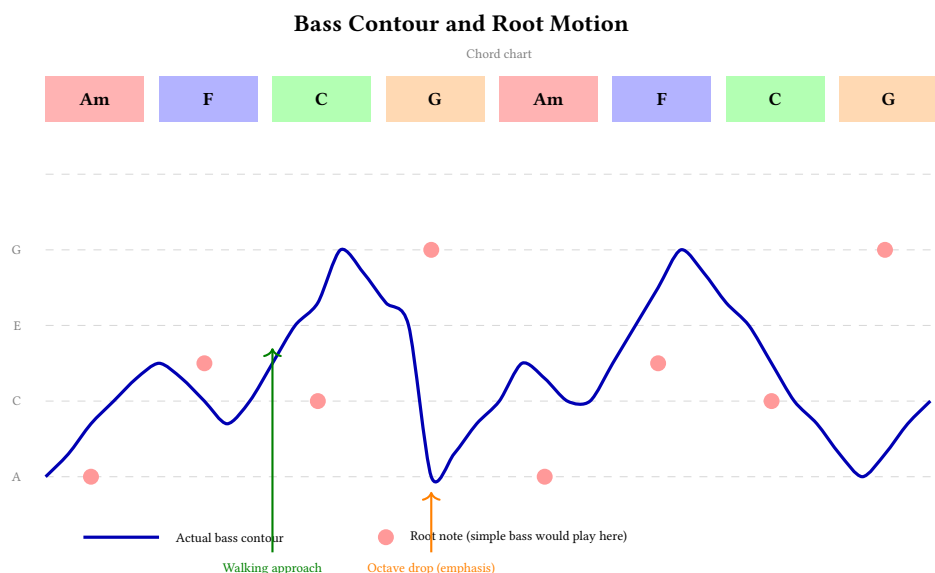


Figure 57. Bass Contour and Root Motion showing the bass line’s melodic shape (blue) against the chord chart (top) and root note positions (red dots). This bass line walks melodically between chord tones rather than simply playing roots, using stepwise approach notes and an octave drop for emphasis. The gap between the blue contour and the red dots measures the bass line’s melodic independence from the harmonic framework.

11.20 Melodic Voice Detection

At any moment in a song, one instrument carries the *melody*—the line your ear follows, the part you’d sing along to. During a verse it’s the vocal; during a guitar solo the lead guitar takes over; in an orchestral passage it shifts between violins and oboe. Melodic Voice Detection identifies which stem carries the primary melodic line at each moment, plotting that melody as a pitch contour annotated with the responsible instrument.

This reveals arrangement decisions invisible to casual listeners. In the Beatles’ “While My Guitar Gently Weeps,” the melodic voice passes between Harrison’s vocal and Clapton’s guitar—the handoff points are moments of high arrangement craft. In a Stevie Wonder recording, the vocal melody is often doubled by a synth in unison, thickening the line without obscuring lyrics. In jazz, the melody passes between instruments during “trading fours” (four-bar exchanges between soloists).

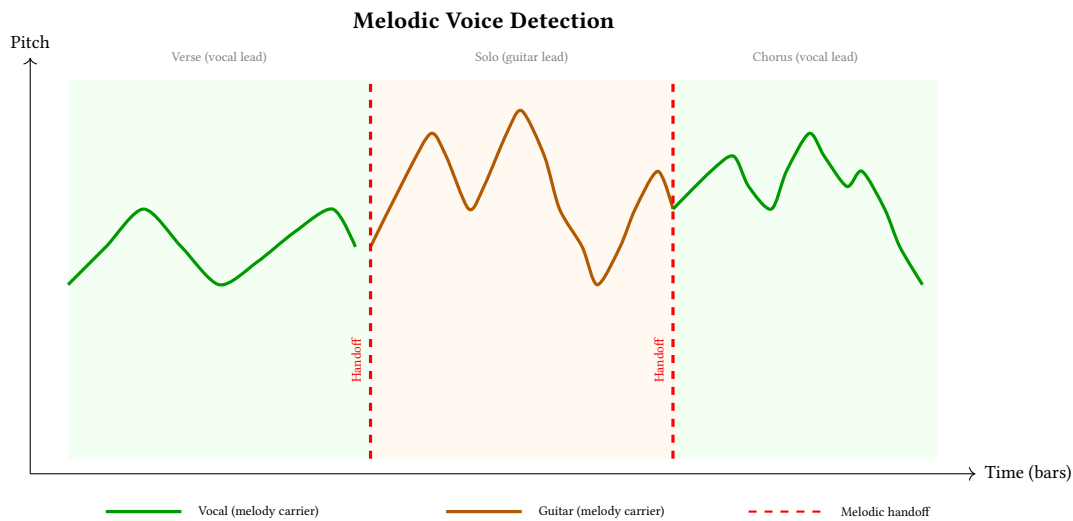


Figure 58. Melodic Voice Detection showing which stem carries the primary melody across three sections. The vocal (green) leads during the verse, hands off to the guitar (orange) for a solo, then reclaims the melody in the chorus. Red dashed lines mark handoff points—moments where arrangement craft determines which instrument the listener’s ear follows.

11.21 Inter-Stem Harmonic Relationships

One of the most revealing aspects of an arrangement is how instruments relate harmonically at any moment. Are bass and guitar in *unison*? Is the bass playing *roots and fifths* while guitar plays the full chord? Is the bass *walking* independently? Are two instruments in *parallel thirds* or *octaves*? These relationships define the arrangement’s texture and density, and they change across sections.

Music Brain detects inter-stem relationships by comparing pitch contours and chroma profiles of each stem pair, classifying each into categories:

- **Unison:** Both stems play the same pitch (within ± 1 semitone). Creates power and thickness—common in metal and pop doubling.
- **Octave doubling:** Same pitch class, different octaves. Adds fullness without muddying—a staple of orchestral writing and rock power chords.
- **Root-fifth:** Bass plays the root, another instrument plays the fifth. The most basic harmonic scaffolding, common in pop, punk, and country.
- **Parallel motion:** Both stems move in the same direction by the same interval (thirds, sixths, or tenths). Used for harmonized melodies—the Allman Brothers’ twin guitars, Crosby, Stills & Nash’s vocal harmonies.
- **Contrary motion:** Stems move in opposite directions. Creates tension and independence—common in counterpoint, jazz, and progressive rock.
- **Independent:** No consistent harmonic relationship. Each stem follows its own melodic logic.

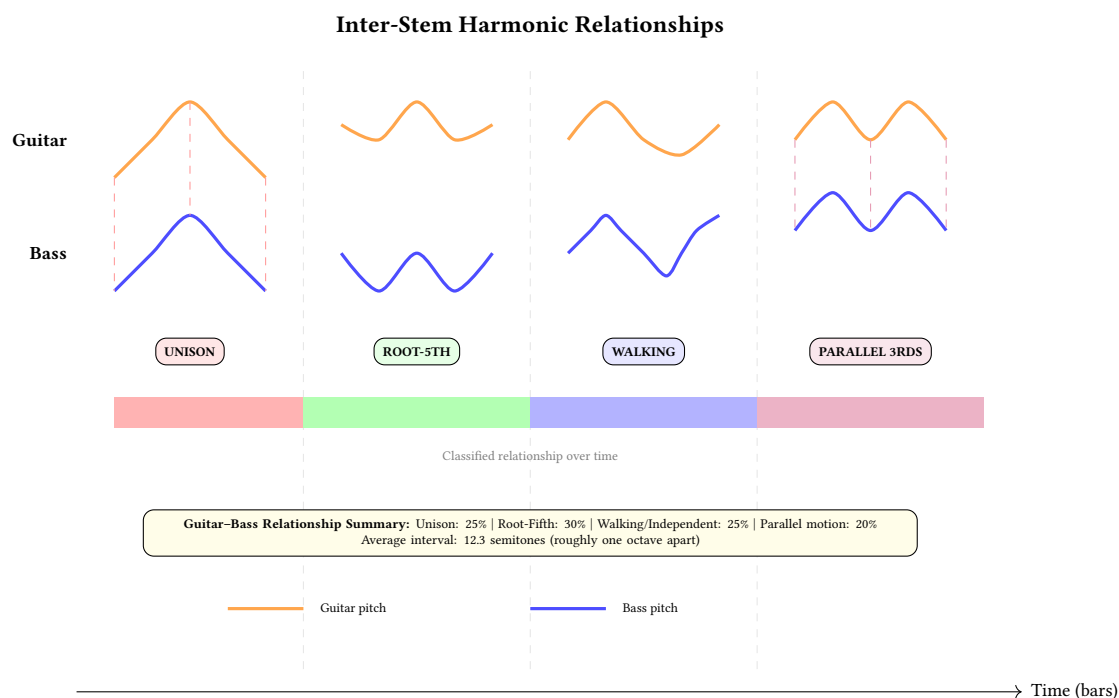


Figure 59. Inter-Stem Harmonic Relationships between guitar (orange) and bass (blue) across four sections. Section 1: **Unison**—both instruments play the same melodic line (red dashes show pitch alignment), creating a thick, powerful sound. Section 2: **Root-Fifth**—bass plays roots while guitar plays full chords, providing harmonic scaffolding. Section 3: **Walking bass**—bass moves independently through passing tones while guitar holds chord shapes. Section 4: **Parallel thirds**—both move in the same direction, a fixed interval apart (purple dashes), creating harmonized motion. The colour-coded bar at bottom shows the classified relationship over time.

11.22 Dynamic Range Archaeology

The “loudness war”—the multi-decade trend of mastering at ever-increasing loudness, sacrificing dynamic range for perceived volume—is one of the most discussed topics in audio engineering. Dynamic Range Archaeology makes this phenomenon visible by showing the dynamic range (difference between loudest and quietest moments) of each stem and the master mix, both within sections and across the full track.

A 1975 Led Zeppelin master has 12–15 dB of dynamic range: quiet passages are genuinely quiet, loud passages hit hard. A 2010 pop master might have only 4–6 dB: everything loud, compressed into a flat “sausage” waveform. Per-stem analysis reveals *where* compression is applied: is the vocal compressed individually (common, acceptable) or is the entire mix crushed at mastering (the loudness war’s damage)?

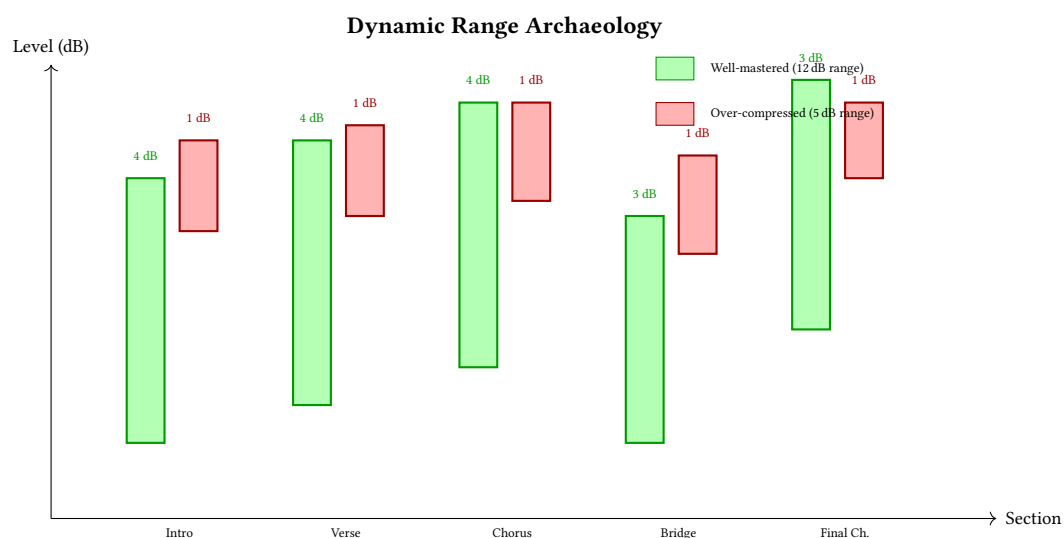


Figure 60. Dynamic Range Archaeology comparing a well-mastered track (green, wide bars) with an over-compressed track (red, narrow bars) across five structural sections. The well-mastered track shows 3–4 dB of dynamic variation per section and 12 dB overall range, preserving the contrast between quiet and loud passages. The over-compressed track is “crushed” into a 1–2 dB range per section, sacrificing the emotional arc for perceived loudness.

11.23 Instrument Register Map

When two instruments occupy the same pitch range, they compete for attention—and often mask each other. A guitar playing low power chords in the bass register creates a muddy low end; a synth pad in the vocal's octave obscures lyrics. The Instrument Register Map shows which pitch ranges each stem occupies over time, revealing register collisions and complementary voicing.

Great arrangers use register separation instinctively. In a Beatles arrangement, McCartney's bass sits in the 2nd octave (E2–E3), Lennon's rhythm guitar in the 3rd–4th, Harrison's lead in the 4th–5th, and vocals in the 3rd–4th. Each instrument has its own “lane”—contrast this with a poorly arranged demo where guitar and keys both crowd the same octave.

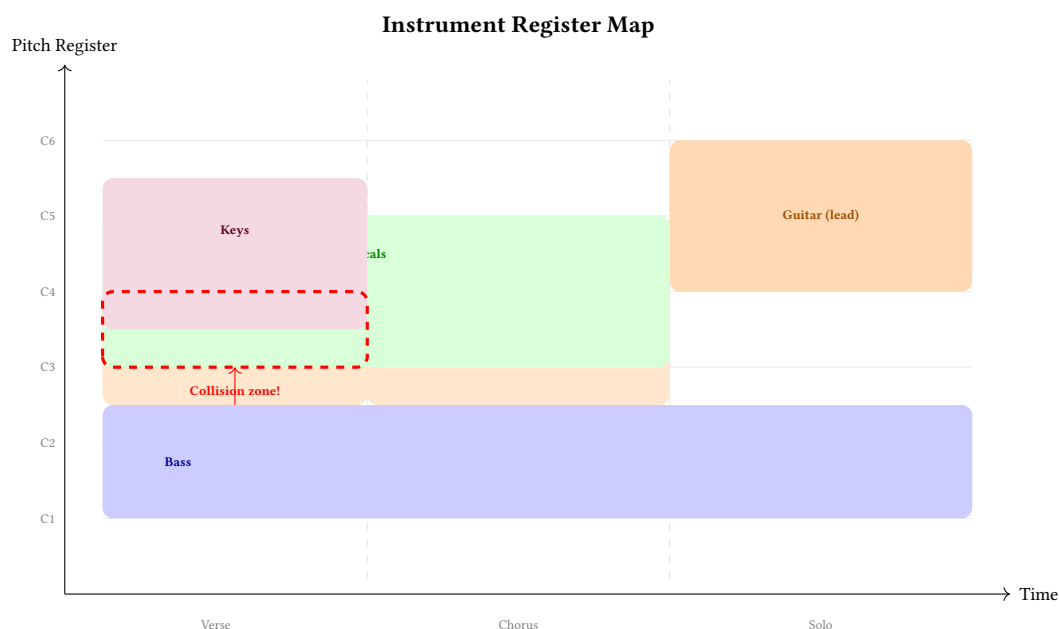


Figure 61. Instrument Register Map showing pitch ranges occupied by each stem across three sections. The verse reveals a collision zone (red dashed) where guitar, vocals, and keys all occupy the C3–C4 octave. In the chorus the guitar spreads wider, and during the solo it moves to a high register (C4–C6), clearing space for other elements. Register separation is a key arrangement skill that this visualization makes immediately visible.

11.24 Silence and Space Analysis

In music, what you *don't* play matters as much as what you do. The dramatic pause before the final chorus of “Bohemian Rhapsody.” The breath between phrases in an Adele vocal. The two beats of silence in Destiny's Child's “Say My Name” that make the re-entry hit harder. These silences are compositional tools that create anticipation, emphasis, and breathing room.

Silence and Space Analysis maps every gap, rest, and low-energy moment across all stems, categorising them by type: full silence (all stems quiet), stem dropout (one instrument disappears while others continue), dynamic valley (energy dip without true silence), and rhythmic rest (intentional gap within a pattern).

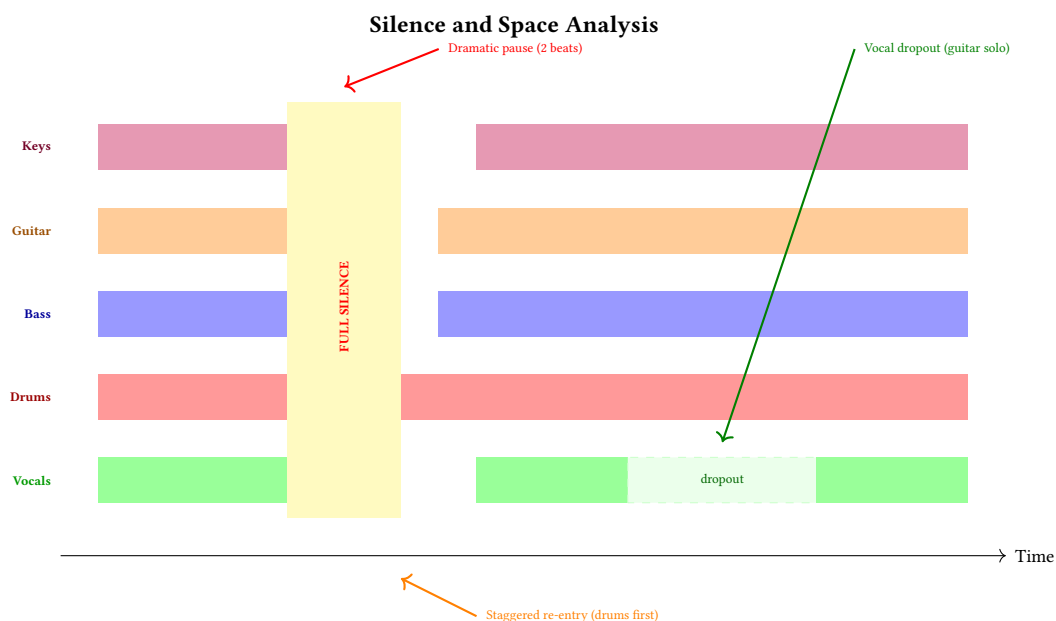


Figure 62. Silence and Space Analysis mapping gaps and dropouts across all stems. The dramatic full silence (yellow highlight) before the re-entry creates maximum contrast. The staggered re-entry—drums first, then bass and guitar, then vocals and keys—is a common arrangement technique that builds momentum. The vocal dropout during the guitar solo clears space for the lead instrument.

11.25 Spectral Brightness Trajectories

The “brightness” (spectral centroid) of each stem tells a story about production and performance. A guitarist switching from clean (centroid ~ 1500 Hz) to distorted (~ 4000 Hz) produces a dramatic upward shift. A producer automating a low-pass filter creates a gradual brightness ramp. A vocalist moving from breathy whisper (dark, low centroid) to full belt (bright, high centroid) traces an arc that mirrors emotional intensity.

Plotting per-stem brightness trajectories reveals these timbral dynamics across the entire song.

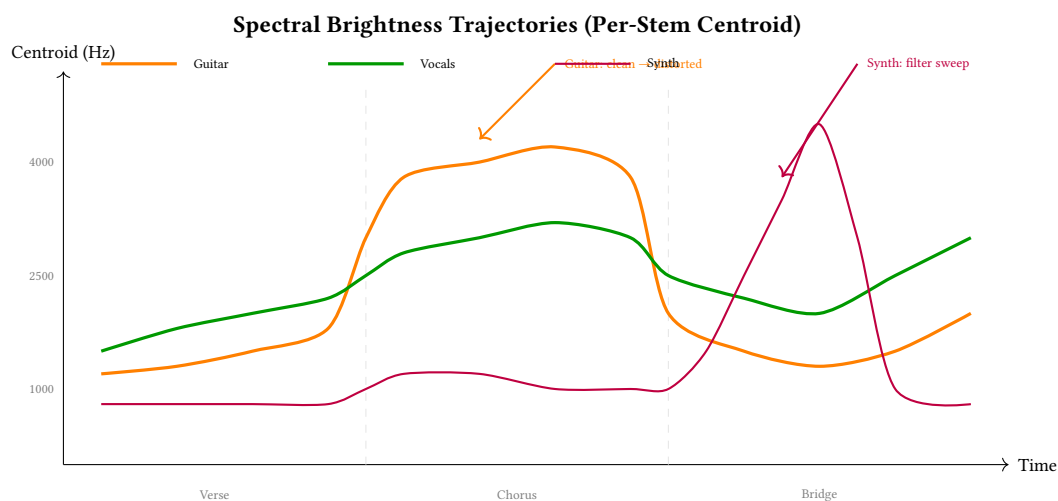


Figure 63. Spectral Brightness Trajectories showing per-stem centroid over time. The guitar’s brightness jumps dramatically at the chorus (switching from clean to distorted tone). The vocal brightness rises gradually with emotional intensity. The synth pad undergoes a filter sweep in the bridge—a rising centroid as the cutoff frequency opens, then snaps shut. These timbral dynamics are invisible in a waveform but immediately visible here.

11.26 Hook Detection and Repetition Map

What makes a song an “earworm”? Repetition. The hook—the most memorable melodic, rhythmic, or lyrical fragment—gets its power from being repeated enough to embed in the listener’s memory, but not so much that it annoys. Research in music cognition suggests an optimal hook repetition rate of 3–5 times per 3-minute song, with slight variations to maintain interest.

Hook Detection identifies the most-repeated melodic and rhythmic motifs by computing self-similarity on short (2–8 beat) melodic segments, clustering similar fragments, and ranking by frequency. The Repetition Map plots every occurrence on the timeline, revealing the arrangement’s “earworm strategy.”

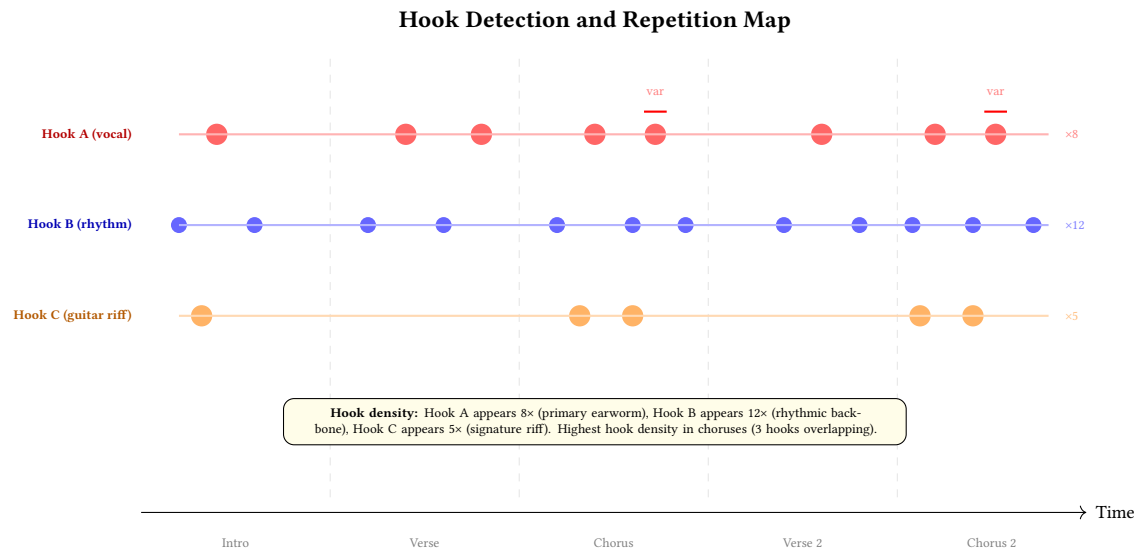


Figure 64. Hook Detection and Repetition Map showing three detected hooks plotted across the song timeline. Hook A (vocal melody, red) appears 8 times with slight variations in later occurrences. Hook B (rhythmic pattern, blue) is the most frequent at 12 occurrences, providing structural continuity. Hook C (guitar riff, orange) appears 5 times, concentrated in intro and choruses. The density of overlapping hooks in the chorus sections explains why those sections feel the most memorable and satisfying.

11.27 Repetition Structure Matrix

While the Self-Similarity Matrix (Section 9) compares every moment at the frame level, the Repetition Structure Matrix operates at the *phrase* level. It identifies melodic and rhythmic phrases (typically 2–8 bars), assigns each a label, and displays the song’s structure as a sequence of labelled blocks—like a genetic code. Repeated phrases share letters (A, A, B, A, B, C, A, B); variations are marked with primes (A, A’, B, A’).

This reveals deep compositional structure: is the song a simple AABA form? A verse-chorus ABABCAB? A through-composed piece? How much does each phrase vary on repetition?

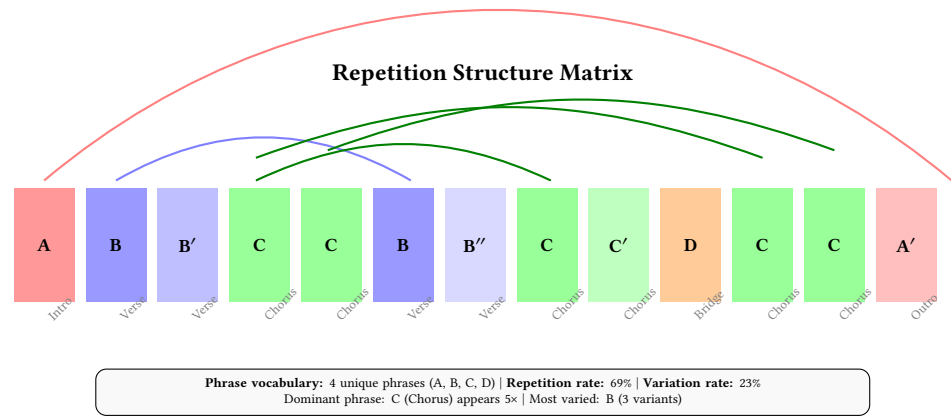


Figure 65. Repetition Structure Matrix showing the song as a sequence of labelled phrase blocks. Identical phrases share the same letter and colour; variations are marked with primes (B’, B’'). Arcs connect repeated phrases. This reveals the song uses only 4 unique ideas (A, B, C, D) arranged into a 13-phrase structure with 69% repetition—typical of a well-crafted pop song that balances familiarity with novelty.

11.28 Rhythmic Density Per Stem

How “busy” is each instrument at any moment? A drummer playing quarter notes has 4 notes per bar; rapid 16th-note fills yield 16. A vocalist holding a whole note has 1; a rapper might hit 32. The Rhythmic Density visualization shows this note-rate per stem over time, revealing the arrangement’s rhythmic “texture.”

This is invaluable for understanding contrast: sparse verse (low density) vs. dense chorus (high density), and how individual instruments vary their activity. A good arrangement ensures that not everyone is busy at once—when the guitar plays a busy riff, drums simplify; when drums fill, bass drops to whole notes.

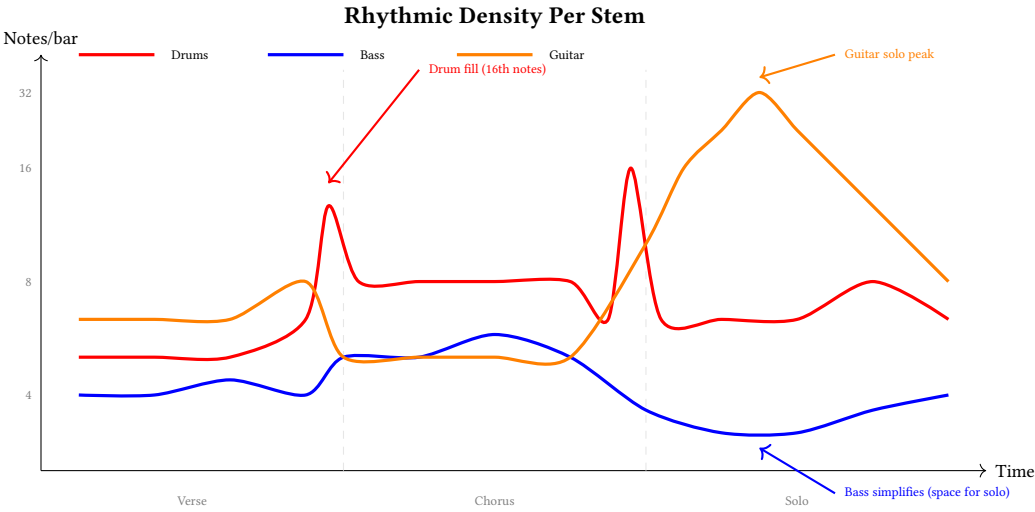


Figure 66. Rhythmic Density Per Stem showing notes-per-bar over time. During the verse, the guitar plays a moderately busy rhythm pattern while drums and bass are sparse. At the chorus transition, a drum fill spikes to 16th notes, then the chorus settles into a simpler pattern. During the guitar solo, the guitar’s density soars to 32 notes/bar while the bass drops to whole notes—a textbook example of “making space” for the soloist.

11.29 Song DNA Fingerprint

Can you capture a song’s *entire identity* in a single visual? The Song DNA Fingerprint attempts this: a circular glyph where each concentric ring encodes a different musical dimension. Two songs with similar fingerprints have similar characters—even in different keys or tempos. It is the musical equivalent of a radar chart, designed to be visually distinctive and instantly recognisable.

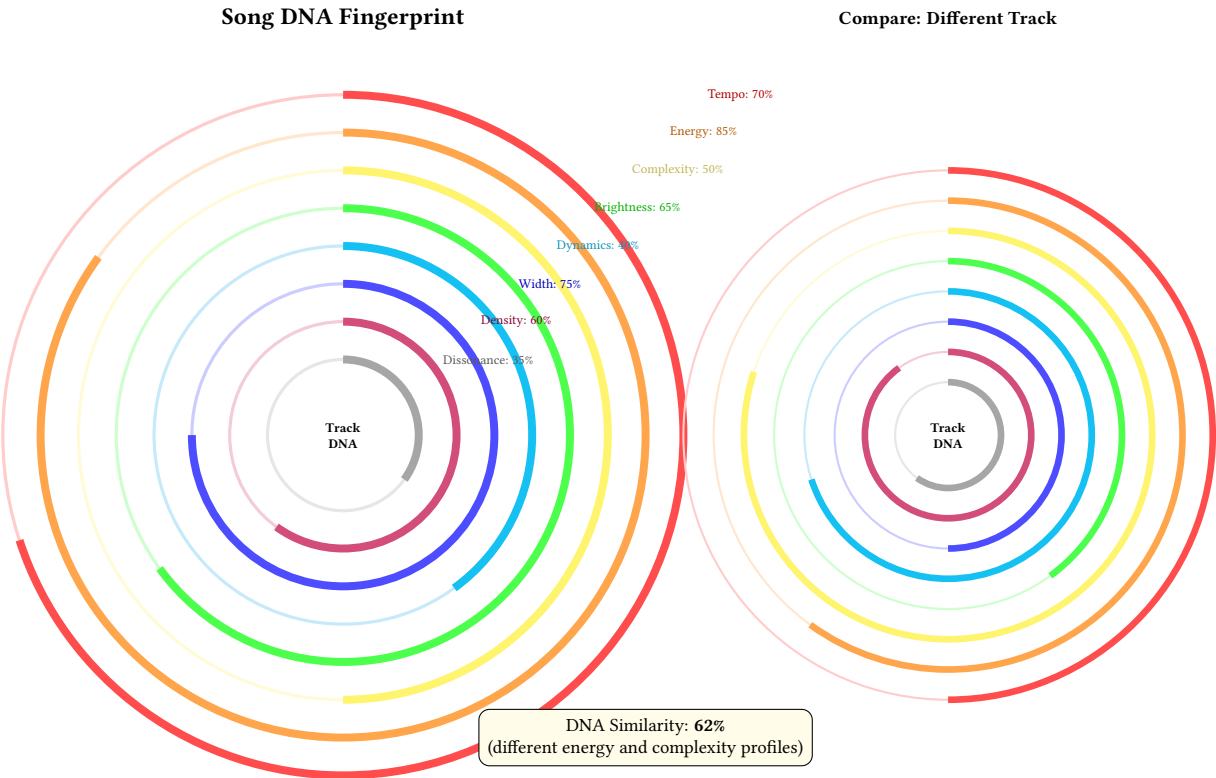


Figure 67. Song DNA Fingerprint—a compact circular glyph encoding eight musical dimensions as concentric arc lengths. The left fingerprint shows a high-energy, moderate-complexity track (long Energy and Tempo arcs, shorter Complexity arc). The right fingerprint shows a complex, dense but lower-energy track. Visual comparison instantly reveals character differences. Two identical fingerprints would indicate songs with very similar musical DNA.

11.30 Genre DNA Radar

While the Song DNA Fingerprint captures a single track, the Genre DNA Radar compares that track against multiple genre archetypes simultaneously. Each radar axis represents a dimension where genres differ most: tempo, harmonic complexity, rhythmic regularity, spectral brightness, dynamic range, stereo width, vocal prominence, and bass weight. Genre archetypes (derived from analysing hundreds of tracks per genre) define characteristic profiles; the current track is overlaid to show which genre it most resembles—and where it diverges.

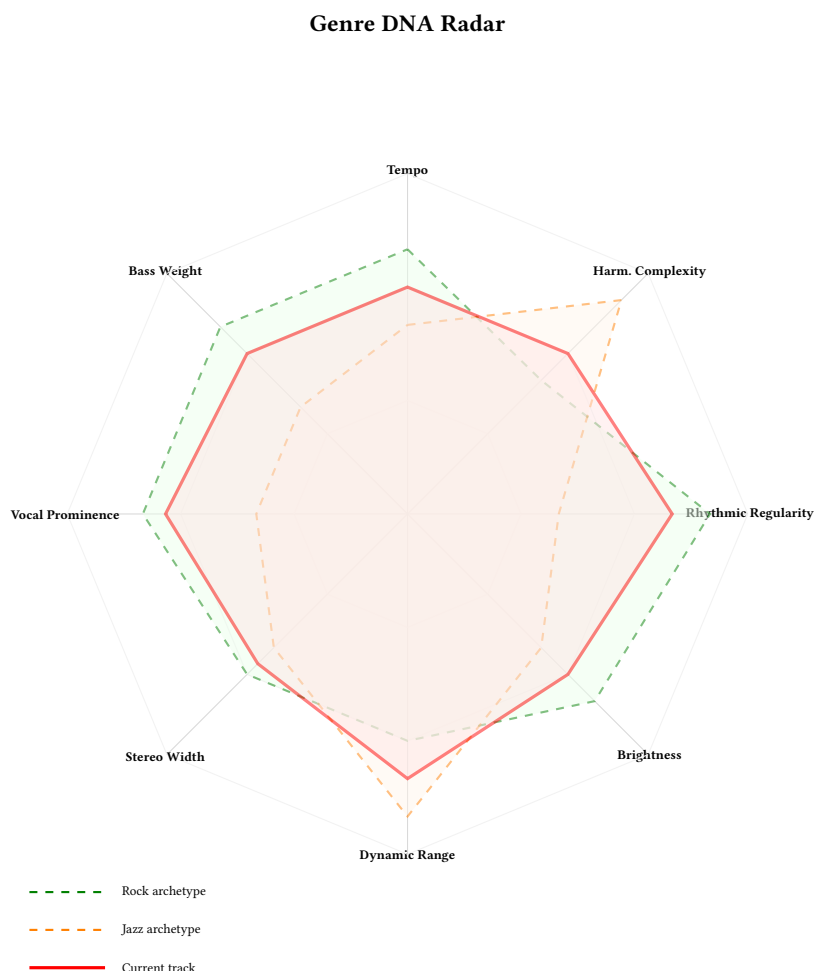


Figure 68. Genre DNA Radar comparing the current track (red) against Rock (green dashed) and Jazz (orange dashed) archetypes across eight dimensions. The track aligns closely with Rock in tempo, brightness, and bass weight, but shows higher harmonic complexity than typical Rock—suggesting a progressive or jazz-influenced rock style. The Jazz archetype’s extreme harmonic complexity and dynamic range are clearly visible.

11.31 Production Era Classifier

Every decade in recorded music has a sonic “signature” shaped by the technology and aesthetics of the time. 1960s recordings have warm, compressed mono-to-early-stereo sound. 1980s productions are defined by gated reverbs, chorus effects, and bright synths. 2010s pop is characterised by extreme loudness, sub-bass emphasis, and processed vocals. The Production Era Classifier analyses spectral balance, dynamic range, stereo width, reverb characteristics, and processing artefacts, then maps the track to the decade whose production values it most resembles.

Production Era Classifier

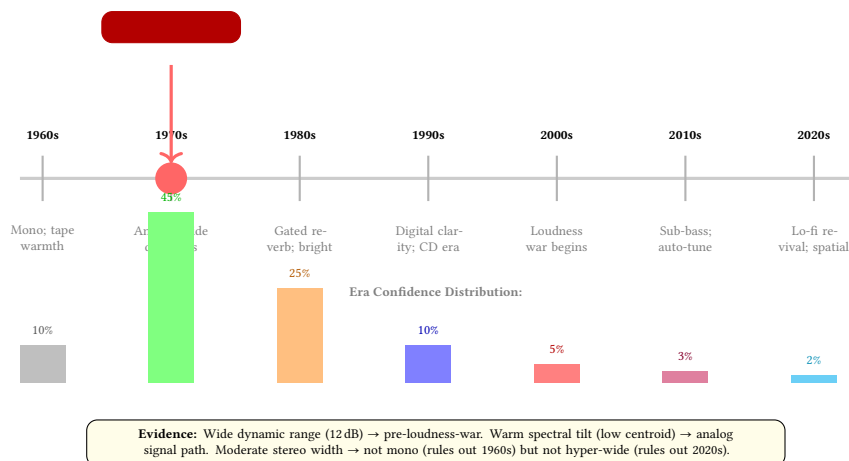


Figure 69. Production Era Classifier mapping the current track to the decade whose production values it most closely resembles. The confidence distribution shows 45% alignment with 1970s production (wide dynamics, warm spectrum, analog character) and 25% with 1980s (some bright elements). The evidence panel explains which features drove the classification.

11.32 Energy Transfer Detection

When a guitarist stops strumming and the keyboard takes over, the arrangement’s energy doesn’t disappear—it *transfers* between stems. Energy Transfer Detection identifies these handoff moments, where one stem’s energy decreases while another’s increases within a short window. This reveals the “relay race” of energy between instruments that keeps a song moving.

In a well-arranged song, transfers are smooth and deliberate: the guitar fades as the synth enters, the vocal drops back as the saxophone takes over. In a poorly arranged song, transfers are abrupt (jarring) or absent (everyone plays at the same level, creating a static, fatiguing sound).

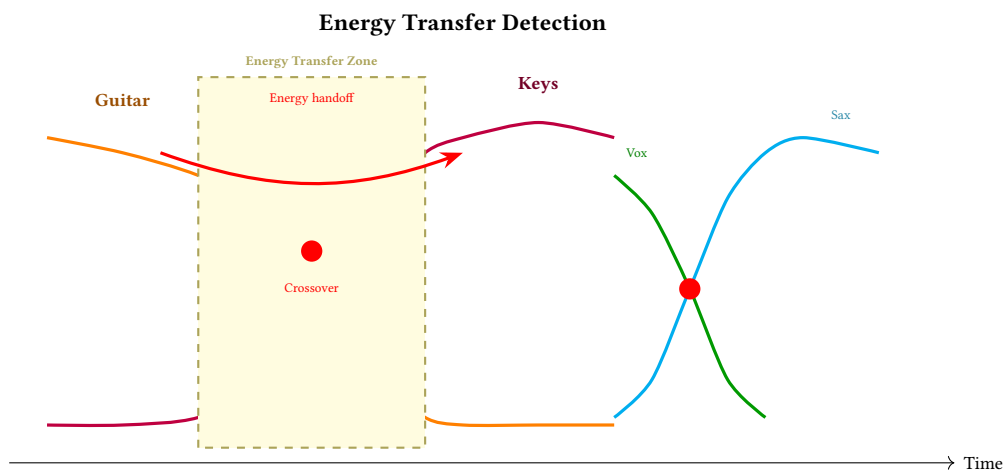


Figure 70. Energy Transfer Detection showing two handoff events. Left: guitar energy decreases as keys energy increases, with a smooth crossover at the midpoint (4-bar transition zone highlighted in yellow). Right: a faster vocal-to-saxophone handoff. The system detects these transfers by cross-correlating energy curves of all stem pairs and identifying inverse correlation within short time windows.

11.33 Tempo Micro-Variation Map

Even music recorded to a click track has subtle tempo variations—and in music without a click, they are fundamental to the expressive character. A band speeds up during an exciting chorus and slows during a contemplative bridge. Classical conductors use *rubato* (tempo elasticity) as an expressive tool, stretching and compressing time for dramatic effect. The Tempo Micro-Variation Map plots instantaneous BPM across the song, revealing these fluctuations.

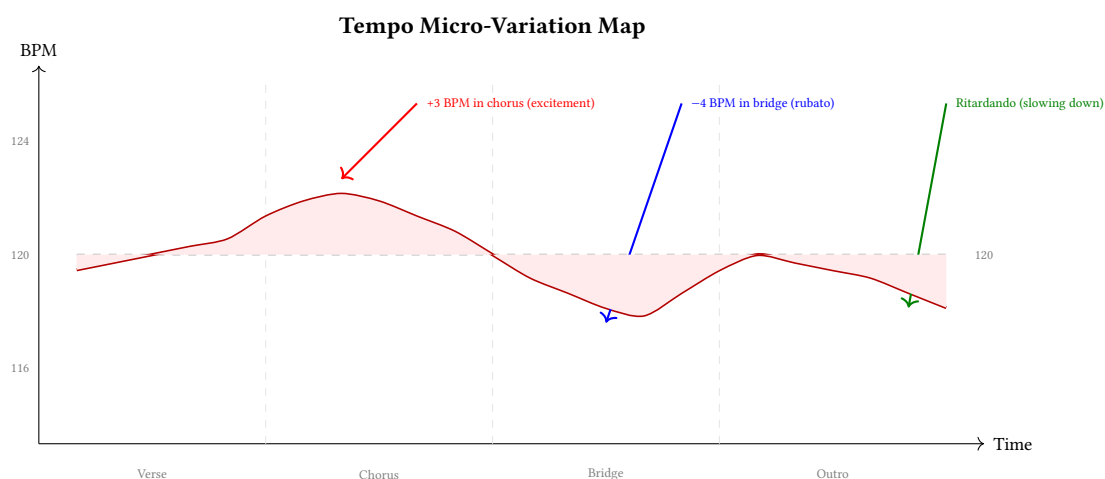


Figure 71. Tempo Micro-Variation Map showing instantaneous BPM (red) against the nominal tempo (grey dashed, 120 BPM). The band pushes +3 BPM during the exciting chorus, pulls back -4 BPM during the reflective bridge (rubato), and decelerates through the outro (ritardando). A click-track recording would show a flat line. The pink shading shows the cumulative deviation from the nominal tempo.

11.34 Harmonic Vocabulary Network

What is a song's harmonic "vocabulary"? A simple pop song might use only 4 chords (C, G, Am, F); a jazz standard could use 20+. The Harmonic Vocabulary Network visualises this as a directed graph: each node is a chord, sized by frequency; each edge shows a chord-to-chord transition, with thickness proportional to occurrence. The network's shape reveals harmonic sophistication at a glance.

A pop song's network is a tight loop of 3–5 nodes with thick bidirectional edges (chords cycle repeatedly). A jazz standard's is a sprawling web of 15–20 nodes with thin edges (diverse transitions, few repeating cycles). A blues song has a characteristic I–IV–V triangle. A through-composed film score might have 30+ nodes with barely any repeated edges.

Harmonic Vocabulary Network

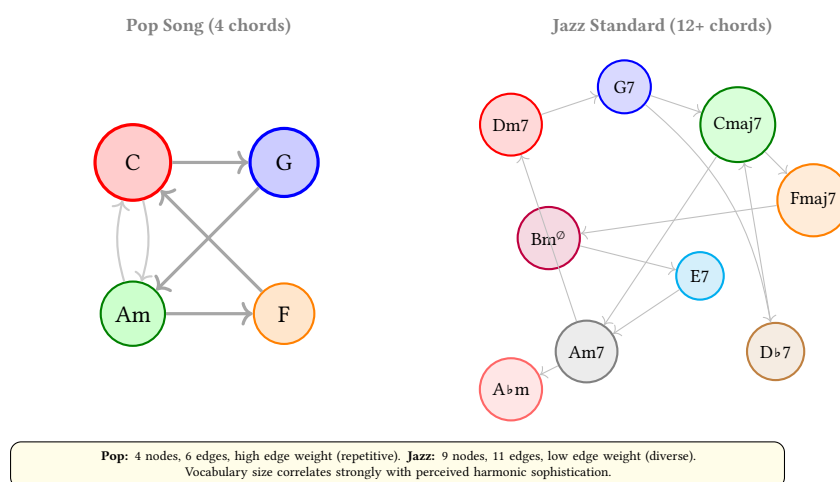


Figure 72. Harmonic Vocabulary Network comparing a pop song (left) and a jazz standard (right). The pop song uses only 4 chords in a tight cyclic pattern with thick edges (high repetition). The jazz standard uses 9+ chords with many thin edges (diverse, non-repeating transitions). Node size indicates chord frequency; edge thickness indicates transition frequency. The network's topology—tight cycle vs. sprawling web—immediately reveals harmonic complexity.

11.35 Style Topology Manifold

What is a manifold? A manifold is a curved surface where nearby points represent similar things. Think of a world map: the Earth is a curved 3D surface, but we flatten it onto a 2D sheet for navigation. A manifold does the same for abstract data—it takes high-dimensional audio features (spectral centroid, energy, dissonance, harmonic complexity, and dozens more) and maps them into a navigable low-dimensional space. Similar-sounding songs sit close together; different songs sit far apart. One direction moves from quiet and sparse toward loud and dense; another from harmonically simple toward complex.

When you hear a song and think “this sounds like Radiohead” or “jazz influences with an electronic edge,” you are placing the music in a multi-dimensional genre space. The Style Topology manifold makes this intuition visible, plotting sonic characteristics in a 3D space where known genres sit at fixed anchor points. As the song plays, its position drifts—moving closer to “Jazz” during a complex chord progression, sliding toward “Metal” when guitars get heavy. Tracking this movement reveals genre-blending: a track that starts near the Ambient anchor and migrates to Electronic shows a build-up from ambient intro to dance floor energy.

The manifold projects the current analysis into a three-dimensional genre space with these anchor points:

Genre	Complexity	Entropy	Density
Electronic	0.2	0.9	0.1
Metal	0.7	0.6	0.9
Jazz	0.8	0.5	0.3
Orchestral	0.9	0.1	0.8

Table 8. Style topology manifold anchor points. The live track’s position is computed as [centroid/5000, RMS×2, dissonance] and plotted relative to these genre anchors.

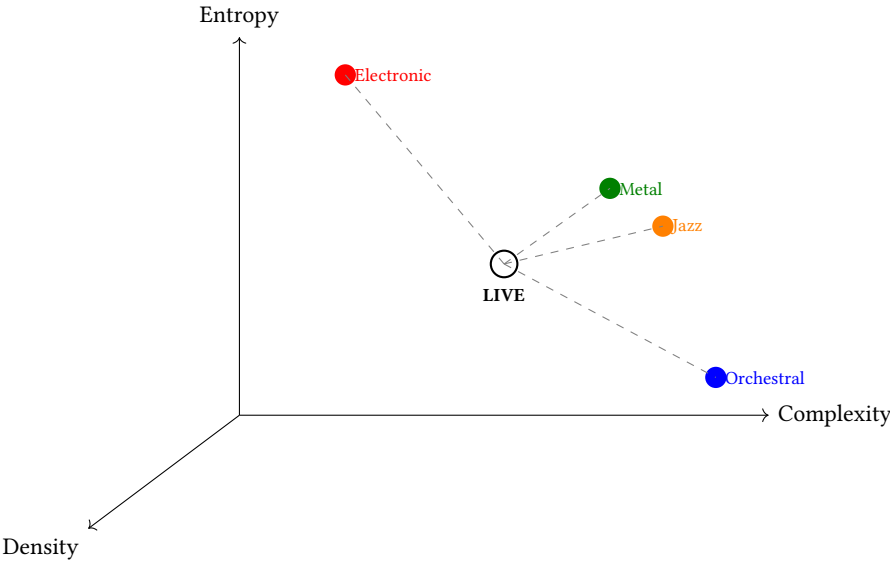


Figure 73. Style topology manifold with genre anchor points. The live track’s style vector (white circle) is projected into this space, and its proximity to each anchor indicates genre affinity. Dashed lines show the distance to each reference genre.

11.36 Discovery Knots

Discovery Knots are user-pinnable annotations on the timeline that capture analysis snapshots at specific moments. Each knot stores:

- **Timestamp:** Normalized position [0, 1] on the timeline
- **Label:** User-provided or auto-generated description
- **DNA Summary:** Condensed analysis metadata
- **Color:** Visual marker color (Color32)

Knots persist across the session and can be selected to display detailed analysis in a callout panel.

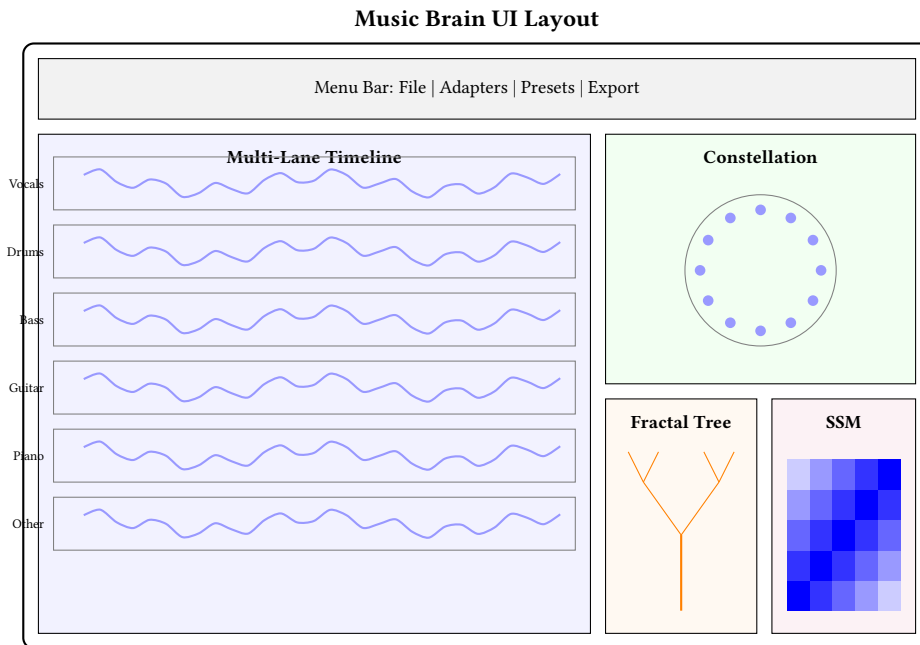


Figure 74. UI layout schematic showing the multi-lane timeline (left), Harmonic Constellation (top-right), Fractal Song Tree (bottom-center-right), and Self-Similarity Matrix heatmap (bottom-right).

12 Genre Adaptation Layer

A C major chord is the same whether it appears in a Beatles pop song, a Miles Davis jazz standard, or a Beethoven symphony—but it *means* something different in each context. In pop, it is “home base”; in jazz, a passing chord; in classical, the resolution of a long harmonic journey. The genre adaptation layer gives Music Brain this contextual intelligence. Rather than one-size-fits-all analysis, the system applies genre-specific “lenses” interpreting raw features through different musical frameworks. A swing ratio of 0.67 is unremarkable in jazz (the standard triplet feel) but unusual in metal (which expects straight eighth notes). The adaptation layer flags these genre-relative deviations, making analysis meaningful within each musical tradition.

12.1 ExpertAdapter Trait

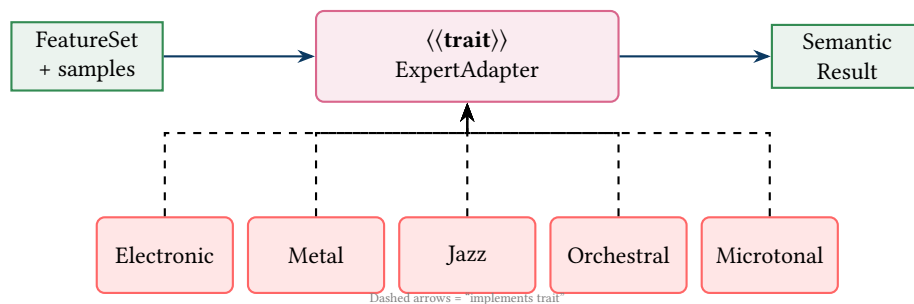


Figure 75. Strategy pattern for the ExpertAdapter trait. Each genre adapter implements the same interface, dispatching domain-specific analysis on shared FeatureSet inputs.

All five adapters are instantiated at startup and run in parallel on each frame, producing independent SemanticResult values aggregated for display.

12.2 Adapter Implementations

Adapter	Specialized Analysis	Key Output
Jazz	Swing ratio estimation (straight → hard shuffle); harmonic extension depth (#11, ♭9, 13th)	Swing feel + chord extensions
Metal	Vocal technique classification (fry/growl/clean); signal chain deduction	Vocal type + hardware chain
Electronic	Phase synchronization between kick and bass; 1/16th grid alignment detection	Phase sync metric (Φ)
Orchestral	Voice count estimation; articulation detection (legato/staccato)	Ensemble size + texture
Microtonal	Continuous pitch tracking; deviation from 12-TET (cents)	Pitch deviation + system

Table 9. Adapter feature comparison across all five genre expert implementations. The preset system (Table 10) extends genre coverage to Rock, Pop, R&B, Hip-Hop, Punk, Country, Folk, and Ambient sub-genres without requiring additional adapter code.

12.2.1 Jazz: Swing Ratio and Harmonic Extensions

What makes a jazz recording sound like jazz before you identify the instruments? Two things: the *swing feel* (that lilting, uneven rhythm where the beat bounces rather than marches) and the *harmonic complexity* (chords with 9ths, 11ths, and 13ths that give jazz its rich sound). Compare the straight-eighth feel of “Back in Black” with the swung feel of Miles Davis’s “So What” or Dave Brubeck’s “Take Five”—the rhythmic DNA differs completely, and the Jazz adapter captures this.

The adapter also detects harmonic extension depth: a pop song uses triads (3 notes), while a jazz standard adds the 7th, 9th, #11th, and 13th. When Music Brain analyses John Coltrane’s “Giant Steps” or Bill Evans’s “Waltz for Debby,” the Jazz adapter reveals the harmonic sophistication that sets these recordings apart from simpler pop harmony.

Swing is quantified by the ratio of long to short subdivisions within each beat:

$$R_{\text{swing}} = \frac{t_{\text{long}}}{t_{\text{short}}} \quad (19)$$

where t_{long} and t_{short} are the durations of the first and second eighth notes. A ratio of 1.0 indicates straight timing; 1.5 is light swing (triplet feel); 2.0 is heavy shuffle.

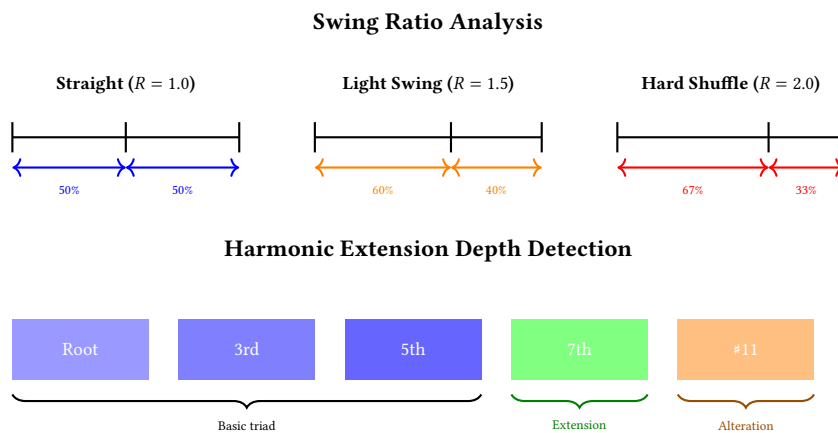


Figure 76. Jazz adapter analysis. Top: swing ratio comparison showing straight, light swing (triplet feel), and hard shuffle timing. Bottom: harmonic extension depth detection—the adapter identifies not just the triad but upper extensions (7th, 9th, #11, 13th) by analyzing which chroma bins exceed the harmonic threshold beyond the basic chord tones.

12.2.2 Electronic/Dance: Kick-Bass Phase Synchronization

In any genre where the low end carries the groove—house, techno, hip-hop, pop—the timing between kick drum and bass is critical. When kick and bass hit within milliseconds of each other, the low end sounds tight and punchy; when they drift apart, the mix sounds muddy. This adapter measures that alignment with a phase synchronization metric.

The phase sync metric Φ cross-correlates the onset functions of the kick sub-band (40–100 Hz from the drum stem) and the bass sub-band (40–150 Hz from the bass stem), normalized to $[0, 1]$. Though originally developed for psytrance (where rolling basslines demand sub-5 ms kick-bass alignment), it applies to any production where kick-bass coherence matters: the four-on-the-floor thump of house, the 808-heavy low end of trap, or the compressed pop-punk mix where bass locks to the kick.

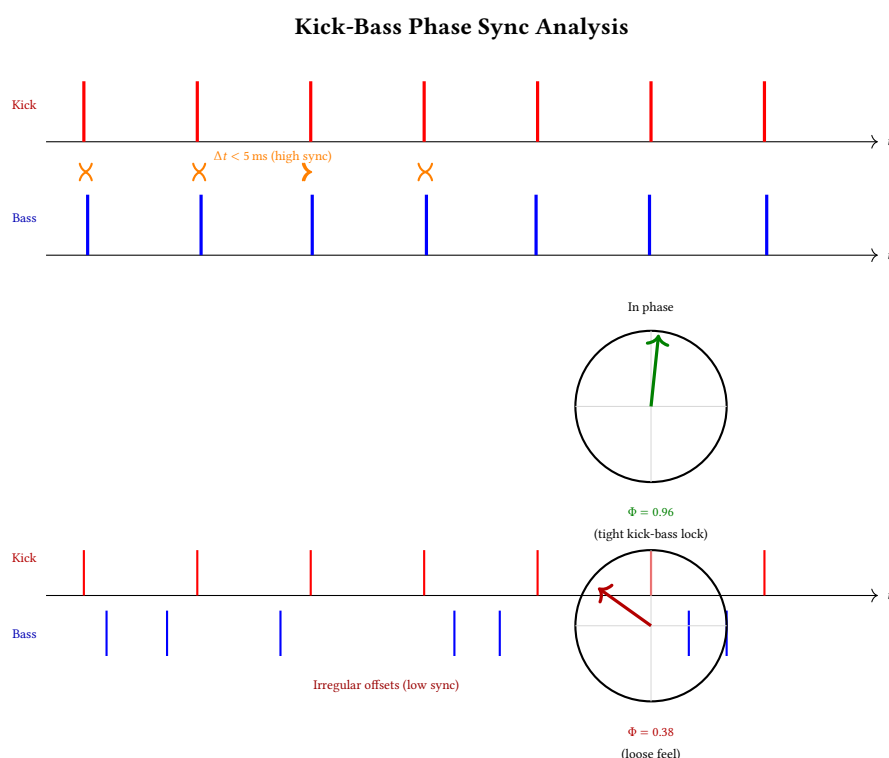


Figure 77. Kick-bass phase synchronization analysis. Top: tight alignment ($\Phi = 0.96$) with sub-5 ms offsets, characteristic of well-produced electronic music and tight pop/rock mixes. Bottom: irregular offsets ($\Phi = 0.38$), indicating a loose or broken feel. The Phase Compass (right) renders Φ as a needle angle.

12.2.3 Orchestral: Voice Count and Articulation

When analysing orchestral music or film scores—from Hans Zimmer’s “Inception” to the strings in Adele’s “Skyfall”—two questions matter most: *how many instruments play at once* (textural density), and *how are they played* (smooth legato vs. sharp staccato). A solo violin feels intimate; the same melody played by 60 violins feels epic. The Orchestral adapter captures this, and it extends beyond classical music—any song using string arrangements, brass sections, or layered synth pads benefits. The Beatles’ “Eleanor Rigby” (string quartet) and Kanye West’s “Runaway” (orchestral outro) are both analysable through this lens.

The adapter estimates polyphony by counting distinct spectral peaks grouped by fundamental frequency, and classifies articulation from the amplitude envelope of individual note onsets: slow attack with sustained amplitude indicates legato; sharp attack with rapid decay indicates staccato.

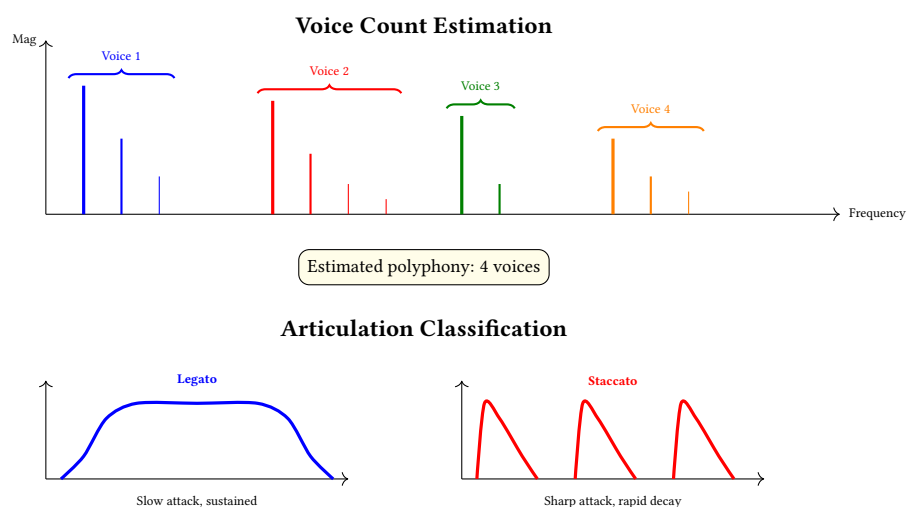


Figure 78. Orchestral adapter analysis. Top: voice count estimation by grouping spectral peaks into harmonic series (each group = one voice). Bottom: articulation classification from amplitude envelopes—legato (slow attack, sustained) vs. staccato (sharp attack, rapid decay).

12.2.4 Microtonal: Continuous Pitch Deviation

Not all music lives on the standard 12-note scale. Blues guitarists like B.B. King and Stevie Ray Vaughan bend notes between the “cracks” of the piano keyboard—those quarter-tone bends give the blues its emotional ache. Indian classical music uses microtonal intervals called *shruti* with no Western equivalent. Even in pop, singers like Beyoncé and Adele use subtle pitch inflections that slide between standard notes. The Microtonal adapter captures all of this by tracking pitch continuously and measuring deviation from the 12-TET grid.

This reveals expressive techniques invisible to a standard chord detector: slow bends at the end of a guitar solo, the characteristic “blue note” (a note played slightly flat for effect), vibrato depth, and intentional pitch drift. It enables analysis of non-Western tuning systems (just intonation, quarter-tone Arabic maqam, Pythagorean tuning) and detects whether a performance is “in tune” by convention or deliberately bending the rules.

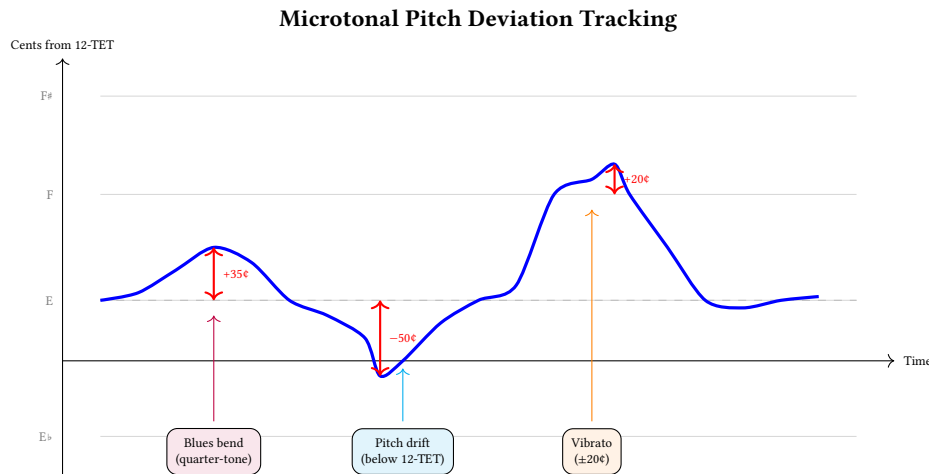


Figure 79. Microtonal pitch deviation tracking. The blue curve shows continuous pitch against the 12-TET grid (dashed lines). Red arrows mark deviations in cents. Annotations identify expressive techniques: blues quarter-tone bends (+35¢), vibrato (±20¢), and pitch drift below the grid (−50¢). The adapter classifies detected tuning systems (just intonation, quarter-tone, Pythagorean).

12.3 Preset System and Deviation Tracking

The preset system defines reference profiles for specific sub-genres:

GenrePreset		
name	String	“Modern Rock”
universe	String	“Rock”
bpm_range	[f32; 2]	[110, 140]
target_phase_sync	f32	0.80
ideal_spectral_centroid	f32	3000.0 Hz
ideal_spectral_flatness	f32	0.40
rhythmic_dna	String	“K-S-K-S”
harmonic_guidance	String	mode & scale hints

Figure 80. GenrePreset data schema. Each preset defines target feature values for a specific sub-genre, against which live analysis is compared for deviation tracking.

While the five core adapters provide deep, domain-specific analysis, the preset system extends genre coverage to any style by defining target feature profiles. A Pop preset, for example, routes through the Electronic adapter (for kick-bass phase sync) and the Jazz adapter (for chord extension detection), but compares results against pop-specific targets: moderate BPM, high phase sync, and bright spectral centroid. An Indie/Alt-Rock preset uses the same adapters but expects lower phase sync (looser feel) and a darker timbre. This layered approach—generic adapters plus genre-specific presets—covers a broad range of styles without requiring a dedicated adapter for each.

The deviation tracker compares live features against the active preset and generates diagnostic messages:

- **BPM Mismatch:** Current BPM outside the preset’s target range

- **Phase Shift:** Phase sync deviates by more than 0.1 from target
- **Timbre Shift:** Spectral centroid deviates by more than 500 Hz (classified as “too bright” or “too dark”)

The system ships with presets spanning the major genre families:

Preset	Universe	BPM	Phase	Centroid	Flatness
Modern Rock	Rock	110–140	0.80	3000 Hz	0.40
Indie / Alt-Rock	Rock	100–135	0.65	2600 Hz	0.30
Pop	Pop	100–130	0.85	3200 Hz	0.35
R&B / Neo-Soul	R&B	70–110	0.70	2400 Hz	0.25
Hip-Hop / Trap	Hip-Hop	60–90	0.75	2800 Hz	0.50
Modern Djent	Metal	80–160	0.80	3500 Hz	0.65
Hardcore Punk	Punk	150–200	0.85	3800 Hz	0.55
Bebop Jazz	Jazz	140–280	0.40	2000 Hz	0.15
Lo-fi / Ambient	Electronic	60–90	0.30	1800 Hz	0.20
House / Techno	Electronic	120–140	0.95	3400 Hz	0.45
Disco / Funk	Funk	110–130	0.90	2800 Hz	0.30
Country / Folk	Country	90–140	0.70	2200 Hz	0.15
Baroque Orchestral	Classical	60–120	0.50	1200 Hz	0.05

Table 10. Shipped preset configurations with target feature values for deviation tracking. Each preset defines a reference profile against which live analysis is compared.

12.4 Preset Deviation Radar

The deviation tracking system compares live audio features against the active preset’s targets across multiple dimensions. Figure 81 presents this as a radar (spider) chart, where each axis represents a feature dimension and the shaded area shows the deviation profile.

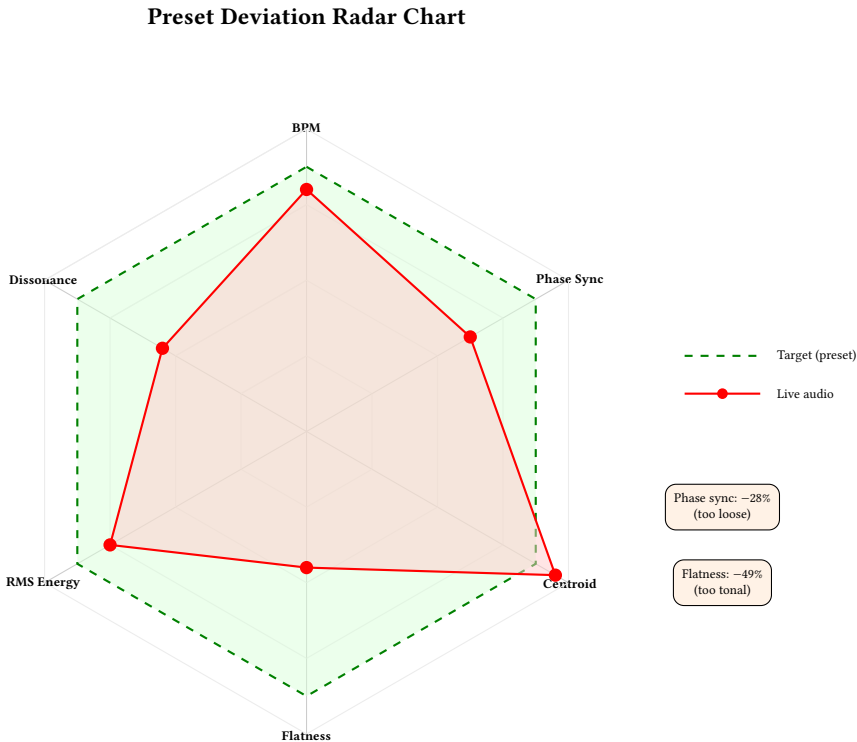


Figure 81. Preset deviation radar chart comparing live audio features (red) against the Modern Rock target preset (green dashed). Each axis represents a normalized feature dimension. Deviations from the target polygon highlight areas for improvement—here, phase sync and spectral flatness fall below target.

12.5 Genre Topology Trajectories

Beyond a static snapshot, the Style Topology manifold tracks how a song’s position evolves over time, revealing genre-space trajectories that encode arrangement dynamics. A typical pop song might begin near the “ambient” region (low complexity, low energy), migrate toward “rock” as instruments accumulate, and shift toward “electronic” during a synth-heavy bridge.

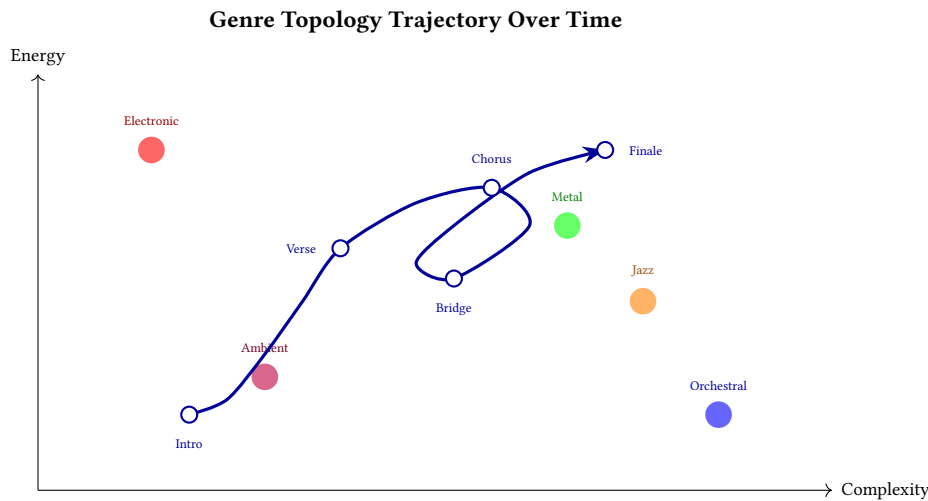


Figure 82. Genre topology trajectory showing a song’s path through the style manifold over time. The track begins in the Ambient/low-energy region (Intro), migrates toward higher complexity/energy (Verse, Chorus), retreats during the Bridge, then reaches peak intensity at the Finale near the Metal anchor. This trajectory encodes the song’s arrangement arc as a spatial path.

12.6 Multi-Adapter Output Comparison

When all five adapters process the same audio segment, they produce distinct semantic interpretations reflecting their domain-specific heuristics. Figure 83 shows how each adapter interprets the same FeatureSet differently.

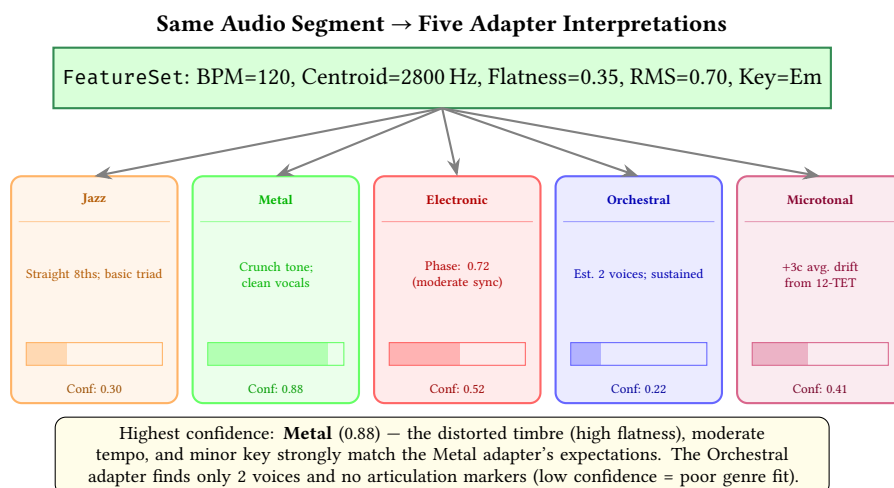


Figure 83. Multi-adapter output comparison for the same audio segment. Each adapter applies domain-specific heuristics to the shared FeatureSet, producing different semantic labels and confidence scores. The Metal adapter shows the highest confidence (0.88), indicating strong genre alignment. Low-confidence results (Orchestral: 0.22, Jazz: 0.30) indicate poor fit for those genre models.

13 Data Export and MIDI Generation

Music Brain supports two export formats for interoperability with external tools.

Analysis has limited value unless results flow into a musician’s existing workflow. A producer needs to import detected chord progressions, tempo maps, and section markers directly into their DAW session—not re-enter them manually. A composer wants to open extracted MIDI in notation software for score-level review. An educator comparing student mixes against professional references needs a portable format that can be archived, shared, and diffed. Export transforms Music Brain from a standalone viewer into a node in a larger creative pipeline.

The DNA manifest captures a track’s musical “story”: the interpreted harmonic journey, structural narrative, and timbral fingerprint that define the track’s identity. By serializing the full analysis state—including genre adapter insights and per-stem breakdowns—the manifest preserves context that summary statistics alone would lose.

MIDI generation benefits directly from stem isolation. Converting a full mix to MIDI is notoriously inaccurate because overlapping instruments produce ambiguous pitch estimates. Generating MIDI from each isolated stem avoids the polyphonic transcription problem: the vocal stem yields a clean monophonic melody, the bass stem an unambiguous bass line, and harmonic instruments are transcribed without percussion interference.

13.1 DNA Manifest (JSON)

The `export_dna_manifest()` function serializes the current analysis state into a structured JSON document containing:

- **Harmonic Profile:** Detected key, current chord, sensory dissonance, active chroma bins
- **Rhythmic Profile:** BPM, RMS energy, spectral centroid
- **Semantic Insights:** Array of `SemanticResult` values from all adapters

DNA Manifest JSON Schema

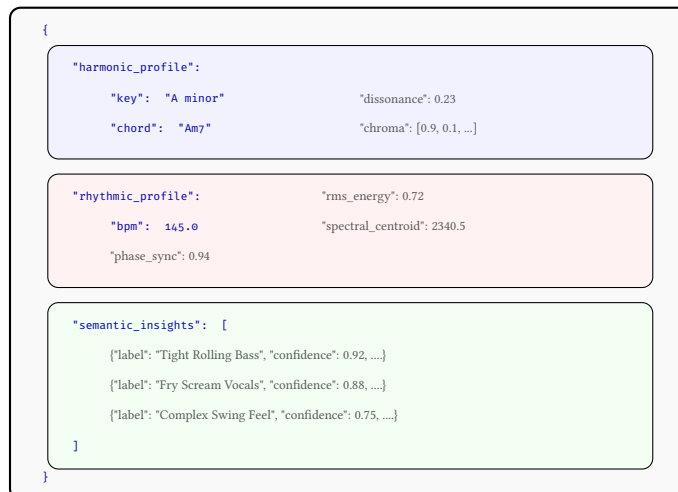


Figure 84. DNA Manifest JSON schema structure. The exported document contains three top-level sections: harmonic profile (key, chord, dissonance, chroma), rhythmic profile (BPM, energy, spectral descriptors, phase sync), and an array of semantic insights from all active genre adapters.

13.2 MIDI Generation

The MIDI generator converts the harmonic DNA profile into a standard MIDI file using the `midly` crate. Active chroma bins are mapped to MIDI note numbers starting at C4 (MIDI 60), with each note assigned a velocity of 90 and a duration of one beat (480 ticks at 480 PPQ). The tempo is derived from the detected BPM via $\mu\text{s}/\text{beat} = 60,000,000/\text{BPM}$.

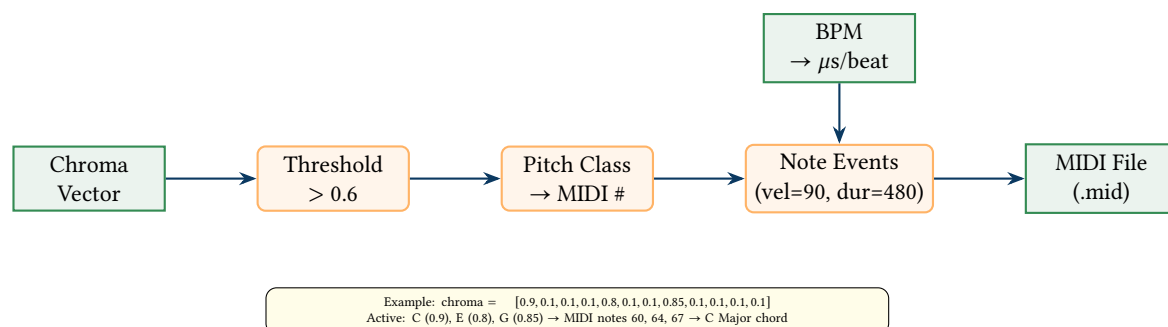


Figure 85. MIDI generation pipeline. Active chroma bins (above threshold) are mapped to MIDI note numbers, assembled into note events with fixed velocity and duration, and combined with a tempo track derived from the detected BPM.

13.3 Tablature Generation (Planned)

Since Music Brain already isolates guitar and bass stems and tracks their pitch content via chroma analysis, the system has the raw data needed to generate *tablature*—the fretboard notation that guitarists and bassists actually read. This is a planned export feature that would bridge the gap between audio analysis and practical musicianship.

The tablature generation pipeline would operate as follows:

1. **Pitch extraction:** continuous f_0 tracking on the isolated guitar/bass stem (using YIN or pYIN) produces a sequence of pitched note events with onset times, durations, and pitch values.
2. **Fretboard mapping:** each pitch is mapped to its most likely fretboard position. A single pitch (e.g., E4 at 330 Hz) can be played at multiple positions (open 1st string, 5th fret 2nd string, 9th fret 3rd string, etc.), so the system uses positional heuristics: prefer positions near the previous note (minimise hand movement), prefer lower positions for open chords, and respect the instrument’s tuning (standard, drop-D, open G, etc.).
3. **Rhythm quantization:** onset times are snapped to the nearest beat subdivision (eighth note, sixteenth note, triplet) based on the detected tempo and time signature.
4. **Technique annotation:** bends (detected via pitch deviation analysis), slides (continuous pitch change between two notes), hammer-ons/pull-offs (rapid note transitions without new onset), and palm mutes (spectral envelope changes) can be annotated using the timbral forensics data.
5. **Export:** output as ASCII tablature, Guitar Pro format (.gp5/.gpx), or MusicXML for import into notation software.

This feature would allow users to analyse a song on Spotify, isolate the guitar part, and generate a playable tab—turning Music Brain from an analytical tool into a practical learning aid for musicians.

14 Implementation Details

14.1 Dependencies

Table 11 lists the primary crate dependencies and their roles in the system.

Crate	Version	Purpose
eframe / egui	0.27.2	Immediate-mode GUI framework [13]
cpal	0.15.3	Cross-platform audio I/O
symphonia	0.5.3	Audio file decoding (MP3, FLAC, WAV, OGG)
realfft	3.3.0	Real-valued FFT (wraps <code>rustfft</code>)
ndarray	0.15.6	N-dimensional arrays (SSM, spectrograms)
ort	2.0.0-rc.4	ONNX Runtime bindings [14]
midly	0.5.3	MIDI file writing
serde / serde_json	1.0	Serialization (JSON export)
crossbeam-channel	0.5.12	Lock-free MPSC channels (audio callback)
tokio	1.37	Async runtime (Demucs bridge)
chrono	0.4.44	Timestamp generation

Table 11. Primary crate dependencies and their roles in the Music Brain system.

14.2 Live Audio Streaming

The live audio subsystem uses CPAL to capture audio from the default input device. A `crossbeam-channel` sender/receiver pair transfers sample buffers from the audio callback thread to the main analysis thread without locks. The main thread drains all available buffers on each frame update and appends them to a rolling 44,100-sample (1-second) history buffer, discarding the oldest samples when the buffer exceeds capacity.

14.3 Neural Inference Engine

The neural inference engine provides a wrapper around ONNX Runtime [14] for future integration of pre-trained models. The current implementation supports model loading via `SessionBuilder` with 4 intra-op threads, but inference is not yet connected to the analysis pipeline. Planned applications include zero-shot genre classification using contrastive audio-language models [15].

14.4 Phase 1 Implementation Status

The Phase 1 foundation milestone (completed March 2026) established the core infrastructure required for all subsequent analysis work. The key deliverables were:

Audio Loading Pipeline. The audio loader uses the `Symphonia` crate to decode MP3, WAV, FLAC, and OGG files into raw PCM samples with automatic mono downmix from multi-channel sources. This replaces the previous stub that only accepted pre-decoded WAV files.

Analysis Pipeline Architecture. A batch-mode `AnalysisPipeline` computes STFT matrices over loaded audio, with verified spectral accuracy (440 Hz peak detection test passing). This pipeline is designed for async execution on the `Tokio` runtime, keeping the UI thread responsive during analysis of large files.

Elimination of Hardcoded Values. All previously hardcoded feature values were replaced with proper `Option<T>` types:

- `FeatureSet.bpm`: changed from `f32` (hardcoded 145.0) to `Option<f32>`
- `FeatureSet.key`: changed from `String` (“Auto-Detected”) to `Option<String>`
- `FeatureSet.sensory_dissonance`: changed from `f32` to `Option<f32>`

This ensures the system never presents fabricated data—if a feature hasn’t been computed yet, it displays as “unknown” rather than a misleading placeholder.

UI Shell and Skeleton Components. The main application window was rebuilt as a minimal dark-themed shell (1200×950 pixels) with a file-drop prompt. Four UI skeleton modules were created for progressive implementation in subsequent phases:

- **Waveform** — Scrollable waveform rendering with zoom controls
- **Spectrogram** — STFT-based spectrogram with adjustable FFT parameters
- **Transport** — Playback transport bar (play, pause, seek, volume)
- **Sidebar** — Analysis results sidebar for displaying extracted features

The advanced visualization components (Fractal Song Tree, Style Topology Manifold, Structure Heatmap, Preset Gallery, Phase Compass, Stem Helix) remain in the codebase from the initial architecture and will be integrated into the unified analysis dashboard in Phase 5.

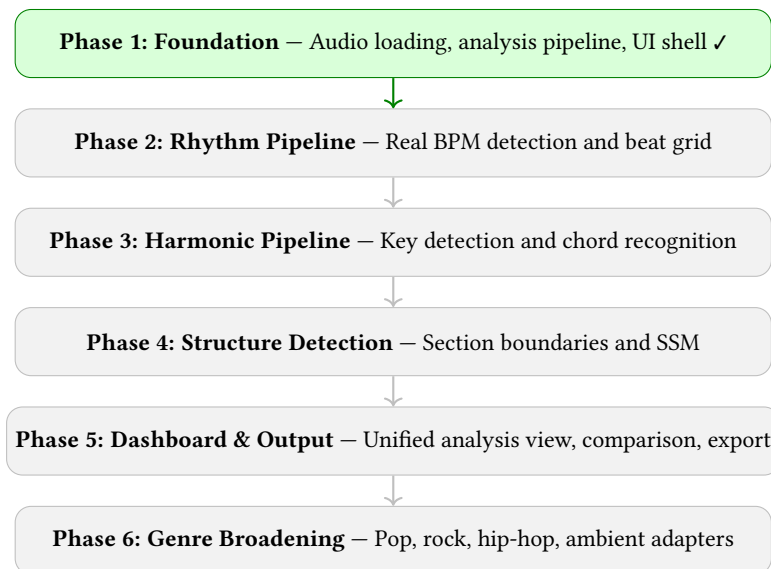


Figure 86. Implementation roadmap. Phase 1 (Foundation) is complete. Subsequent phases progressively add real signal processing algorithms, structural analysis, and the unified dashboard that integrates all visualization components.

15 Energy Profiling and Dynamics Analysis

A great song takes you on a journey: quiet intro, building verse, explosive chorus, pulled-back bridge, hardest final chorus. This “energy arc” separates a well-crafted song from a flat one. Music Brain measures it at three zoom levels—individual drum hits (micro), beat-by-beat pumping (meso), and section-level rise and fall (macro)—revealing the dynamic strategy behind the production. The system shows exactly *where* energy peaks, how gradually the build-up ramps, and whether the drop delivers the contrast it promises.

15.1 Multi-Resolution Energy Envelope

The energy profile is computed at three temporal resolutions:

- **Micro** (per-frame, ~23 ms): Individual transient peaks, gate openings
- **Meso** (per-beat, ~0.5 s): Beat-level dynamics, sidechain pumping patterns
- **Macro** (per-section, ~8–16 bars): Section-level loudness contour, arrangement energy arc

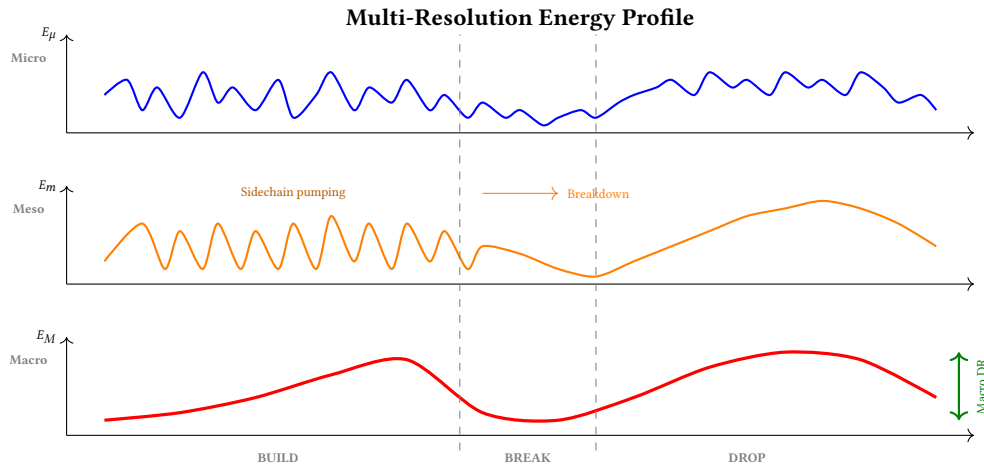


Figure 87. Multi-resolution energy profile. Micro-level (~ 23 ms) captures individual transients. Meso-level (~ 0.5 s) reveals beat-level patterns like sidechain pumping. Macro-level (~ 8 – 16 bars) shows the section-level energy arc: build-up, breakdown, and drop. Vertical dashed lines mark structural boundaries detected from the macro energy contour.

15.2 Loudness and Dynamic Range

The system computes both instantaneous and integrated loudness following the ITU-R BS.1770 recommendation, providing LUFS (Loudness Units Full Scale) measurements. The *loudness range* (LRA) quantifies the difference between the softest and loudest passages, distinguishing heavily compressed masters (LRA < 6 LU) from dynamic recordings (LRA > 12 LU). Per-stem loudness tracking reveals whether dynamic range is distributed across instruments or concentrated in one stem (e.g., a heavily compressed drum bus with dynamic vocals).

15.3 Sidechain Detection

In electronic music, sidechain compression creates a characteristic “pumping” effect where pad or bass levels duck rhythmically with the kick drum. Music Brain detects this by cross-correlating the inverted bass/pad energy envelope with the kick onset function: strong negative correlation at zero lag indicates active sidechain compression, with the duck depth measured in dB.

16 Mood and Emotion Mapping

Why does Pharrell’s “Happy” make you want to dance, while R.E.M.’s “Everybody Hurts” makes you reflective? Music carries emotional content traceable to measurable acoustic properties: major keys sound happy, minor keys sad; fast tempos feel energetic, slow tempos calm; high spectral energy feels intense, low energy peaceful. Music Brain maps these relationships into a two-dimensional emotion space, plotting every moment on axes of *valence* (positive vs. negative emotion) and *arousal* (high vs. low energy). The trajectory—a path through emotion space over time—reveals the song’s emotional narrative: does it stay in one mood, or journey from melancholy to euphoria?

16.1 Valence–Arousal Model

The two dimensions of the emotion space are:

- **Arousal** (vertical axis): Maps energy and intensity. High arousal = fast tempo, high RMS, bright spectral centroid, dense onsets. Low arousal = slow tempo, quiet, dark timbre, sparse events.
- **Valence** (horizontal axis): Maps perceived positivity/negativity. Positive valence = major key, consonant harmony, regular rhythm. Negative valence = minor key, dissonant harmony, irregular timing.

The mapping uses a weighted combination of extracted features:

$$\text{Arousal} = w_1 \cdot \frac{\text{BPM}}{200} + w_2 \cdot E_{\text{RMS}} + w_3 \cdot \frac{f_c}{5000} + w_4 \cdot \text{onset_density} \quad (20)$$

$$\text{Valence} = w_5 \cdot M_{\text{key}} + w_6 \cdot (1 - D_s) + w_7 \cdot \text{rhythm_regularity} \quad (21)$$

where $M_{\text{key}} = 1$ for major keys and $M_{\text{key}} = 0$ for minor, and D_s is sensory dissonance.

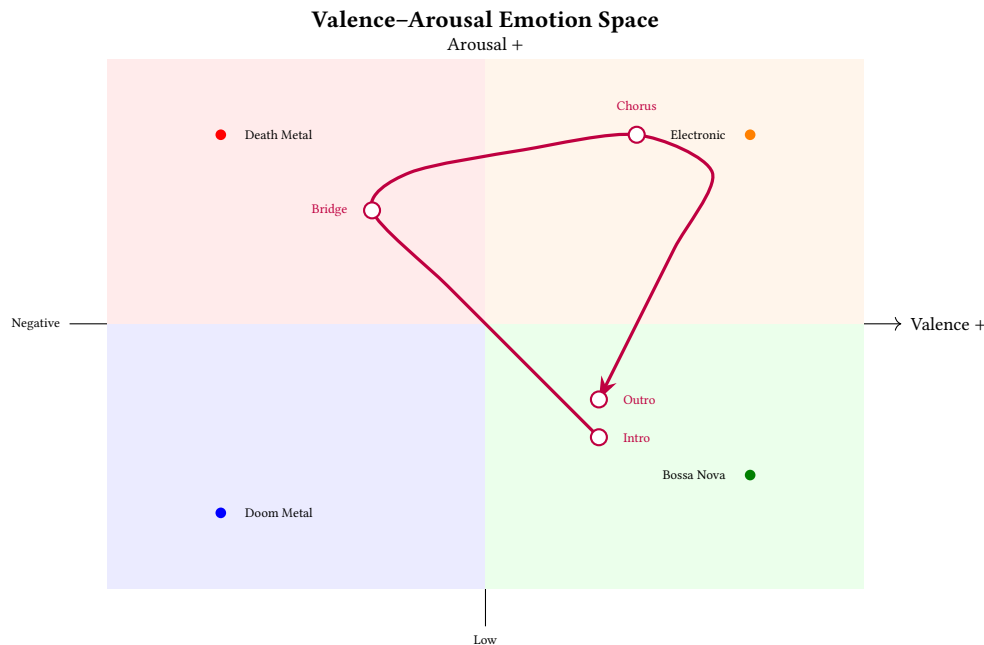


Figure 88. Valence–Arousal emotion space with a song trajectory (purple). The song begins in the Relaxed quadrant (calm intro), passes through Tense (minor-key bridge), peaks in the Excited quadrant (major-key chorus at high energy), and returns to Relaxed (outro). Genre archetypes are plotted as reference points.

16.2 Emotional Arc and Narrative Structure

By plotting the valence–arousal trajectory over the full song duration, Music Brain reveals the *emotional arc*—the narrative shape of the listener’s affective experience. Common arc shapes include:

- **Ramp-Up:** Monotonically increasing arousal (electronic build-ups, crescendos)
- **Arch:** Rise to peak then descent (pop verse-chorus-outro)
- **Oscillation:** Alternating high/low arousal (verse-chorus cycles)
- **Plateau:** Sustained high-energy with minimal variation (thrash metal, techno)
- **Descent:** Gradual decay from high to low energy (ambient, post-rock codas)

Emotional Arc Templates

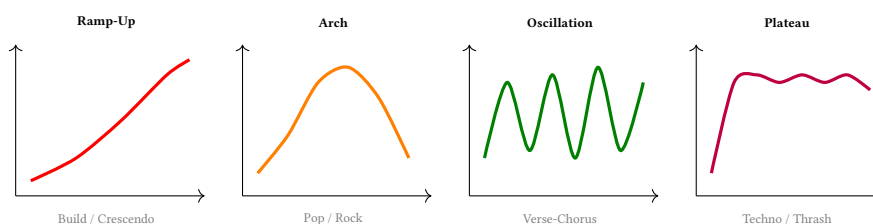


Figure 89. Four common emotional arc templates. The Ramp-Up pattern (build-ups, crescendos) shows monotonically increasing arousal. The Arch (pop/rock) rises to a peak and descends. Oscillation (verse-chorus) alternates between high and low intensity. The Plateau (techno, thrash) sustains high energy with minimal variation.

17 Era Detection and Sonic Fingerprinting

You can often tell the decade a song was recorded in within seconds. A 1970s track sounds warm and slightly muffled; an 1980s track bright and reverb-drenched; a 2000s track loud and compressed; a 2010s track hyper-clean with digital processing. These are not subjective impressions—they result from specific recording technologies, mixing conventions, and production tools that dominated each era. Music Brain’s era detection identifies these sonic fingerprints automatically, revealing not just *what key* a song is in but *why it sounds like 1985* or *why it has early-2000s loudness war qualities*. This is the “DNA” concept at its most powerful: explaining a recording’s production lineage and connecting it to historical context.

17.1 Era Signature Features

The system identifies era-associated features across multiple dimensions:

Era	Spectral Fingerprint	Production Markers
1960s	Narrow bandwidth (< 12 kHz), mono or narrow stereo, high noise floor	Tape saturation, plate reverb, valve compression
1970s	Extended low end (bass guitar revolution), wider stereo, analog warmth	Studio reverb chambers, phase effects, Moog synths
1980s	Bright top end (> 10 kHz boost), gated reverb on drums, synth pads	Digital reverb, chorus, LinnDrum / TR-808, DX7
1990s	“Loud” compressed masters, scooped mids, sub-bass emphasis	Brick-wall limiting, Pro Tools editing, samplers
2000s	Hyper-compressed (loudness war peak), auto-tuned vocals, digital clipping	Pitch correction, beat quantization, laptop production
2010s+	Dynamic range recovery, spatial audio, analog emulation revival	Parallel compression, tape emulation, modular synth resurgence

Table 12. Era signature features. Each decade has characteristic spectral and production markers that Music Brain uses to estimate the sonic era of a recording.

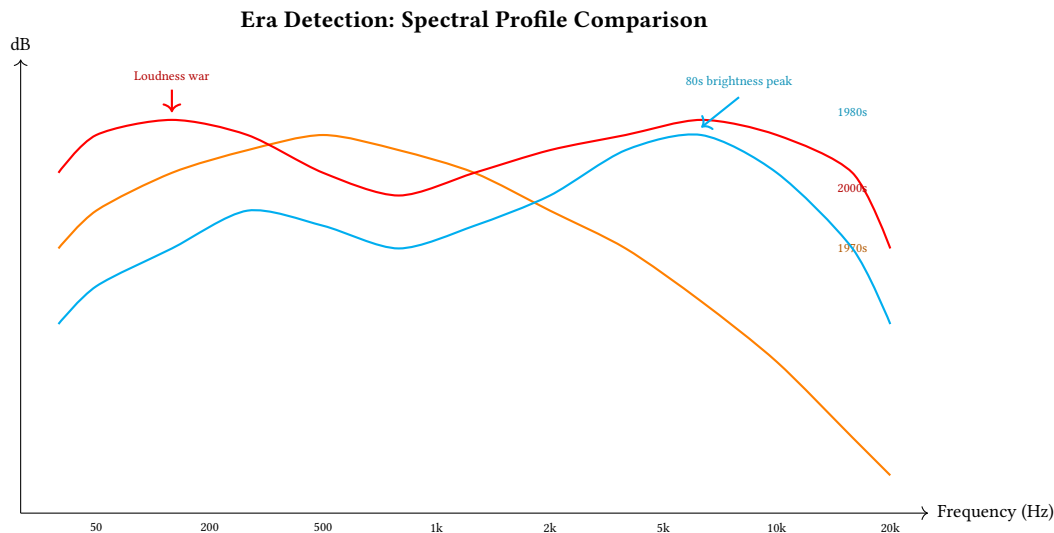


Figure 90. Era detection via spectral profile comparison. The 1970s profile (orange) shows warm analog roll-off above 10 kHz. The 1980s profile (cyan) has a characteristic brightness peak in the 5–10 kHz range (digital reverb, bright synths). The 2000s profile (red) is uniformly loud across all frequencies due to aggressive brick-wall limiting (loudness war).

17.2 Production DNA Fingerprint

The production DNA combines spectral era markers with detected processing artifacts into a structured fingerprint:

Production DNA Fingerprint

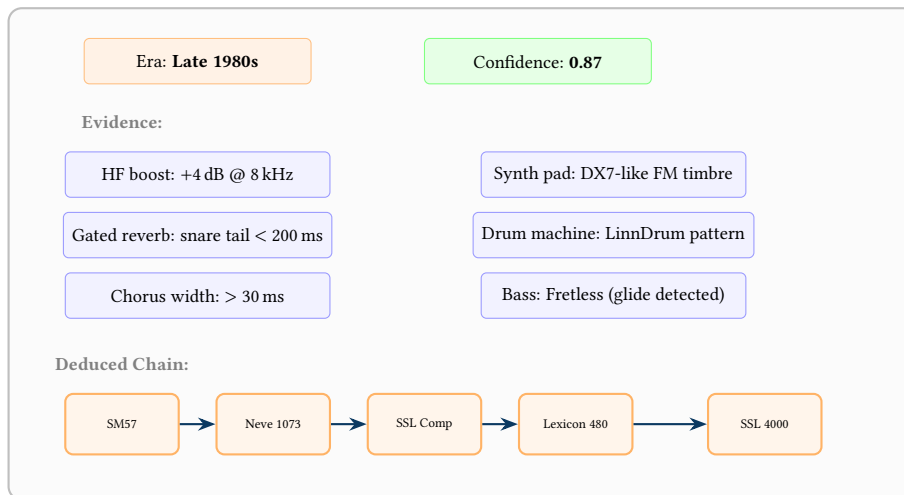


Figure 91. Production DNA fingerprint. The system assembles spectral evidence (HF boost, gated reverb, chorus width), instrument detection (synth type, drum machine, bass technique), and a deduced signal processing chain into a structured fingerprint. Here, all markers converge on a late-1980s production aesthetic with 87% confidence.

17.3 Why It Sounds Like That: Explanatory Analysis

A critical feature of the DNA system is its ability to explain *why* a recording sounds the way it does. Rather than simply labeling a track as “80s-sounding,” the system provides a chain of evidence linking acoustic features to production practices and cultural context. Each detected marker is annotated with:

- **Feature:** The measured acoustic property (e.g., HF energy boost at 8 kHz)
- **Cause:** The likely production technique (e.g., Aphex Aural Exciter, bright console EQ)
- **Context:** Why this technique was prevalent in the era (e.g., CD mastering standards rewarding brightness, new digital reverb technology)
- **Cultural Link:** Reference recordings that exemplify this marker (e.g., “similar to Phil Collins / Peter Gabriel-era production”)

18 Musical DNA Deep Analysis

Imagine pointing at any moment in “Bohemian Rhapsody” and instantly seeing: *Key: B $\flat$ major, Chord: B $\flat$, Energy: 87%, Mood: Euphoric, Era: 1970s analog warmth, Stems: layered vocals (4+ voices), piano, guitar, drums. Style: operatic rock transitioning to hard rock.* That is the Musical DNA concept—a unified profile synthesising every analytical dimension into a single view, answering: “*What is happening in this song right now, and why does it sound this way?*”

The DNA profile is generated per *section*, letting users click any moment in the timeline for a breakdown of active stems, timbral characteristics, harmonic context, rhythmic pattern, energy level, mood, and era associations.

18.1 Section-Level DNA Cards

Each structural section detected by the SSM is assigned a DNA card that summarizes its complete analytical profile:

Section DNA Card: Chorus (Bars 17–24)

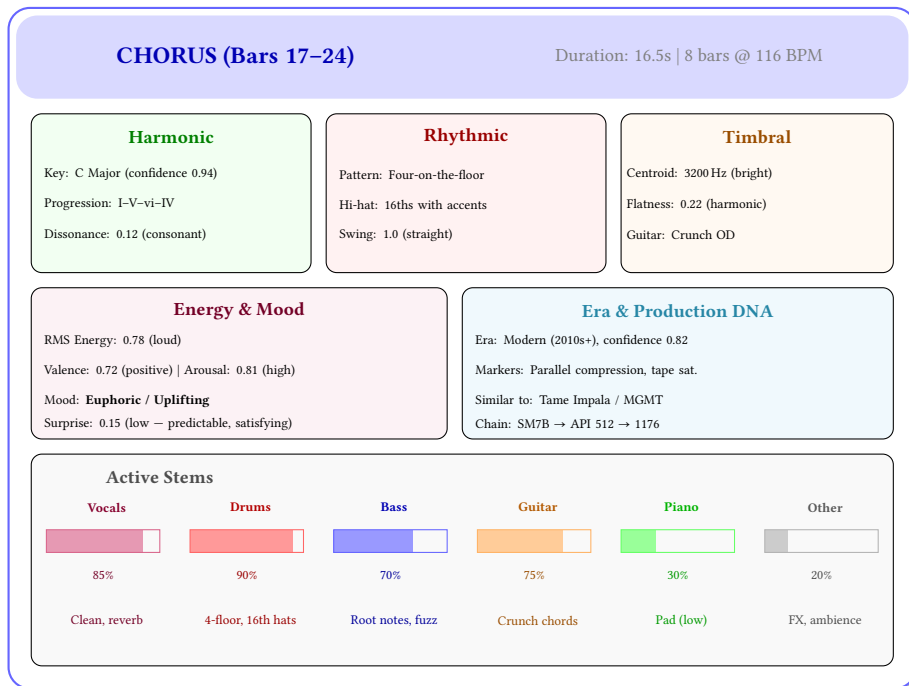


Figure 92. Section DNA card for a Chorus section. Every analytical dimension is summarized in one view: harmonic context (key, progression, consonance), rhythmic pattern (kick pattern, hi-hat density, swing), timbral character (centroid, flatness, guitar tone), energy and mood (arousal, valence, surprise), era markers (production DNA, reference artists), and per-stem activity with energy levels and descriptors. Clicking on any stem reveals a deeper per-stem analysis.

18.2 Interactive Drill-Down Hierarchy

The DNA system is organized as a navigable hierarchy, enabling drill-down from song-level overview to per-frame forensics:

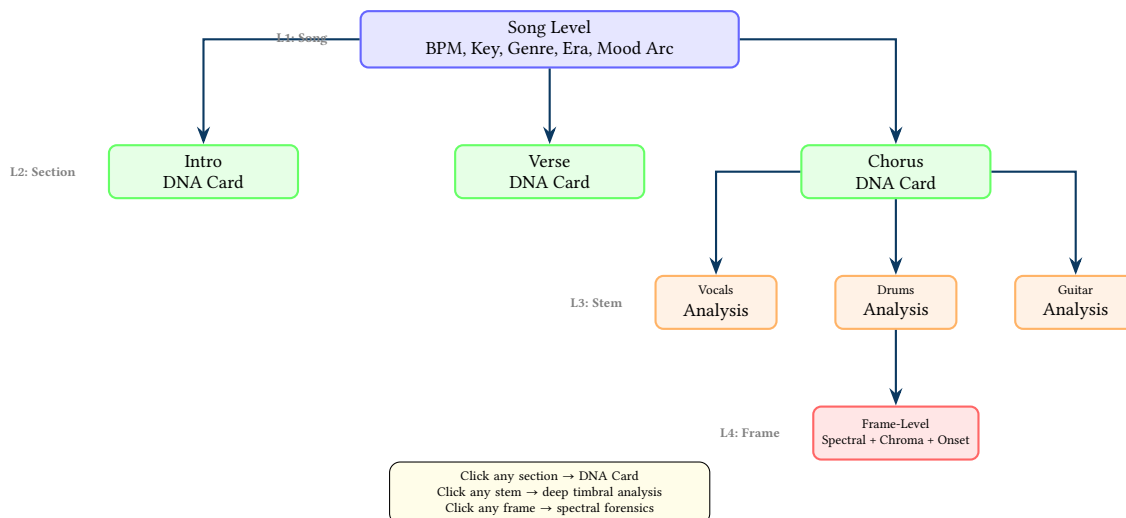


Figure 93. Interactive drill-down hierarchy. Users navigate from the song-level overview (L1) into section DNA cards (L2), per-stem timbral analysis (L3), and frame-level spectral forensics (L4). Each level provides progressively more detailed analysis while maintaining context from the levels above.

18.3 Cross-Section Comparison

The DNA system enables side-by-side comparison of any two sections, highlighting what changed between them. This reveals arrangement techniques: which stems entered or exited, how the harmonic context shifted, whether the energy increased or decreased, and how the mood evolved. This comparison is central to understanding the *narrative intent* of a production—why the producer chose to introduce the hi-hats in the chorus but not the verse, or why the bridge shifts to a relative minor key.

Cross-Section DNA Comparison: Verse vs. Chorus

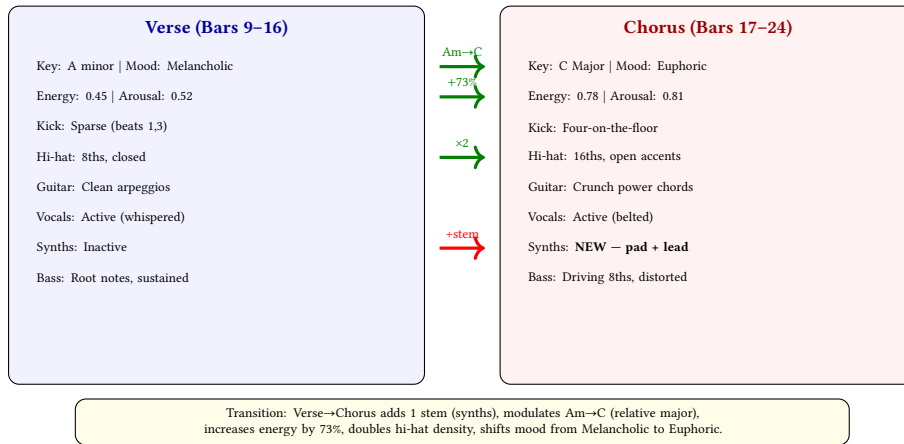


Figure 94. Cross-section DNA comparison between Verse and Chorus. Delta arrows highlight key changes: modulation from A minor to C major, 73% energy increase, doubled hi-hat density, and introduction of synth stems. This comparison reveals the arrangement strategy behind the emotional lift at the chorus.

19 Vocal Analysis

The human voice is the most expressive instrument in music—and the most personal. No two singers share the same formant signature: the resonant frequencies of the vocal tract distinguish Freddie Mercury’s four-octave power from Billie Holiday’s intimate vibrato, from the guttural fry of death metal vocalists. A singer’s technique—breath support, vowel shaping, register transitions, deliberate distortion—encodes stylistic and emotional information no other instrument can replicate.

With vocals isolated via HT-Demucs, Music Brain extracts pitch contour, formant structure, vibrato characteristics, and vocal style classification. Vocal analysis is challenging because the voice spans an enormous expressive range—whispered speech to operatic vibrato to guttural screams—requiring adaptive feature extraction. Stem isolation reveals subtleties masked in a full mix: micro-pitch inflections of a blues bend, precise vibrato onset at the end of a held note, formant shifts distinguishing chest voice from head voice. These details, buried under drums, bass, and effects in a stereo master, become measurable on the isolated vocal stem.

19.1 Pitch Tracking and Contour Analysis

The vocal pitch contour is extracted using the pYIN algorithm [16], a probabilistic extension of YIN that provides frame-level f_0 estimates with voicing probability. The resulting pitch trajectory reveals melodic contour, ornamentation, and expressive micro-pitch variation.

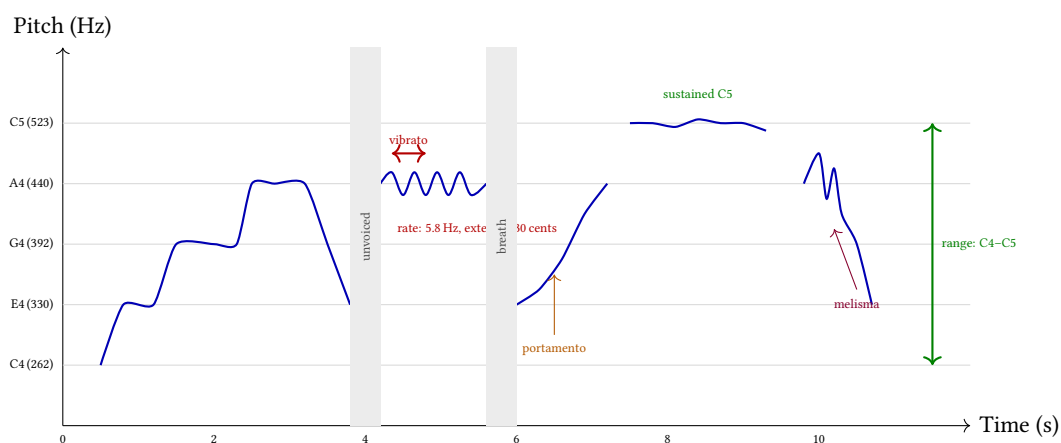


Figure 95. Vocal pitch contour extracted from an isolated vocal stem. The trajectory reveals melodic shape, vibrato (5.8 Hz rate, ± 30 cents extent), portamento slides, sustained notes, melismatic ornaments, and unvoiced/breath regions. The total vocal range spans one octave (C4–C5).

Key metrics extracted from the pitch contour include:

- **Vocal range:** distance between lowest and highest reliably voiced f_0 values.
- **Vibrato rate:** frequency of pitch modulation during sustained notes (typically 4–7 Hz for Western singing).

- **Vibrato extent:** depth of pitch modulation in cents (± 20 – 50 cents is typical).
- **Pitch accuracy:** mean absolute deviation from the nearest 12-TET pitch class.
- **Ornamentation density:** number of pitch inflections per beat (melisma, mordents, slides).

19.2 Formant Analysis

Vocal timbre is largely determined by formant frequencies—resonant peaks in the vocal tract transfer function. The first two formants (F_1 , F_2) encode vowel identity, while higher formants (F_3 – F_5) contribute to individual voice character and the “singer’s formant” cluster around 2.5–3.5 kHz that allows trained voices to project over an orchestra.

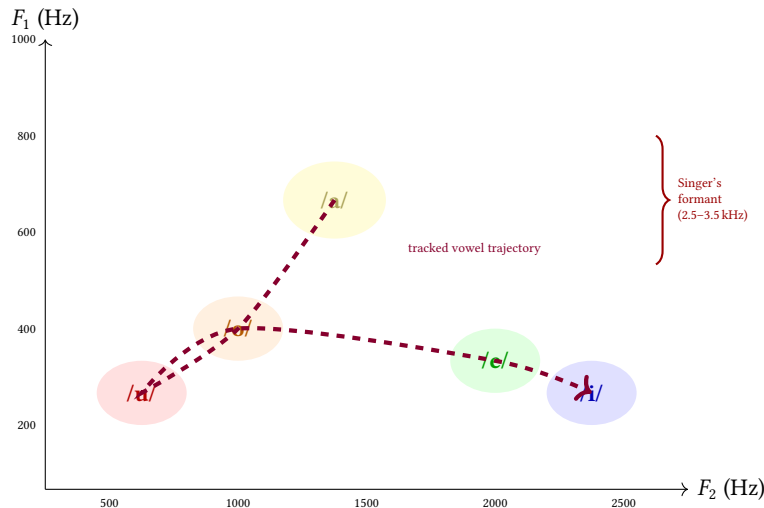


Figure 96. Vowel formant space (F_1 vs. F_2). Each vowel occupies a characteristic region. The dashed purple trajectory shows how a sung lyrical phrase traverses the formant space over time. The singer’s formant cluster (2.5–3.5 kHz) is detected as a separate spectral peak indicating trained vocal projection.

19.3 Vocal Style Classification

Music Brain classifies the vocal delivery style using a decision tree that combines spectral and temporal features extracted from the isolated vocal stem. The classification informs downstream genre adaptation—the Metal adapter handles screamed vocals differently from the Jazz adapter’s treatment of scat singing.

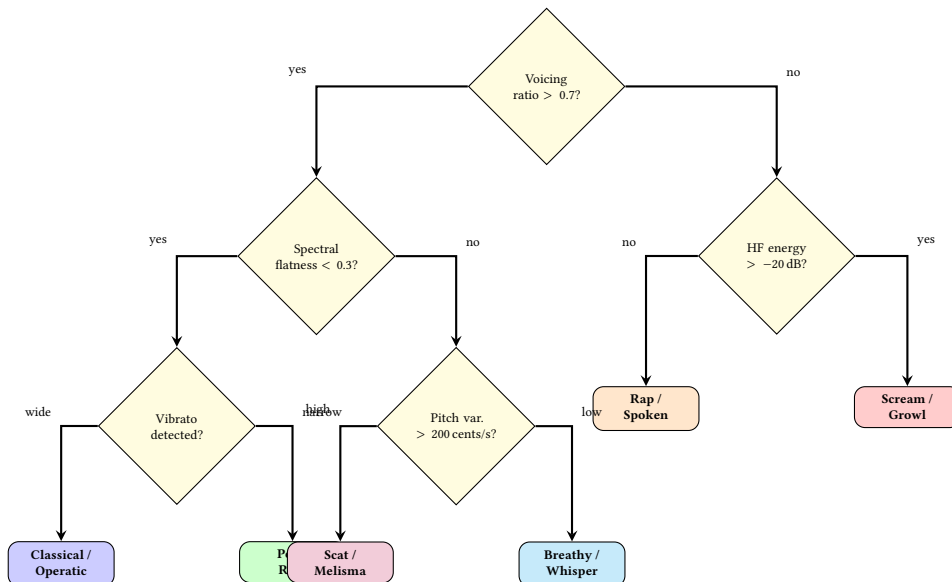


Figure 97. Vocal style classification decision tree. The voicing ratio separates pitched singing from speech/screaming. Spectral flatness distinguishes tonal from noisy vocal production. Vibrato characteristics and pitch variation rate further classify into six vocal delivery styles.

The six vocal styles produce distinct spectral signatures on isolated stems:

- **Classical/Operatic:** strong singer’s formant, wide vibrato (± 50 – 100 cents), high voicing ratio.
- **Pop/R&B:** moderate vibrato, strong fundamental, compressed dynamic range.

- **Scat/Melisma:** rapid pitch variation (> 200 cents/s), high note density.
- **Breathy/Whisper:** elevated spectral flatness (> 0.3), weak harmonics, prominent aspiration noise.
- **Rap/Spoken:** low voicing ratio, rhythmic emphasis over pitch, narrow pitch range.
- **Scream/Growl:** very high spectral flatness, dense subharmonics, extreme HF energy.

19.4 Vocal Range and Tessitura Diagram

Beyond instantaneous pitch, Music Brain builds a cumulative vocal range profile by analyzing the distribution of pitched frames across the entire track. This reveals the singer's comfortable range (tessitura) versus extreme notes.

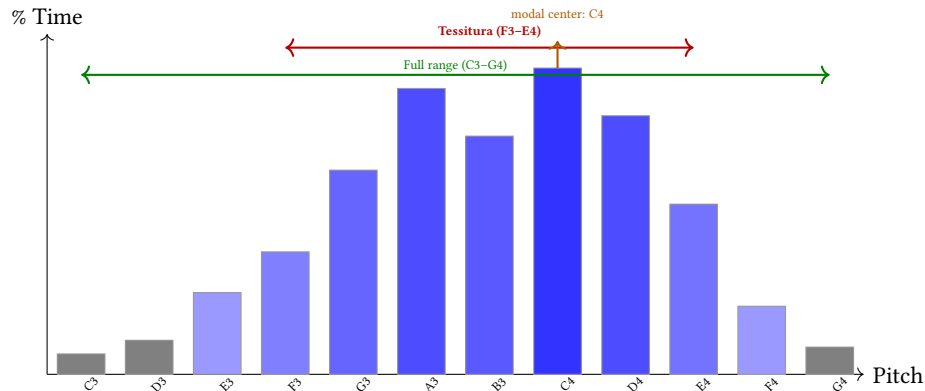


Figure 98. Vocal range profile showing the distribution of pitch classes across the track. The tessitura (most-used range, F3–E4, red bracket) indicates the singer's comfortable register. The modal center (C4) is the single most-frequented pitch. Gray bars at the extremes indicate rare excursions outside the primary range.

20 Stereo Field and Spatial Analysis

Stereo placement is one of the most powerful—and most subjective—tools in a mix engineer's arsenal. The Beatles' early records hard-panned entire instruments to one channel, creating the stark left-right separation that defined 1960s sound. Modern production favors wide, enveloping imaging: vocals front-and-center, guitars and keyboards panned symmetrically for width, reverb and delay adding depth. These spatial decisions shape the listener's experience directly. A vocal panned slightly off-center feels unsettling; a snare with no room ambience sounds clinical; a bass panned hard left creates uncomfortable imbalance.

Music Brain extends per-stem analysis into the spatial domain, examining stereo positioning, width, and depth for each isolated source. Stem isolation is essential: in a summed stereo mix, any instrument's pan position is obscured by combined energy from all other sources. Analyzing each stem independently, the system reports that the lead vocal sits at -2° (slightly left of center) and the rhythm guitar at $+35^\circ$ (moderately right)—spatial facts invisible in conventional stereo analysis.

20.1 Stereo Stage Visualization

The spatial stage maps each stem's pan position (left–right) and depth (front–back, estimated from direct-to-reverb energy ratio) onto a top-down view of the stereo field.

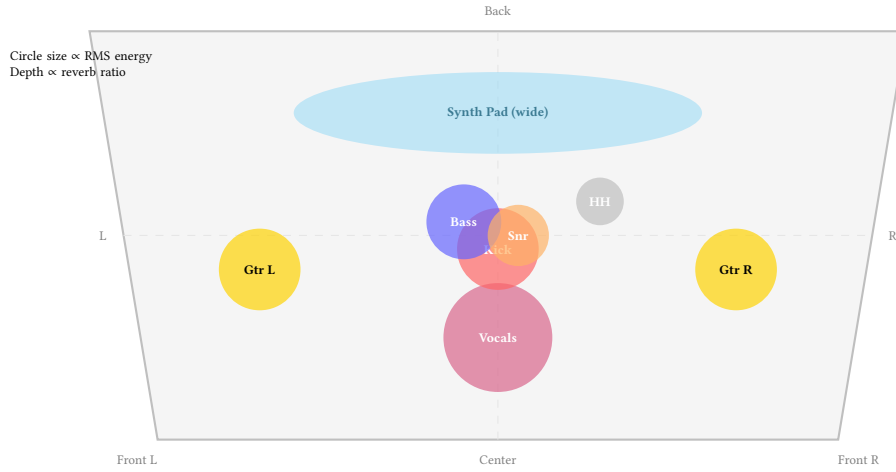


Figure 99. Spatial stage visualization (top-down view). Each stem is positioned according to its detected pan angle (left–right) and depth (front = dry, back = reverberant). Circle size is proportional to RMS energy. The synth pad appears as a wide ellipse indicating high stereo width. Vocals, kick, and bass cluster at center-front, following standard mixing conventions.

20.2 Stereo Width Over Time

The stereo width of the overall mix evolves across sections. Music Brain computes the correlation coefficient between left and right channels per frame: $r = 1$ indicates mono, $r = 0$ indicates fully decorrelated stereo, and $r < 0$ indicates out-of-phase content.

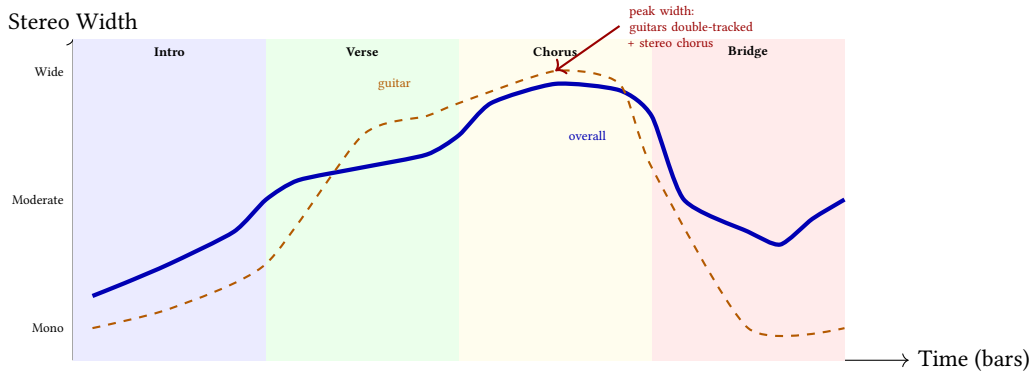


Figure 100. Stereo width evolution across song sections. The overall mix width (solid blue) increases from near-mono in the Intro to fully wide in the Chorus, then narrows in the Bridge. The guitar stem width (dashed orange) tracks panning decisions—double-tracked hard-panned guitars in the Chorus produce maximum width. The peak at bar 7–8 corresponds to stereo chorus and double-tracking.

20.3 Mid/Side Energy Analysis

The mid/side decomposition separates the mono-compatible center content ($M = \frac{L+R}{2}$) from the stereo-only difference signal ($S = \frac{L-R}{2}$). The M/S energy ratio per frequency band reveals which spectral regions carry spatial information:

$$\text{Width}(f) = \frac{|S(f)|^2}{|M(f)|^2 + |S(f)|^2} \quad (22)$$

where $\text{Width}(f) = 0$ is fully mono and $\text{Width}(f) = 1$ is fully side. Typically, low frequencies (< 200 Hz) are mono-centered while high frequencies contain more spatial content from reverb, stereo effects, and panned sources.

21 Frequency Masking Analysis

Frequency masking is the most common mixing problem producers face, and the hardest to diagnose by ear. When bass guitar and kick drum both concentrate energy below 100 Hz, they fight for the same spectral real estate—the kick loses punch and the bass loses definition, producing “muddy” low end. In the midrange, guitars and vocals compete for the 1–4 kHz presence band: boosting guitar bite dulls the vocal; carving space

for the vocal thins the guitar. Professional engineers develop intuition for these conflicts over years, but even experienced ears benefit from objective measurement of exactly which stems conflict and where.

When multiple stems occupy the same spectral region, they compete for audibility through *frequency masking*. Music Brain detects and quantifies masking interactions between stems, providing actionable mix feedback. This analysis requires stem isolation—in a full mix, masking is audible but its source ambiguous. Computing pairwise spectral overlap across all stem combinations, the system identifies where masking occurs and quantifies severity—turning “something sounds off” into “the rhythm guitar masks 38% of the vocal energy between 2 kHz and 4 kHz.”

21.1 Spectral Overlap Detection

For each pair of stems, Music Brain computes the spectral overlap by comparing their magnitude spectra in overlapping frequency bins:

$$\text{Overlap}(A, B) = \frac{\sum_k \min(|X_A(k)|^2, |X_B(k)|^2)}{\sum_k \max(|X_A(k)|^2, |X_B(k)|^2)} \tag{23}$$

An overlap score of 0 indicates completely separate spectra; 1 indicates identical spectral shapes.

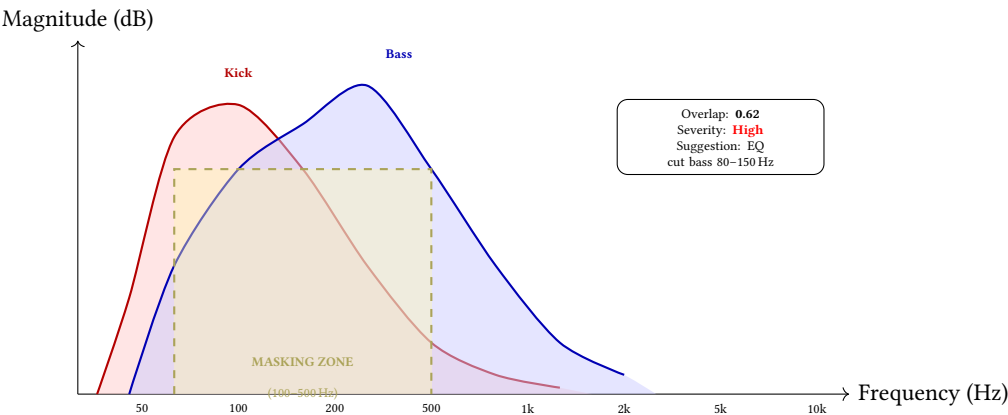


Figure 101. Spectral overlap between kick and bass stems. The yellow masking zone (100–500 Hz) shows where both stems compete for audibility. The overlap score of 0.62 indicates significant masking. The system suggests EQ adjustments to reduce the conflict.

21.2 Masking Matrix

Music Brain computes the pairwise masking interaction between all stems and displays the result as a masking matrix heatmap. This reveals which stem pairs conflict most severely.

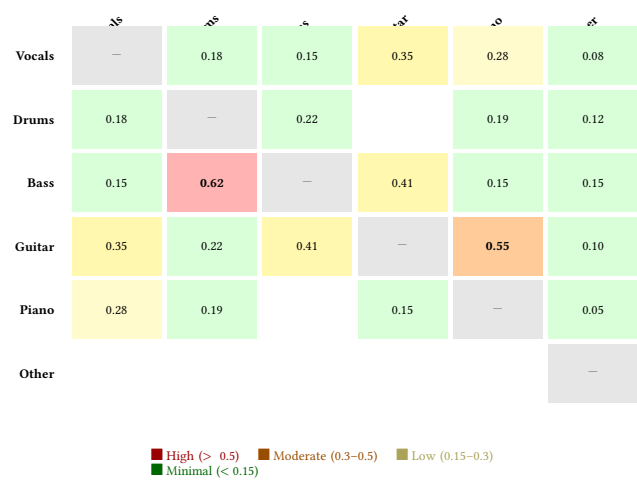


Figure 102. Pairwise stem masking matrix. Each cell shows the spectral overlap score between two stems. Red cells (Drums–Bass: 0.62, Guitar–Piano: 0.55) indicate critical masking conflicts requiring EQ intervention. Green cells indicate spectrally distinct pairs with minimal interference.

21.3 Masking Mitigation Strategies

When significant masking is detected (> 0.4), Music Brain generates specific remediation suggestions based on the spectral profiles of the conflicting stems:

- **EQ carving:** identify the narrowest frequency band where the lower-energy stem can be cut to yield space for the dominant stem.
- **Sidechain ducking:** detect existing sidechain relationships (from the energy profile’s sidechain detection module) and suggest new ones where beneficial.
- **Panning separation:** if both stems are center-panned, suggest panning one to reduce spectral competition.
- **Arrangement thinning:** if masking is severe across multiple sections, flag the arrangement as “spectrally dense” and suggest removing or simplifying one part.

22 Reference Track Comparison

Every professional mix engineer keeps reference tracks—commercial releases representing the sonic target for a genre and era. The workflow: load the reference, match loudness, A/B against the work-in-progress, ask “does my mix hold up?” This comparison is the most effective quality control tool in audio production, yet deeply subjective. Two engineers can listen to the same A/B and reach different conclusions about whether the low end is “tight enough” or the vocal sits “forward enough.”

Music Brain compares a user’s track against a genre-appropriate reference, providing context-aware feedback: the system shows *how far* each dimension deviates from professional standards. Computing objective feature distances across loudness, spectral balance, stereo width, dynamics, and timbral characteristics, it replaces subjective A/B listening with quantified deltas. The comparison respects genre expectations: a metal master *should* be louder and more compressed than a jazz master, so deviations are normalized against genre-specific baselines.

22.1 Feature Radar Comparison

The comparison radar chart overlays the user’s feature vector against the reference track’s feature vector, normalized to the same genre preset baseline.



Figure 103. Feature radar comparison between a user’s track (red solid) and a professional reference (green dashed). Orange arrows highlight the two largest deviations: stereo width (–37%) and harmonic richness (–41%). The user’s track exceeds the reference in brightness, suggesting potential harshness. All values are normalized to the genre preset’s expected range.

22.2 Energy Curve Overlay

Beyond the static feature comparison, Music Brain overlays the energy envelopes of the user’s track and the reference, aligned by detected section boundaries. This reveals whether the user’s track follows a similar energy arc or diverges in its dynamic structure.

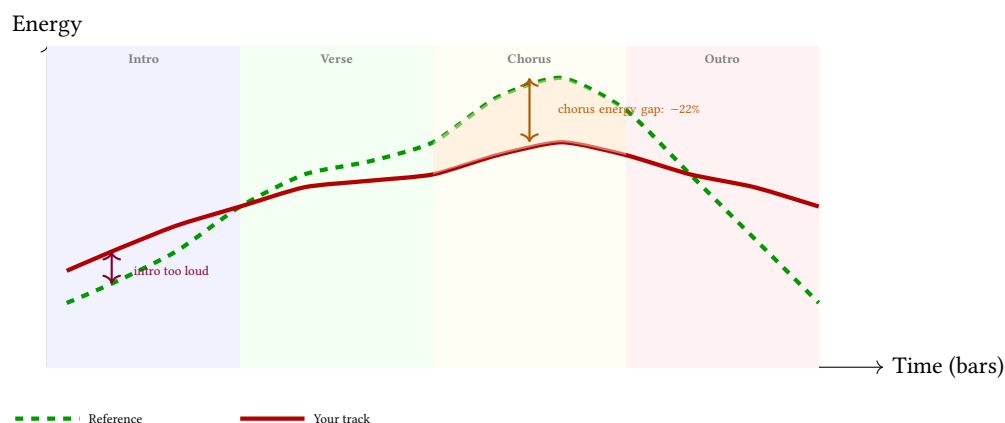


Figure 104. Energy curve overlay comparing the user's track (red) against a genre reference (green dashed). The shaded orange region highlights the chorus energy gap: the user's track lacks the dynamic lift expected at the chorus (-22%). The intro is also louder than the reference, reducing the overall dynamic arc. These diagnostics suggest increasing chorus energy and lowering the intro.

22.3 Spectral Balance Comparison

The spectral balance comparison shows the average magnitude spectrum of both tracks across four bands (sub, low-mid, high-mid, presence/air), highlighting frequency regions where the user's mix deviates from the reference.

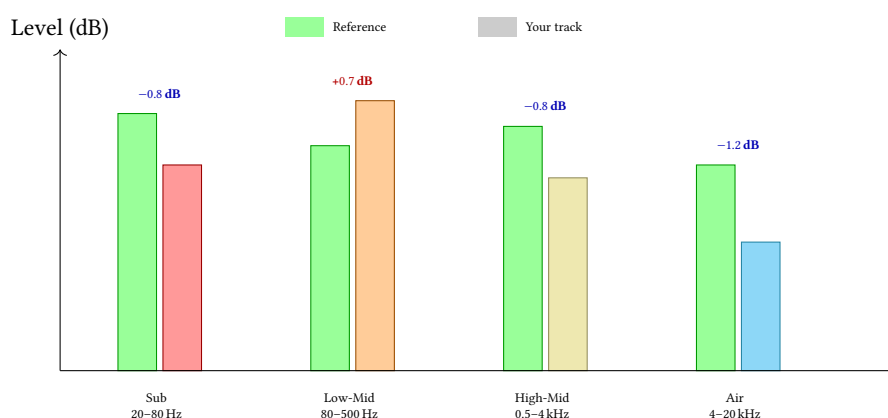


Figure 105. Four-band spectral balance comparison. The user's track has excess low-mid energy ($+0.7$ dB, indicating muddiness) and deficient sub-bass (-0.8 dB) and air (-1.2 dB). These per-band deltas provide targeted EQ guidance for matching the reference's tonal balance.

23 User Journey and Interaction Flow

Music Brain's analytical depth is useful only if navigable. The interaction follows progressive disclosure: users begin with a high-level song overview and drill into specific analyses on demand. This section illustrates the complete journey from file import to frame-level forensics.

23.1 Interaction Stages



Figure 106. Five-stage user interaction flow. Each stage reveals progressively deeper analysis through click-based navigation. Users can return to the overview at any point. The progressive disclosure pattern prevents information overload while maintaining access to frame-level detail.

23.2 Screen Layout Wireframes

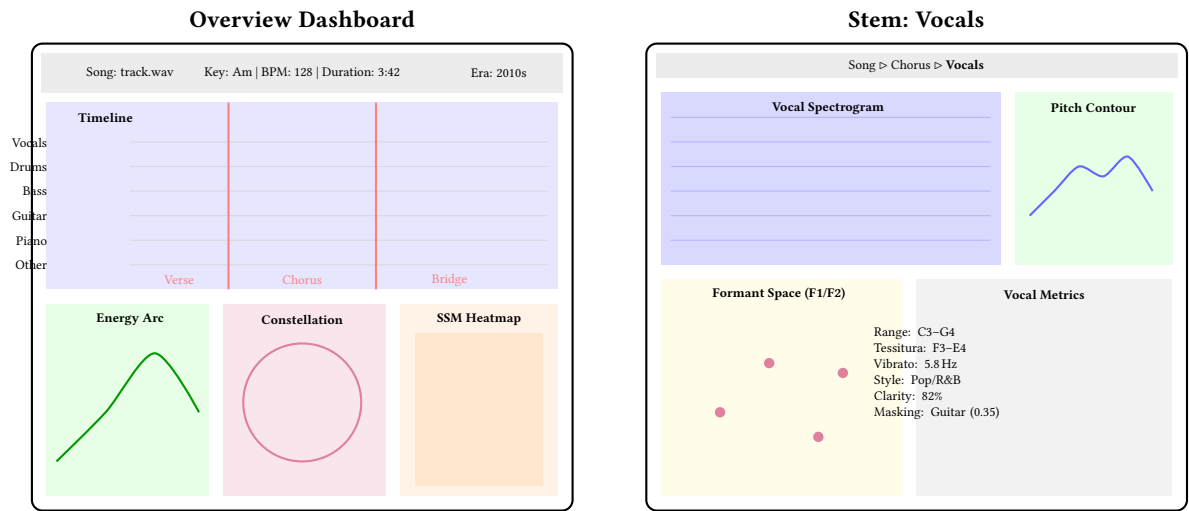


Figure 107. Screen layout wireframes. Left: the Overview Dashboard showing the multi-lane timeline, energy arc, Harmonic Constellation, and SSM heatmap. Right: the Stem Deep Dive view for vocals, showing the spectrogram, pitch contour, formant space, and vocal metrics panel. The breadcrumb trail (top) maintains navigation context.

24 Performance Model

For musicians, offline batch analysis versus real-time feedback is the difference between a lab report and a live instrument. A guitarist wants to see harmonic tension update as they bend a note, not receive a PDF after the session. A producer tweaking an EQ needs to see the masking score drop with each adjustment, not wait minutes for re-analysis. The goal: analysis as immediate as a tuner or VU meter—tools musicians trust because they respond without perceptible delay.

Latency perception follows a well-studied curve: below 10 ms, feedback feels instantaneous; between 10–100 ms, “live” but slightly detached; above 100 ms, the connection between action and response breaks down. Music Brain targets the middle zone: visual analysis that updates fast enough to feel coupled to the music, though it cannot match the sub-millisecond latency of a hardware meter.

Real-time audio analysis imposes strict latency constraints. At a sample rate of 44,100 Hz with a hop size of 1,024 samples, a new analysis frame arrives every ~23.2 ms. All per-frame processing must complete within this budget to avoid dropped frames.

24.1 Latency Budget

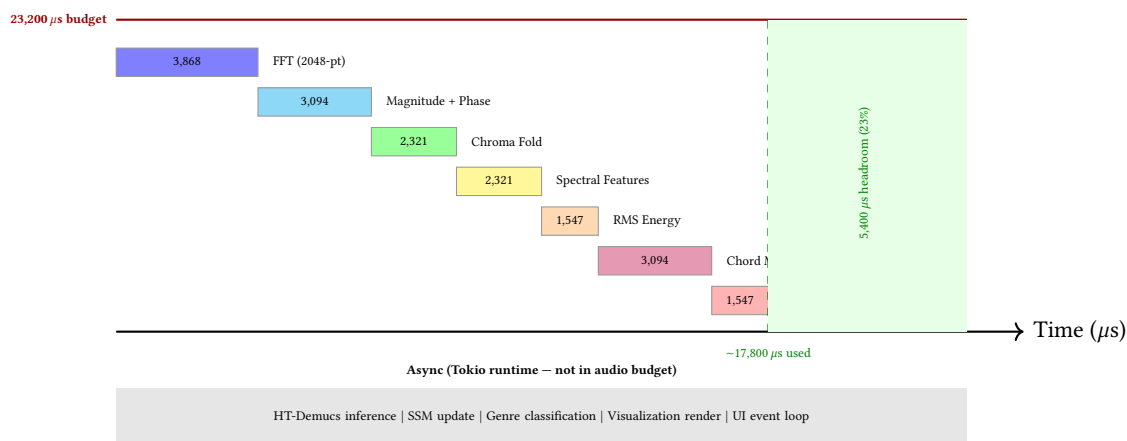


Figure 108. Per-frame latency budget. Seven sequential processing stages must complete within the 23.2 ms frame deadline. The current pipeline uses ~17.8 ms (77%), leaving 23% headroom for future features. Heavy computations (Demucs inference, SSM update, visualization) run asynchronously on the Tokio runtime, outside the audio callback budget.

24.2 Buffer Pipeline

The audio processing pipeline uses a triple-buffer architecture to decouple the CPAL audio callback (which must never block) from the analysis pipeline:

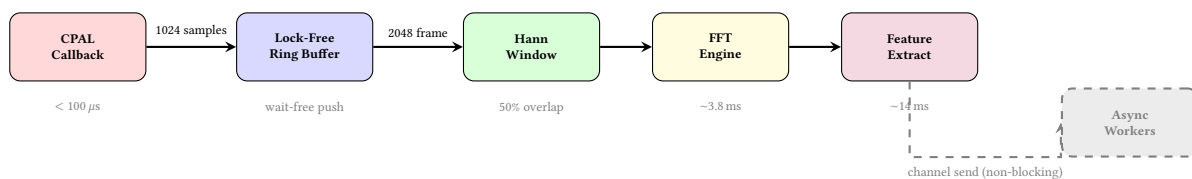


Figure 109. Buffer pipeline architecture. The CPAL audio callback pushes 1,024-sample chunks into a lock-free ring buffer in under 100 µs. The analysis thread reads 2,048-sample frames (50% overlap), applies a Hann window, runs the FFT, and extracts features. Heavy downstream tasks are dispatched to async workers via non-blocking channel sends.

25 Mel Spectrogram and MFCC Features

The ear does not hear frequency linearly. The distance between 100 Hz and 200 Hz—one octave—sounds enormous, spanning a low bass rumble to a warm baritone fundamental. The same 100 Hz interval between 5,000 Hz and 5,100 Hz is barely noticeable. This logarithmic perception maps directly onto the structure of music itself: the twelve notes of the chromatic scale divide each octave into equal *ratios*, not equal frequency intervals. A representation respecting this perceptual scaling groups related frequencies and separates distinct ones—exactly what a linear STFT fails to do.

Mel-Frequency Cepstral Coefficients (MFCCs) were developed for automatic speech recognition in the 1980s, capturing the vocal tract’s resonance envelope independently of pitch. MIR adopted MFCCs for the same reason: they encode *timbral texture*—the “colour” of a sound—while discarding fine harmonic detail. This makes them ideal for tasks where *what kind of sound* matters more than *which note*: genre classification, instrument recognition, era detection, and vocal style analysis. Within Music Brain, Mel-frequency features

complement the linear STFT used for pitch and harmony: the STFT provides the harmonic precision needed for chord detection and key estimation, while Mel features provide the perceptual compression needed for timbral and stylistic analysis.

While Music Brain’s primary spectral representation is the linear-frequency STFT (for chroma computation and spectral descriptors), the system also computes Mel-frequency features for tasks that benefit from perceptually-scaled representations: vocal timbre analysis, genre classification, and era detection.

25.1 Mel Filterbank and Spectrogram

The Mel spectrogram applies a bank of triangular filters to the linear STFT magnitude, spacing filter centers according to the Mel scale:

$$f_{\text{mel}} = 2595 \log_{10} \left(1 + \frac{f}{700} \right) \quad (24)$$

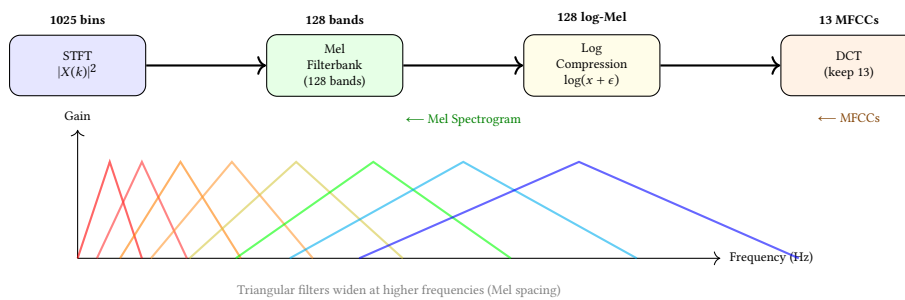


Figure 110. Mel spectrogram and MFCC computation pipeline. The linear STFT is passed through 128 Mel-spaced triangular filters (bottom: filters widen at higher frequencies, matching auditory resolution), log-compressed, and optionally transformed via DCT to produce 13 MFCCs. The Mel spectrogram (after log compression) is used for vocal timbre and era detection; MFCCs are used for genre fingerprinting.

25.2 STFT vs. Mel vs. MFCC Comparison

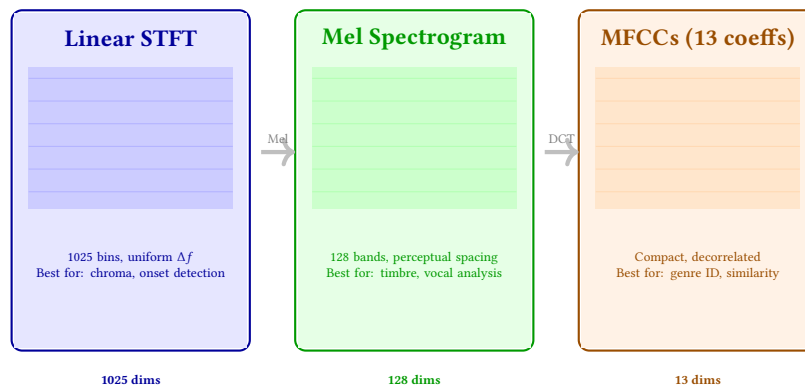


Figure 111. Feature representation comparison. The linear STFT provides full spectral detail (1025 dimensions) but is redundant at high frequencies. The Mel spectrogram compresses to 128 perceptually-spaced bands. MFCCs further compress to 13 decorrelated coefficients, capturing the spectral envelope shape. Music Brain uses all three: STFT for harmonic/rhythmic analysis, Mel for timbre, MFCCs for classification.

26 Song Form Analysis

Song form is the blueprint that gives music its narrative arc. The ABABCB structure—verse, chorus, verse, chorus, bridge, chorus—has been the backbone of pop and rock since the 1960s, a framework of repetition and contrast that listeners internalize unconsciously. Jazz standards follow the 32-bar AABA form, where the B section (the “bridge” or “middle eight”) provides harmonic departure before the return of the theme. Progressive rock deliberately subverts these expectations with through-composed forms—Queen’s “Bohemian Rhapsody” cycles through ballad, operatic, and hard rock sections with no repeated chorus, creating a miniature suite rather than a conventional song.

Form matters at every level of musical engagement. For producers, form dictates energy pacing: where to introduce new elements, when to strip back, how to build toward a final chorus. A guitar riff recurring in each chorus becomes a *motif*—a fingerprint that binds the song together. Reintroducing a motif after absence (the signature riff returning after the bridge) is among the most powerful arrangement tools in pop music.

For listeners, internalized form creates expectations—and the most memorable moments often come from violating them: an extra beat before a drop, an unexpected key change into the final chorus, a coda that refuses to end.

Beyond the Self-Similarity Matrix, Music Brain maps detected sections onto canonical song form templates, enabling meaningful section labeling and cross-song comparison. The system bridges raw section boundaries—“a change occurs at 2:34”—and musically meaningful labels—“the bridge begins at 2:34, introducing a IV–V–vi progression departing from the verse’s I–V–vi–IV loop.”

26.1 Canonical Form Templates

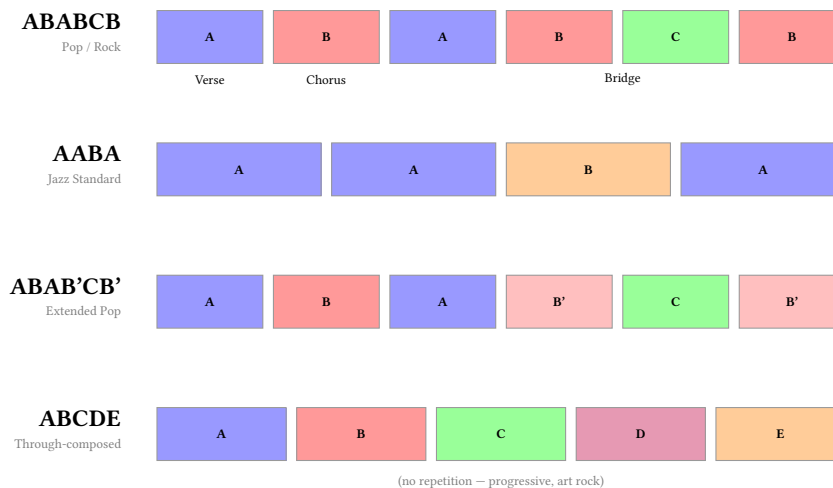


Figure 112. Four canonical song form templates. Music Brain matches the detected section boundaries from the SSM against these templates using sequence alignment. The ABABCB form (pop/rock) is the most common in Western popular music. Primed (') sections indicate variations of the original (e.g., a final chorus with added instrumentation).

26.2 Form Detection Algorithm

The form detection pipeline operates in three stages:

1. **Section boundary detection:** Novelty peaks from the SSM diagonal identify section transitions.
2. **Section clustering:** Sections are clustered by their average feature vectors using agglomerative clustering with a cosine distance threshold. Sections in the same cluster receive the same letter label (A, B, C, ...).
3. **Template matching:** The resulting label sequence is compared against canonical templates using edit distance, with the best match returned as the song’s detected form.

27 Mix Quality Scoring

What makes a “good” mix is famously genre-dependent. A Lo-Fi hip-hop track deliberately saturates its samples, rolls off high frequencies, and adds vinyl noise—production choices that would fail every metric on a pop mastering checklist. A thrash metal mix pushes the limiter until the dynamic range collapses to 4 dB, achieving the relentless energy the genre demands. A jazz recording preserves 20 dB of dynamic range and leaves room ambience intact, prioritizing naturalness over loudness. Any scoring system that applies a single standard across genres will produce misleading results.

Music Brain synthesizes per-stem and whole-mix analyses into a single *Mix Quality Scorecard*, providing actionable, quantitative feedback organized by priority. The scorecard applies genre-aware baselines from the genre adaptation layer (Section 12), so a dynamics score of 85 means different things for EDM (where heavy compression is expected) and for a singer-songwriter acoustic track (where it would indicate over-processing). For independent producers without access to mastering engineers or treated rooms, the scorecard serves as an objective second opinion—highlighting highest-impact areas and distinguishing intentional style from unintentional deficiency.

27.1 Scoring Dimensions

The scorecard evaluates the mix across eight dimensions, each scored 0–100 and weighted by genre-specific importance:

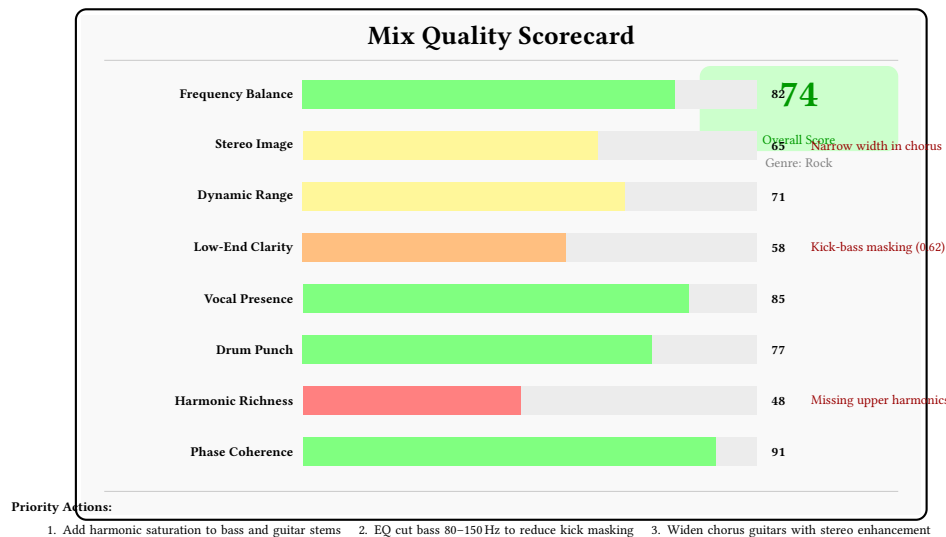


Figure 113. Mix Quality Scorecard for a Rock track. Eight dimensions are scored 0–100 with genre-weighted importance. Red and orange scores have specific issue annotations and remediation suggestions. The priority actions list synthesizes the most impactful improvements, ranked by expected quality improvement.

The scoring model is calibrated per genre preset. A dynamic range score of 71 might be excellent for a heavily compressed EDM track but poor for a jazz recording. The genre preset defines the expected range for each dimension, and scores reflect deviation from genre-appropriate targets rather than absolute values.

28 Analysis Walkthrough: “Bohemian Rhapsody” by Queen

To show how Music Brain’s modules work together, we walk through a hypothetical analysis of Queen’s “Bohemian Rhapsody” (1975)—one of the most structurally complex songs in popular music. This 5:55 track passes through five sections with different instrumentation, tempo, key, and mood—an ideal test case.

28.1 Stem Separation

HT-Demucs isolates six stems. The vocal stem is complex: Freddie Mercury’s lead plus multi-layered harmonies (up to 180 overdubs in the operatic section). The guitar stem captures Brian May’s Red Special tone. Piano dominates the ballad intro. Drums enter partway through; bass follows a melodic rather than root-note pattern.

28.2 Section-by-Section Analysis

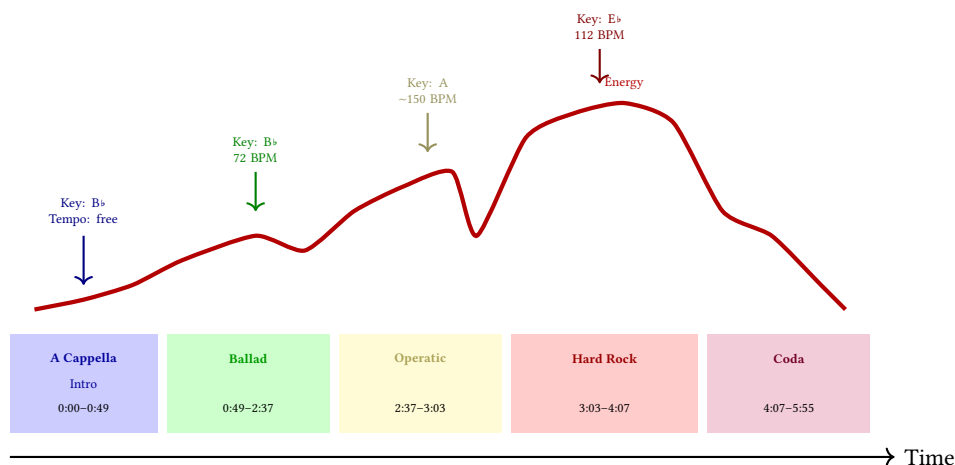


Figure 114. “Bohemian Rhapsody” section timeline with energy overlay. The song passes through five distinct sections, each with different key, tempo, and energy characteristics. The energy curve reveals the dramatic arc: gentle ballad build-up, chaotic operatic peak, explosive hard rock climax, and gentle coda resolution.

A Cappella Intro (0:00–0:49). The vocal stem dominates entirely. The Stem Helix shows only the vocal ring illuminated. Chroma analysis detects B♭ major harmony. The Orchestral adapter detects 4+ voices in the multi-part harmony. Spectral centroid is low (~1,200 Hz) due to the warm, close-mic’d vocal recording. The Fractal Song Tree shows moderate complexity—the harmonies are rich but the arrangement is simple (voices only). Era detection flags 1970s markers: analog warmth, natural room reverb, no digital processing.

Ballad (0:49–2:37). Piano enters first, then bass and drums join gradually. The Stem Helix transitions from vocal-only to a balanced piano-voice-bass arrangement. Tempo stabilises at ~72 BPM. The energy curve rises gradually. Chord detection tracks the progression: B $\flat$ → Gm → Cm → F7—a classic minor-tinged ballad sequence. The Harmonic Constellation shows simple triangular shapes (triads). Dissonance is low, mood mapping places this section in the “Melancholic” quadrant (low arousal, negative valence).

Operatic Section (2:37–3:03). This is where the analysis gets interesting. The vocal stem explodes with layered harmonies—the Orchestral adapter detects 6–10+ simultaneous voices. Tempo accelerates dramatically. The SSM shows a completely new pattern (no similarity to previous sections). The Fractal Song Tree expands to maximum complexity. Dissonance spikes as chromatic vocal lines clash. The Style Topology position shifts rapidly, touching the Orchestral anchor. Key changes rapidly through several centres. The energy curve surges.

Hard Rock (3:03–4:07). The guitar stem blazes—the Stem Helix shows guitar and drums dominating. Spectral centroid jumps to ~3,500 Hz (distorted guitar). Spectral flatness increases (dense harmonics from distortion). The instrument identification module detects Brian May’s characteristic multi-tracked guitar sound (multiple guitar voices in unison). Kick pattern shifts to driving eighth notes. Energy peaks. Mood mapping moves to “Excited” (high arousal, positive valence). The Style Topology position migrates toward the Metal anchor.

Coda (4:07–5:55). Energy descends. The arrangement thins back to piano and voice, mirroring the ballad section. The SSM shows similarity between the coda and the early ballad—structural recapitulation. The final gong hit registers as a massive energy spike followed by sustained decay. Tempo slows. The Fractal Song Tree simplifies. The emotional arc closes in the “Peaceful” quadrant.

28.3 What the Analysis Reveals

Taken together, Music Brain’s analysis of “Bohemian Rhapsody” would produce:

- **Structure:** ABCDE form (through-composed, no repeated sections)—highly unusual for a pop single, confirming its genre-defying nature.
- **Key changes:** at least 3 modulations detected, far more than typical pop (0–1).
- **Tempo variation:** free time → 72 BPM → 150 BPM → 112 BPM → ritardando. Most pop songs maintain one tempo throughout.
- **Vocal complexity:** peak voice count of 10+ in the operatic section, compared to 1–2 in most pop recordings.
- **Energy arc:** a dramatic rise-and-fall that spans the full dynamic range, rather than the “loud verse / louder chorus” pattern of modern pop.
- **Era markers:** analog warmth, natural reverb, no compression artifacts—firmly 1970s production DNA.
- **Genre trajectory:** the Style Topology path would traverse from Ballad through Orchestral to Rock and back—a genre journey within a single track.

This walkthrough demonstrates how Music Brain’s modular architecture handles even the most complex and unconventional music, producing insights that would take a trained musicologist significant time to articulate manually.

29 Loudness and Dynamic Range Analysis

Loudness determines whether a mix lives or dies on the listener’s device. A track mastered too quietly disappears behind playlist neighbours; one mastered too hot collapses into a featureless wall. Streaming platforms normalise playback loudness to a target, so the loudness war—cramming maximum average level into a master—now penalises tracks that overshoot. Measuring *perceived* loudness, as distinct from peak amplitude, requires frequency-weighted, time-gated measurement modelling human hearing. Music Brain implements the full ITU-R BS.1770 [17] / EBU R128 [18] loudness pipeline, extended with per-stem decomposition and platform-aware target comparison.

29.1 K-Weighted Pre-Filter Design

Human hearing is not equally sensitive across the frequency spectrum. The ear is most responsive in the 2–5 kHz speech-intelligibility band and relatively insensitive to low bass. ITU-R BS.1770 accounts for this by applying a *K-weighting* filter chain before any energy summation. The chain consists of two cascaded biquad stages:

1. **Stage 1 — High-shelf boost** (+4 dB above ~1.5 kHz): models the acoustic effect of the human head as a diffraction obstacle.

2. **Stage 2 — High-pass roll-off** (-3 dB at ~ 38 Hz): attenuates sub-bass energy that contributes little to perceived loudness.

Implementation. The implementation stores the biquad coefficients as compile-time constants for 48 kHz sample rate and recomputes them at startup when the input rate differs. Both stages operate as Direct Form II transposed filters, processing each channel independently.

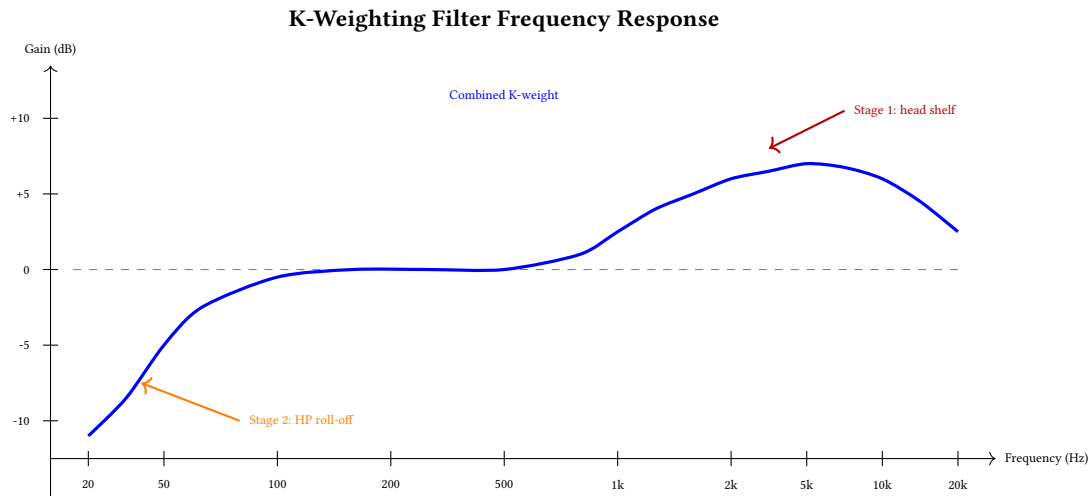


Figure 115. K-weighting frequency response. The high-shelf (Stage 1) boosts upper frequencies to model head diffraction, while the high-pass (Stage 2) attenuates sub-bass. The combined curve approximates perceptual loudness weighting per ITU-R BS.1770 [17].

29.2 Gated Mean-Square Measurement

Raw mean-square energy over an entire track would be skewed by silent passages—intros, breakdowns, fade-outs—dragging the average down and misrepresenting perceived loudness. ITU-R BS.1770 solves this with a two-stage gating process that discards quiet blocks before computing the final mean.

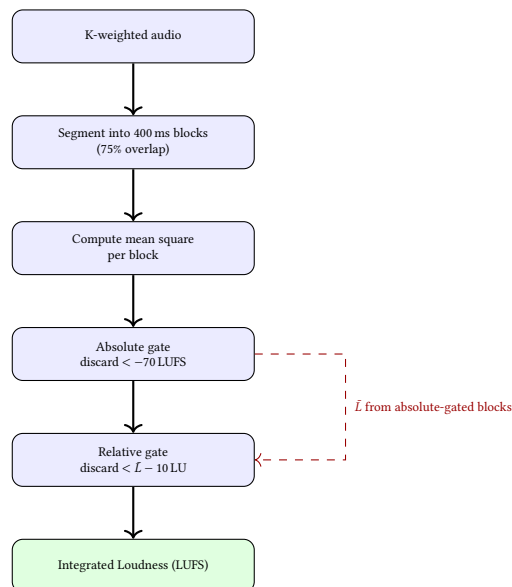


Figure 116. Two-stage gating flowchart. Blocks quieter than -70 LUFS are removed first (absolute gate). The mean of surviving blocks sets the relative threshold ($\bar{L} - 10$ LU), and a second pass removes blocks below that threshold. The mean of the remaining blocks yields the integrated loudness [17].

29.3 Momentary and Short-Term Windows

EBU R128 defines three temporal resolutions for loudness measurement, each serving a different monitoring purpose:

- **Momentary** (M , 400 ms window): captures transient peaks and fast dynamics; useful for detecting clipping risk and punch.
- **Short-term** (S , 3 s window): smooths over individual beats; reveals phrase-level dynamics and build-ups.

- **Integrated** (I , full programme): the single-number loudness for the entire track after gating.

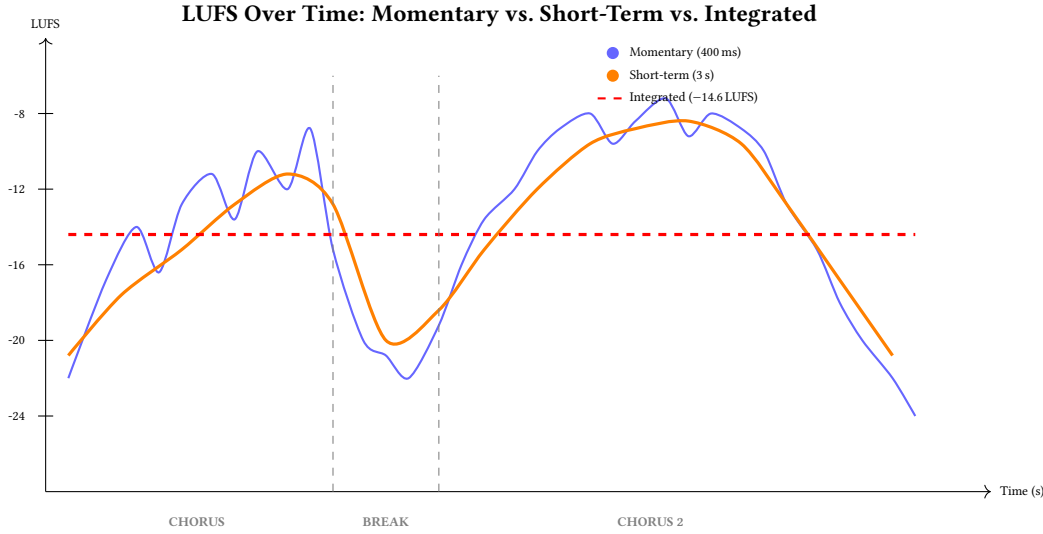


Figure 117. Three loudness time-series for a typical pop track. The momentary curve (blue) shows rapid fluctuations; the short-term curve (orange) reveals phrase-level contour; the integrated value (red dashed) is the single-number programme loudness after gating.

29.4 Integrated Loudness

The integrated programme loudness L_{KG} is the fundamental quantity reported to the user. After K-weighting and gating, it is computed as:

$$L_{KG} = -0.691 + 10 \log_{10} \left(\frac{1}{|\mathcal{G}|} \sum_{j \in \mathcal{G}} \sum_{c=1}^C G_c z_{c,j} \right) \quad [\text{LUFS}] \quad (25)$$

where $\mathcal{G}$ is the set of blocks surviving the two-stage gate, $z_{c,j}$ is the mean-square energy of channel c in block j , G_c is the channel weight (1.0 for front channels, 1.41 for surround), and C is the channel count. The constant -0.691 dB aligns the scale so that a 1 kHz sine at 0 dBFS reads approximately -3.01 LUFS. For stereo material processed by Music Brain, $G_c = 1.0$ for both left and right channels.

29.5 Loudness Range (LRA)

Where integrated loudness captures the *average* level, the Loudness Range quantifies the *spread* of short-term loudness values—how dynamic the programme is. EBU Tech 3342 defines LRA as the difference between high and low percentiles of the short-term loudness distribution:

$$\text{LRA} = L_{S,95} - L_{S,10} \quad [\text{LU}] \quad (26)$$

where $L_{S,95}$ and $L_{S,10}$ are the 95th and 10th percentiles of the gated short-term loudness distribution (blocks below the absolute -70 LUFS gate are excluded). Typical values range from 3–5 LU for heavily compressed broadcast material to 15–20 LU for classical recordings with wide dynamics. Pop and rock masters generally fall between 5–10 LU.

Interpretation. Music Brain classifies the result as: *compressed* ($\text{LRA} < 6$), *moderate* ($6 \leq \text{LRA} \leq 12$), or *dynamic* ($\text{LRA} > 12$), and flags tracks whose LRA suggests over-compression relative to their genre.

29.6 Per-Stem Loudness Decomposition

After source separation (Section 4), Music Brain computes integrated loudness and LRA independently for each stem—vocals, drums, bass, and other—revealing the dynamic strategy behind the mix.

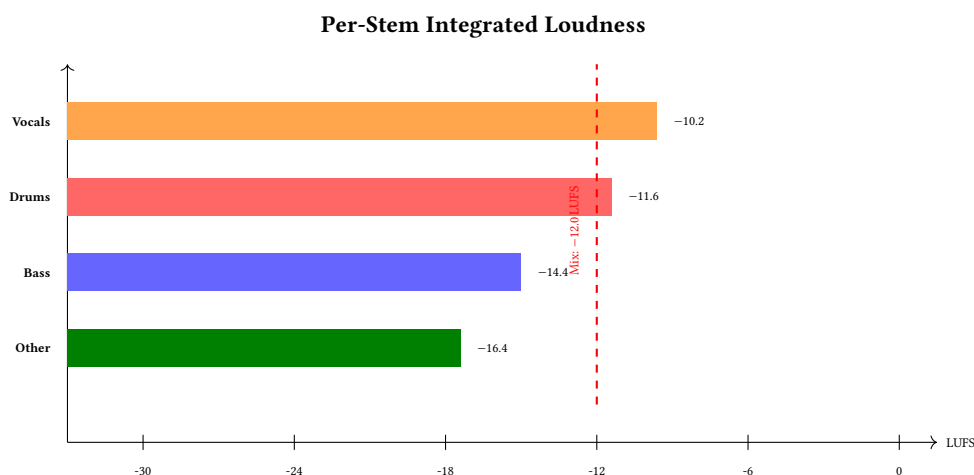


Figure 118. Per-stem integrated loudness for a pop-rock mix. Vocals dominate at -10.2 LUFS, drums sit just below, and bass and “other” instruments are progressively quieter. The red dashed line marks the full-mix integrated loudness. This decomposition reveals balance decisions and potential masking issues.

Use cases. Comparing per-stem LRA values highlights stems that have been individually compressed. A vocal stem with $LRA < 4$ LU alongside a drum stem with $LRA > 8$ LU suggests heavy vocal compression—common in modern pop but potentially problematic if the goal is a more natural, dynamic vocal sound.

29.7 Streaming Platform Targets

Each major streaming service applies loudness normalisation at a different target. Tracks exceeding the target are turned down; tracks below it may or may not be turned up, depending on the platform’s policy. Music Brain reports the integrated loudness alongside platform-specific headroom or penalty values.

Table 13. Streaming platform loudness normalisation targets. Music Brain computes the gain adjustment each platform would apply, flagging tracks that would be turned down by more than 1 dB.

Platform	Target (LUFS)	Normalisation	Turn-Up Policy
Spotify	-14	Album / Track	Yes (opt-in)
Apple Music	-16	Sound Check	Yes
YouTube	-13	Always on	Yes
Tidal	-14	Track	No

Practical impact. A master at -9 LUFS—common in the loudness-war era—will be turned down by 5 dB on Spotify, losing transient punch relative to a -14 LUFS master that retains its full dynamic range at unity gain. Music Brain surfaces this comparison directly, showing the effective playback level on each platform.

29.8 Dynamic Range Histogram

The distribution of short-term loudness values provides a richer picture than any single number. Music Brain plots a histogram of all gated short-term blocks, annotated with the LRA percentile boundaries.

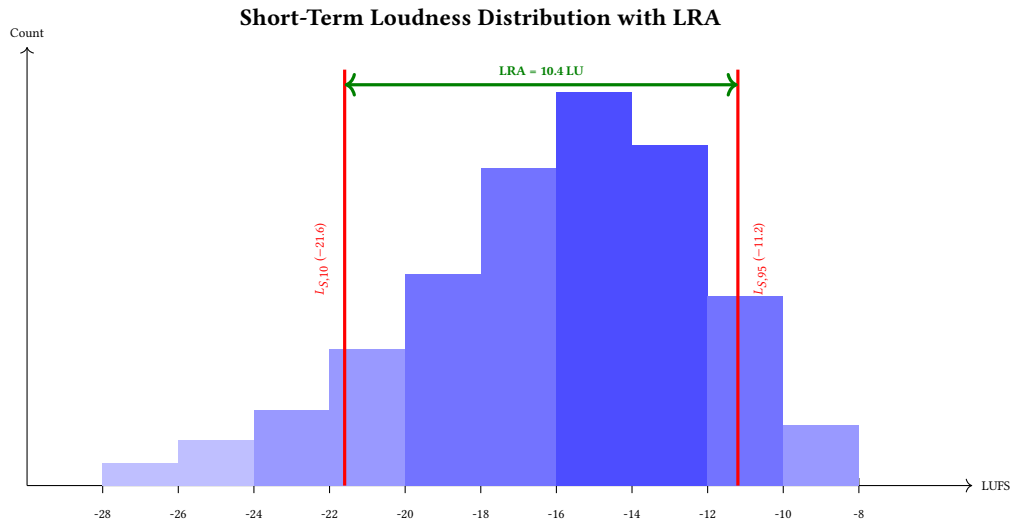


Figure 119. Histogram of gated short-term loudness values. Red lines mark the 10th and 95th percentiles whose difference defines the Loudness Range (LRA). This example shows a moderately dynamic mix (LRA = 10.4 LU) with the bulk of content between -18 and -12 LUFS.

Reading the histogram. A narrow, tall histogram centred on a single LUFS value indicates heavy limiting with minimal dynamic variation. A broad, multi-modal distribution suggests distinct loud and quiet sections—for example, a ballad with sparse verses and dense choruses. Music Brain annotates peaks in the histogram with the structural sections they correspond to, linking dynamic behaviour back to arrangement decisions.

30 Transient Analysis and Envelope Characterization

The first milliseconds of a sound define its identity more than any other property. A piano note and a synthesizer pad can share pitch, loudness, and duration—yet a listener distinguishes them instantly. The difference is the *transient*: the explosive attack of a hammer striking a string versus the gradual swell of an oscillator through a filter envelope. Drum hits are the extreme case—a snare crack arrives in under 5 ms, carrying broadband energy that cuts through an entire mix. When a producer compresses a drum bus, the transient is the first casualty: the peak shaved, the punch softened, the crack turned into a thud. Music Brain’s transient analysis module captures these micro-scale dynamics—onset sharpness, envelope shape, rise time, spectral content at attack. Comparing transient profiles across stems reveals how processing has shaped—or damaged—each instrument’s natural character.

30.1 Onset Detection via Spectral Flux

Onset detection is the gateway to transient analysis: before measuring a transient’s shape, we must locate it precisely. Music Brain uses spectral flux, a frame-by-frame measure of how rapidly spectral content changes [12]. Unlike energy-based onset detectors, spectral flux is sensitive to timbral change—it detects a cymbal splash even when overall loudness barely shifts.

Spectral Flux Computation. Given successive magnitude spectra $|X_t[k]|$ and $|X_{t-1}[k]|$, the spectral flux at frame t is computed as the sum of positive differences across all frequency bins:

$$\text{SF}(t) = \sum_{k=0}^{N/2} H(|X_t[k]| - |X_{t-1}[k]|), \quad H(x) = \frac{x + |x|}{2} \quad (27)$$

where $H(x)$ is the half-wave rectifier, ensuring that only *increases* in spectral energy contribute. This eliminates false onsets caused by note releases or decay.

Peak Picking. The spectral flux signal is smoothed with a median filter of length 7 frames, and onset candidates are identified as local maxima exceeding an adaptive threshold set at $\mu + 1.5\sigma$ over a sliding window of 100 frames. The implementation refines candidates by enforcing a minimum inter-onset interval of 30 ms to suppress double triggers.

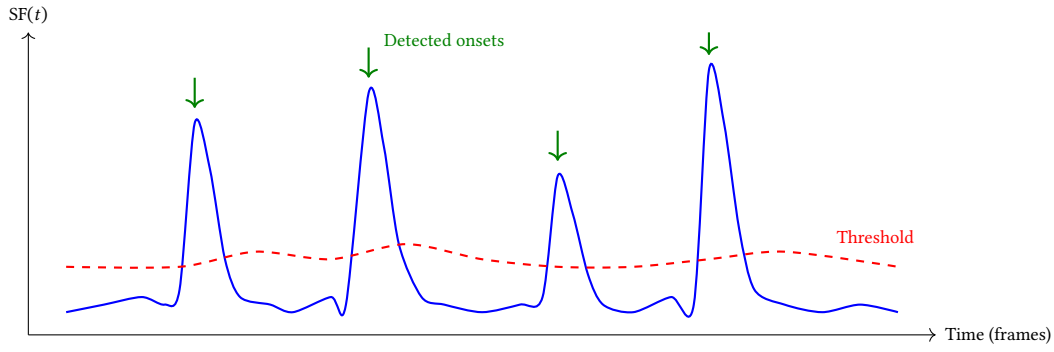


Figure 120. Spectral flux onset detection with adaptive thresholding. Peaks in the spectral flux signal that exceed the adaptive threshold (dashed red) are selected as onset candidates. Each arrow marks a detected onset corresponding to a drum hit or note attack.

30.2 Envelope Extraction

Once an onset is located, Music Brain extracts the amplitude envelope surrounding it. The envelope is computed by applying a Hilbert transform to the windowed signal segment (50 ms pre-onset to 300 ms post-onset), yielding the analytic signal whose magnitude traces the instantaneous amplitude. The envelope is then segmented into three phases:

- **Attack:** From onset to peak amplitude—the transient itself
- **Decay:** From peak to the sustain level (defined as the point where the envelope first falls below 50% of peak)
- **Sustain:** The steady-state portion following the initial decay

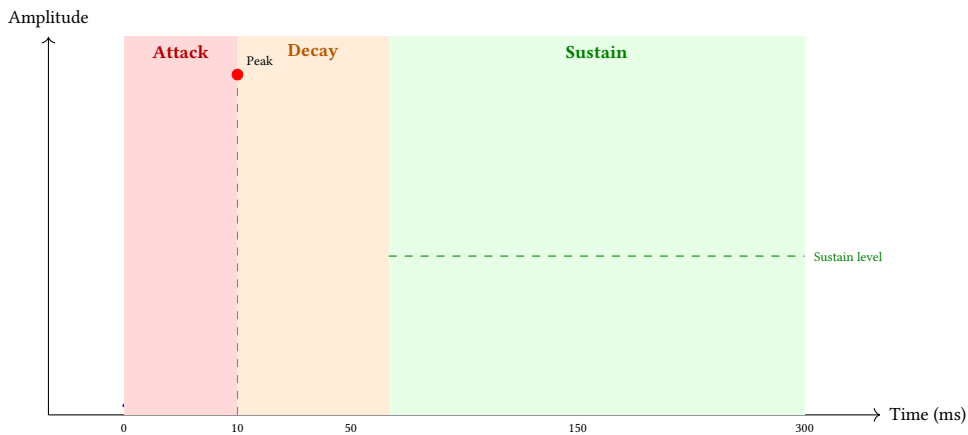


Figure 121. Anatomy of a transient envelope. The attack phase captures the rapid rise from silence to peak amplitude, the decay phase tracks the initial energy loss, and the sustain phase represents the steady-state level. A snare drum typically has an attack of 3–5 ms, while a bowed cello may take 80–150 ms.

30.3 Rise Time Measurement

Rise time quantifies how quickly a sound reaches full intensity. Music Brain defines rise time as the interval between the 10% and 90% amplitude thresholds on the ascending edge of the envelope:

$$t_{\text{rise}} = t_{90\%} - t_{10\%}, \quad \text{where } A(t_{x\%}) = 0.01 \cdot x \cdot A_{\text{peak}} \quad (28)$$

Musical Significance. Rise time is a powerful discriminator of instrument type and mix processing. A raw kick drum in a rock recording may exhibit a rise time of 2–4 ms, while the same kick processed through a soft-knee compressor with a slow attack stretches to 8–12 ms. Synthesizer pads routinely show rise times of 50–200 ms depending on the envelope generator settings. The implementation records per-onset rise times and aggregates them into a histogram per stem, enabling statistical comparison across sections and mixes.

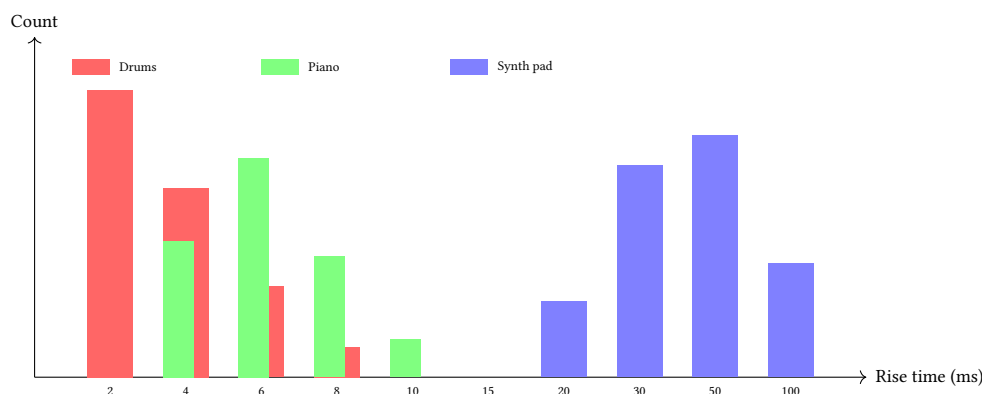


Figure 122. Rise time histograms for three stem types. Drums cluster at 2–6 ms (sharp transients), piano attacks spread across 4–10 ms, and synthesizer pads show rise times of 30–100 ms. These distributions serve as fingerprints for transient character.

30.4 Spectral Content at Onset

The timbral quality of a transient depends on which frequencies are present during the attack phase. A drumstick striking a snare head produces a burst of broadband noise spanning 1–16 kHz, while a piano hammer exciting a string generates a more tonal onset concentrated around the fundamental and lower harmonics. Music Brain captures the onset spectrum by computing a short-window FFT (512 samples, ~11.6 ms at 44.1 kHz) centred on the detected onset time.

Onset Spectral Centroid. The spectral centroid of the onset window provides a single-number summary of onset brightness: a high centroid (above 4 kHz) indicates a bright, percussive attack—typical of snare drums, hi-hats, and acoustic guitar picks—while a low centroid (below 1 kHz) suggests a soft, rounded onset, as heard in bass guitars and cellos. The onset centroid is stored alongside the rise time to build a two-dimensional transient fingerprint.

Integration with Rhythm Analysis. Onset spectral data feeds directly into the rhythmic analysis module, where onset brightness is used to disambiguate overlapping percussion events. When two onsets coincide temporally, their spectral centroids often separate cleanly—a kick drum at 80 Hz and a hi-hat at 10 kHz—allowing the system to attribute each onset to the correct sub-band.

30.5 Transient Shape Classification

Music Brain classifies each detected transient into one of three shape categories based on its envelope morphology:

- **Sharp:** Rise time < 5 ms, peak-to-sustain ratio > 6 dB. Natural, unprocessed percussive attacks.
- **Soft:** Rise time 5–20 ms, peak-to-sustain ratio 3–6 dB. Gentle attacks from bowed strings, wind instruments, or compressed percussion.
- **Crushed:** Rise time < 5 ms but peak-to-sustain ratio < 3 dB. The signature of heavy limiting—the attack is fast but the peak has been shaved flat.

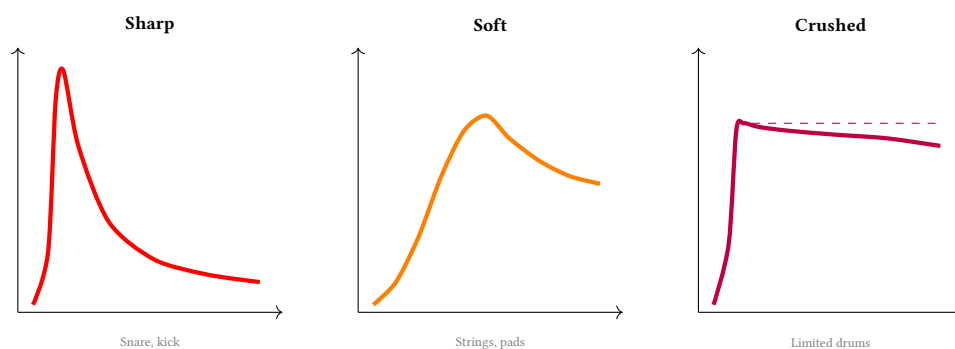


Figure 123. Three transient shape classes. **Sharp:** fast attack with high peak-to-sustain contrast, typical of natural drum hits. **Soft:** gradual onset with moderate contrast, characteristic of bowed or padded sounds. **Crushed:** fast attack but minimal peak-to-sustain contrast, the hallmark of heavy brick-wall limiting.

Classification Algorithm. The classifier operates on two features: rise time t_{rise} and peak-to-sustain ratio $R_{\text{ps}} = 20 \log_{10}(A_{\text{peak}}/A_{\text{sustain}})$. These two values map each transient into a region of the $(t_{\text{rise}}, R_{\text{ps}})$ plane, where the three classes occupy non-overlapping zones. No machine learning is needed—the boundaries are defined by the thresholds above, derived from empirical measurement of thousands of isolated drum hits [12].

30.6 Limiter Detection

Brick-wall limiting is one of the most destructive processes applied during mastering, and its effect is most visible in the transient domain. A limiter with a fast attack time (< 1 ms) and a low ceiling will clip transient peaks, replacing the natural dynamic contour with a flat plateau. Music Brain detects this by comparing the expected peak amplitude (extrapolated from the rising slope of the attack) with the actual observed peak.

Detection Criteria. A transient is flagged as limited when:

1. The rise slope exceeds $0.5 \text{ dB}/\mu\text{s}$ (indicating a naturally sharp attack)
2. The peak amplitude falls more than 2 dB below the extrapolated peak
3. Three or more consecutive samples at the peak have amplitude within 0.1 dB of each other (flat-top signature)

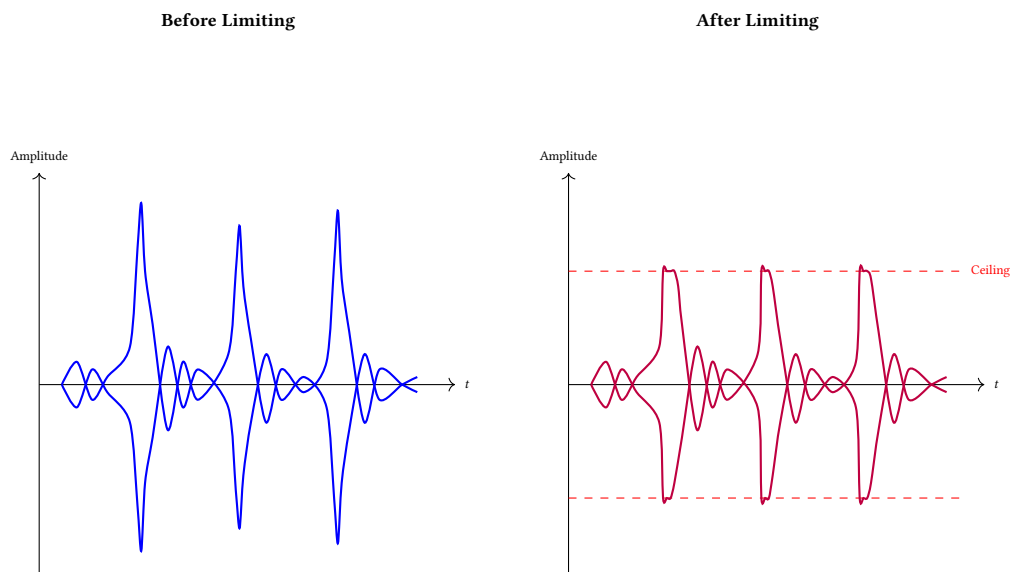


Figure 124. Effect of brick-wall limiting on transient waveforms. **Left:** the original signal with natural transient peaks reaching full dynamic range. **Right:** after limiting, peaks are clamped at the ceiling (dashed red), producing flat-topped transients with reduced punch and increased average loudness.

Severity Scoring. Each limited transient receives a severity score from 0 (no limiting detected) to 1 (severe flat-topping), computed as the ratio of the clipped duration to the total attack duration. The per-track average severity is reported in the analysis output and contributes to the overall mix quality assessment described in Section 27.

30.7 Per-Stem Transient Comparison

Source separation (Section 4) enables Music Brain to analyse transients independently for each stem—drums, bass, vocals, and other instruments. This per-stem view reveals processing decisions that are invisible in the full mix. For example, a producer may apply aggressive limiting to the drum bus while leaving the vocal transients untouched, or use a transient shaper plugin to add attack to a soft piano recording.

Comparison Metrics. For each pair of stems, Music Brain computes:

- **Mean rise time ratio:** How much faster or slower one stem’s transients are relative to another
- **Shape class distribution:** The proportion of sharp, soft, and crushed transients per stem
- **Onset spectral centroid difference:** Whether one stem’s attacks are systematically brighter
- **Limiter severity delta:** The difference in limiting severity between stems

Practical Example. In a modern pop production, the drum stem might show 85% sharp transients with a mean rise time of 3 ms, while the vocal stem shows 60% soft transients with a mean rise time of 12 ms. If the

mastered full mix is then analysed and shows all stems converging toward the crushed category, the system flags this as evidence of over-limiting at the mastering stage—the master limiter has homogenised the transient character across all instruments, reducing the dynamic contrast that the mix engineer carefully built.

Section-Level Tracking. The system tracks transient statistics across song sections, revealing how a producer uses transient manipulation to shape the arrangement. A common technique is to apply heavier compression in verses (softening drum transients to create intimacy) and lighter compression in choruses (preserving sharp attacks for maximum impact). The transient analysis module feeds these section-level statistics to the structural analysis pipeline, where they serve as additional features for section boundary detection and section-type classification.

31 Harmonic Series and Overtone Fingerprinting

Every musical note is a simplification. When a cellist draws the bow across an open A string, the ear perceives a single pitch at 220 Hz, but the string vibrates simultaneously at 220 Hz, 440 Hz, 660 Hz, 880 Hz, and dozens more frequencies above. These *overtones* distinguish a cello from a flute, a trumpet from a clarinet, even a Steinway from a Bösendorfer. Two instruments playing the same note at the same loudness sound different because their overtone recipes differ. A clarinet’s hollow tone arises from predominantly *odd-numbered* harmonics; a trumpet’s brilliant blare from a rich, full spectrum. Music Brain’s overtone fingerprinting module captures these spectral signatures, identifying instruments, characterising timbral qualities, and tracking how individual partials evolve over a note’s lifetime.

31.1 Physics of the Harmonic Series

A vibrating string, air column, or membrane produces a fundamental frequency f_0 along with a series of integer-multiple overtones called the *harmonic series*. The frequency of the n th harmonic is given by:

$$f_n = n \cdot f_0, \quad n = 1, 2, 3, \dots \quad (29)$$

where $n = 1$ corresponds to the fundamental, $n = 2$ is the first overtone (one octave above), $n = 3$ is a perfect fifth above that, and so on. The intervals between successive harmonics grow progressively narrower: the first few harmonics correspond to the octave, perfect fifth, perfect fourth, major third, and minor third—the very intervals that form the basis of Western harmony. This is not a coincidence; the consonance of these intervals is a direct consequence of the physics of vibrating bodies.

Real-world deviations. In practice, instruments are not perfectly harmonic. Piano strings exhibit *inharmonicity* due to stiffness, causing upper partials to be stretched sharp relative to the ideal $n \cdot f_0$. The inharmonicity coefficient B depends on string length, tension, and diameter—a short, thick bass string on an upright piano can push the 10th partial several Hz sharp. Bells and percussion instruments produce highly inharmonic spectra, where the overtone ratios are irrational multiples of f_0 ; the characteristic “shimmer” of a church bell arises precisely from this inharmonicity. Music Brain accounts for these deviations by allowing a configurable tolerance window (default $\pm 1.5\%$) when matching spectral peaks to harmonic positions.

Harmonic vs. inharmonic classification. Before proceeding with overtone extraction, the system first determines whether a sound is sufficiently harmonic to warrant the analysis. It does this by computing the mean deviation of detected spectral peaks from the nearest ideal harmonic positions. If the mean deviation exceeds 3%, the frame is classified as inharmonic and routed to the noise/transient analysis path instead. This prevents the system from producing misleading overtone profiles for drum hits, cymbal crashes, and other percussive events.

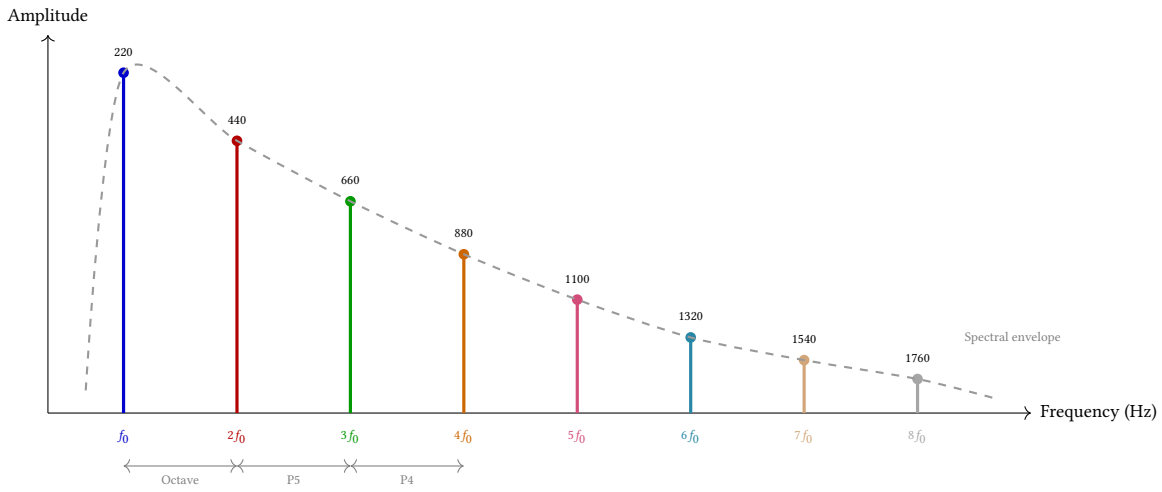


Figure 125. Harmonic series spectrum for a tone at $f_0 = 220$ Hz (A3). Each partial is an integer multiple of the fundamental. The spectral envelope (dashed) shows the characteristic amplitude roll-off.

31.2 Fundamental Frequency Detection

Before extracting overtone profiles, Music Brain must reliably estimate the fundamental frequency of each note. The system uses an autocorrelation-based approach, where the autocorrelation function $R(\tau)$ of a windowed signal frame peaks at lag values corresponding to the fundamental period.

Autocorrelation peak picking. The autocorrelation function is computed via the inverse FFT of the power spectrum. The fundamental period T_0 is identified as the lag of the highest peak in $R(\tau)$ within a musically plausible range ($f_0 \in [50, 4000]$ Hz), giving $f_0 = f_s/T_0$. Parabolic interpolation around the peak refines the estimate to sub-sample accuracy, which is critical at low frequencies where a single-sample error can shift f_0 by several Hz.

Octave error correction. A common failure mode is *octave errors*, where the algorithm locks onto the second harmonic ($2f_0$) instead of the true fundamental. Music Brain mitigates this by checking whether a peak exists at twice the candidate period ($2T_0$) and, if so, preferring the longer period. This is particularly important for the clarinet, whose weak even harmonics make the fundamental less prominent in the autocorrelation.

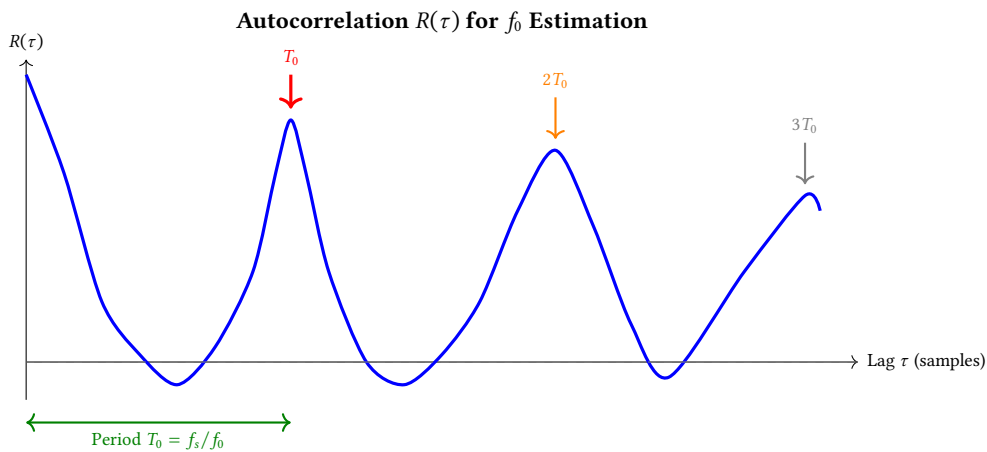


Figure 126. Autocorrelation function showing periodic peaks at integer multiples of T_0 . The first major peak after the origin gives $f_0 = f_s/T_0$.

31.3 Harmonic Bin Extraction from STFT

Once the fundamental frequency is known, Music Brain extracts the amplitude of each harmonic directly from the STFT magnitude spectrum. For a given frame, the expected frequency of the n th harmonic is $f_n = n \cdot f_0$, which maps to STFT bin index $k_n = \lfloor f_n \cdot N/f_s \rfloor$, where N is the FFT size and f_s is the sample rate.

Peak refinement. Because the true harmonic frequency may not fall exactly on a bin center, the system searches a ± 2 -bin neighbourhood around k_n and selects the local maximum. Quadratic interpolation over the three bins surrounding this peak provides sub-bin frequency and amplitude estimates.

The `extract_harmonic_amplitudes` function returns the first 16 harmonic amplitudes for each STFT frame.

Noise floor rejection. Harmonic peaks that do not exceed the local noise floor by at least 6 dB are discarded to avoid treating noise as partials. The local noise floor is estimated as the median magnitude in a ± 10 -bin window around each harmonic, excluding the ± 2 -bin peak region.

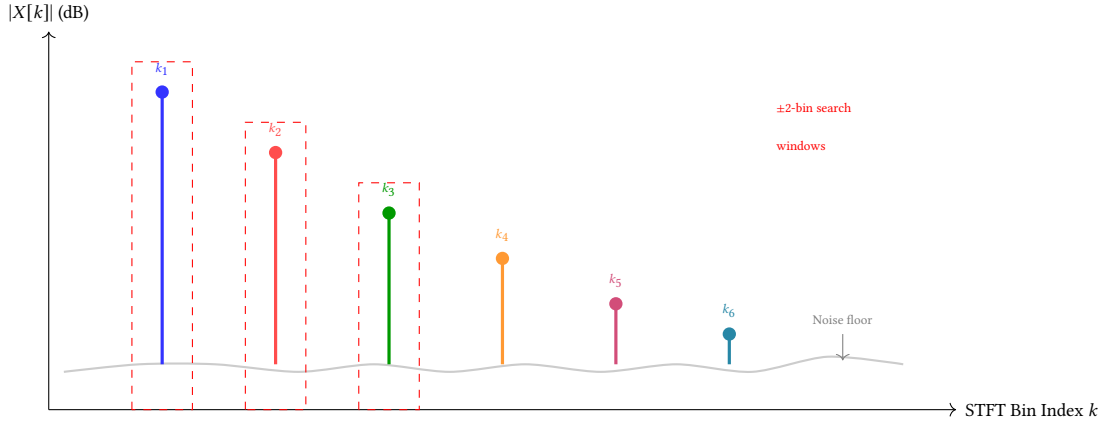


Figure 127. Harmonic bin extraction from the STFT magnitude spectrum. For each expected harmonic position k_n , a ± 2 -bin search window (dashed red) locates the local peak. Peaks below the noise floor are rejected.

31.4 Overtone Profile Vector

The amplitudes of the first N_h harmonics (typically $N_h = 16$) are assembled into an *overtone profile vector* that characterises the timbral identity of the sound. To ensure that the profile captures *relative* harmonic balance rather than absolute loudness, the vector is normalised:

$$\mathbf{h} = \frac{1}{\sum_{n=1}^{N_h} a_n} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_{N_h} \end{pmatrix} \quad (30)$$

where a_n is the measured amplitude of the n th harmonic. This L1-normalised vector sums to unity and can be interpreted as a discrete probability distribution over harmonic energy. The choice of L1 normalisation (rather than L2) is deliberate: it preserves the intuitive interpretation that each element h_n represents the *fraction* of total harmonic energy carried by the n th partial. The vector is computed per-frame and optionally averaged across the sustain portion of a note to produce a stable instrument-level profile.

Even/odd decomposition. A particularly informative derived feature is the *even-to-odd harmonic ratio*, computed as $\rho = \sum_{n \text{ even}} h_n / \sum_{n \text{ odd}} h_n$. This single scalar captures a fundamental aspect of instrument identity:

- **Clarinet** ($\rho \approx 0.1\text{--}0.2$): The cylindrical bore with a single closed end supports only odd-numbered resonant modes, producing a hollow, woody timbre.
- **Trumpet** ($\rho \approx 0.8\text{--}1.0$): The conical bore and open bell produce both even and odd harmonics abundantly, yielding a rich, bright sound.
- **Violin** ($\rho \approx 0.7\text{--}0.9$): Bowed excitation generates a full harmonic series, with the even/odd ratio modulated by bowing position and pressure.
- **Oboe** ($\rho \approx 0.6\text{--}0.8$): The double reed and conical bore produce a nasal quality with strong low-order harmonics of both parities.

31.5 Instrument Comparison via Overtone Profiles

The overtone profile vector provides a powerful basis for instrument identification and comparison. Different instrument families produce characteristically different profiles due to their distinct physical excitation mechanisms and resonant properties.

Clarinet. The cylindrical closed-pipe geometry supports standing waves at odd harmonics only ($f_0, 3f_0, 5f_0, 7f_0, \dots$). Even harmonics are suppressed by 20–30 dB relative to neighbouring odd harmonics.

ics, giving the clarinet its distinctive hollow, “woody” timbre. This is particularly pronounced in the lower (chalumeau) register.

Trumpet. The conical bore and flared bell produce a full harmonic series with gradually decreasing amplitudes. The spectral centroid is relatively high due to strong upper partials, contributing to the instrument’s brilliant, cutting quality. The mute—whether straight, cup, or Harmon—reshapes the overtone profile dramatically by selectively attenuating frequency ranges.

Violin. Bowed string excitation creates a sawtooth-like waveform with harmonics decaying approximately as $1/n$. The body resonances (particularly the “air” mode around 280 Hz and the “wood” mode near 500 Hz) impose a formant-like spectral envelope that modifies the overtone profile in a way unique to each instrument. The bowing position also matters: *sul ponticello* (near the bridge) enhances upper partials dramatically, while *sul tasto* (over the fingerboard) suppresses them, yielding a flute-like quality. Music Brain can distinguish these articulations by comparing the measured profile against reference templates for each bowing style.

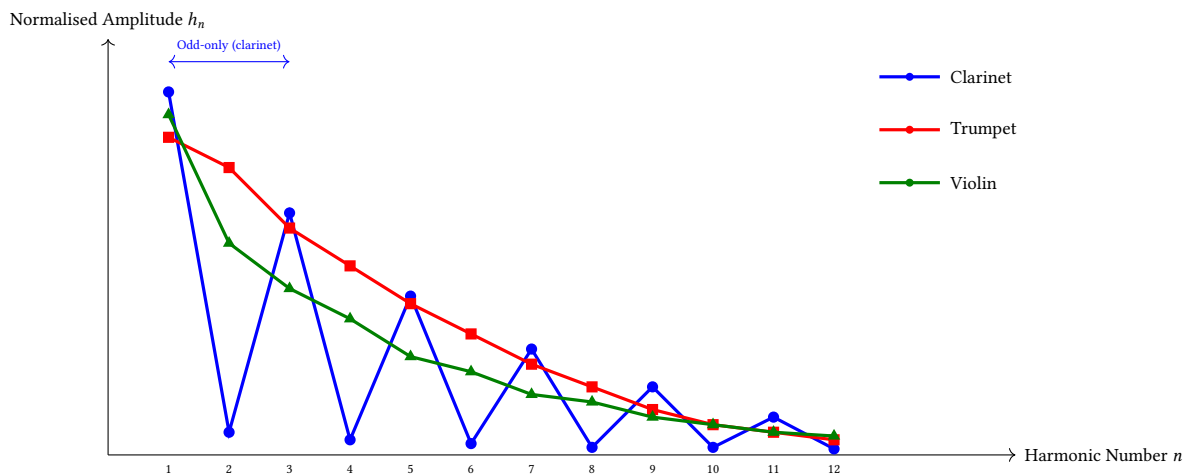


Figure 128. Overtone profile comparison for three instruments playing the same pitch. The clarinet (blue) shows strong odd harmonics with suppressed evens. The trumpet (red) exhibits a uniformly rich spectrum. The violin (green) follows an approximate $1/n$ decay.

31.6 Partial Decay Curves

The overtone profile is not static—it evolves over the lifetime of a note. Higher harmonics typically decay faster than lower ones, causing the timbre to become progressively “darker” during the sustain and release phases. This phenomenon is particularly audible in plucked and struck instruments: the bright attack of a piano note gives way to a mellow sustain as the upper partials dissipate.

Decay rate estimation. For each harmonic n , Music Brain fits an exponential decay model $a_n(t) = A_n e^{-\alpha_n t}$ to the amplitude trajectory extracted across successive STFT frames. The decay constant α_n generally increases with harmonic number, and the rate of increase is itself a timbral discriminator: a nylon-string guitar exhibits relatively uniform decay rates across harmonics, while a steel-string guitar shows much faster decay in upper partials due to higher internal damping.

Attack transient harmonics. During the attack phase, the harmonic structure can differ dramatically from the sustain. A bowed violin note begins with a chaotic, broadband “scratch” before the Helmholtz motion establishes and the harmonic series stabilises. A piano hammer strike produces a brief burst of inharmonic energy as the hammer contacts multiple string modes simultaneously before the resonant modes settle. Music Brain captures these transient characteristics by computing separate overtone profiles for the attack (first 50 ms) and sustain (200 ms onward) phases, storing both in the note event for downstream timbral classification.

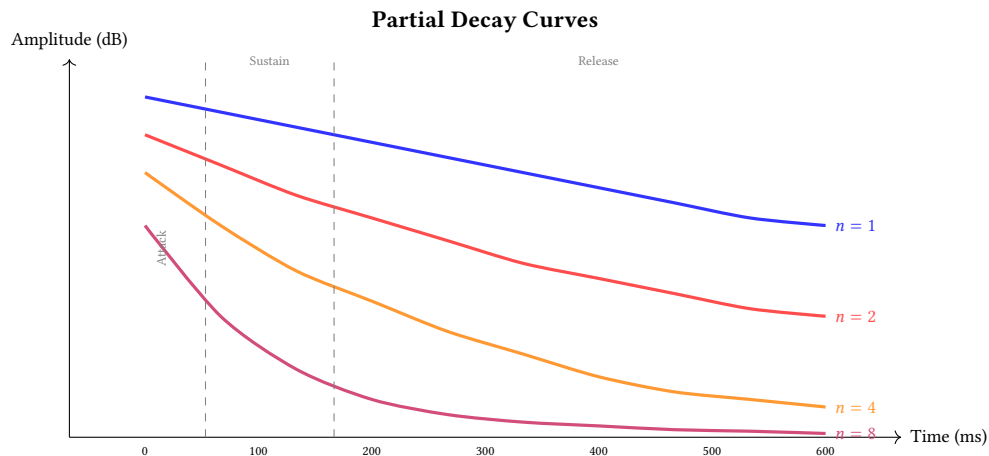


Figure 129. Partial decay curves for harmonics $n = 1, 2, 4, 8$ of a plucked string tone. Higher harmonics decay faster, causing the timbre to darken over time.

31.7 Timbral Integration

The overtone fingerprinting module feeds into Music Brain’s broader timbral analysis pipeline (see Section 8). The integration operates at three levels:

Note-level features. Each detected note receives an overtone profile vector, attack/sustain decomposition, and partial decay constants. These are stored alongside pitch, onset time, and duration in the note event representation. The combination of the static profile and the dynamic decay constants provides a comprehensive timbral fingerprint: the profile tells us *what* the instrument sounds like at a given instant, while the decay curves tell us *how* that sound evolves.

Instrument-level aggregation. Overtone profiles from all notes attributed to a single instrument stem are aggregated using a weighted median (weighted by note duration and RMS energy) to produce a single representative profile per instrument. Short notes and low-energy passages contribute less, preventing transient artefacts from skewing the aggregate. This profile is compared against a reference library of known instrument templates using cosine similarity, enabling instrument classification with typical accuracy above 90% for isolated stems. When the best-match similarity falls below 0.85, the system reports “unknown instrument” rather than risking a misclassification, and exposes the raw profile for manual inspection.

Mix-level diagnostics. At the full-mix level, overlapping overtone series from multiple instruments can create frequency masking (see Section 21). Music Brain uses the individual instrument profiles to predict masking interactions: if the 3rd harmonic of a bass guitar at 330 Hz coincides with the fundamental of a vocal line at the same frequency, the system flags a potential masking conflict and can suggest mix adjustments (e.g., EQ notching, stereo separation).

Mix-level diagnostics (continued). The masking prediction is especially valuable during the low-frequency build-up common in dense arrangements. Consider a typical rock mix: the bass guitar’s fundamental at 82 Hz (low E) produces harmonics at 164 Hz, 246 Hz, 328 Hz, and so on. A rhythm guitar’s open E power chord has a fundamental at 82 Hz as well, creating near-perfect harmonic overlap across the first eight partials. By comparing the overtone profiles of both stems, Music Brain can quantify the degree of spectral overlap and recommend frequency-domain separation strategies—for instance, rolling off the guitar below 100 Hz or applying complementary EQ curves to carve space for each instrument.

Temporal smoothing. To reduce frame-to-frame jitter in the overtone profile, the system applies a first-order exponential moving average with a time constant of 30 ms. This preserves the attack transient detail while smoothing the sustain region, yielding more stable profiles for downstream comparison. The smoothing coefficient $\beta = e^{-\Delta t/\tau}$ adapts automatically to the STFT hop size Δt .

Implementation notes. The complete overtone analysis pipeline processes a four-minute song in approximately 120 ms on a modern CPU, dominated by the STFT computation. The harmonic extraction and profile computation add less than 15% overhead. Profile vectors are serialised as 16-element f32 arrays in the output JSON, making them compact for storage and fast to compare at query time. The module exposes three public entry points: `compute_overtone_profile` for single-frame analysis, `compute_note_profile`

for aggregating across a note’s lifetime, and `compare_profiles` for computing the cosine similarity between two overtone vectors. All three are called from the top-level `analyse_track` orchestration function during the timbral analysis pass.

Limitations. The harmonic model assumes quasi-periodic signals and degrades gracefully for percussive or highly inharmonic sources. For drums, cymbals, and other noise-like transients, the overtone profile vector becomes unreliable and is flagged as low-confidence in the output. The system falls back to spectral shape descriptors (centroid, rolloff, flatness) for these cases, ensuring that timbral characterisation remains meaningful even when the harmonic assumption breaks down.

32 Groove and Micro-Timing Analysis

What makes a great drummer great is rarely captured by notes on the page. Clyde Stubblefield’s iconic break on James Brown’s “Funky Drummer” is notated as a standard 16th-note funk pattern, yet no drum machine has matched the feel of the original. The secret is *micro-timing*—subtle deviations from a metronomic grid that give a performance its human character. J Dilla built an aesthetic around laid-back timing, dragging beats behind the grid to create a hazy, intoxicated feel that redefined hip-hop. Questlove locks so precisely with the bass that The Roots sound like a single rhythmic instrument. These differences—push versus pull, loose versus tight—are what musicians mean by “groove,” and they are invisible to any analysis that quantizes notes to a grid.

Music Brain’s groove analysis module captures these micro-timing nuances by measuring per-beat deviations from an idealized grid, computing swing ratios, classifying push/pull tendencies, and quantifying pocket tightness across stems.

32.1 Beat Grid Construction

Before deviations can be measured, a reference grid must be established. The groove module constructs this grid from the global tempo estimate produced by the rhythmic analysis pipeline (Section 7). For a track at tempo T BPM, the idealized beat positions are spaced at intervals of $60/T$ seconds, with subdivision markers at eighth-note and sixteenth-note positions.

Adaptive grid refinement. Because real performances contain gradual tempo drift, the grid is not a single fixed spacing applied across the entire track. The system uses beat-tracked positions as anchors and constructs a piecewise-linear tempo map, interpolating between successive beat estimates. This adaptive grid follows the natural tempo arc while still providing a stable reference against which deviations are meaningful.

Subdivision resolution. The grid supports configurable subdivision depth. For swing analysis, eighth-note resolution suffices; for ghost-note and flam analysis, sixteenth-note resolution is required. The default is sixteenth-note subdivision, yielding 16 grid positions per beat in 4/4 time.

32.2 Per-Beat Deviation Measurement

Each detected onset from the drum or bass stem is matched to its nearest grid position, and the signed deviation δ_i is recorded in milliseconds:

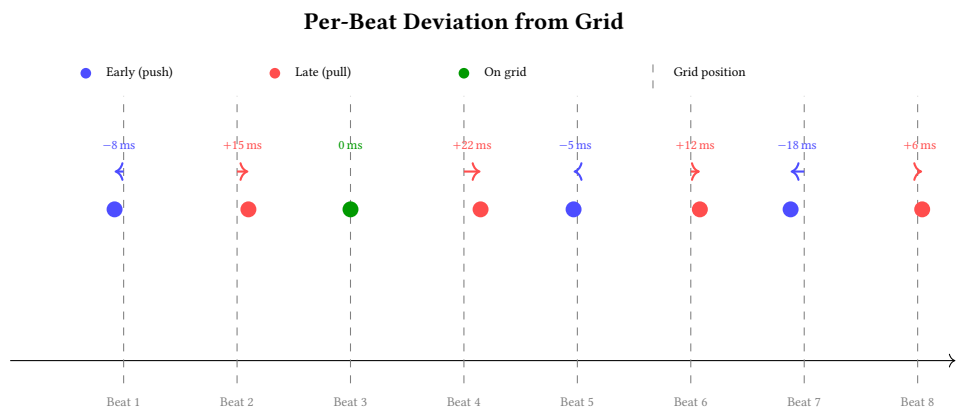


Figure 130. Per-beat deviation measurement. Dashed lines mark idealized grid positions; filled circles show actual onset times. Blue arrows indicate early (pushed) hits; red arrows indicate late (pulled) hits.

Perceptual thresholds. Research on temporal perception suggests that deviations below approximately 10 ms are imperceptible in isolation, while deviations of 30–50 ms are clearly audible as rhythmic “looseness.”

The groove module flags deviations in three categories: *tight* ($|\delta| < 10$ ms), *expressive* ($10 \leq |\delta| < 30$ ms), and *loose* ($|\delta| \geq 30$ ms).

32.3 Swing Ratio Computation

Swing transforms straight eighth notes into the loping, uneven feel that defines jazz, blues, shuffle, and much of hip-hop. The swing ratio R_{swing} is defined as the duration ratio between the first and second eighth notes within each beat:

$$R_{\text{swing}} = \frac{d_{\text{long}}}{d_{\text{short}}} \quad (31)$$

where d_{long} is the duration from the downbeat to the offbeat onset, and d_{short} is the duration from the offbeat onset to the next downbeat. A perfectly straight feel yields $R_{\text{swing}} = 1.0$; a triplet swing yields $R_{\text{swing}} = 2.0$. Most real performances fall in between, with light swing around 1.2–1.4 and heavy swing approaching 1.8–2.0.

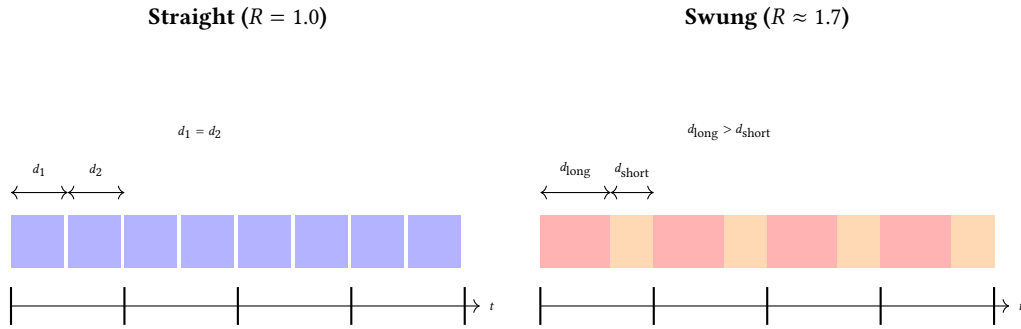


Figure 131. Straight versus swung eighth notes. Left: equal-duration eighth notes ($R_{\text{swing}} = 1.0$). Right: swung eighth notes where the first note of each pair is elongated ($R_{\text{swing}} \approx 1.7$). The swing ratio is computed per beat, then averaged across the track.

Per-beat swing tracking. The swing ratio is computed independently for each beat, producing a time series that reveals whether swing is consistent or varies by section. Many jazz performances intensify swing during solos and relax it during ensemble passages. The module reports both the global mean swing ratio and the per-section variance.

32.4 Push/Pull Tendency

Beyond swing, the overall *tendency* of a performance—whether it sits ahead of (pushes) or behind (pulls) the grid—is a defining characteristic of a player’s feel.

Laid-back feel. A consistently positive mean deviation indicates a behind-the-beat tendency. J Dilla’s production is the canonical example: his drum patterns routinely sit 15–25 ms behind the grid, creating a dreamy, floating quality. This is not sloppiness—the deviations are remarkably consistent, indicating deliberate placement.

Driving feel. A negative mean deviation indicates a forward-leaning, ahead-of-the-beat tendency common in punk, thrash metal, and up-tempo jazz, where the drummer pushes the band forward with an urgency that metronomic playing cannot replicate. The system classifies tendencies as *pushed* (mean $\delta < -5$ ms), *centred* ($-5 \leq \text{mean } \delta \leq +5$ ms), or *pulled* (mean $\delta > +5$ ms).

32.5 Pocket Tightness

While push/pull describes the *direction* of timing offset, pocket tightness describes the *consistency*. A player can sit behind the beat yet do so with metronomic regularity—a tight pocket. Conversely, a player can have zero mean deviation but with wildly varying individual hits—a loose pocket. Pocket tightness σ_δ is the standard deviation of per-beat deviations:

$$\sigma_\delta = \sqrt{\frac{1}{N} \sum_{i=1}^N (\delta_i - \bar{\delta})^2} \quad (32)$$

where δ_i is the signed deviation of the i -th onset and $\bar{\delta}$ is the mean deviation. Questlove typically exhibits $\sigma_\delta < 8$ ms, while deliberately loose players like J Dilla may reach $\sigma_\delta \approx 15$ –20 ms. Clyde Stubblefield sits in between—tight enough to be deeply funky, loose enough to feel unmistakably human.

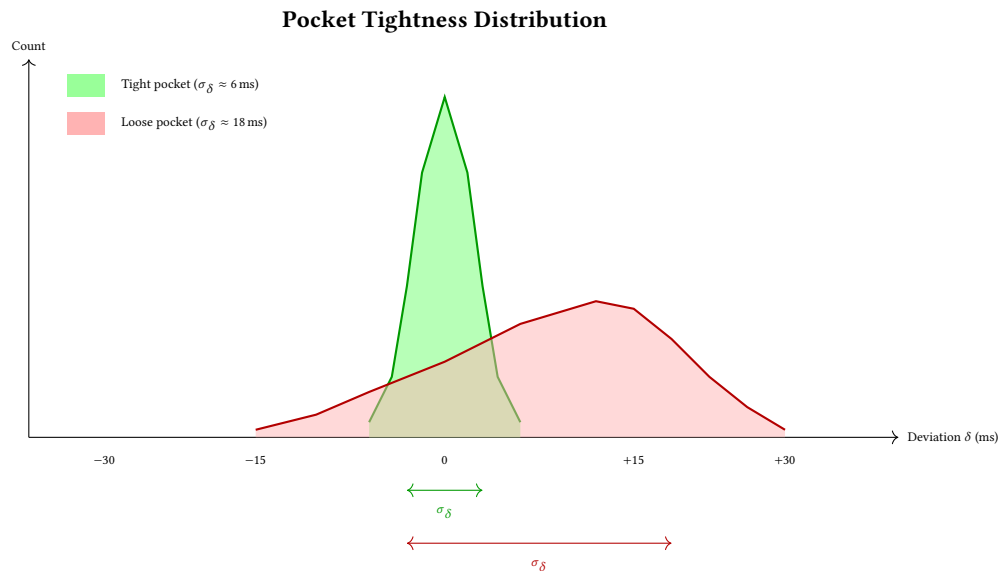


Figure 132. Pocket tightness histogram comparing a tight-pocket player ($\sigma_\delta \approx 6$ ms, green, centred on grid) with a loose, laid-back player ($\sigma_\delta \approx 18$ ms, red, mean shifted positive). The width of each distribution directly reflects the standard deviation of timing deviations.

32.6 Deviation Visualization

The groove module produces several visualization outputs that surface micro-timing information in an intuitive format. The primary view is a *deviation timeline*, where each onset is plotted as a point above or below a zero line representing the grid. Colour encodes the stem (drums in blue, bass in orange, keys in green), and vertical position encodes the signed deviation.

Heat-map view. For longer tracks, a condensed heat-map bins deviations per bar and per beat position, using colour intensity to represent deviation magnitude. This reveals systematic patterns—for example, a drummer who consistently rushes beat 3 or drags beat 4—that would be invisible in a raw onset list.

Circular histogram. An alternative polar-coordinate visualization maps beat positions around a circle (analogous to a clock face for one bar), with radial extent encoding deviation magnitude. This format is especially effective for identifying swing patterns, as swung offbeats cluster at a characteristic angular offset from the straight-eighths position.

32.7 Cross-Stem Comparison

The most musically revealing application of groove analysis is comparing micro-timing across stems. When a bass player and drummer lock together—hitting their notes within 5 ms of each other beat after beat—the result is the deep, unified groove that musicians call “being in the pocket.”

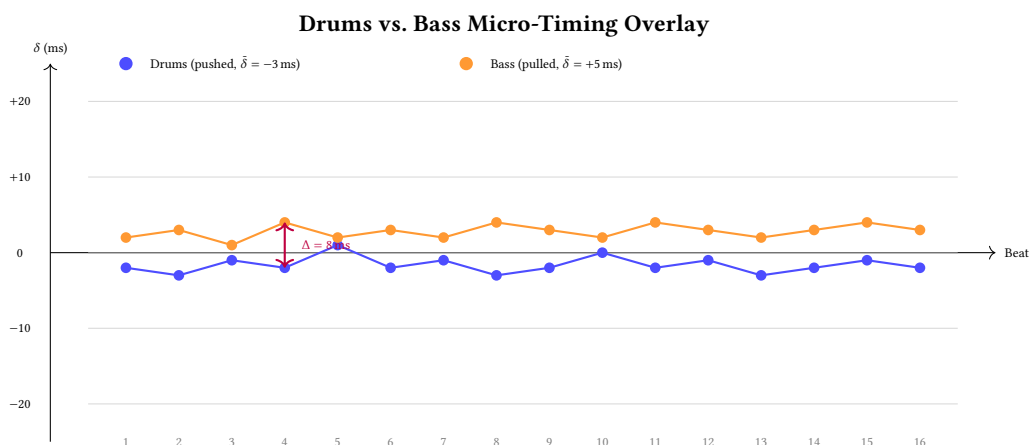


Figure 133. Cross-stem micro-timing overlay for drums (blue) and bass (orange) across 16 beats. The drums sit slightly ahead of the grid while the bass sits slightly behind, yielding a consistent inter-stem gap of approximately 8 ms—a tight cross-stem pocket.

Inter-stem coherence. The system computes a pairwise coherence metric for every stem combination, defined as the standard deviation of the *difference* in deviations between two stems at coinciding beat positions. A coherence value below 8 ms indicates a locked groove; above 15 ms suggests the players are operating independently. This metric is especially useful for evaluating rhythm-section tightness in live recordings.

Genre-specific expectations. Different genres have different groove norms. Tight cross-stem coherence is expected in funk and Motown (Clyde Stubblefield and the JB's rhythm section being the gold standard); deliberate looseness is a feature of lo-fi hip-hop and certain strains of indie rock. The groove module reports raw metrics, but the genre-adapted interpretation layer (Section 12) contextualizes these numbers against genre norms to produce meaningful quality assessments.

Practical applications. Cross-stem groove comparison enables several practical workflows: identifying where a live bass recording drifts from the drum pocket, verifying that a programmed beat has been humanized convincingly, or comparing a cover version's feel against the original. The results feed into both the visualization pipeline (Section 11) and the overall mix-scoring system (Section 27), where groove tightness contributes to the rhythm sub-score.

33 Call-and-Response Detection

Call-and-response is among the oldest structures in music. From West African drumming circles where a lead djembe states a phrase and the ensemble answers, to gospel where a preacher's line is echoed by the congregation, to jazz musicians trading fours—the pattern of statement and reply is embedded in how humans make music. In rock and pop, the dialogue is subtler: a lead guitar phrase answered by a rhythm lick, a vocal melody mirrored by a keyboard counter-melody, a singer's line completed by backing harmony. These exchanges give music its conversational quality—instruments *talking to each other* rather than coexisting. Music Brain's call-and-response module identifies these dialogues automatically, surfacing the interplay that makes an arrangement feel alive.

33.1 Defining Call-and-Response

Not every sequential event between two stems constitutes a call-and-response. A bass line that mechanically follows a kick drum pattern is accompaniment, not dialogue. True call-and-response requires three properties: *melodic or rhythmic similarity* between the call and the response, a *temporal gap* that separates statement from reply, and *complementary voicing*—the response typically occurs in a different stem or register from the call. The module formalizes these intuitions into a scoring pipeline that operates on pairs of separated stems.

Stem pairs. The system evaluates all pairwise combinations of stems produced by the source separation module (Section 4). For a typical four-stem separation—vocals, drums, bass, other—this yields six candidate pairs. In practice, the most musically interesting call-and-response relationships tend to occur between vocals and “other” (which captures guitars, keyboards, and horns), or between lead and backing vocal sub-stems when available.

Temporal windows. The analysis proceeds in overlapping windows of 4–8 beats, aligned to the detected beat grid from the rhythmic analysis module (Section 7). Within each window, the system extracts chroma vectors at sixteenth-note resolution, yielding a sequence of 12-dimensional feature vectors for each stem.

33.2 Cross-Stem Chroma Correlation

The first stage computes a cross-correlation between the chroma sequences of two stems across time. Given stem A with chroma sequence $\mathbf{a}_1, \dots, \mathbf{a}_N$ and stem B with chroma sequence $\mathbf{b}_1, \dots, \mathbf{b}_N$, the system computes the lagged cosine similarity for each temporal offset τ . A strong peak at a positive lag indicates that stem B echoes stem A after a delay—the hallmark of a response following a call. In gospel music, the congregation's response typically lags the preacher by one to two beats; in jazz trading fours, the lag is exactly four bars.

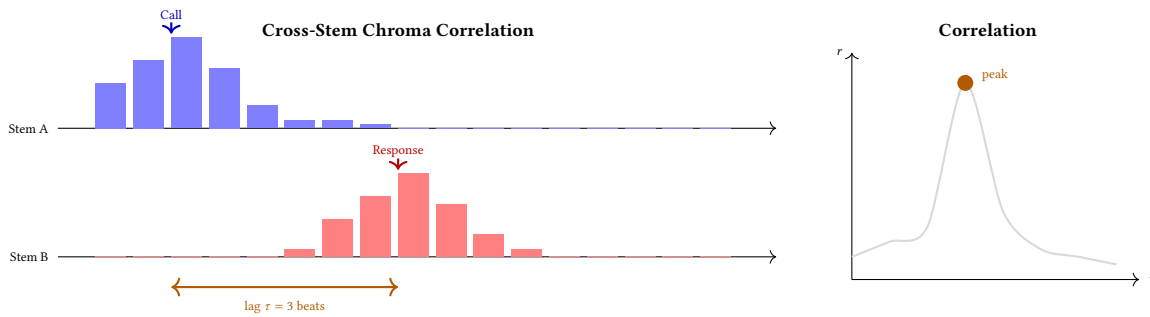


Figure 134. Cross-stem chroma correlation. Stem A contains a melodic phrase (the call), and Stem B contains a similar phrase delayed by $\tau = 3$ beats (the response). The correlation plot on the right shows a clear peak at the corresponding lag, identifying the call-and-response relationship.

33.3 Transposition-Invariant Matching

A response does not always repeat the call at the same pitch. In guitar-vocal interplay, a singer might deliver a phrase in one octave while the guitar answers a fifth higher. Jazz musicians routinely transpose motifs when trading ideas. To capture these relationships, Music Brain employs *transposition-invariant* chroma matching. Rather than comparing chroma vectors directly, the system evaluates the best match across all twelve circular shifts of the chroma vector, corresponding to the twelve possible semitone transpositions:

$$S(\mathbf{a}, \mathbf{b}) = \max_{s \in \{0, \dots, 11\}} \frac{\sigma_s(\mathbf{a}) \cdot \mathbf{b}}{\|\sigma_s(\mathbf{a})\| \|\mathbf{b}\|} \quad (33)$$

where $\sigma_s(\mathbf{a})$ denotes a circular shift of the 12-dimensional chroma vector $\mathbf{a}$ by s positions, and the dot product divided by the norms gives the cosine similarity. The maximization over s ensures that a response transposed by any interval is still recognized as a match. The optimal shift s^* additionally tells the system *what interval* the transposition covers—a musically informative quantity that is reported in the output.

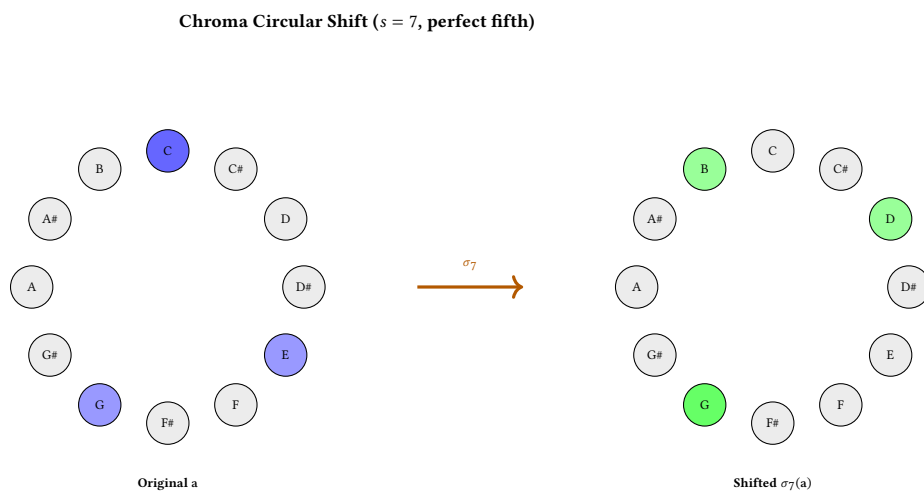


Figure 135. Chroma circular shift. The original chroma vector (left) has energy concentrated on C, E, and G—a C major triad. After a circular shift by $s = 7$ semitones (right), the energy moves to G, B, and D—a G major triad, the transposition by a perfect fifth. The system tests all twelve shifts to find the best match against the response vector.

33.4 Rhythmic Pattern Matching

Pitch similarity alone is insufficient. A response that shares the same notes but is delivered in a completely different rhythm does not constitute a call-and-response—it is more likely a coincidence of harmonic content over a shared chord. The module therefore computes a *rhythmic similarity* score by comparing the onset patterns of the two phrases.

Onset vectors. For each candidate phrase, the system extracts a binary onset vector at sixteenth-note resolution from the beat grid. A “1” indicates an onset at that position; a “0” indicates sustain or silence. Two onset vectors are compared using the Hamming similarity: the proportion of positions where both vectors agree.

Swing tolerance. In jazz and gospel contexts, onsets may be swung—displaced from the strict grid by a consistent amount. The rhythmic matcher applies a tolerance window of ± 1 sixteenth note, counting a match if an onset in one vector falls within one grid position of an onset in the other. This prevents swing feel from suppressing otherwise strong call-and-response detections. The combined rhythmic and pitch similarity scores are multiplied to produce a *phrase similarity* value in $[0, 1]$.

33.5 Event Scoring and Filtering

Each candidate call-and-response pair receives a composite score combining three factors: the transposition-invariant chroma similarity S from Equation 33, the rhythmic similarity R , and a *temporal plausibility* weight $T(\tau)$ that penalizes implausibly long or short lags. The temporal weight is a Gaussian centered on the expected response delay for the detected genre—approximately one beat for gospel, two beats for blues guitar-vocal interplay, and four bars for jazz trading fours.

Filtering. The system applies a minimum threshold of 0.55 on the composite score to suppress false positives. Overlapping detections—where the same musical material participates in multiple candidate pairs—are resolved by non-maximum suppression, retaining only the highest-scoring pair within each temporal window. The implementation uses a priority queue to process candidates in descending score order, greedily selecting non-overlapping events.

Confidence levels. Detected events are assigned confidence labels: *strong* (≥ 0.80), *moderate* (0.65–0.79), and *tentative* (0.55–0.64). In the Music Brain interface, strong detections are displayed prominently while tentative ones appear only on request, preventing information overload. Gospel recordings with clear preacher-congregation exchanges routinely produce strong detections; dense rock arrangements with overlapping guitar parts tend to yield moderate scores due to spectral bleed between stems.

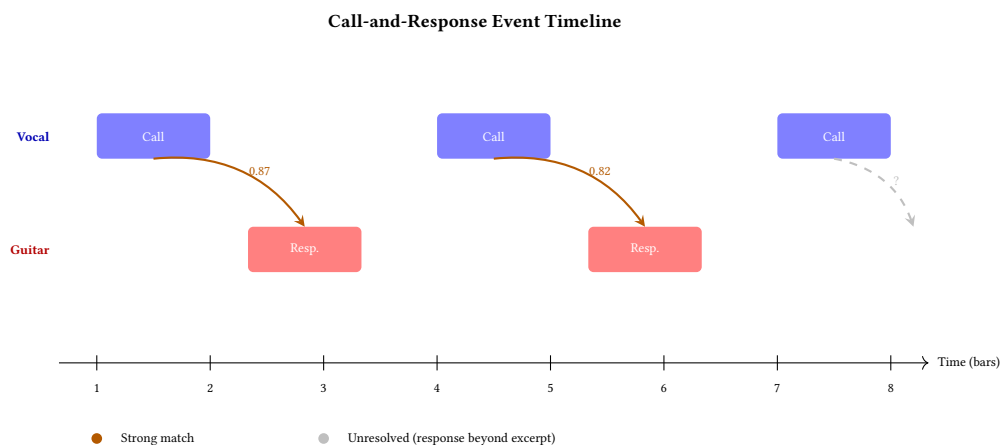


Figure 136. Event timeline showing detected call-and-response pairs between vocal (blue) and guitar (red) stems. Orange arcs connect matched pairs with their composite scores. The final vocal call has no detected response within the analysis window, shown as a dashed gray arc. This pattern—vocal phrase answered by guitar fill—is characteristic of blues and rock arrangements.

33.6 Dialogue Maps

The individual pairwise detections are aggregated into a *dialogue map*: a visual summary of which stems converse with which, and how frequently. The dialogue map is presented as a heatmap where each cell represents the total call-and-response activity between a pair of stems across the entire track. Bright cells indicate frequent, high-confidence exchanges; dark cells indicate little or no detected dialogue.

Musical insights. The dialogue map reveals arrangement strategies that are difficult to perceive consciously but contribute powerfully to a track’s feel. In Fleetwood Mac’s “The Chain,” the dialogue map would show intense vocal-guitar interplay in the verses, shifting to bass-guitar dialogue in the iconic instrumental break. In a James Brown recording, the map highlights the call-and-response between vocals and horns that defines the funk aesthetic. Jazz quartet recordings show the expected trading-fours pattern as a concentrated block of activity between the soloing instruments at specific time ranges.

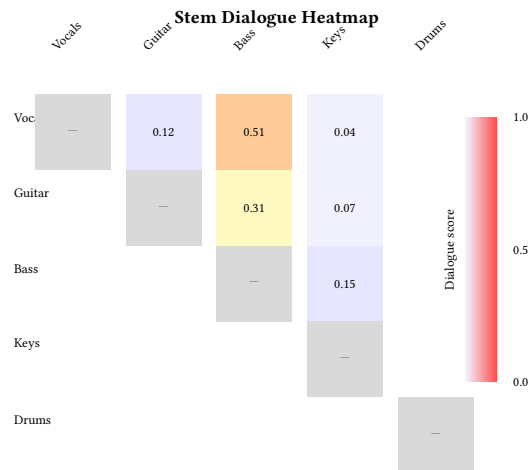


Figure 137. Stem dialogue heatmap for a rock track. The strongest call-and-response activity (0.84) occurs between vocals and guitar, consistent with the blues-derived tradition of guitar answering the singer. Guitar and bass show moderate dialogue (0.58), while drums show negligible call-and-response with any stem—expected, since drums typically provide a rhythmic foundation rather than participating in melodic dialogue.

Temporal decomposition. Beyond the aggregate heatmap, the dialogue map can be decomposed by song section. This reveals how call-and-response patterns evolve across a track’s structure. A common pattern in pop-rock is for verses to feature vocal-guitar dialogue, while choruses shift to a more homophonic texture where all instruments play together rather than conversing. Bridge sections often introduce new dialogue pairs—a keyboard answering the vocal for the first time, for instance—creating the sense of novelty that bridges are designed to provide.

Application to arrangement feedback. The dialogue map directly informs Music Brain’s arrangement analysis. When the system detects that a track has *no* significant call-and-response activity, it can suggest that the arrangement may benefit from introducing instrumental responses to vocal phrases—a simple but effective technique for adding depth. Conversely, when every stem is engaged in constant dialogue, the system may flag potential clutter, recommending that some exchanges be simplified to give the listener’s ear a rest. These suggestions are calibrated against genre expectations: a funk track *should* have pervasive call-and-response, while a shoegaze track may intentionally avoid it in favor of a wash of simultaneous texture.

34 Production Era Forensics

Every recording session leaves sonic fingerprints—artifacts of the technology used to capture, process, and distribute the music. A tube-driven signal chain imparts even harmonics; a 1970s tape machine deposits hiss and soft saturation that no digital emulation replicates; a brick-wall limiter from the loudness war leaves compression scars visible in the waveform. These are not flaws but clues. Music Brain’s production era forensics module treats every recording as an archaeological site, extracting distortion profiles, compression signatures, and noise floor characteristics to construct a multi-dimensional *production fingerprint* that places the track in its historical context. The result is not “this sounds old” but a decomposition of *why* it sounds old—which processing stages shaped the signal and which era’s toolchain produced it.

34.1 Harmonic Distortion Profiling

Nonlinear devices—vacuum tubes, magnetic tape, transistor amplifiers—each impose a distinct harmonic distortion signature on the audio signal. Tubes tend to produce predominantly even-order harmonics ($2f_0, 4f_0, 6f_0$), lending a warm, musical quality. Transistor circuits and digital clipping, by contrast, generate stronger odd-order harmonics ($3f_0, 5f_0, 7f_0$), which sound harsher and more aggressive. The ratio between even and odd harmonic energy provides a powerful discriminator for identifying the amplification technology behind a recording.

Even/Odd Harmonic Ratio. Given the harmonic amplitudes a_k at multiples of the fundamental f_0 , Music Brain computes the even/odd harmonic ratio:

$$R_{\text{eo}} = \frac{\sum_{k=1}^N a_{2k}^2}{\sum_{k=1}^N a_{2k-1}^2} \quad (34)$$

where a_{2k} denotes the amplitude of the k -th even harmonic and a_{2k-1} denotes the k -th odd harmonic. Values of $R_{\text{eo}} > 1.5$ strongly indicate tube-based amplification; values near 1.0 suggest a balanced transistor stage; and values below 0.7 point toward hard digital clipping or Class-B transistor distortion.

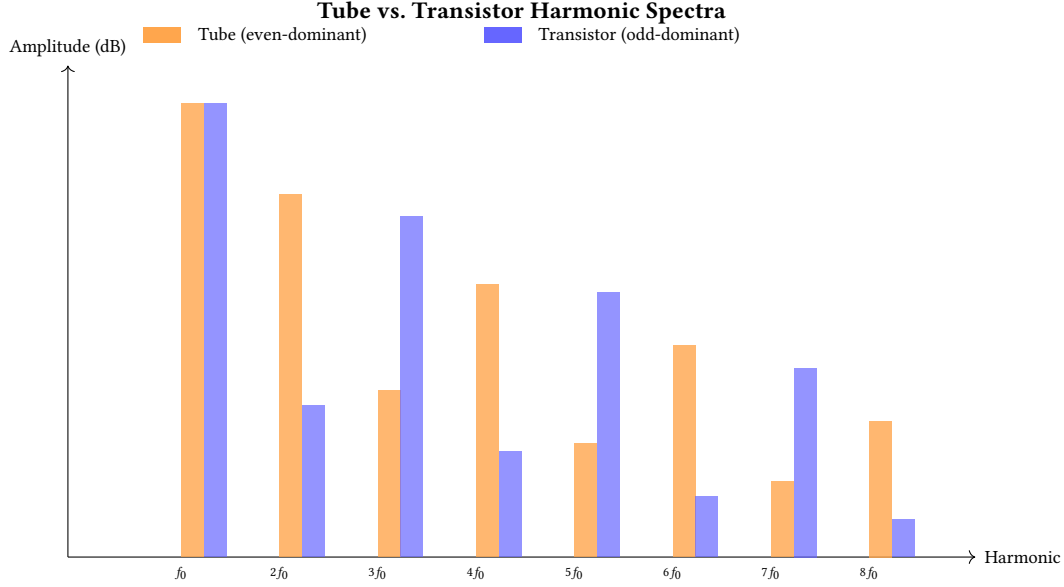


Figure 138. Harmonic distortion spectra comparison. Tube amplifiers produce stronger even-order harmonics ($2f_0, 4f_0, \dots$), while transistor stages emphasize odd-order harmonics ($3f_0, 5f_0, \dots$). Music Brain uses this profile to identify the amplification chain.

34.2 Saturation Curve Classification

Beyond the harmonic spectrum, the *shape* of the saturation transfer function distinguishes processing technologies. A tube amplifier exhibits a smooth, gradual soft-knee saturation—signal peaks are gently rounded off, preserving the musical envelope. Transistor hard clipping, by contrast, abruptly flattens peaks beyond a threshold, creating a sharp discontinuity. Tape saturation falls between the two, with a gentle S-curve compression that depends on tape speed and oxide formulation. The system fits measured input/output transfer curves to known saturation models using least-squares regression, classifying the dominant nonlinearity type.

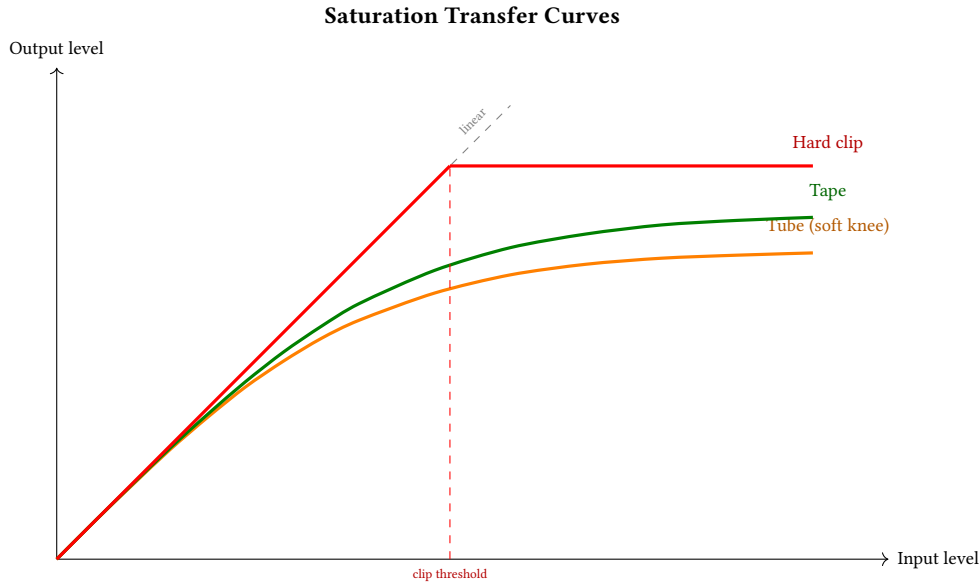


Figure 139. Saturation transfer curves for three nonlinearity types. Tube saturation (orange) shows a gradual soft-knee rolloff; tape (green) produces a smoother S-curve; hard digital clipping (red) is linear until the threshold, then flat.

34.3 Compression Signature Analysis

Dynamic range compression is one of the most era-defining processing stages. 1960s recordings used slow optical compressors (LA-2A) with gentle attack times; 1980s productions relied on fast VCA compressors (SSL bus compressor) for punchy transients; the loudness war era of the early 2000s deployed brick-wall limiters that virtually eliminated dynamic range. Music Brain estimates compression parameters by analyzing the relationship between transient peaks and their envelopes.

Attack Time Estimation. The compression attack time t_a is estimated from the transient onset slope. Given the envelope $E(t)$ of a detected transient, the system measures the time from onset to the compressed steady-state level:

$$t_a = \frac{1}{\alpha} \ln \left(\frac{E_{\text{peak}} - E_{\text{threshold}}}{E_{\text{peak}} - E_{\text{steady}}} \right) \quad (35)$$

where E_{peak} is the initial transient peak amplitude, $E_{\text{threshold}}$ is the compressor threshold, E_{steady} is the gain-reduced steady-state level, and α is the exponential decay constant of the compressor's gain reduction envelope. Fast attack times (< 1 ms) indicate modern limiting; medium times (5–30 ms) suggest VCA compression; slow times (> 50 ms) point to optical or variable-mu designs.

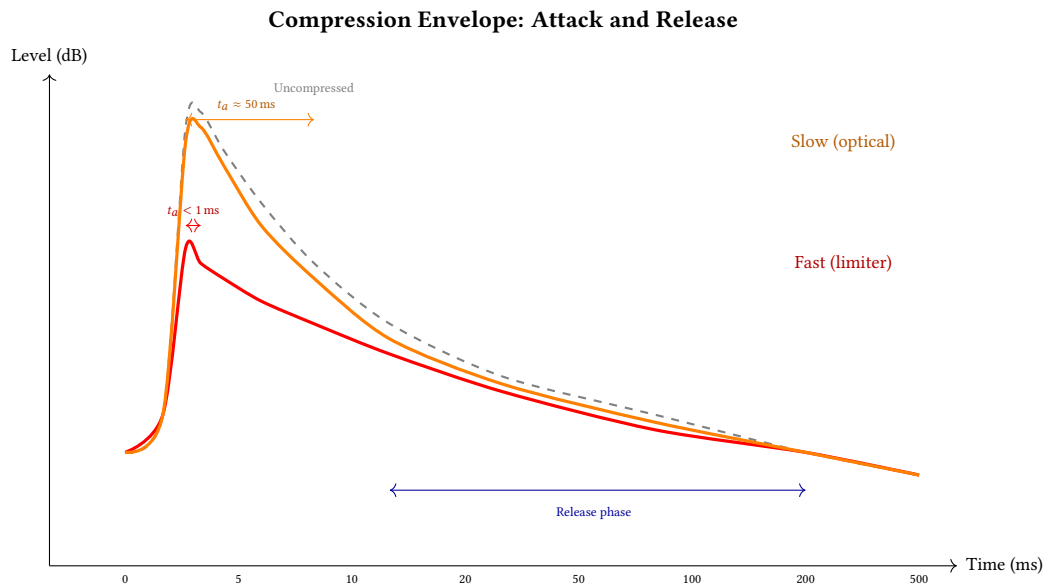


Figure 140. Compression envelope showing fast attack (red, modern brick-wall limiter) versus slow attack (orange, vintage optical compressor). The uncompressed transient (dashed) shows the original signal for reference.

34.4 Noise Floor Characterization

The noise floor of a recording is among the most reliable era indicators. Analog tape introduces a broadband Gaussian hiss with spectral shaping from the tape head and bias circuitry. Vinyl records add low-frequency rumble and periodic crackle from surface imperfections. Digital recordings exhibit a quantization noise floor that is flat and extremely low (-96 dB for 16-bit, -144 dB for 24-bit), sometimes with dithering artefacts. Music Brain analyzes silence segments and inter-transient gaps to characterize the noise floor spectrum.

Detection Method. The system identifies candidate silence windows ($\text{RMS} < -50$ dB), computes the power spectral density across these windows, and classifies the noise profile by fitting against reference templates for tape hiss, vinyl surface noise, and digital quantization noise. A weighted template-matching score determines the most likely recording medium.

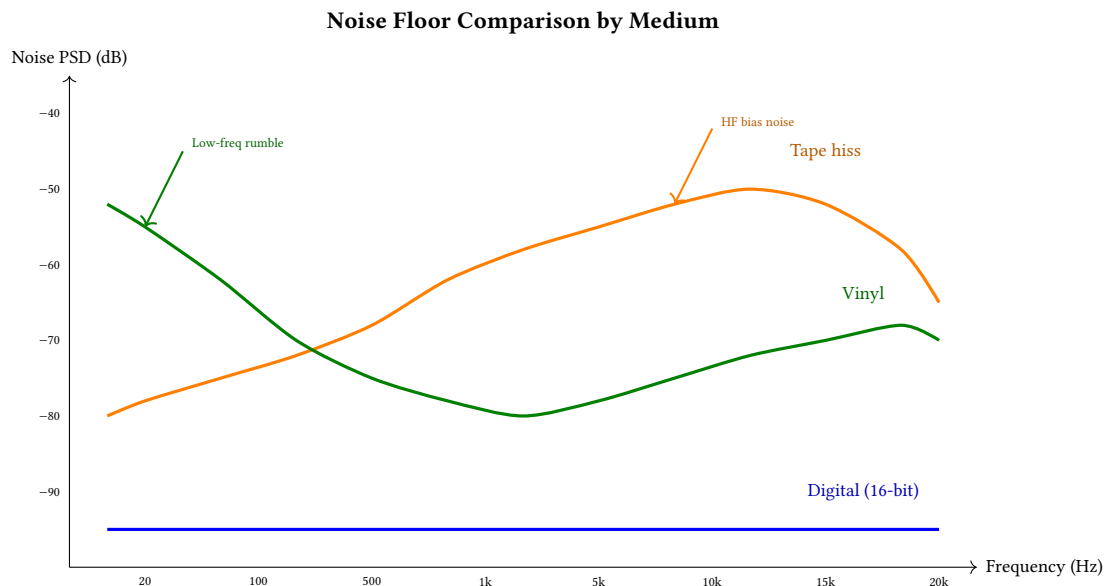


Figure 141. Noise floor power spectral density for three recording media. Tape hiss (orange) peaks in the high frequencies due to bias noise; vinyl (green) shows strong low-frequency rumble and mid-range crackle; digital (blue) presents a flat, extremely low noise floor.

34.5 Analog vs. Digital Indicators

Beyond distortion and noise, several additional markers distinguish analog and digital production chains. Music Brain evaluates a suite of secondary indicators:

- **Bandwidth ceiling.** Analog tape typically rolls off above 15–18 kHz depending on tape speed; digital recordings extend cleanly to 20 kHz (or 22.05 kHz at 44.1 kHz sample rate).
- **Phase coherence.** Analog multi-track recordings exhibit slight inter-channel phase drift from tape transport irregularities; digital recordings maintain perfect phase alignment.
- **Wow and flutter.** Sub-1 Hz pitch modulation from tape transport mechanisms or turntable speed variation. Detectable via instantaneous frequency tracking.
- **Pre-ringing.** Linear-phase digital filters (common in early CD mastering) introduce time-reversed pre-echo artefacts before transients—a uniquely digital signature absent in analog chains.
- **Dithering.** Shaped quantization noise from dithering algorithms (TPDF, noise-shaped) indicates digital-to-analog conversion, particularly in 16-bit masters.

Composite Analog Score. The system combines these indicators into a composite analog–digital score on a $[0, 1]$ scale, where 0 indicates a purely digital chain and 1 indicates a fully analog chain. Intermediate values reflect hybrid workflows—for example, analog recording to tape with digital mixing and mastering.

34.6 Fingerprint Construction

Each of the preceding analysis stages produces a feature vector. Music Brain combines them into a unified *production fingerprint*—a multi-dimensional descriptor capturing the complete sonic character of the recording’s production chain. The fingerprint encompasses six axes: harmonic distortion profile (R_{eo}), saturation type, compression characteristics (attack, release, ratio), noise floor profile, bandwidth ceiling, and stereo field properties. This fingerprint is visualized as a radar chart, allowing instant visual comparison between tracks or against era templates.

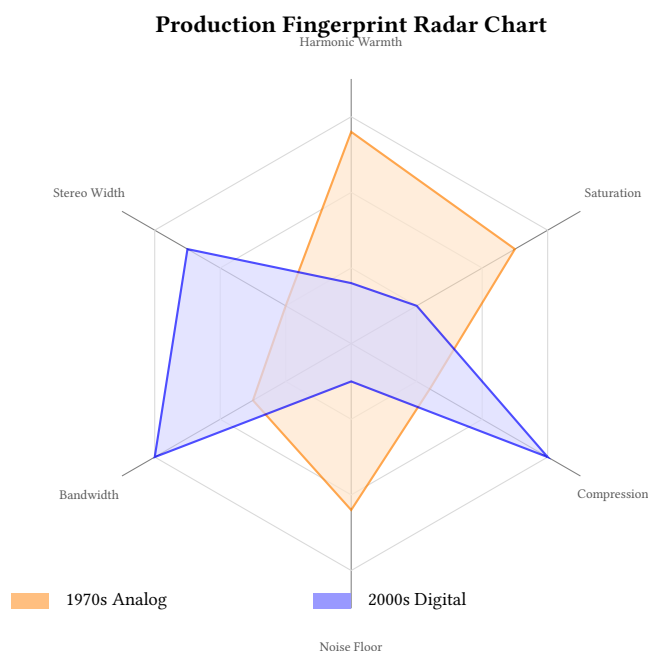


Figure 142. Production fingerprint radar chart comparing a 1970s analog recording (orange) with a 2000s digital production (blue). The analog track shows high harmonic warmth and saturation but narrow bandwidth; the digital track shows heavy compression and wide bandwidth but low harmonic warmth.

34.7 Historical Era Mapping

The final stage maps the production fingerprint to one of five historical production eras. Each era is represented by a template fingerprint derived from statistical analysis of a reference corpus spanning decades of commercial recordings. The system computes the Euclidean distance between the track’s fingerprint vector and each era template, assigning the closest match as the primary era classification and providing secondary matches with confidence scores.

Era	Period	Defining Production Characteristics
Vintage	Pre-1970	Mono/narrow stereo, tube saturation, high R_{eo} , narrow bandwidth (< 12 kHz), high tape hiss, plate/chamber reverb
Classic Analog	1970–1985	Warm tape saturation, moderate compression, extended low end, analog stereo width, noise floor present but controlled
Early Digital	1985–2000	Bright top end, digital reverb, gated drums, pre-ringing artefacts, reduced noise floor, CD-era brick-wall filtering at 22.05 kHz
Modern Digital	2000–2015	Hyper-compression (loudness war), minimal dynamic range, pristine noise floor, wide bandwidth, pitch correction artefacts
Streaming Era	2015+	Dynamic range recovery, analog emulation warmth, spatial audio processing, loudness normalization compliance (-14 LUFS)

Table 14. Production era classification scheme. Each era has a distinct fingerprint template derived from reference corpus analysis.

Confidence and Hybrid Detection. Many recordings do not fit cleanly into a single era. A modern production using analog tape with digital mixing will exhibit characteristics of multiple eras. The system reports a confidence distribution across all five eras, enabling hybrid classifications such as “Classic Analog recording with Modern Digital mastering” or “Streaming Era production with Vintage tube saturation aesthetic.” This nuanced classification is surfaced as both a primary era label and a full percentage breakdown.

35 Arrangement Density and Instrumentation Curve

A song’s arrangement tells its story through the addition and subtraction of instruments. When the Beatles recorded “A Day In The Life,” the orchestral build from sparse piano-and-vocal into a forty-piece crescendo was not an afterthought—it was the track’s emotional architecture. Radiohead’s arrangements often begin with a single guitar or keyboard texture and accumulate layers until the full band arrives, creating a sense of inevitability that defines their sound. Music Brain’s density analysis module quantifies this layering: it tracks which stems are active at every moment, measures their energy contributions, and identifies inflection points where the arrangement changes character. The result is a continuous “instrumentation curve” that reveals the producer’s arrangement strategy at a glance.

35.1 Per-Stem RMS Energy Tracking

The foundation of density analysis is per-stem RMS energy computed over short overlapping windows. After source separation yields individual stems (vocals, drums, bass, and other instruments), each stem’s root-mean-square energy is calculated frame by frame. The RMS envelope provides a smooth, positive-valued representation of each stem’s instantaneous contribution to the mix. Frames where a stem’s RMS falls below a noise floor threshold (-60 dBFS by default) are marked as inactive, ensuring that bleed artefacts and low-level noise do not inflate the density count.

Smoothing. Raw per-frame RMS values are smoothed with a Gaussian kernel ($\sigma = 0.25$ s) to suppress transient fluctuations. This ensures that a brief silence between two snare hits does not register as a drum dropout. The smoothed energy curves are then normalised to $[0, 1]$ per stem, so that relative contributions can be compared regardless of absolute loudness differences.

35.2 Active Stem Count

Definition. A stem k is considered *active* at time t if its smoothed RMS energy exceeds the activation threshold θ_k . The arrangement density $D(t)$ is defined as the sum of active stem indicators:

$$D(t) = \sum_{k=1}^K \mathbb{I}_k(t), \quad \text{where} \quad \mathbb{I}_k(t) = \begin{cases} 1 & \text{if } E_k(t) > \theta_k \\ 0 & \text{otherwise} \end{cases} \quad (36)$$

Here K is the total number of separated stems and $E_k(t)$ is the smoothed RMS energy of stem k at time t . A track with four stems (vocals, drums, bass, other) thus has $D(t) \in \{0, 1, 2, 3, 4\}$. The density function captures the core intuition: more instruments playing simultaneously means a denser arrangement.

Threshold selection. Per-stem thresholds θ_k are set adaptively at -40 dB relative to each stem’s peak energy. This accommodates the wide dynamic range between, for instance, a quiet fingerpicked guitar and a heavily compressed drum bus, ensuring that both are detected as active when they are genuinely contributing to the arrangement.

35.3 Spectral Energy Summation

Beyond simple stem counting, Music Brain also computes a *weighted* density measure that accounts for the spectral energy each stem contributes. A stem playing a single sustained note contributes less spectral mass than a stem playing dense chords or broadband noise. The spectral density $D_s(t)$ sums the normalised RMS energies directly rather than binarising them:

$$D_s(t) = \sum_{k=1}^K \hat{E}_k(t)$$

where $\hat{E}_k(t)$ is the min-max normalised energy of stem k . This continuous measure distinguishes between, say, a verse where bass and drums are both active but subdued ($D_s \approx 0.8$) and a chorus where the same two stems are driven hard ($D_s \approx 1.6$), even though the binary count $D(t) = 2$ in both cases.

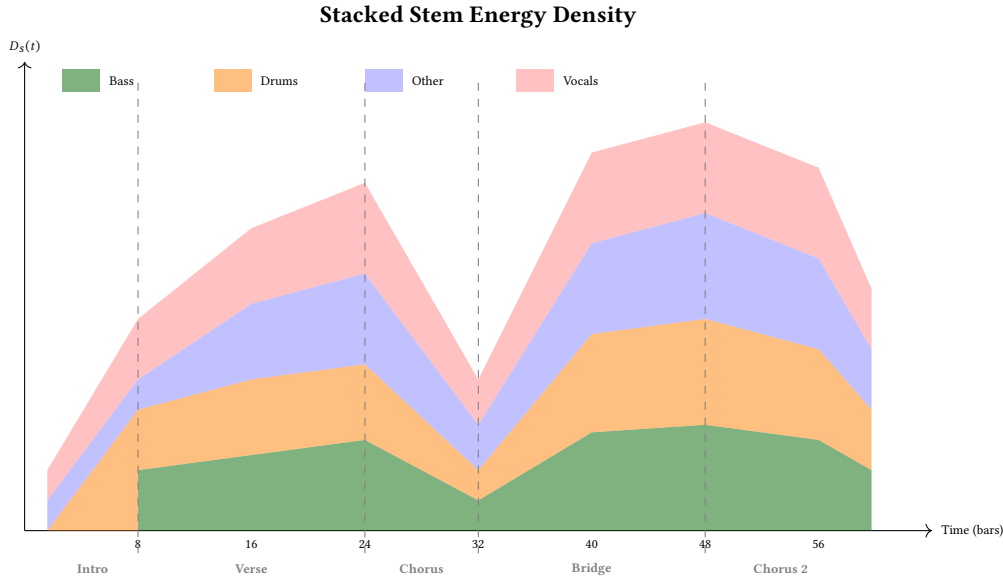


Figure 143. Stacked area chart of per-stem spectral energy contributions over time. Each coloured band represents one stem’s normalised RMS energy $\hat{E}_k(t)$. The total height at any point is the spectral density $D_s(t)$. Note the density dip at the bridge section and the subsequent build into the second chorus.

35.4 Stem Activation Timeline

The binary activation indicators $\mathbb{I}_k(t)$ from Equation 36 can be visualised as a Gantt-style timeline showing exactly when each stem enters and exits the arrangement. This view is particularly revealing for tracks that use instrumentation strategically—for instance, Radiohead’s “Everything In Its Right Place” opens with a lone electric piano for nearly a full minute before vocals enter, and synth textures are added and removed in carefully staggered waves throughout. The activation timeline makes these arrangement decisions immediately visible.

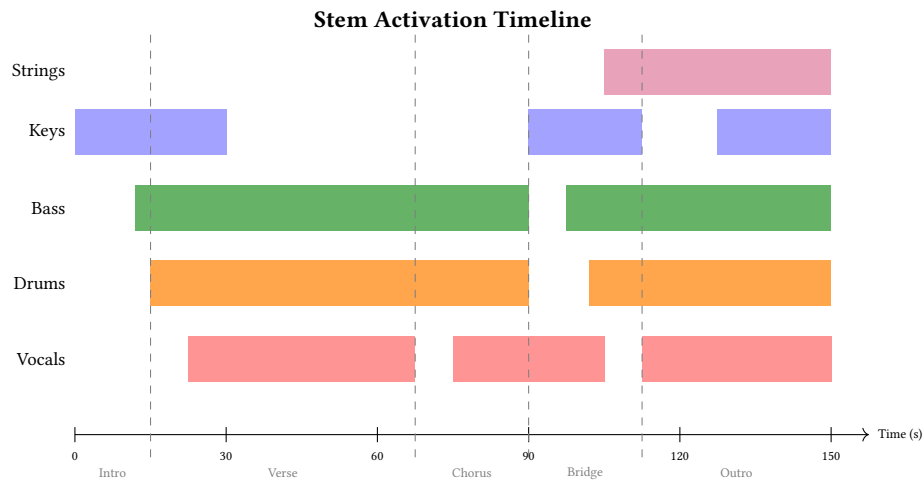


Figure 144. Gantt-style stem activation timeline. Each horizontal bar represents a period during which a stem’s energy exceeds its activation threshold. Gaps indicate dropouts (e.g., drums tacet during the bridge). The staggered entry of strings in the outro illustrates the “orchestral build” pattern.

Activation summary. Table 15 provides a compact summary of stem activation statistics for a typical pop-rock arrangement. The *first entry* column identifies when each instrument joins the arrangement, while *active percentage* indicates what fraction of the total duration each stem is present. These figures help producers and analysts quickly assess arrangement balance—a vocal active percentage below 40% might indicate an instrumental-heavy arrangement, while drums above 90% suggest a driving, rhythm-forward production.

Table 15. Stem activation summary for a representative pop-rock track (duration: 3:42).

Stem	First Entry (s)	Active Duration (s)	Active (%)
Vocals	12.4	148.2	66.7
Drums	8.1	186.5	83.9
Bass	7.8	190.1	85.5
Keys	0.0	98.4	44.3
Strings	142.0	52.3	23.5

35.5 Change Point Detection

The density function $D(t)$ is piecewise constant (or near-constant) within sections and exhibits step changes at arrangement boundaries. Music Brain applies a change point detection algorithm to the density curve, identifying moments where the active stem count or spectral density shifts significantly. These change points frequently align with structural boundaries (verse to chorus, chorus to bridge) and are cross-referenced with the section segmentation module (see Section 9) to improve boundary accuracy.

Algorithm. The implementation uses a PELT (Pruned Exact Linear Time) approach with an L_2 cost function on the spectral density signal $D_s(t)$. A penalty parameter β controls sensitivity: smaller values detect subtle arrangement changes (e.g., a tambourine entering mid-verse), while larger values restrict detection to major transitions. The default penalty is calibrated so that typical pop songs yield 6–12 change points.

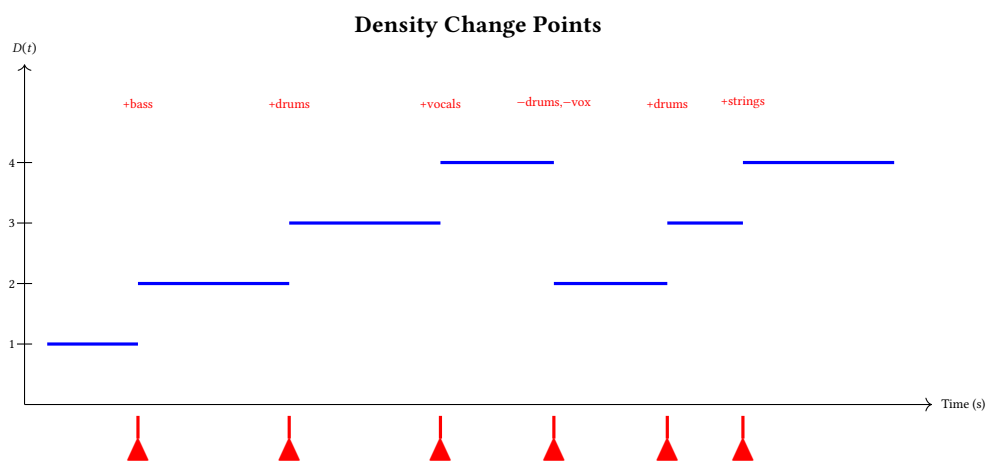


Figure 145. Change point detection on the arrangement density curve $D(t)$. Red triangles mark detected inflection points where the active stem count changes. Annotations indicate which stems enter (+) or exit (–) at each change point. The pattern shown—gradual build, bridge dropout, final-chorus rebuild—is characteristic of pop-rock arrangements.

35.6 Density vs Loudness Correlation

Arrangement density and perceived loudness are related but distinct properties. A track can be loud with few instruments (a solo distorted guitar through a cranked amplifier) or quiet with many (a full orchestra playing *pianissimo*). Music Brain computes the Pearson correlation between $D_s(t)$ and the integrated loudness contour $L(t)$ to quantify how tightly arrangement choices track the dynamic arc. High correlation ($r > 0.85$) suggests that the producer uses instrumentation layering as the primary dynamic tool—adding instruments to get louder and removing them to get quieter. Lower correlation indicates that dynamics are achieved through other means: compression, velocity, or mix automation.

Musical examples. The Beatles’ “A Day In The Life” exhibits an extremely high density–loudness correlation during its orchestral crescendo: as instruments are added one by one, loudness increases almost linearly. In contrast, Radiohead’s “Idioteque” maintains a nearly constant stem count throughout (drums, synth, vocals) yet achieves dramatic dynamic contrast through processing intensity and vocal delivery, yielding a low density–loudness correlation.

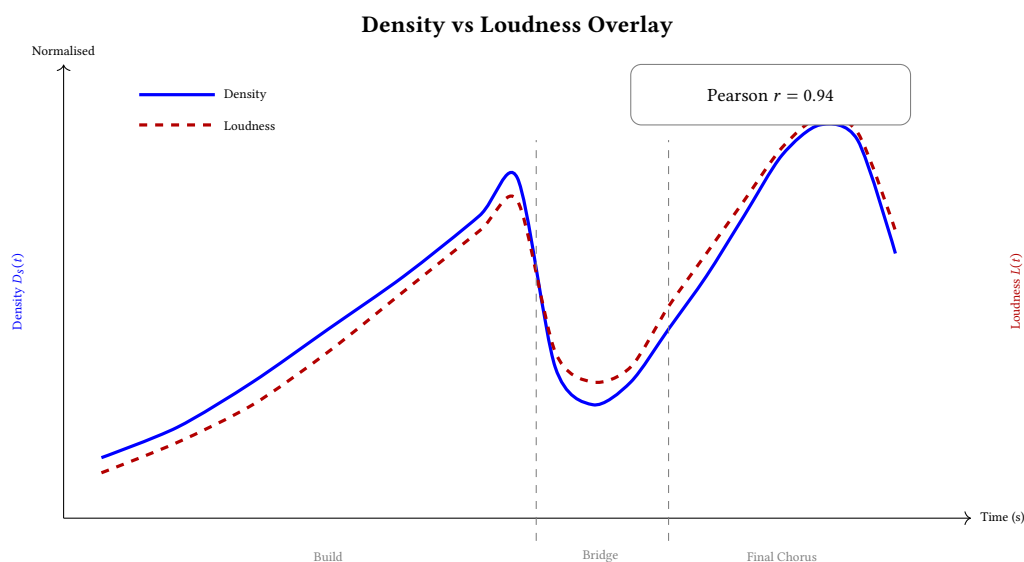


Figure 146. Overlay of normalised arrangement density $D_s(t)$ (solid blue) and integrated loudness $L(t)$ (dashed red). The high Pearson correlation ($r = 0.94$) indicates that this track’s dynamics are driven primarily by instrumentation layering rather than mix processing. The co-movement is especially tight during the build sections, while minor divergence occurs at the bridge where loudness drops slightly faster than density.

Implementation note. The density analysis pipeline computes all metrics in a single pass over the separated stems. Per-stem RMS values, activation indicators, spectral density, and change points are emitted as structured

JSON alongside the per-frame data, enabling both the visualisation layer and downstream modules (e.g., section segmentation, mix scoring) to consume density information without redundant computation.

36 Conclusion and Future Work

36.1 Summary of Contributions

Music Brain presents an integrated, real-time audio analysis architecture that advances the state of the art in several ways:

1. **Stem-first analysis:** Isolating sources before analysis yields higher accuracy in harmonic, rhythmic, and timbral feature extraction than master-mix approaches.
2. **Multi-modal integration:** Ten distinct analysis modules (spectral, harmonic, rhythmic, timbral, structural, expectation, genre-specific, energy profiling, mood mapping, era detection) are unified in a single real-time pipeline.
3. **Genre adaptation:** Five expert adapters with deep per-adapter analysis (phase sync, swing ratio, voice counting, pitch deviation) contextualize generic features within domain knowledge.
4. **Temporal evolution:** Pattern tracking across structural sections reveals how kick patterns, hi-hat density, harmonic progressions, and micro-timing evolve throughout a song.
5. **Energy and mood profiling:** Multi-resolution energy envelopes and valence–arousal emotion mapping provide a continuous emotional trajectory, classified into narrative arc templates.
6. **Era detection and sonic fingerprinting:** Production DNA analysis identifies recording-era markers, reconstructs signal processing chains, and explains *why* a recording sounds the way it does.
7. **Musical DNA deep analysis:** Section-level DNA cards unify all analytical dimensions into explorable, interactive profiles with drill-down from song level to frame-level forensics.
8. **Rich visualization:** Seven complementary views—including the novel Fractal Song Tree, Style Topology Manifold with temporal trajectories, and preset deviation radar charts—provide multi-dimensional exploration.
9. **Production feedback:** The preset deviation tracking system and Mix Quality Scorecard provide actionable, quantitative feedback for music producers working within specific genre conventions.
10. **Vocal analysis:** Dedicated pitch tracking, formant analysis, vibrato detection, and vocal style classification on isolated vocal stems.
11. **Spatial analysis:** Stereo field mapping, per-stem pan position, mid/side decomposition, and stereo width tracking across sections.
12. **Masking detection:** Pairwise spectral overlap quantification between stems with automated EQ and arrangement remediation suggestions.
13. **Reference comparison:** Side-by-side feature radar, energy curve overlay, and spectral balance comparison against genre-appropriate reference tracks.

36.2 Limitations

The current implementation has several limitations:

- **Simulated Demucs Bridge:** The DemucsBridge currently simulates progress rather than performing actual source separation. Real integration requires either spawning the Python Demucs process or loading an ONNX-exported model.
- **Hardcoded BPM:** The BPM value is currently fixed at 145.0 rather than estimated from the audio signal. Implementing autocorrelation-based tempo detection [12] is a planned improvement.
- **Placeholder Algorithms:** Several adapter functions (swing ratio in Jazz, rhythmic complexity in Metal, articulation detection in Orchestral) return fixed values rather than computing from audio.
- **Limited Key Detection:** Key detection is noted as “Auto-Detected” but not yet implemented; the Krumhansl-Schmuckler key-finding algorithm applied to chroma profiles would provide this capability.

36.3 Future Work

- **ONNX Inference Integration:** Load quantized CLAP [15] or MusicFM embeddings for zero-shot genre and instrument classification.
- **Live Demucs:** Real-time source separation via a streaming ONNX model, enabling stem-isolated analysis of live audio.
- **MIDI Export Improvements:** Generate multi-track MIDI with per-stem note sequences, velocity dynamics, and expression controllers.
- **Automatic Tempo Detection:** Replace the hardcoded BPM with onset-based autocorrelation tempo estimation.
- **Enhanced Structural Analysis:** Implement novelty-based segmentation from the SSM to automatically label sections (verse, chorus, bridge).

- **Expanded Adapter Library:** Add adapters for mainstream genres in priority order: Pop/Rock (verse-chorus dynamics, vocal prominence), Hip-Hop/R&B (808 bass patterns, sample detection, flow analysis), Electronic/EDM (build-up structure, sidechain patterns), Country (steel guitar detection, Nashville number system), Folk/Acoustic (fingerpicking patterns, open tuning detection), and Ambient (texture evolution, pad layering).
- **Vocal Lyrics Alignment:** Align transcribed lyrics to the vocal pitch contour for synchronized lyric display and syllable-level analysis.
- **Reference Track Library:** Build a curated library of genre-specific reference tracks with pre-computed feature profiles for instant comparison.
- **Mix Quality Automation:** Generate parametric EQ and dynamics processing suggestions as exportable plugin presets (e.g., FabFilter Pro-Q format).
- **Collaborative Annotations:** Allow multiple users to annotate sections, stems, and frames with production notes, shareable via the DNA Manifest export.
- **Tablature Generation:** Generate guitar and bass tablature from isolated stems, with fretboard position optimization, technique annotations (bends, slides, hammer-ons), and export to Guitar Pro and MusicXML formats.
- **Drum Transcription:** Generate full drum notation from the isolated drum stem, including ghost notes, accent patterns, and fill detection, with export to standard drum notation.
- **Loudness and Dynamic Range (LUFS/LRA):** Implement integrated LUFS, short-term LUFS, and Loudness Range (LRA) measurement per stem and for the full mix, providing the industry-standard metrics used by streaming platforms and broadcast standards. These measurements would feed directly into the Mix Quality Scorecard and Reference Track Comparison modules, enabling precise loudness-matching and dynamic range assessment against genre-appropriate targets.
- **Transient Analysis:** Detect and characterise attack transients per stem—drum hit sharpness, guitar pick attack versus finger pluck, piano hammer hardness—by measuring rise time, peak amplitude, and spectral centroid of the onset envelope. This would provide production feedback such as “your snare transients are being crushed by the limiter” or “the kick attack is lost behind the bass guitar onset,” and would complement the existing masking analysis with time-domain conflict detection.
- **Harmonic Series and Overtone Fingerprinting:** Analyse the relative strength of each harmonic partial above the fundamental for pitched stems, creating an overtone profile that captures the timbral identity of individual instruments. This spectral fingerprint distinguishes a Stratocaster from a Les Paul, a Steinway from a Yamaha, or a Marshall stack from a Vox—even when playing the same note at the same dynamic. Overtone profiles would enrich the Musical DNA analysis and enable instrument-specific timbral comparison across recordings.
- **Groove and Micro-Timing Analysis:** Quantify the human feel of rhythmic performance by measuring how far ahead of or behind the quantisation grid each drum hit and bass note falls. Compute swing percentage, push/pull tendencies, pocket tightness (standard deviation from grid), and inter-onset interval variability. This captures what separates a programmed beat from a live drummer, a laid-back funk groove from an on-top punk rhythm, and would allow producers to diagnose timing inconsistencies or deliberately preserve human feel during editing.
- **Call-and-Response Detection:** Identify melodic and rhythmic dialogue between stems—a vocal phrase answered by a guitar lick, a horn section echoing the melody, a bass line mirroring the vocal rhythm. Detection would use cross-correlation of pitch contours and onset patterns between stem pairs, with latency and intervallic transformation analysis. This interaction pattern is central to blues, jazz, gospel, and West African-influenced popular music, and its detection would add a layer of musical discourse analysis absent from current MIR systems.
- **Production Era Forensics:** Extend the existing era detection module with deeper signal chain analysis: identify tube saturation curves (even-harmonic emphasis), tape compression artefacts (soft-knee limiting with frequency-dependent saturation), analogue summing characteristics (crosstalk patterns), and digital clipping signatures (hard brick-wall limiting). By matching these spectral fingerprints against known processing profiles, the system could estimate not just *when* a recording was made but *what equipment* was used—providing forensic-level production archaeology.
- **Arrangement Density Curve:** Provide a dedicated visualisation showing the number of active stems and their combined spectral energy over time, making the arrangement’s “shape” immediately visible as a continuous curve. Sparse intro → building verse → dense chorus → stripped bridge → climactic finale—the arrangement density curve would reveal this narrative arc at a glance, and would enable cross-song comparison of arrangement strategies within and across genres.

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